

0902.0431: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ0515 [conf 0.002]	85	■0.002	$\begin{aligned} & \lambda_0 - \lambda_1 = 1 \quad \lambda_0 - \lambda_2 = 1 \quad \lambda_0 - \lambda_3 = 1 \\ & \lambda_1 - \lambda_2 = 0 \quad \lambda_1 - \lambda_3 = 0 \quad \lambda_2 - \lambda_3 = 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \lambda_1 &= 1 & 0 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_2 &= 1 & 1 & 0 & 0 & 0 & 0 \\ \lambda_0 - \lambda_3 &= 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_1 - \lambda_2 &= 0 & 1 & 0 & 0 & 0 & 0 \\ \lambda_1 - \lambda_3 &= 0 & 1 & 1 & 0 & 0 & 0 \\ \lambda_2 - \lambda_3 &= 0 & 0 & 1 & 0 & 0 & 0 \end{aligned}$	$\begin{aligned} \lambda_0 + \lambda_1 &= 1 & 2 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_2 &= 1 & 1 & 2 & 1 & 1 & 1 \\ \lambda_0 + \lambda_3 &= 1 & 1 & 1 & 1 & 1 & 1 \\ \lambda_1 + \lambda_2 &= 0 & 1 & 2 & 1 & 1 & 1 \\ \lambda_1 + \lambda_3 &= 0 & 1 & 1 & 1 & 1 & 1 \\ \lambda_2 + \lambda_3 &= 0 & 0 & 1 & 1 & 1 & 1 \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ0517 [conf 0.002]	85	■ 0.002	$\begin{aligned} & \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 3 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \end{aligned}$	$\begin{aligned} & \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 3 \quad 1 \quad 1 \quad 2 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 2 \quad 2 \quad 1 \quad 2 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 2 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0, 0 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 1 \end{aligned}$	$\begin{aligned} & \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 3 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 2 \quad 2 \quad 1 \quad 2 \quad 1 \\ & \frac{1}{2} \left(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 \right) + \frac{1}{2} \left(\mu_3 - \mu_1 \right) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 2 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3 \right) - \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) + \frac{1}{2} \left(\mu_1 - \mu_2 \right) = 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ E01026 [conf 0.018]	171	■ 0.018	$\begin{array}{l} \backslash left.\backslash begin{array}{llllllllrllllll} \backslash lambda_{0}-\\ \backslash lambda_{1}=2 \ \& \ 3 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 1 \ \& \ 0 \ \& \ 2 \ \& \ \backslash lambda_{0}+\\ \backslash lambda_{1}=2 \ \& \ 4 \ \& \ 6 \ \& \ 5 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \backslash end{array}\backslash right],\\ \backslash begin{array}{llllllllllll} \backslash lambda_{0}-\backslash lambda_{2}=2 \ \& \ 3\\ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 1 \ \& \ 1 \ \& \ 2 \ \& \ \backslash lambda_{0}+\backslash lambda_{2}=2 \ \& \ 4\\ \& \ 6 \ \backslash \backslash \backslash lambda_{0}-\backslash lambda_{3}=2 \ \& \ 3 \ \& \ 4 \ \& \ 3 \ \& \ 2 \ \& \ 2 \ \& \ 1\\ \& \ 2 \ \& \ \backslash lambda_{0}+\backslash lambda_{3}=2 \ \& \ 4 \ \& \ 6 \ \backslash \backslash \backslash lambda_{1}-\\ \backslash lambda_{2}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 4 \ \& \ 2 \ \& \ 1\\ \backslash \backslash \backslash lambda_{1}-\backslash lambda_{3}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 1 \ \& \ 1 \ \& \\ 0 \ \& \ \backslash lambda_{1}+\backslash lambda_{2}=0 \ \& \ 1 \ \& \ 2 \ \backslash \backslash \backslash lambda_{2}-\\ \backslash lambda_{3}=0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 0 \ \& \ 1 \ \& \ 0 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \backslash \\ \backslash \backslash \backslash lambda_{2}+\backslash lambda_{3}=0 \ \& \ 1 \ \& \ 2 \ \& \ 2 \ \& \ 2 \ \& \ 1 \ \& \ 0 \ \& \ 1 \\ \backslash end{array} \end{array}$	$\begin{array}{l} \lambda_0 - \lambda_1 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 0 \ 2 \ \lambda_0 + \lambda_1 = 2 \ 4 \ 6 \ 5 \ 4 \ 3 \ 2 \\ \lambda_0 - \lambda_2 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 1 \ 2 \ \lambda_0 + \lambda_2 = 2 \ 4 \ 6 \\ \lambda_0 - \lambda_3 = 2 \ 3 \ 4 \ 3 \ 2 \ 2 \ 1 \ 2 \ \lambda_0 + \lambda_3 = 2 \ 4 \ 6 \\ \lambda_1 - \lambda_2 = 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ \lambda_1 + \lambda_2 = 0 \ 1 \ 2 \\ \lambda_1 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ \lambda_1 + \lambda_3 = 0 \ 1 \ 2 \\ \lambda_2 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ \lambda_2 + \lambda_3 = 0 \ 1 \ 2 \end{array}$	$\begin{array}{l} \lambda_0 - \lambda_1 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 0 \ 2 \ \lambda_0 + \lambda_1 = 2 \ 4 \ 6 \ 5 \ 4 \ 3 \ 2 \\ \lambda_0 - \lambda_2 = 2 \ 3 \ 4 \ 3 \ 2 \ 1 \ 1 \ 2 \ \lambda_0 + \lambda_2 = 2 \ 4 \ 6 \\ \lambda_0 - \lambda_3 = 2 \ 3 \ 4 \ 3 \ 2 \ 2 \ 1 \ 2 \ \lambda_0 + \lambda_3 = 2 \ 4 \ 6 \\ \lambda_1 - \lambda_2 = 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ \lambda_1 + \lambda_2 = 0 \ 1 \ 2 \\ \lambda_1 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 1 \ 0 \ \lambda_1 + \lambda_3 = 0 \ 1 \ 2 \\ \lambda_2 - \lambda_3 = 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ \lambda_2 + \lambda_3 = 0 \ 1 \ 2 \end{array}$
0902.0431_ E00518 [conf 0.021]	86	■ 0.021	$\begin{array}{l} \backslash begin{aligned} \ \& \ \frac{1}{2}\backslash left(\backslash lambda_{0}+\\ \backslash lambda_{1}-\backslash lambda_{2}-\backslash lambda_{3}\backslash right)-\frac{1}{2}\\ \backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=1 \quad 2 \quad 2 \quad \backslash quad \\ 2 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad c \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(-\\ \backslash lambda_{0}+\backslash lambda_{1}-\backslash lambda_{2}+\backslash lambda_{3}\backslash right)-\\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=0 \quad \backslash quad 1 \\ \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(-\\ \backslash lambda_{0}+\backslash lambda_{1}+\backslash lambda_{2}-\backslash lambda_{3}\backslash right)-\\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=0 \quad \backslash quad 1 \\ \backslash quad 2 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash quad 1 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(\\ \backslash lambda_{0}+\backslash lambda_{1}+\backslash lambda_{2}+\backslash lambda_{3}\backslash right)-\\ \frac{1}{2}\backslash left(\backslash mu_{1}-\backslash mu_{2}\backslash right)=1 \quad \backslash quad 2 \quad \backslash quad 3 \\ \backslash quad 2 \quad \backslash \backslash \ \& \ \frac{1}{2}\backslash left(-\backslash lambda_{0}-\backslash lambda_{1}+\\ \backslash lambda_{2}+\backslash lambda_{3}\backslash right)-\frac{1}{2}\backslash left(\backslash mu_{1}-\\ \backslash mu_{2}\backslash right)=0 \quad \backslash quad 0 \quad \backslash end{aligned} \end{array}$	$\begin{array}{l} \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \ 2 \ 2 \ 2 \ 1 \ 1 \ c \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 1 \ 1 \ 1 \ 1 \ 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 1 \ 2 \ 1 \ 1 \ 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \ 2 \ 3 \ 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 0 \end{array}$	$\begin{array}{l} \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \ 2 \ 2 \ 1 \ 1 \ 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 1 \ 1 \ 1 \ 1 \ 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 1 \ 2 \ 1 \ 1 \ 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 1 \ 2 \ 3 \ 2 \ 2 \ 1 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) = 0 \ 0 \ 1 \ 1 \ 1 \ 0. \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ E01033 [conf 0.026]	174	0.026	$\begin{array}{llllllll} \backslash\lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=2 & 4 & 6 & 4 & 3 & 2 & 1 & 3 \\ \backslash\lambda_0+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=2 & 4 & 5 & 4 & 3 & 2 & 1 & 3 \\ \backslash\lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=2 & 3 & 5 & 4 & 3 & 2 & 1 & 2 \\ \backslash\lambda_0-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=2 & 3 & 4 & 4 & 3 & 2 & 1 & 2 \\ \backslash\lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 1 & 1 & 1 \\ \backslash\lambda_1+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ \backslash\lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \backslash\lambda_1-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ \backslash\lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 1 & 0 & 1 \\ \backslash\lambda_2+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ \backslash\lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \backslash\lambda_2-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ \backslash\lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu+r=0 & 1 & 2 & 1 & 1 & 0 & 0 & 1 \\ \backslash\lambda_3+\frac{1}{2}\mu_1+\frac{1}{3}\nu-r=0 & 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ \backslash\lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu+r=0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \backslash\lambda_3-\frac{1}{2}\mu_1-\frac{1}{3}\nu-r=0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1+\lambda_2-\lambda_3+\frac{1}{2}\mu_2+\frac{1}{3}\nu+r=1 & 2 & 4 & 3 & 2 & 2 & 1 & 2 \\ 1+\lambda_2-\lambda_3+\frac{1}{2}\mu_2+\frac{1}{3}\nu-r=1 & 2 & 3 & 3 & 2 & 2 & 1 & 2 \end{array}$		

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Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ1031 [conf 0.036]	173	■ 0.036	$\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \ \& \ \backslash \text{lambda}_{\{2\}}+\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}- \\ & \backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{2\}}-\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}+ \\ & \backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{3\}}+\backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}-\backslash \text{frac}\{2\}{\{3\}} \\ & \backslash \text{nu}=0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \backslash \ \& \ \backslash \text{lambda}_{\{3\}}- \\ & \backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{1\}}+\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \\ & \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ 0 \ \backslash \text{end}\{\text{aligned}\} \end{aligned}$	$\begin{aligned} \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 1 & 0 & \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & & & \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 0 & & \\ 0 & & & & & & & & \end{aligned}$	$\begin{aligned} \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{aligned}$
0902.0431_ EQ0761 [conf 0.036]	127	■ 0.036	$\begin{aligned} & \backslash \text{begin}\{\text{aligned}\} \ \& \ \backslash \text{frac}\{1\}{\{2\}}\backslash \text{left}(-\backslash \text{lambda}_{\{0\}}- \\ & \backslash \text{lambda}_{\{1\}}+\backslash \text{lambda}_{\{2\}}-\backslash \text{lambda}_{\{3\}}\backslash \text{right})+\backslash \text{frac}\{1\}{\{2\}} \\ & \backslash \text{mu}_{\{2\}}-\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=0 \ \backslash \text{quad} \ 0 \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 1 \\ & \backslash \text{quad} \ \textbf{[U+0930]} \ \textbf{[U+094D]} \ 0 \ \backslash \text{quad} \ 0 \ \backslash \backslash \ \& \ \backslash \text{frac}\{1\}{\{2\}}\backslash \text{left} \\ & \backslash \text{lambda}_{\{0\}}+\backslash \text{lambda}_{\{1\}}+\backslash \text{lambda}_{\{2\}}-\backslash \text{lambda}_{\{3\}}\backslash \text{right})+ \\ & \backslash \text{frac}\{1\}{\{2\}} \ \backslash \text{mu}_{\{2\}}-\backslash \text{frac}\{2\}{\{3\}} \ \backslash \text{nu}=1 \ \backslash \text{quad} \ 2 \ \backslash \text{quad} \ 3 \\ & \backslash \text{quad} \ 3 \ \backslash \text{quad} \ 2 \ \backslash \text{end}\{\text{aligned}\} \ \backslash \text{quad} \ 1 \ \backslash \text{quad} \ 2 \ . \end{aligned}$	$\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & \textbf{[U+0930][U+094D]} & 0 & 0 & \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 2 & \end{aligned}$	$\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 1 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 2 & 2 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 1 & 1 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_2 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 1 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{2}{3}\nu &= 1 & 1 & 1 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 1 & 2 & 3 & 4 & 3 & 1 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 1 & 2 & 1 & 0 & 1. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ EQ0516 [conf 0.041]	85	■ 0.041	$\begin{aligned} & \begin{aligned} & \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad a \quad 0 \\ & \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \\ & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \\ & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \end{aligned} \\ & \begin{aligned} & \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \\ & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \end{aligned} \end{aligned}$		
0902.0431_ EQ0425 [conf 0.082]	70	■ 0.082	$\begin{aligned} & AXA^* \times AYA^* = A(X \times Y)A^*, \\ & (PMA^*)^*(PNA^*) = AM^*P^*PNA = A(M^*N)A, \\ & (PMA^*)(AYA^*) = P(MY)A^*, \\ & \overline{PMA^* \times PNA^*} = \overline{{}^t\tilde{P}(M \times N){}^t\tilde{A}^*} = \overline{PM \times NA^*}, \text{ etc.} \end{aligned}$		
0902.0431_ EQ1059 [conf 0.093]	180	■ 0.093	$\begin{aligned} & = \{\alpha \in (E_8^C)_{1,1-,1-} \mid \tau\tilde{\lambda}\alpha = \alpha\tau\tilde{\lambda}\} \\ & = \{\alpha \in E_7^C \mid \tau\tilde{\lambda}\alpha = \alpha\tau\tilde{\lambda}\} \text{ (Proposition 5.7.1)} \\ & = \{\alpha \in E_7^C \mid \tau\lambda\alpha = \alpha\tau\lambda\} \text{ (by the correspondence to Proposition 5.7.1)} \\ & = E_7 \text{ (Lemma 4.3.3.(4)).} \end{aligned}$		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_ E00757 [conf 0.095]	126	■0.095	$\begin{aligned} & \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \\ & \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad a \\ & \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 1 \quad 1 \quad 1 \\ & \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) = 0 \quad 0 \quad 0 \\ & 0 \end{aligned}$	$\begin{aligned} \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{aligned}$	
0902.0431_E00922	153	■0.102	$f^{-1} \left(\sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \left(\begin{array}{ccc} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{array} \right),$	$f^{-1} \left(\sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \left(\begin{array}{ccc} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{array} \right),$	$f^{-1} \left(\sum_{i < j < k} x_{ijk} \mathbf{e}_i \wedge \mathbf{e}_j \wedge \mathbf{e}_k \right) = \left(\begin{array}{ccc} h(x_{156}) & h(x_{164}, \bar{x}_{256}) & h(x_{145}, \bar{x}_{356}) \\ h(x_{256}, \bar{x}_{164}) & h(x_{264}) & h(x_{245}, \bar{x}_{364}) \\ h(x_{356}, \bar{x}_{145}) & h(x_{364}, \bar{x}_{245}) & h(x_{345}) \\ h(x_{423}) & h(x_{431}, \bar{x}_{523}) & h(x_{412}, \bar{x}_{623}) \\ h(x_{523}, \bar{x}_{431}) & h(x_{531}) & h(x_{512}, \bar{x}_{631}) \\ h(x_{623}, \bar{x}_{412}) & h(x_{631}, \bar{x}_{512}) & h(x_{612}) \end{array} \right),$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01117	189	■ 0.108	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) \\ (X_2, Y_1) & (X_2, Y_2) \\ (X_3, Y_1) & (X_3, Y_2) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X}Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$	$\begin{aligned} & (\boldsymbol{X}, \boldsymbol{Y}) = (X_1, Y_1) + (X_2, Y_2) + (X_3, Y_3) \in C, \\ & \langle \boldsymbol{X}, \boldsymbol{Y} \rangle = \langle X_1, Y_1 \rangle + \langle X_2, Y_2 \rangle + \langle X_3, Y_3 \rangle \in C, \\ & \boldsymbol{X} \times \boldsymbol{Y} = \begin{pmatrix} X_2 \times Y_3 - Y_2 \times X_3 \\ X_3 \times Y_1 - Y_3 \times X_1 \\ X_1 \times Y_2 - Y_1 \times X_2 \end{pmatrix} \in (\mathfrak{J}^C)^3, \\ & \boldsymbol{X} \cdot \boldsymbol{Y} = \begin{pmatrix} (X_1, Y_1) & (X_1, Y_2) & (X_1, Y_3) \\ (X_2, Y_1) & (X_2, Y_2) & (X_2, Y_3) \\ (X_3, Y_1) & (X_3, Y_2) & (X_3, Y_3) \end{pmatrix} - \frac{1}{3}(\boldsymbol{X}, \boldsymbol{Y})E \in \mathfrak{sl}(3, C), \\ & \boldsymbol{X} \vee \boldsymbol{Y} = X_1 \vee Y_1 + X_2 \vee Y_2 + X_3 \vee Y_3 \in \mathfrak{e}_6^C, \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00756	125	■ 0.119	$\begin{array}{l} \backslash begin{array}{llllllllll} \backslash lambda_{\{0\}}-\backslash lambda_{\{1\}}=1 \\ \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{1\}}=1 \& \\ 2 \& 2 \& 2 \& 2 \& 2 \& 1 \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{2\}}=1 \& 1 \& \\ 0 \& 0 \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{2\}}=1 \& 1 \& 2 \\ \& 2 \& 2 \& 1 \backslash \backslash \backslash lambda_{\{0\}}-\backslash lambda_{\{3\}}=1 \& 1 \& 1 \& 0 \\ \& 0 \& 0 \& 0 \& 0 \& \backslash lambda_{\{0\}}+\backslash lambda_{\{3\}}=1 \& 1 \& 1 \& 2 \& \\ 2 \& 1 \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{2\}}=0 \& 1 \& 0 \& 0 \& 0 \& 0 \& \\ 0 \& 0 \& \backslash lambda_{\{1\}}+\backslash lambda_{\{2\}}=0 \& 1 \& 2 \& 2 \& 2 \& 1 \\ \backslash \backslash \backslash lambda_{\{1\}}-\backslash lambda_{\{3\}}=0 \& 1 \& 1 \& 0 \& 0 \& 0 \& 0 \& 0 \\ \& \backslash lambda_{\{1\}}+\backslash lambda_{\{3\}}=0 \& 1 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \\ \backslash lambda_{\{2\}}-\backslash lambda_{\{3\}}=0 \& 0 \& 1 \& 0 \& 0 \& 0 \& 0 \& 0 \& \\ \backslash lambda_{\{2\}}+\backslash lambda_{\{3\}}=0 \& 0 \& 1 \& 2 \& 2 \& 1 \backslash \backslash \& \& \\ \backslash lambda_{\{0\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=1 \& 1 \\ \& 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& \backslash lambda_{\{1\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 1 \& 1 \& 1 \& 1 \& 1 \& 1 \& \\ 1 \& \& \& \backslash \backslash \& \& \backslash lambda_{\{2\}}+\frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 1 \& 1 \& 1 \& 1 \& 1 \& \& \& \backslash \backslash \& \& + \\ \frac{1}{2}(\mu_{\{2\}}-\mu_{\{3\}})\left(\mu_{\{2\}}-\mu_{\{3\}}\right)=0 \& 0 \& 0 \& 1 \& 1 \& \\ \& 1 \& 1 \& \& \& \end{array}$	$\begin{array}{llllll} \lambda_0 - \lambda_1 = 1 & 0 & 0 & & & \\ \lambda_0 - \lambda_2 = 1 & 1 & 0 & & & \\ \lambda_0 - \lambda_3 = 1 & 1 & 1 & & & \\ \lambda_1 - \lambda_2 = 0 & 1 & 0 & & & \\ \lambda_1 - \lambda_3 = 0 & 1 & 1 & & & \\ \lambda_2 - \lambda_3 = 0 & 0 & 1 & & & \\ & & & \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 & 1 & 1 & 1 & 1 & 1 \\ & & & \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 1 & 1 & 1 & 1 & 1 \\ & & & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 & 1 & 1 & 1 & 1 \\ & & & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 & 0 & 1 & 1 & 1 \end{array}$	$\begin{array}{llllll} \lambda_0 - \lambda_1 = 1 & 0 & 0 & 0 & 0 & 0 & 0 & \lambda_0 + \lambda_1 = 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0 - \lambda_2 = 1 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_0 + \lambda_2 = 1 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_0 - \lambda_3 = 1 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_0 + \lambda_3 = 1 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_1 - \lambda_2 = 0 & 1 & 0 & 0 & 0 & 0 & 0 & \lambda_1 + \lambda_2 = 0 & 1 & 2 & 2 & 2 & 1 & 1 \\ \lambda_1 - \lambda_3 = 0 & 1 & 1 & 0 & 0 & 0 & 0 & \lambda_1 + \lambda_3 = 0 & 1 & 1 & 2 & 2 & 1 & 1 \\ \lambda_2 - \lambda_3 = 0 & 0 & 1 & 0 & 0 & 0 & 0 & \lambda_2 + \lambda_3 = 0 & 0 & 1 & 2 & 2 & 1 & 1 \\ & & & & & & & \lambda_0 + \frac{1}{2}(\mu_2 - \mu_3) = 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ & & & & & & & \lambda_1 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ & & & & & & & \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ & & & & & & & \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) = 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01187	200	■ 0.183	\begin{aligned} & \&= (\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{a}, *(\boldsymbol{b} \wedge \boldsymbol{x})) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_2, *(\boldsymbol{b} \wedge \boldsymbol{x})) \\ & + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{w} \wedge \boldsymbol{a}_1, *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ & = \frac{2}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) - (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ & \quad + (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5), \\ & ((\boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}))\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{v})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{v}, \boldsymbol{w})(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{v}) + (\boldsymbol{a}_1, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_2, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) \\ & - (\boldsymbol{a}_2, \boldsymbol{w})(\boldsymbol{b} \wedge \boldsymbol{x} \wedge \boldsymbol{w} \wedge \boldsymbol{a}_1, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_5) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}). \end{aligned}	(not rendered)	$ \begin{aligned} &= (v, w)(a, *(b \wedge x)) - (a_1, w)(w \wedge a_2, *(b \wedge x)) \\ &+ (a_2, w)(w \wedge a_1, *(b \wedge x)) - \frac{3}{5}(*(a \wedge b), x)(v, w) \\ &= \frac{2}{5}(*(a \wedge b), x)(v, w) - (a_1, w)(b \wedge x \wedge w \wedge a_2, e_1 \wedge \cdots \wedge e_5) \\ &\quad + (a_2, w)(b \wedge x \wedge w \wedge a_1, e_1 \wedge \cdots \wedge e_5), \\ &((b \times *(a \wedge x))v, w) \\ &= (x \wedge a, w \wedge *(b \wedge v)) - \frac{3}{5}(*(a \wedge b), x)(v, w) \\ &= (v, w)(*(a \wedge b), v) + (a_1, w)(b \wedge x \wedge w \wedge a_2, e_1 \wedge \cdots \wedge e_5) \\ &- (a_2, w)(b \wedge x \wedge w \wedge a_1, e_1 \wedge \cdots \wedge e_5) - \frac{3}{5}(*(a \wedge b), x)(v, w). \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00830	138	■ 0.186	$\begin{aligned} \kappa(X, Y, \xi, \eta) = & \kappa \left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \right) \\ & = \left(\begin{array}{ccc} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{array} \right), -\xi, \eta \right), \\ \mu(X, Y, \xi, \eta) = & \left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{array} \right), \left(\begin{array}{ccc} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{array} \right), \eta_1, \xi_1 \right). \end{aligned}$	$\begin{aligned} \kappa(X, Y, \xi, \eta) &= \kappa \left(\begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ &= \left(\begin{pmatrix} -\xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & -\eta_2 & -y_1 \\ 0 & -\bar{y}_1 & -\eta_3 \end{pmatrix}, -\xi, \eta \right), \\ \mu(X, Y, \xi, \eta) &= \left(\begin{pmatrix} \eta & 0 & 0 \\ 0 & \eta_3 & -y_1 \\ 0 & -\bar{y}_1 & \eta_2 \end{pmatrix}, \begin{pmatrix} \xi & 0 & 0 \\ 0 & \xi_3 & -x_1 \\ 0 & -\bar{x}_1 & \xi_2 \end{pmatrix}, \eta_1, \xi_1 \right). \end{aligned}$	
0902.0431_E01041	177	■ 0.204	$\begin{aligned} & \left(\alpha_i, \alpha_i \right) = \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8, \\ & \left(\alpha_i, \alpha_{i+1} \right) = -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad \left(\alpha_3, \alpha_8 \right) = -\frac{1}{60}, \\ & \left(\alpha_i, \alpha_j \right) = 0, \quad \text{otherwise} \\ & \left(-\mu, -\mu \right) = \frac{1}{30}, \quad \left(-\mu, \alpha_7 \right) = -\frac{1}{60}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$	$\begin{aligned} \left(\alpha_i, \alpha_i \right) &= \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8 \\ \left(\alpha_i, \alpha_{i+1} \right) &= -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad \left(\alpha_3, \alpha_8 \right) = -\frac{1}{60} \\ \left(\alpha_i, \alpha_j \right) &= 0, \quad \text{otherwise} \\ \left(-\mu, -\mu \right) &= \frac{1}{30}, \quad \left(-\mu, \alpha_7 \right) = -\frac{1}{60}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$	$\begin{aligned} \left(\alpha_i, \alpha_i \right) &= \frac{1}{30}, \quad i = 1, 2, 3, 4, 5, 6, 7, 8, \\ \left(\alpha_i, \alpha_{i+1} \right) &= -\frac{1}{60}, \quad i = 1, 2, \cdots, 6, \quad \left(\alpha_3, \alpha_8 \right) = -\frac{1}{60}, \\ \left(\alpha_i, \alpha_j \right) &= 0, \quad \text{otherwise}, \\ \left(-\mu, -\mu \right) &= \frac{1}{30}, \quad \left(-\mu, \alpha_7 \right) = -\frac{1}{60}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 1, 2, 3, 4, 5, 6, 8, \end{aligned}$
0902.0431_E00340	54	■ 0.226	$\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$	$\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$	$\begin{aligned} \lambda_0 &= \alpha_1 + \alpha_2 + \alpha_3 \\ \lambda_1 &= \alpha_2 + \alpha_3 \\ \lambda_3 &= \alpha_3 \\ \lambda_3 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + 2\alpha_4, \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902..0431_E00937	155	■ 0.238	$\begin{aligned} & \begin{array}{l} \text{\texttt{\textbackslash begin{aligned}}} \& P=(0,0,1,0) \text{\texttt{\textbackslash \& \xrightarrow{f}}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \text{\texttt{\textbackslash \& \stackrel{D}{\longrightarrow}}} \\ D \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \\ \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} + \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} \\ D \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} + \\ \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} D \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \text{\texttt{\textbackslash \& \quad + \text{\texttt{\textbackslash left(b_{11}} + \text{\texttt{\textbackslash frac{\nu}{3}}}}} \\ \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \\ \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} - \text{\texttt{\textbackslash bar{l}_{11}}} \text{\texttt{\textbackslash boldsymbol{e}_{4}}} \\ \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} - \\ \text{\texttt{\textbackslash bar{l}_{12}}} \text{\texttt{\textbackslash boldsymbol{e}_{5}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \\ \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} - \text{\texttt{\textbackslash bar{l}_{13}}} \text{\texttt{\textbackslash boldsymbol{e}_{6}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \\ \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \\ \text{\texttt{\textbackslash \& \quad + \text{\texttt{\textbackslash left(b_{33}} + \text{\texttt{\textbackslash frac{\nu}{3}}}}} \\ \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} - \text{\texttt{\textbackslash bar{l}_{21}}} \text{\texttt{\textbackslash boldsymbol{e}_{4}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{2}}} - \text{\texttt{\textbackslash bar{l}_{32}}} \text{\texttt{\textbackslash boldsymbol{e}_{5}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{2}}} - \text{\texttt{\textbackslash bar{l}_{33}}} \text{\texttt{\textbackslash boldsymbol{e}_{6}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{1}}} \text{\texttt{\textbackslash wedge}} \text{\texttt{\textbackslash boldsymbol{e}_{6}}} \text{\texttt{\textbackslash wedge}} \\ \text{\texttt{\textbackslash boldsymbol{e}_{3}}} \text{\texttt{\textbackslash \& \quad \text{\texttt{\textbackslash left(\begin{array}{ccc} \\ \\ \end{array}}}}} \\ \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash \& \xrightarrow{f^{-1}}} \text{\texttt{\textbackslash left(\begin{array}{ccc} \\ \\ \end{array}}} \\ \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{11}}}}} \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{21}}}, \text{\texttt{\textbackslash l_{12}}}}} \\ \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{31}}}, \text{\texttt{\textbackslash l_{13}}}}} \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash \& -h}} \\ \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{12}}}, \text{\texttt{\textbackslash l_{21}}}}} \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{22}}}}} \\ \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{32}}}, \text{\texttt{\textbackslash l_{23}}}}} \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash \& -h}} \\ \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{13}}}, \text{\texttt{\textbackslash l_{31}}}}} \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{123}}}, \\ \text{\texttt{\textbackslash l_{32}}}}} \text{\texttt{\textbackslash right)}} \& \text{\texttt{\textbackslash left(\text{\texttt{\textbackslash bar{l}_{133}}}}} \text{\texttt{\textbackslash right)}} \text{\texttt{\textbackslash end{array}}} \\ \text{\texttt{\textbackslash right)}} = \text{\texttt{\textbackslash left(\begin{array}{c} h(\nu) \\ \\ \end{array}}} \\ \text{\texttt{\textbackslash right)}} = \text{\texttt{\textbackslash left(\begin{array}{ccc} \phi_C(B,C) + \frac{i\nu e_1}{3} & -2\tau h(L) & 0 \\ 2h(L) & \tau\phi_C(B,C)\tau - \frac{i\nu e_1}{3} & -\tau h(L) \\ 0 & h(L) & 0 \\ -\tau h(L) & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}}} \\ \text{\texttt{\textbackslash right)}} = \Phi(\phi_C(B,C), h(L), -\tau h(L), -i\nu e_1) P. \end{array}$	$P = (0, 0, 1, 0)$ $\xrightarrow{f} \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3$ $\xrightarrow{D} D\boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge D\boldsymbol{e}_2 \wedge \boldsymbol{e}_3 + \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge D\boldsymbol{e}_3$ $= \left(b_{11} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{11} \boldsymbol{e}_4 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{12} \boldsymbol{e}_5 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{13} \boldsymbol{e}_6 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3$ $+ \left(b_{22} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{21} \boldsymbol{e}_4 \wedge \boldsymbol{e}_3 \wedge \boldsymbol{e}_1 - \bar{l}_{22} \boldsymbol{e}_1 \wedge \boldsymbol{e}_5 \wedge \boldsymbol{e}_3 - \bar{l}_{23} \boldsymbol{e}_1 \wedge \boldsymbol{e}_6 \wedge \boldsymbol{e}_3$ $+ \left(b_{33} + \frac{\nu}{3}\right) \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_3 - \bar{l}_{31} \boldsymbol{e}_4 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{32} \boldsymbol{e}_5 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 - \bar{l}_{33} \boldsymbol{e}_6 \wedge \boldsymbol{e}_1 \wedge \boldsymbol{e}_2$ $\xrightarrow{f^{-1}} \left(\begin{array}{ccc} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & & \\ \begin{pmatrix} -h(\bar{l}_{11}) & -h(\bar{l}_{21}, l_{12}) & -h(\bar{l}_{31}, l_{13}) \\ -h(\bar{l}_{12}, l_{21}) & -h(\bar{l}_{22}) & -h(\bar{l}_{32}, l_{23}) \\ -h(\bar{l}_{13}, l_{31}) & -h(\bar{l}_{23}, l_{32}) & -h(\bar{l}_{33}) \end{pmatrix} & & \\ & h(\nu) & \\ & 0 & \end{array} \right) = \begin{pmatrix} 0 \\ -\tau h(L) \\ -i\nu e_1 \\ 0 \end{pmatrix}$ $= \begin{pmatrix} \phi_C(B, C) + \frac{i\nu e_1}{3} & -2\tau h(L) & 0 & h(L) \\ 2h(L) & \tau\phi_C(B, C)\tau - \frac{i\nu e_1}{3} & -\tau h(L) & 0 \\ 0 & h(L) & -i\nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i\nu e_1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ $= \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1)P.$	

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Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00206	35	■ 0.294	$\begin{aligned} & \& \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_2, \kappa \alpha_3\right), \\ & \& \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \& \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$	$\begin{aligned} & \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ & \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$	$\begin{aligned} & \kappa\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_1, \kappa \alpha_3, \kappa \alpha_2\right), \\ & \pi\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\kappa \alpha_3, \kappa \alpha_2, \kappa \alpha_1\right), \\ & \nu\left(\alpha_1, \alpha_2, \alpha_3\right)=\left(\alpha_2, \alpha_3, \alpha_1\right), \end{aligned}$
0902.0431_E00655	108	■ 0.299	$\begin{aligned} & \& \left(P M \tau h(A, B)^*\right)^*(P N \tau h(A, B)^*)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & \& =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \& \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}=\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\bar{t} \bar{P}(M \times N) \tau^t(h(A, B)^*)} \\ & \& =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$	$\begin{aligned} & (P M \tau h(A, B)^*)^*(P N \tau h(A, B)^*)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}=\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\bar{t} \bar{P}(M \times N) \tau^t(h(A, B)^*)} \\ & =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$	$\begin{aligned} & (P M \tau h(A, B)^*)^*(P N \tau h(A, B)^*)=\tau h(A, B) M^* P^* P N \tau h(A, B)^* \\ & =\tau\left(h(A, B) \tau\left(M^* N\right) h(A, B)^*\right), \\ & \frac{(P M \tau h(A, B)^*)(h(A, B) Y h(A, B)^*)}{(P M \tau h(A, B)^*) \times(P N \tau h(A, B)^*)}=\frac{\tau(P \tau(M Y) \tau h(A, B)^*)}{\bar{t} \bar{P}(M \times N) \tau^t(h(A, B)^*)} \\ & =P(\overline{M \times N}) h(A, B)^*=\tau(P(\overline{M \times N}) \tau h(A, B)^*). \end{aligned}$
0902.0431_E00101	20	■ 0.306	$\begin{aligned} & \& \lambda_1-\lambda_2=\alpha_1, \quad \lambda_1-\lambda_3=2 \alpha_1+3 \alpha_2, \quad \lambda_2-\lambda_3=\alpha_1+3 \alpha_2, \\ & \& \lambda_1=\alpha_1+\alpha_2, \quad \lambda_2=\alpha_2, \quad \lambda_3=2 \alpha_2 \end{aligned}$	$\begin{aligned} & \lambda_1-\lambda_2=\alpha_1, \quad \lambda_1-\lambda_3=2 \alpha_1+3 \alpha_2, \quad \lambda_2-\lambda_3=\alpha_1+3 \alpha_2, \\ & \lambda_1=\alpha_1+\alpha_2, \quad \lambda_2=\alpha_2, \quad \lambda_3=2 \alpha_2 \end{aligned}$	$\begin{aligned} & \lambda_1-\lambda_2=\alpha_1, \quad \lambda_1-\lambda_3=2 \alpha_1+3 \alpha_2, \quad \lambda_2-\lambda_3=\alpha_1+3 \alpha_2, \\ & \lambda_1=\alpha_1+\alpha_2, \quad \lambda_2=\alpha_2, \quad -\lambda_3=\alpha_1+2 \alpha_2. \end{aligned}$
0902.0431_EQ1154	195	■ 0.318	$\begin{aligned} & \& +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & \& =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$	$\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right) \end{aligned}$	$\begin{aligned} & +(\boldsymbol{v}, \boldsymbol{w})(\boldsymbol{u}, \boldsymbol{a})-\sum_{i=1}^3 \sum_{j=1}^3\left(\boldsymbol{u}_i \wedge \boldsymbol{v}_j \wedge \boldsymbol{v}_{j+1}, \boldsymbol{w}\right)\left(\boldsymbol{u}_{i+1} \wedge \boldsymbol{u}_{i+2} \wedge \boldsymbol{v}_{j+2}, \boldsymbol{a}\right) \\ & =-(\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{w} \wedge \boldsymbol{a})=-\left(*\left(*(\boldsymbol{u} \wedge \boldsymbol{v}) \wedge \boldsymbol{w}\right), \boldsymbol{a}\right). \end{aligned}$

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0902.0431_EQ1189	200	■ 0.322	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$	$\begin{aligned} & ((\boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y})) \boldsymbol{v}, \boldsymbol{w}) = (\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{v} \wedge \boldsymbol{a}) - \frac{3}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (\boldsymbol{y} \wedge \boldsymbol{w}, \boldsymbol{a}) - (\boldsymbol{y}, \boldsymbol{v}) (\boldsymbol{x} \wedge \boldsymbol{w}, \boldsymbol{a}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}) \\ & = (\boldsymbol{x}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{y}), \boldsymbol{w}) - (\boldsymbol{y}, \boldsymbol{v}) (*(\boldsymbol{a} \wedge \boldsymbol{x}), \boldsymbol{w}) + \frac{2}{5} (\boldsymbol{a}, \boldsymbol{x} \wedge \boldsymbol{y}) (\boldsymbol{v}, \boldsymbol{w}). \end{aligned}$
0902.0431_EQ1157	195	■ 0.323	$\begin{aligned} & \& \boldsymbol{u} = \sum_{\boldsymbol{I}} \boldsymbol{u}_{\boldsymbol{I}} \boldsymbol{D} \boldsymbol{e}_{\boldsymbol{I}} = 3 \sum_{\boldsymbol{I} \not\supseteq 9} \boldsymbol{u}_{\boldsymbol{I}} \boldsymbol{e}_{\boldsymbol{I}} - 6 \sum_{\boldsymbol{I} \supseteq 9} \boldsymbol{u}_{\boldsymbol{I}} \boldsymbol{e}_{\boldsymbol{I}}, \\ & \& \boldsymbol{v} = -\sum_{\boldsymbol{J} \not\supseteq 9} \boldsymbol{v}_{\boldsymbol{J}} \boldsymbol{e}_{\boldsymbol{J}} + 6 \sum_{\boldsymbol{J} \supseteq 9} \boldsymbol{v}_{\boldsymbol{J}} \boldsymbol{e}_{\boldsymbol{J}}, \end{aligned}$	$\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -{}^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$	$\begin{aligned} 0 &= D\boldsymbol{u} = \sum_I u_I D\boldsymbol{e}_I = 3 \sum_{I \not\supseteq 9} u_I \boldsymbol{e}_I - 6 \sum_{I \supseteq 9} u_I \boldsymbol{e}_I, \\ 0 &= -{}^t D\boldsymbol{v} = -3 \sum_{J \not\supseteq 9} v_J \boldsymbol{e}_J + 6 \sum_{J \supseteq 9} v_J \boldsymbol{e}_J, \end{aligned}$
0902.0431_EQ0460	75	■ 0.325	$\begin{aligned} & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)).} \\ & \mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)).} \end{aligned}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)).}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_6^C] \supset [\mathfrak{f}_4^C, \widetilde{\mathfrak{J}}_0^C] = \mathfrak{f}_4^C \widetilde{\mathfrak{J}}_0^C \text{ (Lemma 3.2.2) } = \widetilde{\mathfrak{J}}_0^C \text{ (Proposition 2.4.6.(2)).}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E08802	134	■0.338	$\left(\begin{array}{c} \left(\begin{array}{ccc} \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } \\ 0 \\ 0 \end{array} \right) \\ \left(\begin{array}{ccc} a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \cos a_3 \\ 0 \\ 0 \end{array} \right) \\ \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \\ 0 \end{array} \right) \\ \left(\begin{array}{ccc} \cos a_1 \cos a_2 \cos a_3 \\ a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } \end{array} \right)$	$\left(\begin{array}{ccc} \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 } & 0 & 0 \\ 0 & a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } & 0 \\ 0 & 0 & a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 \end{array} \right) \\ \left(\begin{array}{ccc} a_1 \frac{\sin a_1 }{ a_1 } \cos a_2 \cos a_3 & 0 & 0 \\ 0 & \cos a_1 a_2 \frac{\sin a_2 }{ a_2 } \cos a_3 & 0 \\ 0 & 0 & \cos a_1 \cos a_2 a_3 \frac{\sin a_3 }{ a_3 } \end{array} \right) \\ \cos a_1 \cos a_2 \cos a_3 \\ a_1 \frac{\sin a_1 }{ a_1 } a_2 \frac{\sin a_2 }{ a_2 } a_3 \frac{\sin a_3 }{ a_3 }$	
0902.0431_E08875	145	■0.339	$\begin{aligned} &\varphi_2(A) \left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \right) \\ &= \left(\left(\begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left(\begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \right) \end{aligned}$	$\varphi_2(A) \left(\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array} \right), \left(\begin{array}{ccc} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{array} \right), \xi, \eta \right) \\ = \left(\left(\begin{array}{ccc} \xi'_1 & x'_3 & \bar{x}'_2 \\ \bar{x}'_3 & \xi'_2 & x'_1 \\ x'_2 & \bar{x}'_1 & \xi'_3 \end{array} \right), \left(\begin{array}{ccc} \eta'_1 & y'_3 & \bar{y}'_2 \\ \bar{y}'_3 & \eta'_2 & y'_1 \\ y'_2 & \bar{y}'_1 & \eta'_3 \end{array} \right), \xi', \eta' \right),$	

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0902.0431_E00400	66	■ 0.351	$\backslash\mathrm{beta}_{\{1\}}\mathrm{m}=\backslash\mathrm{operatorname{prm}}\backslash\mathrm{bar}\{r\},\quad\backslash\mathrm{beta}_{\{2\}}\mathrm{m}=\mathrm{r}\mathrm{m}\backslash\mathrm{bar}\{r\}\mathrm{q},\quad\backslash\mathrm{beta}_{\{3\}}\mathrm{m}=\backslash\mathrm{bar}\{q\}\mathrm{r}\mathrm{m}\backslash\mathrm{bar}\{r\}\backslash\mathrm{bar}\{p\},\quad\mathrm{m}\in\boldsymbol{\mathrm{H}}.$	$\beta_1 m = \mathrm{prm} \bar{r}, \quad \beta_2 m = r m \bar{r} q, \quad \beta_3 m = \bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H}.$	$\beta_1 m = \mathrm{prm} \bar{r}, \quad \beta_2 m = r m \bar{r} q, \quad \beta_3 m = \bar{q} r m \bar{r} \bar{p}, \quad m \in \boldsymbol{H}.$
0902.0431_E00860	143	■ 0.354	$\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{cccccccc}\}\mathrm{1}\&\backslash\mathrm{longrightarrow}\&\backslash\mathrm{operatorname{Spin}}(10)\&\backslash\mathrm{longrightarrow}\&\backslash\mathrm{left}(\backslash\mathrm{left}(\mathrm{E}_{\{7\}}\backslash\mathrm{right})^{\{\kappa,\mu\}}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\mathrm{E}_{\{1\}},0,1\backslash\mathrm{right})}\&\backslash\mathrm{longrightarrow}\&\mathrm{S}^{\{10\}}\&\backslash\mathrm{longrightarrow}\backslash\&\&\backslash\mathrm{downarrow}\mathrm{p}^{\{\mathrm{prime}\}}\&\&\backslash\mathrm{downarrow}\mathrm{p}\&\&\backslash\mathrm{downarrow}=\&\backslash\&\mathrm{1}\&\backslash\mathrm{longrightarrow}\&\mathrm{S}\mathrm{0}(10)\&\backslash\mathrm{longrightarrow}\&\mathrm{S}\mathrm{0}(11)\&\backslash\mathrm{longrightarrow}\&\mathrm{S}^{\{10\}}\&\backslash\mathrm{longrightarrow}\backslash\mathrm{end}\{\mathrm{array}\}$	$\begin{array}{ccccccc} 1 & \longrightarrow & \mathrm{Spin}(10) & \longrightarrow & ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} & \longrightarrow & S^{10} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(10) & \longrightarrow & SO(11) & \longrightarrow & S^{10} \longrightarrow * \end{array}$	$\begin{array}{ccccccc} 1 & \longrightarrow & Spin(10) & \longrightarrow & ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} & \longrightarrow & S^{10} \longrightarrow * \\ & & \downarrow p' & & \downarrow p & & \downarrow = \\ 1 & \longrightarrow & SO(10) & \longrightarrow & SO(11) & \longrightarrow & S^{10} \longrightarrow * \end{array}$
0902.0431_E00876	145	■ 0.358	$\backslash\mathrm{begin}\{\mathrm{gathered}\}\backslash\mathrm{binom}\{\backslash\mathrm{x}_1^{\{\mathrm{prime}\}}\}\{\mathrm{eta}^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\backslash\mathrm{x}_1\}\{\mathrm{eta}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\backslash\mathrm{x}_1^{\{\mathrm{prime}\}}\}\{\mathrm{eta}_{\{1\}}^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\backslash\mathrm{x}_1\}\{\mathrm{eta}_{\{1\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{eta}_{\{2\}}^{\{\mathrm{prime}\}}\}\{\backslash\mathrm{x}_3^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\mathrm{eta}_{\{2\}}\}\{\backslash\mathrm{x}_3\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{eta}_{\{3\}}^{\{\mathrm{prime}\}}\}\{\backslash\mathrm{x}_2^{\{\mathrm{prime}\}}\}=A\backslash\mathrm{binom}\{\mathrm{eta}_{\{3\}}\}\{\backslash\mathrm{x}_2\}\backslash\backslash\backslash\mathrm{binom}\{\mathrm{x}_{\{1\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{1\}}^{\{\mathrm{prime}\}}\}=\tau A\backslash\mathrm{binom}\{\mathrm{x}_{\{1\}}\}\{\mathrm{y}_{\{1\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{x}_{\{2\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{2\}}^{\{\mathrm{prime}\}}\}=\backslash\mathrm{binom}\{\mathrm{x}_{\{2\}}\}\{\mathrm{y}_{\{2\}}\},\quad\backslash\mathrm{quad}\backslash\mathrm{binom}\{\mathrm{x}_{\{3\}}^{\{\mathrm{prime}\}}\}\{\mathrm{y}_{\{3\}}^{\{\mathrm{prime}\}}\}=\backslash\mathrm{binom}\{\mathrm{x}_{\{3\}}\}\{\mathrm{y}_{\{3\}}\}.\quad\backslash\mathrm{end}\{\mathrm{gathered}\}$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = A \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_1' \\ \eta_1' \end{pmatrix} = A \begin{pmatrix} \xi \\ \eta_1 \end{pmatrix}, \quad \begin{pmatrix} \eta_2' \\ \xi_3' \end{pmatrix} = A \begin{pmatrix} \eta_2 \\ \xi_3 \end{pmatrix}, \quad \begin{pmatrix} \eta_3' \\ \xi_2' \end{pmatrix} = A \begin{pmatrix} \eta_3 \\ \xi_2 \end{pmatrix}$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \tau A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = A \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_1' \\ \eta_1' \end{pmatrix} = A \begin{pmatrix} \xi \\ \eta_1 \end{pmatrix}, \quad \begin{pmatrix} \eta_2' \\ \xi_3' \end{pmatrix} = A \begin{pmatrix} \eta_2 \\ \xi_3 \end{pmatrix}, \quad \begin{pmatrix} \eta_3' \\ \xi_2' \end{pmatrix} = A \begin{pmatrix} \eta_3 \\ \xi_2 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \tau A \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$

[illegible]

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0902.0431_E00093	19	■ 0.382	$\begin{aligned} H_1 H_1 e_2 &= H_1 e_3 = -e_2, \\ &\quad \end{aligned}$	$H_1 H_1 e_2 = H_1 e_3 = -e_2,$	$\begin{aligned} H_1 H_1 e_2 &= H_1 e_3 = -e_2, & H_1 H_1 e_3 &= -H_1 e_2 = -e_3, \\ H_1 H_1 e_4 &= -H_1 e_5 = -e_4, & H_1 H_1 e_5 &= H_1 e_4 = -e, \\ H_1 H_1 e_i &= 0 & \text{otherwise.} \end{aligned}$
0902.0431_E00371	61	■ 0.388	$\begin{aligned} &\operatorname{tr}(X) = \operatorname{tr}(\alpha X) (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \operatorname{tr}(\alpha X) (\text{Lemma 2.2.1.(2)}) = \sum_{i=1}^3 \xi_i, \\ (X, X) &= (\alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^2, \\ \operatorname{tr}(X, X, X) &= \operatorname{tr}(\alpha X, \alpha X, \alpha X) (\text{Lemma 2.2.4}) = \sum_{i=1}^3 \xi_i^3. \end{aligned}$
0902.0431_E00602	100	■ 0.396	$\begin{aligned} &\left. \begin{array}{l} \left(\begin{array}{ccc} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) \\ \mid \\ \left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \end{array} \right \begin{array}{l} \mu_i \in \mathbf{R} \\ m_i, a_i \in \mathbf{H} \end{array} \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left(\begin{array}{ccc} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \right\} \end{aligned}$	$\begin{aligned} (\mathfrak{J}^C)_{\tau\gamma} &= \{X \in \mathfrak{J}^C \mid \tau\gamma X = X\} \\ &= \left\{ \left(\begin{array}{ccc} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{array} \right) + i \left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array} \right) \right\} \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00900	149	■ 0.397	$\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \quad + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ & = k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ & = k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ & = k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & \quad + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ & = \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ & = \chi(\Phi(0, A, -\gamma A, 0)P). \end{aligned}$	$\begin{aligned} P = & (X, Y, \xi, \eta) \\ & \stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J \\ & \quad + e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T) \\ & = k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J \\ & = k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J \\ & = k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J \\ & \quad + e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)} \\ & = \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right) \\ & = \chi(\Phi(0, A, -\gamma A, 0)P) \end{aligned}$	$P = (X, Y, \xi, \eta)$ $\stackrel{\chi}{\rightarrow} k\left(gX - \frac{\xi}{2}E\right)J + e_1k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $\longrightarrow e_1k(T)k\left(gX - \frac{\xi}{2}E\right)J - k(T)k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J$ $+ e_1k\left(gX - \frac{\xi}{2}E\right)J^tk(T) - k\left(g(\gamma Y) - \frac{\eta}{2}E\right)J^tk(T)$ $= k(-Tg(\gamma Y) - g(\gamma Y)T + \eta T)J + e_1k(T(gX) + (gX)T - \xi T)J$ $= k(-2gA \circ g(\gamma Y) + \eta gA)J + e_1k(2gA \circ gX - \xi gA)J$ $= k\left(-2g(\gamma A \times Y) - \frac{1}{2}(A, Y)E + \eta gA\right)J$ $+ e_1k\left(2g(\gamma A \times \gamma X) + \frac{1}{2}(\gamma A, X)E - \xi gA\right)J \text{ (Lemma 3.12.1)}$ $= \chi\left(\begin{array}{c} -2\gamma A \times Y + \eta A \\ 2A \times X - \xi \gamma A \\ (A, Y) \\ (-\gamma A, X) \end{array}\right) = \chi\left(\left(\begin{array}{cccc} 0 & -2\gamma A & 0 & A \\ 2A & 0 & -\gamma A & 0 \\ 0 & A & 0 & 0 \\ -\gamma A & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} X \\ Y \\ \xi \\ \eta \end{array}\right)\right)$ $= \chi(\Phi(0, A, -\gamma A, 0)P).$
0902.0431_E01000	183	■ 0.403	$S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}} - \frac{1}{8}(S_{\{1\}} \tau S_{\{2\}} - S_{\{2\}} \tau S_{\{1\}}) \in \mathfrak{su}(8),$	$S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$	$S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8}(S_1 \tau S_2 - S_2 \tau S_1) \in \mathfrak{su}(8),$
0902.0431_E01030	173	■ 0.410	$\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 2 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$	$\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$	$\begin{aligned} & \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 2 \quad 4 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 2 \\ & \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 2 \quad 3 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 3 \\ & \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ & \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{aligned}$

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0902.0431_E00492	81	■0.411	$\delta\left(\begin{array}{ccc}\xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & \delta_3x_{12} & \bar{\delta}_2\bar{x}_{13} \\ \bar{\delta}_3\bar{x}_{21} & 0 & \delta_1x_{23} \\ \delta_2x_{31} & \bar{\delta}_1\bar{x}_{32} & 0\end{array}\right).$	$\delta\left(\begin{array}{ccc}\xi_1 & x_{12} & x_{13} \\ x_{21} & \xi_2 & x_{23} \\ x_{31} & x_{32} & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & \delta_3x_{12} & \bar{\delta}_2\bar{x}_{13} \\ \bar{\delta}_3\bar{x}_{21} & 0 & \delta_1x_{23} \\ \delta_2x_{31} & \bar{\delta}_1\bar{x}_{32} & 0\end{array}\right).$	
0902.0431_E01097	186	■0.420	$\begin{aligned} \zeta[(icE),l(S)I]&=\zeta(2\,l(i\,c\,S)I)\\ &\quad$		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00332	53	■ 0.438	$\begin{aligned} & \pm \left(\lambda_k - \lambda_l \right), \pm \left(\lambda_k + \lambda_l \right), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right), \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right), \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right), \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right), \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right). \end{aligned}$	$\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_0, \quad \pm \lambda_1, \quad \pm \lambda_2, \quad \pm \lambda_3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3), \quad \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3), \quad \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3), \quad \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3), \quad \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3). \end{aligned}$	
0902.0431_EQ1211	203	■ 0.448	$\begin{aligned} & \exp(\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & = (\exp(\operatorname{ad}(Z))C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, ((\exp(-^t Z))\boldsymbol{c} \otimes \boldsymbol{z}, ((\exp(-^t Z))\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = (\operatorname{Ad}(\exp Z)C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, (^t(\exp Z)^{-1}\boldsymbol{c}) \otimes \boldsymbol{z}, (^t(\exp Z)^{-1}\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = \varphi_1(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}). \end{aligned}$	$\begin{aligned} & \exp(\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & = (\exp(\operatorname{ad}(Z))C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, ((\exp(-^t Z))\boldsymbol{c} \otimes \boldsymbol{z}, ((\exp(-^t Z))\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = (\operatorname{Ad}(\exp Z)C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, (^t(\exp Z)^{-1}\boldsymbol{c}) \otimes \boldsymbol{z}, (^t(\exp Z)^{-1}\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = \varphi_1(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}). \end{aligned}$	$\begin{aligned} & \exp(\operatorname{ad}(Z, 0))(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}) \\ & = (\exp(\operatorname{ad}(Z))C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, ((\exp(-^t Z))\boldsymbol{c} \otimes \boldsymbol{z}, ((\exp(-^t Z))\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = (\operatorname{Ad}(\exp Z)C, D, ((\exp Z)\boldsymbol{x}) \otimes \boldsymbol{a}, \\ & \quad ((\exp Z)\boldsymbol{b}) \otimes \boldsymbol{y}, (^t(\exp Z)^{-1}\boldsymbol{c}) \otimes \boldsymbol{z}, (^t(\exp Z)^{-1}\boldsymbol{w}) \otimes \boldsymbol{d}) \\ & = \varphi_1(\exp Z)(C, D, \boldsymbol{x} \otimes \boldsymbol{a}, \boldsymbol{b} \otimes \boldsymbol{y}, \boldsymbol{c} \otimes \boldsymbol{z}, \boldsymbol{w} \otimes \boldsymbol{d}). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01039	177	■0.452	$\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$	$\begin{aligned} H_{\alpha_3} &= \left(0, 0, 0, \frac{1}{60}, 0, 0\right), \\ H_{\alpha_4} &= \left(\Phi\left(\frac{1}{90}(-E_1 + 2E_2 - E_3)^\sim, 0, 0, -\frac{1}{120}\right), 0, 0, -\frac{1}{120}, 0, 0\right), \\ H_{\alpha_5} &= \left(\Phi\left(\frac{1}{60}(H_3 - (E_2 - E_3)^\sim), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_6} &= \left(\Phi\left(\frac{1}{60}(H_2 - H_3), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_7} &= \left(\Phi\left(\frac{1}{60}(H_1 - H_2), 0, 0, 0\right), 0, 0, 0, 0, 0\right), \\ H_{\alpha_8} &= \left(\Phi\left(0, 0, 0, \frac{1}{40}\right), 0, 0, -\frac{1}{120}, 0, 0\right). \end{aligned}$	
0902.0431_E00939	156	■0.471	$\left.\begin{array}{l} \xrightarrow{f^{-1}} \left(\begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ = \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right).$	$\xrightarrow{f^{-1}} \left(\begin{array}{ccc} \left(\begin{array}{ccc} h(b_{11} - c_{11} - \frac{\nu}{3}) & -h(c_{12}, b_{12}) & -h(c_{13}, b_{13}) \\ -h(\bar{b}_{12}, \bar{c}_{12}) & 0 & 0 \\ -h(\bar{b}_{13}, \bar{c}_{13}) & 0 & 0 \end{array} \right) \\ \left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & h(l_{33}) & -h(l_{23}, \bar{l}_{32}) \\ 0 & -h(l_{32}, \bar{l}_{23}) & h(l_{22}) \end{array} \right) \\ 0 \\ -h(\bar{l}_{11}) \end{array} \right)$ $= \begin{pmatrix} \phi_C(B, C)E_1 + \frac{i\nu e_1}{3}E_1 \\ 2h(L) \times E_1 \\ 0 \\ -(\tau h(L), E_1) \end{pmatrix} = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} E_1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$	

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0902.0431_E01196	201	■ 0.477	$\begin{aligned} \mathfrak{g}_{\{0\}} &= \left\{ (C, 0) \in \mathfrak{g}_0 \mid C \in \mathfrak{sl}(5, C) \right\} \cong \mathfrak{sl}(5, C), \\ \mathfrak{g}_{\{0\}} \mid C &\in \mathfrak{mathfrak{sl}}(5, C) \right\} \cong \mathfrak{sl}(5, C) \\ \mathfrak{g}_{\{0\}} &\mid D \in \mathfrak{sl}(5, C) \} \cong \mathfrak{sl}(5, C) \\ \mathfrak{q} &= \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}. \end{aligned}$	$\begin{aligned} \mathfrak{g}_{01} &= \{(C, 0) \in \mathfrak{g}_0 \mid C \in \mathfrak{sl}(5, C)\} \cong \mathfrak{sl}(5, C) \\ \mathfrak{g}_{02} &= \{(0, D) \in \mathfrak{g}_0 \mid D \in \mathfrak{sl}(5, C)\} \cong \mathfrak{sl}(5, C) \\ \mathfrak{q} &= \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}. \end{aligned}$	$\begin{aligned} \mathfrak{g}_{01} &= \{(C, 0) \in \mathfrak{g}_0 \mid C \in \mathfrak{sl}(5, C)\} \cong \mathfrak{sl}(5, C), \\ \mathfrak{g}_{02} &= \{(0, D) \in \mathfrak{g}_0 \mid D \in \mathfrak{sl}(5, C)\} \cong \mathfrak{sl}(5, C), \\ \mathfrak{q} &= \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}. \end{aligned}$
0902.0431_E01195	201	■ 0.485	$\begin{aligned} & \& (-*(y \wedge *(x \wedge a)), b) = -(x \wedge a, y \wedge b) \\ & \& \quad = (y, a_1)(x \wedge a_2, b) - (y, a_2)(x \wedge a_1, b) - (x, y)(a, b). \end{aligned}$	$\begin{aligned} & (-*(y \wedge *(x \wedge a)), b) = -(x \wedge a, y \wedge b) \\ & = (y, a_1)(x \wedge a_2, b) - (y, a_2)(x \wedge a_1, b) - (x, y)(a, b). \end{aligned}$	$\begin{aligned} & (-*(\mathbf{y} \wedge *(\mathbf{x} \wedge \mathbf{a})), \mathbf{b}) = -(\mathbf{x} \wedge \mathbf{a}, \mathbf{y} \wedge \mathbf{b}) \\ & = (\mathbf{y}, \mathbf{a}_1)(\mathbf{x} \wedge \mathbf{a}_2, \mathbf{b}) - (\mathbf{y}, \mathbf{a}_2)(\mathbf{x} \wedge \mathbf{a}_1, \mathbf{b}) - (\mathbf{x}, \mathbf{y})(\mathbf{a}, \mathbf{b}). \end{aligned}$
0902.0431_E01104	187	■ 0.490	$\varphi(R)R_1 = [R, R_1], \quad R \in (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, R_1 \in (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$	$\varphi(R)R_1 = [R, R_1], \quad R \in (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, R_1 \in (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$	$\varphi(R)R_1 = [R, R_1], \quad R \in (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, R_1 \in (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$
0902.0431_E01010	168	■ 0.494	$\begin{aligned} \mathfrak{e}_8 &= \left\{ R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R \right\} \\ &= \{ (\Phi, P, -\tau \lambda P, r, s, -\tau s) \in \mathfrak{e}_8^C \mid \Phi \in \mathfrak{e}_7, P \in \mathfrak{P}^C, r \in i\mathbf{R}, s \in C \}. \end{aligned}$	$\begin{aligned} \mathfrak{e}_8 &= \left\{ R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R \right\} \\ &= \{ (\Phi, P, -\tau \lambda P, r, s, -\tau s) \in \mathfrak{e}_8^C \mid \Phi \in \mathfrak{e}_7, P \in \mathfrak{P}^C, r \in i\mathbf{R}, s \in C \}. \end{aligned}$	$\begin{aligned} \mathfrak{e}_8 &= \{ R \in \mathfrak{e}_8^C \mid \tau \tilde{\lambda} R = R \} \\ &= \{ (\Phi, P, -\tau \lambda P, r, s, -\tau s) \in \mathfrak{e}_8^C \mid \Phi \in \mathfrak{e}_7, P \in \mathfrak{P}^C, r \in i\mathbf{R}, s \in C \}. \end{aligned}$
0902.0431_E00952	158	■ 0.496	$\begin{aligned} D &= \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right) \in \mathfrak{su}(6), \\ \widetilde{M} &= \left(\begin{array}{cc} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{array} \right), \quad M_i, N_i \in M(3, C), \\ M &= M_1 + iM_2, N = N_1 + iN_2. \end{aligned}$	$\begin{aligned} D &= \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right) \in \mathfrak{su}(6), \\ \widetilde{M} &= \left(\begin{array}{cc} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{array} \right), \quad M_i, N_i \in M(3, C), \\ M &= M_1 + iM_2, N = N_1 + iN_2. \end{aligned}$	$\begin{aligned} D &= \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right) \in \mathfrak{su}(6), \\ \widetilde{M} &= \left(\begin{array}{cc} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{array} \right), \quad M_i, N_i \in M(3, C), \\ M &= M_1 + iM_2, N = N_1 + iN_2. \end{aligned}$
0902.0431_E01091 (1)	185	■ 0.503	$\begin{aligned} \zeta[l(D_1), l(D_2)] &= \zeta[l(D_1), l(D_2)] \\ &= \Theta(\varphi_*[D_1, D_2], 0, 0, 0, 0, 0) = \Theta([\varphi_* D_1, \varphi_* D_2], 0, 0, 0, 0, 0) \\ &= [\Theta(\varphi_* D_1, 0, 0, 0, 0, 0), \Theta(\varphi_* D_2, 0, 0, 0, 0, 0)] \\ &= [\zeta l(D_1), \zeta l(D_2)]. \end{aligned}$	$\begin{aligned} \zeta[l(D_1), l(D_2)] &= \zeta l[D_1, D_2] \\ &= \Theta(\varphi_*[D_1, D_2], 0, 0, 0, 0, 0) = \Theta([\varphi_* D_1, \varphi_* D_2], 0, 0, 0, 0, 0) \\ &= [\Theta(\varphi_* D_1, 0, 0, 0, 0, 0), \Theta(\varphi_* D_2, 0, 0, 0, 0, 0)] \\ &= [\zeta l(D_1), \zeta l(D_2)]. \end{aligned}$	$\begin{aligned} \zeta[l(D_1), l(D_2)] &= \zeta l[D_1, D_2] \\ &= \Theta(\varphi_*[D_1, D_2], 0, 0, 0, 0, 0) = \Theta([\varphi_* D_1, \varphi_* D_2], 0, 0, 0, 0, 0) \\ &= [\Theta(\varphi_* D_1, 0, 0, 0, 0, 0), \Theta(\varphi_* D_2, 0, 0, 0, 0, 0)] \\ &= [\zeta l(D_1), \zeta l(D_2)]. \end{aligned}$

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0902.0431_EQ1150	194	0.504	$\begin{aligned} & (\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v}))\boldsymbol{x} \\ & = \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j \\ & \quad - (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = (\text{ii}) \end{aligned}$	$(\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v}))\boldsymbol{x}$ $= \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j$ $- (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = (\text{ii})$	$(\boldsymbol{u} \times *(\boldsymbol{v} \wedge \boldsymbol{w}) + \boldsymbol{v} \times *(\boldsymbol{w} \wedge \boldsymbol{u}) + \boldsymbol{w} \times *(\boldsymbol{u} \wedge \boldsymbol{v}))\boldsymbol{x}$ $= \sum_{j=1}^9 (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{u}_j$ $- (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_9) \boldsymbol{x} = (\text{ii}).$
0902.0431_EQ0317	51	0.506	$\left(\left(\left[\delta_1, \delta_2\right], \delta_2\right)_4 + \left(\delta_1, \left[\delta, \delta_2\right]\right)_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$	$([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$	$([\delta, \delta_1], \delta_2)_4 + (\delta_1, [\delta, \delta_2])_4 = 0, \quad \delta, \delta_i \in \mathfrak{f}_4^C.$
0902.0431_EQ0593	99	0.507	$\begin{aligned} & \left\langle \varphi(A)M, \varphi(A)N \right\rangle = \left\langle k_J(\varphi(A)M), k_J(\varphi(A)N) \right\rangle \quad (\text{Lemma 3.11.1}) \\ & = \left\langle A(k_J(M))^t A, A(k_J(N))^t A \right\rangle = \left\langle k_J(M), k_J(N) \right\rangle \quad (\text{because } A \in SU(6)) \\ & = 2\langle M, N \rangle \end{aligned}$	$2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \left\langle A(k_J(M))^t A, A(k_J(N))^t A \right\rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle$	$2\langle \varphi(A)M, \varphi(A)N \rangle = \langle k_J(\varphi(A)M), k_J(\varphi(A)N) \rangle \quad (\text{Lemma 3.11.1})$ $= \langle A(k_J(M))^t A, A(k_J(N))^t A \rangle = \langle k_J(M), k_J(N) \rangle \quad (\text{because } A \in SU(6))$ $= 2\langle M, N \rangle,$
0902.0431_EQ0764	128	0.507	$\begin{aligned} & B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu' \\ & = \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5}) \\ & = 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right). \end{aligned}$	$B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$	$B_7(h, h') = \frac{3}{2} B_6 \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, \sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^{\sim} \right) + 24\nu\nu'$ $= \frac{3}{2} 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right) + 24\nu\nu' \quad (\text{Theorem 3.6.5})$ $= 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda_k' + 3 \sum_{j=1}^3 \mu_j \mu_j' + 4\nu\nu' \right).$
0902.0431_EQ0051	13	0.507	$\left(D_2 x \right) y + x \left(D_3^{\prime} y \right) = (\kappa D_1^{\prime})(xy), \quad x, y \in \mathfrak{C},$	$(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$	$(D_2 x) y + x (D_3' y) = (\kappa D_1')(xy), \quad x, y \in \mathfrak{C},$

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0902.0431_E00702 (iii)	116	■ 0.508	$\begin{aligned} & 3(\phi X, X, X) = 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ & \quad = (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ & \quad = 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0 \end{aligned}$	$\begin{aligned} 3(\phi X, X, X) &= 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ &= (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ &= 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0 \end{aligned}$	$\begin{aligned} 3(\phi X, X, X) &= 3(g(X), X, X) - (\nu + 2\mu)(X, X, X) \\ &= (\mu X \times X + h(X \times X), X) + (g(X), X \times X) - 3(\nu + 2\mu) \det X \\ &= 3\mu \det X + 3(\nu + \mu) \det X - 3(\nu + 2\mu) \det X = 0. \end{aligned}$
0902.0431_E00903	150	■ 0.510	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are real diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are real diagonal, } \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \text{ are real diagonal, } \xi > 0.$
0902.0431_EQ1101	187	■ 0.512	$\begin{aligned} & \mathfrak{e}_8^C = \left(\mathfrak{e}_8^C \right)_{\tilde{\lambda}\gamma} \oplus \left(\mathfrak{e}_8^C \right)_{-\tilde{\lambda}\gamma}, \\ & \left(\mathfrak{e}_8^C \right)_{\tilde{\lambda}\gamma} = \left\{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \right\} \cong \left\{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = R \right\} = (\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}, \\ & \left(\mathfrak{e}_8^C \right)_{-\tilde{\lambda}\gamma} = \left\{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \right\} \cong \left\{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = -R \right\}, \end{aligned}$	$\begin{aligned} & \mathfrak{e}_8^C = (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}, \\ & (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = R \} = (\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}, \\ & (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = -R \}, \end{aligned}$	$\begin{aligned} & \mathfrak{e}_8^C = (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}, \\ & (\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = \Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = R \} = (\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}, \\ & (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma} = \{ \Theta \in \text{Der}(\mathfrak{e}_8^C) \mid \tilde{\lambda}\gamma\Theta = -\Theta\tilde{\lambda}\gamma \} \cong \{ R \in \mathfrak{e}_8^C \mid \tilde{\lambda}\gamma R = -R \}, \end{aligned}$

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0902.0431_EQ0878	145	■ 0.528	$\begin{aligned} & \binom{\xi_1'}{\eta'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \binom{\xi_2'}{\eta_2'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix}, \\ & \binom{\xi_3'}{\eta_3'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \binom{\xi_4'}{\eta_4'} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix}, \\ & \binom{x_1'}{y_1'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \binom{x_2'}{y_2'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \binom{x_3'}{y_3'} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}. \end{aligned}$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$	$\begin{pmatrix} \xi_1' \\ \eta' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_1 \\ \eta \end{pmatrix}, \quad \begin{pmatrix} \xi_2' \\ \eta_2' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_2 \\ \eta_2 \end{pmatrix},$ $\begin{pmatrix} \xi_3' \\ \eta_3' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_3 \\ \eta_3 \end{pmatrix}, \quad \begin{pmatrix} \xi_4' \\ \eta_4' \end{pmatrix} = \begin{pmatrix} \nu & a \\ -\tau a & -\nu \end{pmatrix} \begin{pmatrix} \xi_4 \\ \eta_4 \end{pmatrix},$ $\begin{pmatrix} x_1' \\ y_1' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad \begin{pmatrix} x_2' \\ y_2' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, \quad \begin{pmatrix} x_3' \\ y_3' \end{pmatrix} = \begin{pmatrix} -\nu & \tau a \\ -a & \nu \end{pmatrix} \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}.$
0902.0431_EQ0945	157	■ 0.528	$\begin{aligned} & \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ & = \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ & = \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ & = f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})). \end{aligned}$	$\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ &= \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ &= \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ &= f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})) \end{aligned}$	$\begin{aligned} \overline{f^{-1}(D(fP))} &= \overline{\Phi(\phi_C(B, C), h(L), -\tau h(L), -ive_1)P} \\ &= \Phi(\overline{\phi_C(B, C)}, \overline{h(L)}, -\tau \overline{h(L)}, -\overline{ive_1})\overline{P} \\ &= \Phi(\phi_C(\overline{B}, \overline{C}), h(\overline{L}^*), -\tau h(\overline{L}^*), -ive_1)\overline{P} \\ &= f^{-1}((\overline{(\text{Ad } J_3)D})(f\overline{P})). \end{aligned}$
0902.0431_EQ1082	184	■ 0.533	$\begin{aligned} & \begin{aligned} & \left(\left(gX_1 - \frac{\xi_1}{2}E \right) + ik \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left(k \left(gX_2 - \frac{\xi_2}{2}E \right) - ik \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left(\left(gX_1 - \frac{\xi_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) - \left(gX_1 - \frac{\xi_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned} \end{aligned}$	$\begin{aligned} & \left(k \left(gX_1 - \frac{\xi_1}{2}E \right) + ik \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left(k \left(gX_2 - \frac{\xi_2}{2}E \right) - ik \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left(\left(gX_1 - \frac{\xi_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) - \left(gX_1 - \frac{\xi_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned}$	$\begin{aligned} & \left(k \left(gX_1 - \frac{\xi_1}{2}E \right) + ik \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \right) \left(k \left(gX_2 - \frac{\xi_2}{2}E \right) - ik \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) J^2 \\ & = -k \left(\left(gX_1 - \frac{\xi_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) + \left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right) \\ & \quad - ik \left(\left(g(\gamma Y_1) - \frac{\eta_1}{2}E \right) \left(gX_2 - \frac{\xi_2}{2}E \right) - \left(gX_1 - \frac{\xi_1}{2}E \right) \left(g(\gamma Y_2) - \frac{\eta_2}{2}E \right) \right). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E09989	165	■ 0.535	<pre> \begin{aligned} & \left.=\left(\begin{array}{c} \left[\Phi, \Phi_1 \right]+P \times Q_1-P_1 \times Q \\ \Phi P_1-\Phi_1 P+r P_1-r_1 P+s Q_1-s_1 Q \\ \Phi Q_1-\Phi_1 Q-r Q_1+r_1 Q+t P_1-t_1 P \\ -\frac{1}{8}\{P, Q_1\}+\frac{1}{8}\{P_1, Q\}+s t_1-s_1 t \\ \frac{1}{4}\{P, P_1\}+2 r s_1-2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\}-2 r t_1+2 r_1 t \end{array}\right) \left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array}\right)_8 \\ & =\cdots\left(\text { using }([\Phi, \Phi_1], \Phi_2)_7+(\Phi_1,[\Phi, \Phi_2])_7=0 \text { (Lemma 4.5.1.(1))},\right. \\ & \left.-(\Phi, P \times Q)_7=\{\Phi P, Q\} \text { (Lemma 4.5.1.(2)) etc.) } \cdots\right. \end{aligned} </pre>	$ \begin{aligned} & = \left(\begin{array}{c} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8}\{P, Q_1\} + \frac{1}{8}\{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4}\{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{array} \right), \left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array} \right)_8 \\ & = \cdots \left(\text{ using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))}, \right. \\ & \left. = -(\Phi, P \times Q)_7 = \{\Phi P, Q\} \text{ (Lemma 4.5.1.(2)) etc.) } \cdots \right. \end{aligned} $	$ \begin{aligned} & = \left(\begin{array}{c} [\Phi, \Phi_1] + P \times Q_1 - P_1 \times Q \\ \Phi P_1 - \Phi_1 P + r P_1 - r_1 P + s Q_1 - s_1 Q \\ \Phi Q_1 - \Phi_1 Q - r Q_1 + r_1 Q + t P_1 - t_1 P \\ -\frac{1}{8}\{P, Q_1\} + \frac{1}{8}\{P_1, Q\} + s t_1 - s_1 t \\ \frac{1}{4}\{P, P_1\} + 2 r s_1 - 2 r_1 s \\ -\frac{1}{4}\{Q, Q_1\} - 2 r t_1 + 2 r_1 t \end{array} \right), \left(\begin{array}{c} \Phi_2 \\ P_2 \\ Q_2 \\ r_2 \\ s_2 \\ t_2 \end{array} \right)_8 \\ & = \cdots \left(\text{using } ([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0 \text{ (Lemma 4.5.1.(1))}, \right. \\ & \quad \left. (\Phi, P \times Q)_7 = \{\Phi P, Q\} \text{ (Lemma 4.5.1.(2)) etc.) } \cdots \right. \\ & = -(R_1, [R, R_2])_8. \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00221	36	0.542	$\begin{aligned} (X, Y) &= \sum_{i=1}^3 \left(\xi_i \eta_i + 2(x_i, y_i) \right), \\ \operatorname{tr}(X, Y, Z) &= \sum_{i=1}^3 \left(\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. + \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2})) \right. \\ &\quad \left. + \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2})) \right), \end{aligned}$	$(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$	$(X, Y) = \sum_{i=1}^3 (\xi_i \eta_i + 2(x_i, y_i)),$ $\operatorname{tr}(X, Y, Z) = \sum_{i=1}^3 (\xi_i \eta_i \zeta_i + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2})$ $+ \xi_i((y_{i+1}, z_{i+1}) + (y_{i+2}, z_{i+2})) + \eta_i((z_{i+1}, x_{i+1}) + (z_{i+2}, x_{i+2}))$ $+ \zeta_i((x_{i+1}, y_{i+1}) + (x_{i+2}, y_{i+2}))),$
0902.0431_E00960	159	0.542	$\begin{aligned} &\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN) \\ &= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C \end{aligned}$	$\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$	$\beta(P_C + (M + N)) = P_C + Q(M + N) = P_C + (QM + QN)$ $= \psi(Q, E)(P_C + (M + N)), \quad P_C + (M + N) \in \mathfrak{P}^C$
0902.0431_E00005	6	0.542	$\begin{array}{l} \overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i, \\ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0. \end{array}$	$\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i$ $ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$	$\overline{x_0 + \sum_{i=1}^7 x_i e_i} = x_0 - \sum_{i=1}^7 x_i e_i, \quad \left(\sum_{i=0}^7 x_i e_i, \sum_{i=0}^7 y_i e_i \right) = \sum_{i=0}^7 x_i y_i,$ $ x = \sqrt{(x, x)}, \quad R\left(x_0 + \sum_{i=1}^7 x_i e_i\right) = x_0.$
0902.0431_E00097	19	0.544	$\begin{array}{l} \pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}), \\ \pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}), \\ \pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}). \end{array}$	$\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$	$\pm(\lambda_1 - \lambda_2) : \pm(G_{24} + G_{35}) + i(-G_{25} + G_{34}),$ $\pm(\lambda_1 - \lambda_3) : \pm(G_{26} + G_{37}) + i(-G_{27} + G_{36}),$ $\pm(\lambda_2 - \lambda_3) : \pm(G_{46} + G_{57}) + i(-G_{47} + G_{56}).$
0902.0431_E00949	158	0.546	$\begin{aligned} &\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right) \\ &= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right), \end{aligned}$	$\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right)$ $= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right)$	$\mu \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array} \right)$ $= \left(\frac{(M_{21} - M_{12})e_1}{2} + i \frac{M_{21} + M_{12}}{2}, \frac{(M_{22} + M_{11})e_1}{2} + i \frac{M_{22} - M_{11}}{2} \right),$
0902.0431_E00297	47	0.546	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$	$x \xrightarrow{G_{i0}} x_i e_0 - x_0 e_i \xrightarrow{G_{j0}(j \neq i)} x_i e_j \xrightarrow{x_j^{-1} G_{0j}} e_0 = 1.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E09953	158	■ 0.553	$\begin{aligned} & \backslash \text{begin}{aligned} \& \ \psi_{*}(D)(\mu \widetilde{M}) \backslash \& = \\ & \Phi \left(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & \backslash \& = \left(\begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left(\begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$	$\begin{aligned} & \psi_{*}(D)(\mu \widetilde{M}) \\ & = \Phi \left(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & = \left(\begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left(\begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$	$\begin{aligned} & \psi_{*}(D)(\mu \widetilde{M}) \\ & = \Phi \left(\phi_{\boldsymbol{C}}(B, C), h(L), -\tau h(L), -i \nu e_1 \right) (M, N) \\ & = \left(\begin{array}{cccc} \phi_{\boldsymbol{C}}(B, C) + \frac{1}{3} i \nu e_1 & -2 \tau h(L) & 0 & h(L) \\ 2 h(L) & \tau \phi_{\boldsymbol{C}}(B, C) \tau - \frac{1}{3} i \nu e_1 & -\tau h(L) & 0 \\ 0 & h(L) & -i \nu e_1 & 0 \\ -\tau h(L) & 0 & 0 & i \nu e_1 \end{array} \right) \left(\begin{array}{c} M \\ N \\ 0 \\ 0 \end{array} \right) \\ & = \left(\begin{array}{c} \phi_{\boldsymbol{C}}(B, C) M + \frac{1}{3} i \nu e_1 M - 2 \tau h(L) \times N \\ 2 h(l) \times M + \tau \phi_{\boldsymbol{C}}(B, C) \tau N - \frac{1}{3} i \nu e_1 N \\ (h(L), N) \\ -(\tau h(L), M) \end{array} \right) \\ & = \left(\begin{array}{c} -M h(B, C) + N \tau h(L) + \frac{1}{3} i \nu e_1 M \\ -M h(L) - N \tau h(B, C) - \frac{1}{3} i \nu e_1 N \\ 0 \\ 0 \end{array} \right) \end{aligned}$
0902.0431_E09440	72	■ 0.554	$F_{4(4)} = \left(F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau \sigma}.$	$F_{4(4)} = \left(F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau \sigma}.$	$F_{4(4)} = \left(F_4^C \right)^{\tau \gamma}, \quad F_{4(-20)} = \left(F_4^C \right)^{\tau \sigma}.$
0902.0431_E01174	197	■ 0.554	$\begin{aligned} & \backslash \text{begin}{gathered} \mathfrak{g}_{-0} = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \oplus \mathfrak{sl}(5, C), \backslash \& \mathfrak{g}_{-1} = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \\ & \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \\ & \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$	$\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$	$\begin{aligned} & \mathfrak{g}_0 = \mathfrak{sl}(5, C) \oplus \mathfrak{sl}(5, C), \\ & \mathfrak{g}_1 = C^5 \otimes \Lambda^2(C^5) = \mathfrak{g}_{-1}, \quad \mathfrak{g}_2 = \Lambda^2(C^5) \otimes C^5 = \mathfrak{g}_{-2}, \end{aligned}$
0902.0431_E09037	11	■ 0.555	$\begin{aligned} & \backslash \text{begin}{array}{lll} 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$	$\begin{aligned} & 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$	$\begin{aligned} & 1 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right), \quad \kappa \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 1 & 3 \end{array} \right), \quad \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 2 & 1 \end{array} \right), \\ & \nu \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 3 & 1 \end{array} \right), \quad \nu^2 \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 3 & 1 & 2 \end{array} \right), \quad \nu \pi \rightarrow \left(\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00370	61	0.557	$\begin{aligned} & \begin{aligned} & \eta_1(t)^2 + \eta_2(t)^2 + \eta_3(t)^2 + \\ & \eta_4(t)^2 \quad \& = \xi_1^2 + \left(\frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t + x_1 \sin 2t \right)^2 \\ & + \left(\frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t - x_1 \sin 2t \right)^2 \\ & + \left(\frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t + x_1 \sin 2t \right)^2 \\ & + \left(\frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t - x_1 \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\frac{\xi_2 - \xi_3}{2} \cos 2t + x_1 \sin 2t \right)^2 \\ & + 2 \left(\frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t - x_1 \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\left(\frac{\xi_2 - \xi_3}{2} \right)^2 + x_1 ^2 \right) \sin^2(2t + t_0) \text{ (for some } t_0 \in \boldsymbol{R} \text{)} \\ & \leq \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\left(\frac{\xi_2 - \xi_3}{2} \right)^2 + x_1 ^2 \right) \\ & = \xi_1^2 + \xi_2^2 + \xi_3^2 + 2 x_1 ^2 \end{aligned} \end{aligned}$	$\begin{aligned} & \eta_1(t)^2 + \eta_2(t)^2 + \eta_3(t)^2 \\ & = \xi_1^2 + \left(\frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2t + x_1 \sin 2t \right)^2 \\ & + \left(\frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t - x_1 \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\frac{\xi_2 - \xi_3}{2} \cos 2t + x_1 \sin 2t \right)^2 \\ & + 2 \left(\frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2t - x_1 \sin 2t \right)^2 \\ & = \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\left(\frac{\xi_2 - \xi_3}{2} \right)^2 + x_1 ^2 \right) \sin^2(2t + t_0) \text{ (for some } t_0 \in \boldsymbol{R} \text{)} \\ & \leq \xi_1^2 + 2 \left(\frac{\xi_2 + \xi_3}{2} \right)^2 + 2 \left(\left(\frac{\xi_2 - \xi_3}{2} \right)^2 + x_1 ^2 \right) \\ & = \xi_1^2 + \xi_2^2 + \xi_3^2 + 2 x_1 ^2 \end{aligned}$	
0902.0431_E00674 (i)	112	0.558	$2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$	$2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$	$2(P \times R)Q - (P \times Q)R - (R \times Q)P + \frac{3}{8}\{Q, R\}P - \frac{3}{8}\{P, Q\}R = 0.$
0902.0431_E01188	200	0.562	$\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$	$\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$	$\begin{aligned} & (a \times (b \wedge x))v + (b \times (a \wedge x))v = (x, w) * (a \wedge b) - \frac{1}{5}(* (a \wedge b), x)v \\ & = (x \times *(a \wedge b))v. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01095	186	■ 0.566	$\begin{aligned} & \zeta[l \left(S_1 \right) I, l \left(S_2 \right) I] = \zeta[l \left(S_1 \tau S_2 - S_2 \tau S_1 \right) \\ & I, l \left(S_2 \tau S_1 - S_1 \tau S_2 \right) I] = \zeta[l \left(S_1 \tau S_2 - S_2 \tau S_1 \right) E \\ & + l \left(S_2 \tau S_1 - S_1 \tau S_2 \right) E] \\ & = \zeta \left(l \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right. \\ & \left. + l \left(\frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right) \\ & = \Theta \left(\varphi_* \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right), 0, 0, 0 \right. \\ & \left. - \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1), \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) \right) \\ & = \Theta (4 (\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1), 0, 0, 0, \\ & \{ \chi^{-1} S_1, \chi^{-1} S_2 \} - \{ \chi^{-1} S_1, \chi^{-1} S_2 \}) \text{ (Lemma 5.8.3).} \end{aligned}$	$\begin{aligned} & \zeta[l \left(S_1 \right) I, l \left(S_2 \right) I] = \zeta[l \left(S_1 \tau S_2 - S_2 \tau S_1 \right) \\ & = \zeta \left(l \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right. \\ & \left. + l \left(\frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right) \\ & = \Theta \left(\varphi_* \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right), 0, 0, 0 \right. \\ & \left. - \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1), \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) \right) \\ & = \Theta (4 (\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1), 0, 0, 0 \\ & \{ \chi^{-1} S_1, \chi^{-1} S_2 \} - \{ \chi^{-1} S_1, \chi^{-1} S_2 \}) \text{ (Lemma 5.8.3).} \end{aligned}$	$\begin{aligned} & \zeta[l \left(S_1 \right) I, l \left(S_2 \right) I] = \zeta[l \left(S_1 \tau S_2 - S_2 \tau S_1 \right) \\ & = \zeta \left(l \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right. \\ & \left. + l \left(\frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right) \right) \\ & = \Theta \left(\varphi_* \left(S_1 \tau S_2 - S_2 \tau S_1 - \frac{1}{8} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) E \right), 0, 0, 0, \right. \\ & \left. - \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1), \frac{i}{4} \text{tr} (S_1 \tau S_2 - S_2 \tau S_1) \right) \\ & = \Theta (4 (\lambda \gamma \chi^{-1} S_1 \times \chi^{-1} S_2 - \lambda \gamma \chi^{-1} S_2 \times \chi^{-1} S_1), 0, 0, 0, \\ & \{ \chi^{-1} S_1, \chi^{-1} S_2 \} - \{ \chi^{-1} S_1, \chi^{-1} S_2 \}) \text{ (Lemma 5.8.3).} \end{aligned}$
0902.0431_E01179	199	■ 0.568	$\begin{aligned} & (*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) + *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) + \boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{c}) \\ & = (*(\boldsymbol{b} \wedge \boldsymbol{x}, \boldsymbol{c} \wedge \boldsymbol{a}) + (*(\boldsymbol{a} \wedge \boldsymbol{x}, \boldsymbol{b} \wedge \boldsymbol{c}) + (*(\boldsymbol{c} \wedge \boldsymbol{x}, \boldsymbol{a} \wedge \boldsymbol{b}) \\ & = (*\boldsymbol{x}, *\boldsymbol{b} \wedge *(\boldsymbol{c} \wedge \boldsymbol{a}) + *\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{c}) + *\boldsymbol{c} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b})) = 0. \end{aligned}$	$\begin{aligned} & (*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) + *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) + \boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{c}) \\ & = (*(\boldsymbol{b} \wedge \boldsymbol{x}, \boldsymbol{c} \wedge \boldsymbol{a}) + (*(\boldsymbol{a} \wedge \boldsymbol{x}, \boldsymbol{b} \wedge \boldsymbol{c}) + (*(\boldsymbol{c} \wedge \boldsymbol{x}, \boldsymbol{a} \wedge \boldsymbol{b}) \\ & = (*\boldsymbol{x}, *\boldsymbol{b} \wedge *(\boldsymbol{c} \wedge \boldsymbol{a}) + *\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{c}) + *\boldsymbol{c} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b})) = 0. \end{aligned}$	$\begin{aligned} & (*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) + *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) + \boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{c}) \\ & = (*(\boldsymbol{b} \wedge \boldsymbol{x}, \boldsymbol{c} \wedge \boldsymbol{a}) + (*(\boldsymbol{a} \wedge \boldsymbol{x}, \boldsymbol{b} \wedge \boldsymbol{c}) + (*(\boldsymbol{c} \wedge \boldsymbol{x}, \boldsymbol{a} \wedge \boldsymbol{b}) \\ & = (*\boldsymbol{x}, *\boldsymbol{b} \wedge *(\boldsymbol{c} \wedge \boldsymbol{a}) + *\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{c}) + *\boldsymbol{c} \wedge *(\boldsymbol{a} \wedge \boldsymbol{b})) = 0. \end{aligned}$
0902.0431_E00454	74	■ 0.570	$=(\delta X, Y) - (\widetilde{T} X, Y) = ((\delta - \widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C.$	$= (\delta X, Y) - (\widetilde{T} X, Y) = ((\delta - \widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C.$	$= (\delta X, Y) - (\widetilde{T} X, Y) = ((\delta - \widetilde{T}) X, Y) \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E01000	166	■ 0.572	$\begin{aligned} & \lambda(\Phi, P, Q, r, s, t) = (\lambda \Phi \lambda^{-1}, \lambda P, \lambda Q, r, s, t), \\ & \lambda'(\Phi, P, Q, r, s, t) = (\Phi, Q, -P, -r, -t, -s), \end{aligned}$	$\begin{aligned} & \lambda(\Phi, P, Q, r, s, t) = (\lambda \Phi \lambda^{-1}, \lambda P, \lambda Q, r, s, t), \\ & \lambda'(\Phi, P, Q, r, s, t) = (\Phi, Q, -P, -r, -t, -s), \end{aligned}$	$\begin{aligned} & \lambda(\Phi, P, Q, r, s, t) = (\lambda \Phi \lambda^{-1}, \lambda P, \lambda Q, r, s, t), \\ & \lambda'(\Phi, P, Q, r, s, t) = (\Phi, Q, -P, -r, -t, -s), \end{aligned}$

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0902.0431_EQ0067	15	■ 0.573	$a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$	$a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$	$a+\left(\begin{array}{l} m_1 \\ m_2 \\ m_3 \end{array}\right) \cdot$
0902.0431_EQ1027	171	■ 0.577	$\begin{array}{l} \backslash begin{array}{l} \backslash lambda_{\{0\}}+\frac{1}{2}\backslash left(\\ \backslash mu_{\{2\}}-\backslash mu_{\{3\}}\backslash right)=2 \& 4 \& 6 \& 5 \& 3 \& 2 \& 1 \& 3 \backslash \backslash \\ \backslash lambda_{\{1\}}+\frac{1}{2}\backslash left(\backslash mu_{\{2\}}-\backslash mu_{\{3\}}\backslash right)=0 \& 1 \\ \& 2 \& 2 \& 1 \& 1 \& 1 \& 1 \end{array}$	$\begin{array}{l} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=2 \quad 4 \quad 6 \quad 5 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$	$\begin{array}{l} \lambda_0+\frac{1}{2}(\mu_2-\mu_3)=2 \quad 4 \quad 6 \quad 5 \quad 3 \quad 2 \quad 1 \quad 3 \\ \lambda_1+\frac{1}{2}(\mu_2-\mu_3)=0 \quad 1 \quad 2 \quad 2 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$
0902.0431_EQ1047	178	■ 0.578	$\begin{array}{l} \backslash begin{aligned} \& -\backslash left(\backslash beta_{\{12\}} P, \backslash beta_{\{2\}} P, \\ \backslash beta_{\{32\}} P, b_{\{1\}} P, b_{\{2\}} P, b_{\{3\}} P\backslash right) \backslash \& \\ \backslash quad=\backslash left[\backslash left(\backslash beta_{\{12\}} P, \backslash beta_{\{2\}} P, \backslash beta_{\{32\}} \\ P, b_{\{1\}} P, b_{\{2\}} P, b_{\{3\}} P\backslash right), (0,0,0,1,0,0)\backslash right] \\ \backslash \& \backslash quad=\backslash left(0,-\backslash beta_{\{2\}} P, \backslash beta_{\{32\}} P, 0,-2 b_{\{2\}} \\ P, 2 b_{\{3\}} P\backslash right) \end{array}$	$\begin{array}{l} -(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P) \\ =[(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P), (0,0,0,1,0,0)] \\ =(0, -\beta_2P, \beta_{32}P, 0, -2b_2P, 2b_3P) \end{array}$	$\begin{array}{l} -(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P) \\ =[(\beta_{12}P, \beta_2P, \beta_{32}P, b_1P, b_2P, b_3P), (0,0,0,1,0,0)] \\ =(0, -\beta_2P, \beta_{32}P, 0, -2b_2P, 2b_3P), \end{array}$
0902.0431_EQ0075	16	■ 0.579	$\begin{array}{l} \backslash begin{aligned} S_{\{1\}} \& =2 G_{\{12\}}-G_{\{47\}}-G_{\{56\}}, \\ \backslash quad S_{\{2\}}=2 G_{\{13\}}-G_{\{46\}}+G_{\{57\}}, \backslash \backslash S_{\{3\}} \& =2 \\ G_{\{14\}}+G_{\{27\}}+G_{\{36\}}, \backslash quad S_{\{4\}}=2 G_{\{15\}}+G_{\{26\}}-G_{\{37\}}, \\ \backslash \backslash S_{\{5\}} \& =2 G_{\{16\}}-G_{\{25\}}-G_{\{34\}}, \backslash quad S_{\{6\}}=2 \\ G_{\{17\}}-G_{\{24\}}+G_{\{35\}}, \end{array}$	$\begin{array}{l} S_1=2G_{12}-G_{47}-G_{56}, \quad S_2=2G_{13}-G_{46}+G_{57}, \\ S_3=2G_{14}+G_{27}+G_{36}, \quad S_4=2G_{15}+G_{26}-G_{37}, \\ S_5=2G_{16}-G_{25}-G_{34}, \quad S_6=2G_{17}-G_{24}+G_{35}, \end{array}$	$\begin{array}{l} S_1=2G_{12}-G_{47}-G_{56}, \quad S_2=2G_{13}-G_{46}+G_{57}, \\ S_3=2G_{14}+G_{27}+G_{36}, \quad S_4=2G_{15}+G_{26}-G_{37}, \\ S_5=2G_{16}-G_{25}-G_{34}, \quad S_6=2G_{17}-G_{24}+G_{35}, \end{array}$
0902.0431_EQ1176	198	■ 0.582	$\begin{array}{l} \backslash begin{aligned} (*(* \backslash boldsymbol{a} \backslash wedge \& \\ *(\backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c}))), \\ \backslash boldsymbol{x})=(\backslash boldsymbol{a}, *(\backslash boldsymbol{b} \\ \backslash wedge \backslash boldsymbol{c})) \backslash wedge \backslash boldsymbol{x}) \backslash \& = \\ \backslash left(\backslash boldsymbol{a}_{\{1\}}, *(\backslash boldsymbol{b} \backslash wedge \\ \backslash boldsymbol{c})\backslash right)\backslash left(\backslash boldsymbol{a}_{\{2\}}, \\ \backslash boldsymbol{x}\backslash right)-\backslash left(\backslash boldsymbol{a}_{\{2\}}, * \\ \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c})\backslash right)\backslash left(\\ \backslash boldsymbol{a}_{\{1\}}, \backslash boldsymbol{x}\backslash right) \backslash \& = \\ \backslash left(\backslash boldsymbol{a}_{\{1\}} \backslash wedge \backslash boldsymbol{b} \backslash wedge \\ \backslash boldsymbol{c}, \backslash boldsymbol{e}_{\{1\}} \backslash wedge \backslash cdots \backslash wedge \\ \backslash boldsymbol{e}_{\{5\}}\backslash right)\backslash left(\backslash boldsymbol{a}_{\{2\}}, \\ \backslash boldsymbol{x}\backslash right)-\backslash left(\backslash boldsymbol{a}_{\{2\}} \backslash wedge \\ \backslash boldsymbol{b} \backslash wedge \backslash boldsymbol{c}, \backslash boldsymbol{e}_{\{1\}} \\ \backslash wedge \backslash cdots \backslash wedge \backslash boldsymbol{e}_{\{5\}}\backslash right)\backslash left(\\ \backslash boldsymbol{a}_{\{1\}}, \backslash boldsymbol{x}\backslash right), \end{array}$	$\begin{array}{l} (*(*\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})),\boldsymbol{x})=(\boldsymbol{a},*(\boldsymbol{b}\wedge\boldsymbol{c})\wedge\boldsymbol{x}) \\ =(\boldsymbol{a}_1,*(\boldsymbol{b}\wedge\boldsymbol{c}))(\boldsymbol{a}_2,\boldsymbol{x})-(\boldsymbol{a}_2,*(\boldsymbol{b}\wedge\boldsymbol{c}))(\boldsymbol{a}_1,\boldsymbol{x}) \\ =(\boldsymbol{a}_1\wedge\boldsymbol{b}\wedge\boldsymbol{c},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5)(\boldsymbol{a}_2,\boldsymbol{x})-(\boldsymbol{a}_2\wedge\boldsymbol{b}\wedge\boldsymbol{c},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5)(\boldsymbol{a}_1,\boldsymbol{x}), \end{array}$	$\begin{array}{l} (*(*\boldsymbol{a}\wedge*(\boldsymbol{b}\wedge\boldsymbol{c})),\boldsymbol{x})=(\boldsymbol{a},*(\boldsymbol{b}\wedge\boldsymbol{c})\wedge\boldsymbol{x}) \\ =(\boldsymbol{a}_1,*(\boldsymbol{b}\wedge\boldsymbol{c}))(\boldsymbol{a}_2,\boldsymbol{x})-(\boldsymbol{a}_2,*(\boldsymbol{b}\wedge\boldsymbol{c}))(\boldsymbol{a}_1,\boldsymbol{x}) \\ =(\boldsymbol{a}_1\wedge\boldsymbol{b}\wedge\boldsymbol{c},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5)(\boldsymbol{a}_2,\boldsymbol{x})-(\boldsymbol{a}_2\wedge\boldsymbol{b}\wedge\boldsymbol{c},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_5)(\boldsymbol{a}_1,\boldsymbol{x}), \end{array}$
0902.0431_EQ0728 (b)	120	■ 0.589	$\begin{array}{l} \backslash begin{aligned} \& W \backslash ni \backslash Phi(0, X, 0,0)(0,0,0,1)=(X, \\ 0,0,0) \backslash \& W \backslash ni \backslash Phi\backslash left(0, E_{\{2\}}, 0,0\backslash right) \\ \backslash left(E_{\{3\}}, 0,0,0\backslash right)=\backslash left(0, E_{\{1\}}, 0,0\backslash right) \backslash \& \\ W \backslash ni \backslash Phi\backslash left(0, E_{\{1\}}, 0,0\backslash right)\backslash left(0, E_{\{1\}}, 0,0 \\ \backslash right)=(0,0,1,0) \backslash \& W \backslash ni \backslash Phi(0,0, Y, 0)(0,0,1,0)=(0, \\ Y, 0,0) \end{array}$	$\begin{array}{l} W \ni \Phi(0, X, 0, 0)(0, 0, 0, 1) = (X, 0, 0, 0) \\ W \ni \Phi(0, E_2, 0, 0)(E_3, 0, 0, 0) = (0, E_1, 0, 0) \\ W \ni \Phi(0, E_1, 0, 0)(0, E_1, 0, 0) = (0, 0, 1, 0) \\ W \ni \Phi(0, 0, Y, 0)(0, 0, 1, 0) = (0, Y, 0, 0) \end{array}$	$\begin{array}{l} W \ni \Phi(0, X, 0, 0)(0, 0, 0, 1) = (X, 0, 0, 0), \\ W \ni \Phi(0, E_2, 0, 0)(E_3, 0, 0, 0) = (0, E_1, 0, 0), \\ W \ni \Phi(0, E_1, 0, 0)(0, E_1, 0, 0) = (0, 0, 1, 0), \\ W \ni \Phi(0, 0, Y, 0)(0, 0, 1, 0) = (0, Y, 0, 0). \end{array}$

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0902.0431_E08688	114	■ 0.589	$\begin{aligned} & \left[\Phi_1, \Phi_2 \right] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P \\ & = \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots \\ & = \Phi P \end{aligned}$	$[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P$	$[\Phi_1, \Phi_2] P = \Phi_1 \Phi_2 P - \Phi_2 \Phi_1 P$ $= \cdots \text{ (using the formula of } A \vee B \text{ (Lemma 3.4.1) etc.) } \cdots$ $= \Phi P.$
0902.0431_E08350	57	■ 0.589	$\begin{aligned} & R(\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right) \\ & =\left(\alpha_3\left(\overline{x_1 x_2}\right), \alpha_3 x_3\right)=\left(\overline{x_1 x_2}, x_3\right)=R\left(x_1 x_2 x_3\right) \end{aligned}$	$R\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right)$ $=\left(\alpha_3\left(\overline{x_1 x_2}\right), \alpha_3 x_3\right)=\left(\overline{x_1 x_2}, x_3\right)=R\left(x_1 x_2 x_3\right)$	$R\left(\left(\alpha_1 x_1\right)\left(\alpha_2 x_2\right)\left(\alpha_3 x_3\right)\right)=R\left(\left(\overline{\alpha_3\left(x_1 x_2\right)}\right) \alpha_3 x_3\right)$ $=\left(\alpha_3\left(\overline{x_1 x_2}\right), \alpha_3 x_3\right)=\left(\overline{x_1 x_2}, x_3\right)=R\left(x_1 x_2 x_3\right),$
0902.0431_E08870	144	■ 0.590	$\begin{array}{l} \text{\texttt{\textbackslash begin{array}{clcccccc} 1 \& \text{\texttt{\textbackslash longrightarrow} \& } \\ \text{\texttt{\textbackslash operatorname{Spin}}(11) \& \text{\texttt{\textbackslash longrightarrow} \& \text{\texttt{\textbackslash left(E_{7}} \\ \text{\texttt{\textbackslash right)}^{\texttt{\textbackslash kappa}, \texttt{\textbackslash mu} \& \text{\texttt{\textbackslash longrightarrow} \& S^{\texttt{\textbackslash 11} \& } \\ \text{\texttt{\textbackslash longrightarrow} \& * \texttt{\textbackslash \& \& \texttt{\textbackslash downarrow} p^{\texttt{\textbackslash prime} \& \& } \\ \text{\texttt{\textbackslash downarrow} p \& \& \texttt{\textbackslash downarrow}=\& \& \texttt{\textbackslash 1 \& \texttt{\textbackslash longrightarrow} \& } \\ \text{\texttt{\textbackslash S 0(11) \& \texttt{\textbackslash longrightarrow} \& S 0(12) \& \texttt{\textbackslash longrightarrow} \& } \\ \text{\texttt{\textbackslash S^{\texttt{\textbackslash 11} \& \texttt{\textbackslash longrightarrow} \& * \texttt{\textbackslash end{array}}} \end{array}$	$\begin{array}{ccccccc} 1 & \longrightarrow & \text{Spin}(11) & \longrightarrow & (E_7)^{\kappa, \mu} & \longrightarrow & S^{11} & \longrightarrow & * \\ & & \downarrow p' & & \downarrow p & & \downarrow = & & \\ 1 & \longrightarrow & SO(11) & \longrightarrow & SO(12) & \longrightarrow & S^{11} & \longrightarrow & * \end{array}$	$\begin{array}{ccccccc} 1 & \longrightarrow & Spin(11) & \longrightarrow & (E_7)^{\kappa, \mu} & \longrightarrow & S^{11} & \longrightarrow & * \\ & & \downarrow p' & & \downarrow p & & \downarrow = & & \\ 1 & \longrightarrow & SO(11) & \longrightarrow & SO(12) & \longrightarrow & S^{11} & \longrightarrow & * \end{array}$
0902.0431_E08324	52	■ 0.591	$\begin{aligned} & \widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i), \\ & \widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i), \\ & \text{the others} = 0. \end{aligned}$	$\widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{4} F_2(e_i),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i),$ $\text{the others} = 0.$	$\widetilde{A}_1(1) \widetilde{A}_1(1) E_2 = -\frac{1}{2} \widetilde{A}_1(1) F_1(1) = -\frac{1}{2} (E_2 - E_3),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) E_3 = \frac{1}{2} \widetilde{A}_1(1) F_1(1) = \frac{1}{2} (E_2 - E_3),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_1(1) = \widetilde{A}_1(1) (E_2 - E_3) = -F_1(1),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_2(e_i) = \frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_2(e_i),$ $\widetilde{A}_1(1) \widetilde{A}_1(1) F_3(e_i) = -\frac{1}{2} \widetilde{A}_1(1) F_3(\overline{e_i}) = -\frac{1}{4} F_3(e_i),$ $\text{the others} = 0.$

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0902.0431_E01083	184	■ 0.593	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k \left(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1) \right) \\ &\quad - i k \left(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1) \right) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + i k (g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k \left(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1) \right) \\ &\quad - i k \left(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1) \right) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + i k (g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 - S_2 \tau S_1 &= -k \left(g X_1 g X_2 - g X_2 g X_1 + g(\gamma Y_1) g(\gamma Y_2) - g(\gamma Y_2) g(\gamma Y_1) \right) \\ &\quad - i k \left(g(\gamma Y_1) g X_2 - g(\gamma Y_2) g X_1 - g X_1 g(\gamma Y_2) + g X_2 g(\gamma Y_1) \right) \\ &\quad - \eta_1 g X_2 + \eta_2 g X_1 + \xi_1 g(\gamma Y_2) - \xi_2 g(\gamma Y_1) + \frac{i}{2} (\xi_1 \eta_2 - \eta_2 \xi_1) E \\ &= k(-[g X_1, g X_2] - [g(\gamma Y_1), g(\gamma Y_2)]) \\ &\quad + i k (g(2\gamma X_1 \times Y_2 - 2\gamma X_2 \times Y_1 + \eta_1 X_2 - \eta_2 X_1 - \xi_1 \gamma Y_2 + \xi_2 \gamma Y_1)) \\ &\quad + \frac{i}{2} ((X_1, Y_2) - (X_2, Y_1) + \xi_1 \eta_2 - \xi_2 \eta_1) E \quad (\text{Lemma 3.12.1}) \end{aligned}$
0902.0431_E00678	112	■ 0.595	$\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$	$\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$	$\left(\begin{array}{c} X \\ \frac{1}{\eta} (X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right), \quad \left(\begin{array}{c} \frac{1}{\xi} (Y \times Y) \\ Y \\ \xi \\ \frac{1}{\xi^2} \det Y \end{array} \right), \quad \mathbf{i} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

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0902.0431_E0671	111	■ 0.606	$P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$	$P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$	$P \times Q = \Phi(\phi, A, B, \nu), \left\{ \begin{array}{l} \phi = -\frac{1}{2}(X \vee W + Z \vee Y) \\ A = -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ B = \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \nu = \frac{1}{8}((X, W) + (Z, Y) - 3(\xi\omega + \zeta\eta)). \end{array} \right.$
0902.0431_E0924	153	■ 0.608	$f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$	$f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$	$f(h(a)P) = a(fP), \quad a \in C, P \in (\mathfrak{P}_C)^C.$
0902.0431_E01103	187	■ 0.608	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma} \oplus (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}.$
0902.0431_E0648	107	■ 0.608	$\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$	$\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$	$\begin{aligned} & \varphi((A, B), \epsilon) \varphi((C, D), 1) X \\ &= \varphi((A, B), \epsilon) (h(C, D) X h(C, D)^*) \\ &= h(A, B) \overline{h(C, D) X h(C, D)^*} h(A, B)^* \\ &= h(A, B) h(\overline{D}, \overline{C}) \overline{X} h(\overline{D}, \overline{C})^* h(A, B)^* \quad (\text{Lemma 3.13.3.(3)}) \\ &= h(A \overline{D}, B \overline{C}) \overline{X} h(A \overline{D}, B \overline{C})^* = \varphi((A \overline{D}, B \overline{C}), \epsilon) X \\ &= \varphi((A, B) \epsilon(C, D), \epsilon) X, \quad X \in (\mathfrak{I}_C)^C. \end{aligned}$
0902.0431_E0837	54	■ 0.622	$\begin{aligned} & \kappa \pi H = \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3, \end{aligned}$	$\begin{aligned} & \kappa \pi H = \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3 \end{aligned}$	$\begin{aligned} \kappa \pi H &= \frac{1}{2} (-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) H_0 + \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) H_1 \\ & \quad + \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) H_2 + \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) H_3, \end{aligned}$
0902.0431_E0863	143	■ 0.623	$\begin{aligned} & V^{12} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$	$\begin{aligned} & V^{12} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$	$\begin{aligned} V^{12} &= \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P\} \\ &= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi, \eta \in C \right\} \end{aligned}$

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0902.0431_E01155	195	■ 0.623	$\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{sl}(9,C)\oplus \mathfrak{q},\quad \mathfrak{q}=\Lambda^3(C^9)\oplus \Lambda^3(C^9).$	$\mathfrak{e}_8^C=\mathfrak{sl}(9,C)\oplus \mathfrak{q},\quad \mathfrak{q}=\Lambda^3(C^9)\oplus \Lambda^3(C^9).$	$\mathfrak{e}_8^C=\mathfrak{sl}(9,C)\oplus \mathfrak{q},\quad \mathfrak{q}=\Lambda^3(C^9)\oplus \Lambda^3(C^9).$
0902.0431_E00319	51	■ 0.625	$B_{\{4\}}(\delta_1,\delta_2,\delta_3)=9\left(\delta_1,\delta_2,\delta_3\right)_4=3\operatorname{tr}\left(\delta_1\delta_2\delta_3\right),\quad \delta_1,\delta_2,\delta_3\in\mathfrak{f}_4^C.$	$B_4(\delta_1,\delta_2)=9(\delta_1,\delta_2)_4=3\operatorname{tr}(\delta_1\delta_2),\quad \delta_1,\delta_2\in\mathfrak{f}_4^C.$	$B_4(\delta_1,\delta_2)=9(\delta_1,\delta_2)_4=3\operatorname{tr}(\delta_1\delta_2),\quad \delta_1,\delta_2\in\mathfrak{f}_4^C.$
0902.0431_E00677	112	■ 0.625	$\begin{aligned} \mathfrak{M}^C&=\left\{P\in\mathfrak{P}^C\mid P\times P=0,P\neq 0\right\}\\ &=\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C\\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta,\\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}.\end{aligned}$	$\mathfrak{M}^C=\left\{P\in\mathfrak{P}^C\mid P\times P=0,P\neq 0\right\}\\ =\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C\\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta,\\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}.$	$\mathfrak{M}^C=\left\{P\in\mathfrak{P}^C\mid P\times P=0,P\neq 0\right\}\\ =\left\{\begin{array}{l} P=(X,Y,\xi,\eta)\in\mathfrak{P}^C\\ P\neq 0 \end{array}\left \begin{array}{l} X\vee Y=0,(X,Y)=3\xi\eta,\\ X\times X=\eta Y,Y\times Y=\xi X \end{array}\right.\right\}.$
0902.0431_E00472	78	■ 0.629	$\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X)\\ &\quad -2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X)\\ &\quad +\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X)\\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C.\end{aligned}$	$\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X)\\ &\quad -2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X)\\ &\quad +\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X)\\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C.\end{aligned}$	$\begin{aligned} &=\phi\left(\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X)\right)-(A\vee B)\phi X\text{ (Lemma 3.4.1)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(A,B)\phi X-2\phi'B\times(A\times X)-2B\times(\phi A\times X)\\ &\quad -2B\times(A\times\phi X)-\frac{1}{2}(B,\phi X)A-\frac{1}{6}(A,B)\phi X+2B\times(A\times\phi X)\text{ (Lemma 3.4.3)}\\ &=\frac{1}{2}(B,X)\phi A+\frac{1}{6}(\phi A,B)X-2B\times(\phi A\times X)\\ &\quad +\frac{1}{2}(\phi'B,X)A+\frac{1}{6}(A,\phi'B)X-2\phi'B\times(A\times X)\\ &=(\phi A\vee B)X+(A\vee\phi'B)X,\quad X\in\mathfrak{J}^C.\end{aligned}$
0902.0431_E00244 (5 Rightarrow4)	40	■ 0.631	$\begin{aligned} &\&(\operatorname{det}(\alpha X))\alpha X=(\alpha X\times\alpha X)\times(\alpha X\times\alpha X)\text{ (Lemma 2.1.1.}\\ &=\alpha((X\times X)\times(X\times X))=(\operatorname{det}X)\alpha X\text{ (Lemma 2.1.1.}\\ &\end{aligned}$	$\begin{aligned} &(\det(\alpha X))\alpha X=(\alpha X\times\alpha X)\times(\alpha X\times\alpha X)\text{ (Lemma 2.1.1.}\\ &=\alpha((X\times X)\times(X\times X))=(\det X)\alpha X\text{ (Lemma 2.1.1.}\\ &\end{aligned}$	$\begin{aligned} &(\det(\alpha X))\alpha X=(\alpha X\times\alpha X)\times(\alpha X\times\alpha X)\text{ (Lemma 2.1.1.}\\ &=\alpha((X\times X)\times(X\times X))=(\det X)\alpha X\text{ (Lemma 2.1.1.}\\ &\end{aligned}$
0902.0431_E01093	185	■ 0.633	$\begin{aligned} &\&{\left[\zeta l(D),\zeta l(S)I\right]}\&\\ &=\left[\Theta\left(\varphi_*D,0,0,0,0,0\right),\Theta\left(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0\right)\right] \\ &=\Theta\left(0,2\left(\varphi_*D\right)\lambda\gamma\chi^{-1}S,2\left(\varphi_*D\right)\chi^{-1}S,0,0,0\right) \end{aligned}$	$\begin{aligned} &[\zeta l(D),\zeta l(S)I] \\ &=\left[\Theta\left(\varphi_*D,0,0,0,0,0\right),\Theta\left(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0\right)\right] \\ &=\Theta\left(0,2\left(\varphi_*D\right)\lambda\gamma\chi^{-1}S,2\left(\varphi_*D\right)\chi^{-1}S,0,0,0\right) \end{aligned}$	$\begin{aligned} &[\zeta l(D),\zeta l(S)I] \\ &=\left[\Theta\left(\varphi_*D,0,0,0,0,0\right),\Theta\left(0,2\lambda\gamma\chi^{-1}S,2\chi^{-1}S,0,0,0\right)\right] \\ &=\Theta\left(0,2\left(\varphi_*D\right)\lambda\gamma\chi^{-1}S,2\left(\varphi_*D\right)\chi^{-1}S,0,0,0\right). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00436	72	■ 0.637	$\begin{aligned} \mathfrak{J}\left(3, \mathfrak{C}'\right) &= \left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\}, \\ \mathfrak{J}\left(1,2, \mathfrak{C}\right) &= \left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\}, \end{aligned}$	$\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$	$\mathfrak{J}\left(3, \mathfrak{C}'\right)=\left\{X \in M\left(3, \mathfrak{C}'\right) \mid X^*=X\right\},$ $\mathfrak{J}\left(1,2, \mathfrak{C}\right)=\left\{X \in M\left(3, \mathfrak{C}\right) \mid I_1 X^* I_1=X\right\},$
0902.0431_E01067	181	■ 0.642	$\begin{aligned} \left(E_8\right)^{\widetilde{\lambda} \gamma} &=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\} \\ &=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} . \end{aligned}$	$\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$	$\left(E_8\right)^{\widetilde{\lambda} \gamma}=\left\{\alpha \in E_8 \mid \widetilde{\lambda} \gamma \alpha=\alpha \widetilde{\lambda} \gamma\right\}$ $=\left\{\alpha \in E_8 \mid \tau \gamma \alpha=\alpha \tau \gamma\right\}=\left(E_8\right)^{\tau \gamma} .$
0902.0431_E00811	135	■ 0.647	$\begin{aligned} &\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0) \\ &=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right) \end{aligned}$	$\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right)$	$\left(0, \omega' X, 0,0\right)=-\lambda\left(\omega' X, 0,0,0\right)=-\lambda \alpha(X, 0,0,0)$ $=-\alpha \lambda(X, 0,0,0)=\alpha(0, X, 0,0)=\left(0, \omega'^{-1} X, 0,0\right),$
0902.0431_E00477	79	■ 0.651	$\begin{aligned} &\left(\left[\delta, \delta_1\right]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 \end{aligned}$	$\begin{aligned} &=\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 . \end{aligned}$	$\begin{aligned} &=\left(\left([\delta, \delta_1]+\left[\widetilde{T}, \widetilde{T}_1\right]\right)+\left(\widetilde{\delta T_1}-\widetilde{\delta_1 T}\right), \delta_2+\widetilde{T}_2\right)_6 \quad(\text { Theorem 3.2.2 }) \\ &=\left([\delta, \delta_1], \delta_2\right)_4+\left([\widetilde{T}, \widetilde{T}_1], \delta_2\right)_4+\left(\delta T_1-\delta_1 T, T_2\right) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4+\left(\delta_2 T, T_1\right)+\left(\delta T_1, T_2\right)-\left(\delta_1 T, T_2\right) \quad(\text { Lemma 2.5.2 }) \\ &=-\left(\delta_1,\left[\delta, \delta_2\right]\right)_4-\left(\delta_1 T, T_2\right)-\left(T_1, \delta T_2\right)+\left(T_1, \delta_2 T\right) \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta, \delta_2\right]+\left[\widetilde{T}, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta T_2}-\widetilde{\delta_2 T}\right)_6 \\ &=-\left(\delta_1+\widetilde{T}_1,\left[\delta+\widetilde{T}, \delta_2+\widetilde{T}_2\right]\right)_6 \\ &=-\left(\phi_1,\left[\phi, \phi_2\right]\right)_6 . \end{aligned}$

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0902.0431_E00522	86	0.651	$\begin{aligned} & \& H_{\alpha_1}=\frac{1}{24}\left(H_{\{0\}}-H_{\{1\}}\right), \quad H_{\alpha_2}=\frac{1}{24}\left(H_{\{1\}}-H_{\{2\}}\right), \quad H_{\alpha_3}=\frac{1}{24}\left(H_{\{2\}}-H_{\{3\}}\right), \\ & H_{\alpha_4}=\frac{1}{24}\left(H_{\{3\}}+\left(E_2-E_3\right)^{\sim}\right), \\ & H_{\alpha_5}=\frac{1}{48}\left(\left(-H_0-H_1-H_2+H_3\right)+2\left(E_3-E_1\right)^{\sim}\right), \\ & H_{\alpha_6}=\frac{1}{48}\left(\left(H_0+H_1+H_2+H_3\right)+2\left(E_1-E_2\right)^{\sim}\right). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{24} (H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24} (H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24} (H_2 - H_3), \\ H_{\alpha_4} &= \frac{1}{24} (H_3 + (E_2 - E_3)^{\sim}), \\ H_{\alpha_5} &= \frac{1}{48} ((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^{\sim}), \\ H_{\alpha_6} &= \frac{1}{48} ((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^{\sim}). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{24} (H_0 - H_1), \quad H_{\alpha_2} = \frac{1}{24} (H_1 - H_2), \quad H_{\alpha_3} = \frac{1}{24} (H_2 - H_3), \\ H_{\alpha_4} &= \frac{1}{24} (H_3 + (E_2 - E_3)^{\sim}), \\ H_{\alpha_5} &= \frac{1}{48} ((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^{\sim}), \\ H_{\alpha_6} &= \frac{1}{48} ((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^{\sim}). \end{aligned}$
0902.0431_E00024	9	0.655	$\begin{aligned} & \& \left[F_{i0}, F_{0k}\right] x = \left(F_{i0} F_{0k} - F_{0k} F_{i0}\right) x = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \end{aligned}$	$\begin{aligned} [F_{i0}, F_{0k}] x &= (F_{i0} F_{0k} - F_{0k} F_{i0}) x = -\frac{1}{4} e_i (e_k x) + \frac{1}{4} e_k (e_i x) \\ &= \frac{1}{2} e_k (e_i x) = F_{ik} x, \end{aligned}$	$\begin{aligned} [F_{i0}, F_{0k}] x &= (F_{i0} F_{0k} - F_{0k} F_{i0}) x = -\frac{1}{4} e_i (e_k x) + \frac{1}{4} e_k (e_i x) \\ &= \frac{1}{2} e_k (e_i x) = F_{ik} x, \end{aligned}$
0902.0431_E00021	9	0.656	$\begin{aligned} & \& \left[\kappa D_1, \kappa D_2\right] x = \left(\kappa D_1\right)\left(\kappa D_2 x\right) - \left(\kappa D_2\right)\left(\kappa D_1 x\right) = \overline{D_1\left(\kappa D_2 x\right)} - \overline{D_2\left(\kappa D_1 x\right)} \\ & = \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa\left(D_1 D_2 - D_2 D_1\right) x = \kappa\left[D_1, D_2\right] x, \quad x \in \mathfrak{C} \end{aligned}$	$\begin{aligned} [\kappa D_1, \kappa D_2] x &= (\kappa D_1) (\kappa D_2 x) - (\kappa D_2) (\kappa D_1 x) = \overline{D_1 (\kappa D_2 x)} - \overline{D_2 (\kappa D_1 x)} \\ &= \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa (D_1 D_2 - D_2 D_1) x = \kappa [D_1, D_2] x, \quad x \in \mathfrak{C}. \end{aligned}$	$\begin{aligned} [\kappa D_1, \kappa D_2] x &= (\kappa D_1) (\kappa D_2 x) - (\kappa D_2) (\kappa D_1 x) = \overline{D_1 (\kappa D_2 x)} - \overline{D_2 (\kappa D_1 x)} \\ &= \overline{D_1 D_2 \bar{x}} - \overline{D_2 D_1 \bar{x}} = \kappa (D_1 D_2 - D_2 D_1) x = \kappa [D_1, D_2] x, \quad x \in \mathfrak{C}. \end{aligned}$
0902.0431_EQ1122	190	0.658	$\begin{aligned} & \& f\left(\Phi\left(\phi, A, B, \nu\right),\left(X, Y, \xi, \eta\right),\left(Z, W, \zeta, \omega\right), r, s, t\right) \\ & =\left(\begin{array}{ccc} \frac{2}{3} \nu & -\frac{1}{2} \xi & \frac{1}{2} \zeta \\ \frac{1}{2} \omega & -\frac{1}{3} \nu-r & t \\ \frac{1}{2} \eta & s & -\frac{1}{3} \nu+r \end{array}\right), \phi,\left(\begin{array}{c} -2 A \\ Z \\ X \end{array}\right),\left(\begin{array}{c} -2 B \\ Y \\ -W \end{array}\right)\right), \end{aligned}$	$\begin{aligned} f(\Phi(\phi, A, B, \nu), (X, Y, \xi, \eta), (Z, W, \zeta, \omega), r, s, t)) \\ = \left(\begin{pmatrix} \frac{2}{3}\nu & -\frac{1}{2}\xi & \frac{1}{2}\zeta \\ \frac{1}{2}\omega & -\frac{1}{3}\nu - r & t \\ \frac{1}{2}\eta & s & -\frac{1}{3}\nu + r \end{pmatrix}, \phi, \begin{pmatrix} -2A \\ Z \\ X \end{pmatrix}, \begin{pmatrix} -2B \\ Y \\ -W \end{pmatrix} \right), \end{aligned}$	$\begin{aligned} f(\Phi(\phi, A, B, \nu), (X, Y, \xi, \eta), (Z, W, \zeta, \omega), r, s, t)) \\ = \left(\begin{pmatrix} \frac{2}{3}\nu & -\frac{1}{2}\xi & \frac{1}{2}\zeta \\ \frac{1}{2}\omega & -\frac{1}{3}\nu - r & t \\ \frac{1}{2}\eta & s & -\frac{1}{3}\nu + r \end{pmatrix}, \phi, \begin{pmatrix} -2A \\ Z \\ X \end{pmatrix}, \begin{pmatrix} -2B \\ Y \\ -W \end{pmatrix} \right), \end{aligned}$

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0902.0431_E00546	92	0.661	$\begin{aligned} & \& \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E \\ & \left(x_1\right)=\alpha E=\alpha(E \times E)=\tau \\ & \alpha \tau E \times \tau \alpha \tau E \\ & \tau \left(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)\right) \times \tau \left(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)\right) \\ & \left(\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)\right) E_1. \end{aligned}$	$\xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E$ $= \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)) E_1$	$\xi_2 E_2 + \xi_3 E_3 + F_1(x_1) = \alpha E = \alpha(E \times E) = \tau \alpha \tau E \times \tau \alpha \tau E$ $= \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times \tau(\xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\tau \xi_2 \tau \xi_3 - (\tau x_1, \tau x_1)) E_1.$
0902.0431_E00675 (ii)	112	0.662	$2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$	$2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$	$2(Q \times R)P - (Q \times P)R - (R \times P)Q + \frac{3}{8}\{P, R\}Q - \frac{3}{8}\{Q, P\}R = 0.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_EQ1015	169	■0.662	$\begin{gathered} \pm\left(\lambda_k-\lambda_l\right), \quad \pm\left(\lambda_k+\lambda_l\right), \quad 0 \leq k< l \leq 3, \\ \pm \lambda_k \pm \frac{1}{2}\left(\mu_2-\mu_3\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_3-\mu_1\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm \frac{1}{2}\left(\mu_1-\mu_2\right), \\ \pm\left(\mu_k+\frac{2}{3} \nu\right), \quad 0 \leq k< l \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1-\frac{2}{3} \nu\right), \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_2-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0-\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0+\lambda_1+\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm \frac{1}{2}\left(-\lambda_0-\lambda_1+\lambda_2+\lambda_3\right) \pm\left(\frac{1}{2} \mu_3-\frac{2}{3} \nu\right), \\ \pm\left(\mu_j-\frac{1}{3} \nu+r\right), \quad 1 \leq j \leq 3, \\ \pm \lambda_k \pm\left(\frac{1}{2} \mu_1+\frac{1}{3} \nu\right) \pm r, \quad 0 \leq k \leq 3, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \\ \pm \frac{1}{2}\left(\lambda_0+\lambda_1-\lambda_2-\lambda_3\right) \pm\left(\frac{1}{2} \mu_2+\frac{1}{3} \nu\right) \pm r, \end{gathered}$	$\begin{aligned}&\pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3,\\&\pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2),\\&\pm \left(\mu_k + \frac{2}{3} \nu \right), \quad 0 \leq k < l \leq 3,\\&\pm \lambda_k \pm \left(\frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2} \mu_3 - \frac{2}{3} \nu \right),\\&\pm \left(\mu_j - \frac{1}{3} \nu + r \right), \quad 1 \leq j \leq 3,\\&\pm \lambda_k \pm \left(\frac{1}{2} \mu_1 + \frac{1}{3} \nu \right) \pm r, \quad 0 \leq k \leq 3,\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 + \frac{1}{3} \nu \right) \pm r,\\&\pm \frac{1}{2} (\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2} \mu_2 + \frac{1}{3} \nu \right) \pm r,\end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00131	25	0.662	$\begin{aligned} & (\alpha x)(\alpha y) = \left(q m \bar{q} + (p a \bar{q}) e_4 \right) \left(q n \bar{q} + (p b \bar{q}) e_4 \right) \\ & = ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ & = q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ & = \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (q m \bar{q} + (p a \bar{q}) e_4) (q n \bar{q} + (p b \bar{q}) e_4) \\ &= ((q m \bar{q})(q n \bar{q}) - (\overline{p b \bar{q}})(p a \bar{q})) + ((p a \bar{q})(\overline{q n \bar{q}}) + (p b \bar{q})(q m \bar{q})) e_4 \\ &= q(m n - \bar{b} a) \bar{q} + (p(a \bar{n} + b m) \bar{q}) e_4 \\ &= \varphi(p, q) ((m + a e_4)(n + b e_4)) = \alpha(x y). \end{aligned}$
0902.0431_E00734	121	0.666	$\left(\left(\left[\Phi, \Phi_1 \right], \Phi_2 \right)_7 + \left(\Phi_1, \left[\Phi, \Phi_2 \right] \right)_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$	$([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$	$([\Phi, \Phi_1], \Phi_2)_7 + (\Phi_1, [\Phi, \Phi_2])_7 = 0, \quad \Phi, \Phi_i \in \mathfrak{e}_7^C.$
0902.0431_E00873	145	0.666	$\begin{aligned} & \operatorname{Spin}(12)/\{1, \sigma\} \cong SO(12) \\ & \operatorname{Spin}(12)/\{1, -1\} \cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12) \end{aligned}$	$\operatorname{Spin}(12)/\{1, \sigma\} \cong SO(12)$ $\operatorname{Spin}(12)/\{1, -1\} \cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12)$	$\begin{aligned} \operatorname{Spin}(12)/\{1, \sigma\} &\cong SO(12), \\ \operatorname{Spin}(12)/\{1, -1\} &\cong \operatorname{Spin}(12)/\{1, -\sigma\} \cong \operatorname{Ss}(12). \end{aligned}$
0902.0431_E00649	107	0.672	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$	$\varphi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}.$
0902.0431_E00774	130	0.675	$\begin{aligned} & \langle \alpha \dot{X}, \dot{\mathbf{i}} \rangle = \langle \alpha \dot{X}, \alpha \dot{\mathbf{i}} \rangle = \langle \dot{X}, \dot{\mathbf{i}} \rangle = 0 \\ & \langle \alpha \dot{X}, \underline{1} \rangle = \langle \alpha \dot{X}, \alpha \underline{1} \rangle = \langle \dot{X}, \underline{1} \rangle = 0. \end{aligned}$	$\begin{aligned} \langle \alpha \dot{X}, \dot{\mathbf{i}} \rangle &= \langle \alpha \dot{X}, \alpha \dot{\mathbf{i}} \rangle = \langle \dot{X}, \dot{\mathbf{i}} \rangle = 0 \\ \langle \alpha \dot{X}, \underline{1} \rangle &= \langle \alpha \dot{X}, \alpha \underline{1} \rangle = \langle \dot{X}, \underline{1} \rangle = 0. \end{aligned}$	$\begin{aligned} \langle \alpha \dot{X}, \dot{\mathbf{i}} \rangle &= \langle \alpha \dot{X}, \alpha \dot{\mathbf{i}} \rangle = \langle \dot{X}, \dot{\mathbf{i}} \rangle = 0, \\ \langle \alpha \dot{X}, \underline{1} \rangle &= \langle \alpha \dot{X}, \alpha \underline{1} \rangle = \langle \dot{X}, \underline{1} \rangle = 0. \end{aligned}$
0902.0431_E00142	26	0.683	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$	$\overline{\alpha x} = \alpha \bar{x}, \quad x \in \mathfrak{C}^C.$
0902.0431_E01206	203	0.686	$\begin{aligned} & \left(\operatorname{tr} \left(C_1 \tau^t C_2 \right) + \operatorname{tr} \left(D_1 \tau^t D_2 \right) + (\mathbf{x}_1, \tau \mathbf{x}_2) (\mathbf{a}_1, \tau \mathbf{a}_2) + (\mathbf{y}_1, \tau \mathbf{y}_2) (\mathbf{b}_1, \tau \mathbf{b}_2) \right. \\ & \quad \left. + (\mathbf{z}_1, \tau \mathbf{z}_2) (\mathbf{c}_1, \tau \mathbf{c}_2) + (\mathbf{w}_1, \tau \mathbf{w}_2) (\mathbf{d}_1, \tau \mathbf{d}_2) \right). \end{aligned}$	$\begin{aligned} \langle R_1, R_2 \rangle &= 60 \left(\operatorname{tr} \left(C_1 \tau^t C_2 \right) + \operatorname{tr} \left(D_1 \tau^t D_2 \right) + (\mathbf{x}_1, \tau \mathbf{x}_2) (\mathbf{a}_1, \tau \mathbf{a}_2) + (\mathbf{y}_1, \tau \mathbf{y}_2) (\mathbf{b}_1, \tau \mathbf{b}_2) \right. \\ & \quad \left. + (\mathbf{z}_1, \tau \mathbf{z}_2) (\mathbf{c}_1, \tau \mathbf{c}_2) + (\mathbf{w}_1, \tau \mathbf{w}_2) (\mathbf{d}_1, \tau \mathbf{d}_2) \right). \end{aligned}$	$\begin{aligned} \langle R_1, R_2 \rangle &= 60 \left(\operatorname{tr} \left(C_1 \tau^t C_2 \right) + \operatorname{tr} \left(D_1 \tau^t D_2 \right) + (\mathbf{x}_1, \tau \mathbf{x}_2) (\mathbf{a}_1, \tau \mathbf{a}_2) + (\mathbf{y}_1, \tau \mathbf{y}_2) (\mathbf{b}_1, \tau \mathbf{b}_2) \right. \\ & \quad \left. + (\mathbf{z}_1, \tau \mathbf{z}_2) (\mathbf{c}_1, \tau \mathbf{c}_2) + (\mathbf{w}_1, \tau \mathbf{w}_2) (\mathbf{d}_1, \tau \mathbf{d}_2) \right). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00765	128	■ 0.686	$\begin{aligned} & \text{\texttt{\textbackslash begin{aligned}}} \ \& \text{\texttt{H_{\alpha_1}}}=\frac{1}{36} \ \Phi \text{\texttt{\textbackslash left(H_{0}-H_{1}, \ 0,0,0\right)}, \ \& \text{\texttt{H_{\alpha_2}}}=\frac{1}{36} \ \Phi \text{\texttt{\textbackslash left(H_{1}-H_{2}, \ 0,0,0\right)}, \ \& \text{\texttt{H_{\alpha_3}}}=\frac{1}{36} \ \Phi \text{\texttt{\textbackslash left(H_{2}-H_{3}, \ 0,0,0\right)}, \ \& \text{\texttt{H_{\alpha_4}}}=\frac{1}{72} \ \Phi \text{\texttt{\textbackslash left(\textbackslash left(-H_{0}-H_{1}-H_{2}+H_{3}\right)+2\textbackslash left(E_{3}-E_{1}\right)^{\textbackslash sim}, \ 0,0,0\right)}, \ \& \text{\texttt{H_{\alpha_5}}}=\frac{1}{72} \ \Phi \text{\texttt{\textbackslash left(\textbackslash left(H_{0}+H_{1}+H_{2}+H_{3}\right)+2\textbackslash left(E_{1}-E_{2}\right)^{\textbackslash sim}, \ 0,0,0\right)}, \ \& \text{\texttt{H_{\alpha_6}}}=\frac{1}{54} \ \Phi \text{\texttt{\textbackslash left(\textbackslash left(-E_{1}+2E_{2}-E_{3}\right)^{\textbackslash sim}, \ 0,0, \ \frac{3}{2}\right)}, \ \& \text{\texttt{H_{\alpha_7}}}=\frac{1}{54} \ \Phi \text{\texttt{\textbackslash left(\textbackslash left(E_{1}+E_{2}-2E_{3}\right)^{\textbackslash sim}, \ 0,0,-\frac{3}{2}\right)} . \\ & \text{\texttt{\textbackslash end{aligned}}} \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi(H_0 - H_1, 0, 0, 0), \\ H_{\alpha_2} &= \frac{1}{36} \Phi(H_1 - H_2, 0, 0, 0), \\ H_{\alpha_3} &= \frac{1}{36} \Phi(H_2 - H_3, 0, 0, 0), \\ H_{\alpha_4} &= \frac{1}{72} \Phi((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim, 0, 0, 0), \\ H_{\alpha_5} &= \frac{1}{72} \Phi((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim, 0, 0, 0), \\ H_{\alpha_6} &= \frac{1}{54} \Phi\left((-E_1 + 2E_2 - E_3)^\sim, 0, 0, \frac{3}{2}\right), \\ H_{\alpha_7} &= \frac{1}{54} \Phi\left((E_1 + E_2 - 2E_3)^\sim, 0, 0, -\frac{3}{2}\right). \end{aligned}$	$\begin{aligned} H_{\alpha_1} &= \frac{1}{36} \Phi(H_0 - H_1, 0, 0, 0), \\ H_{\alpha_2} &= \frac{1}{36} \Phi(H_1 - H_2, 0, 0, 0), \\ H_{\alpha_3} &= \frac{1}{36} \Phi(H_2 - H_3, 0, 0, 0), \\ H_{\alpha_4} &= \frac{1}{72} \Phi((-H_0 - H_1 - H_2 + H_3) + 2(E_3 - E_1)^\sim, 0, 0, 0), \\ H_{\alpha_5} &= \frac{1}{72} \Phi((H_0 + H_1 + H_2 + H_3) + 2(E_1 - E_2)^\sim, 0, 0, 0), \\ H_{\alpha_6} &= \frac{1}{54} \Phi\left((-E_1 + 2E_2 - E_3)^\sim, 0, 0, \frac{3}{2}\right), \\ H_{\alpha_7} &= \frac{1}{54} \Phi\left((E_1 + E_2 - 2E_3)^\sim, 0, 0, -\frac{3}{2}\right). \end{aligned}$
0902.0431_E00616	102	■ 0.688	$\begin{aligned} & \text{\texttt{\textbackslash begin{gathered}}} \ g(\gamma(X \ \text{\texttt{\textbackslash times}} \ Y))=\text{\texttt{g(\gamma(M+}} \\ & \text{\texttt{\textbackslash boldsymbol{a}}) \ \text{\texttt{\textbackslash times(N+}}\text{\texttt{\textbackslash boldsymbol{b}}))}} \ \& \text{\texttt{=g((M-}} \\ & \text{\texttt{\textbackslash boldsymbol{a}}) \ \text{\texttt{\textbackslash times(N-}}\text{\texttt{\textbackslash boldsymbol{b}}))}} \ \& \text{\texttt{=g\textbackslash left(}} \\ & \text{\texttt{\textbackslash left(M \ \text{\texttt{\textbackslash times}} \ N-\frac{1}{2}\text{\texttt{\textbackslash left(\textbackslash boldsymbol{a}}^{\textbackslash *} \text{\texttt{\textbackslash boldsymbol{b}}+}}\text{\texttt{\textbackslash boldsymbol{b}}^{\textbackslash *} \text{\texttt{\textbackslash boldsymbol{a}}\right)}} \\ & \text{\texttt{\textbackslash right)+}\frac{1}{2}(\text{\texttt{\textbackslash boldsymbol{a}} \ N+\text{\texttt{\textbackslash boldsymbol{b}}}} \\ & \text{\texttt{M)\right)}} \ \& \text{\texttt{=\binom{\frac{1}{2} \ \operatorname{tr}(} \\ & \text{\texttt{\textbackslash boldsymbol{a}} \ N+\text{\texttt{\textbackslash boldsymbol{b}}}} \ M)\}\frac{i}{2}(\text{\texttt{\textbackslash boldsymbol{a}} \ N+\text{\texttt{\textbackslash boldsymbol{b}}}} \ M)^{\textbackslash *} \ \text{\texttt{\textbackslash quad}} \ M \ \text{\texttt{\textbackslash times}} \\ & \text{\texttt{N-\frac{1}{2}\text{\texttt{\textbackslash left(\textbackslash boldsymbol{a}}^{\textbackslash *} \ \text{\texttt{\textbackslash boldsymbol{b}}+}}\text{\texttt{\textbackslash boldsymbol{b}}^{\textbackslash *} \ \text{\texttt{\textbackslash boldsymbol{a}}\right)}}-\frac{1}{2}(\text{\texttt{\textbackslash operatorname{tr}}(M \ \text{\texttt{\textbackslash times}} \ N)-(\text{\texttt{\textbackslash boldsymbol{a}}}, \\ & \text{\texttt{\textbackslash boldsymbol{b}}))} \ E)} \ \& \text{\texttt{=\cdots}} \ \& \text{\texttt{=g(M+\textbackslash boldsymbol{a}})} \\ & \text{\texttt{\textbackslash circ}} \ g(\text{\texttt{N+\textbackslash boldsymbol{b}}})-\text{\texttt{\textbackslash left(\frac{1}{4}(M, \ N)-}} \\ & \text{\texttt{\textbackslash frac{1}{2}(\textbackslash boldsymbol{a}, \ \textbackslash boldsymbol{b})\right)}} \\ & \text{\texttt{E}} \ \& \text{\texttt{=g(M+\textbackslash boldsymbol{a}}) \ \text{\texttt{\textbackslash circ}} \ g(\text{\texttt{N+\textbackslash boldsymbol{b}}})-} \\ & \text{\texttt{\textbackslash frac{1}{4}(\gamma(M+\textbackslash boldsymbol{a}), (\text{\texttt{N+\textbackslash boldsymbol{b}}))}}} \\ & \text{\texttt{E}} \ \& \text{\texttt{=g \ X \textbackslash circ} \ g \ Y-\frac{1}{4}(\gamma X, \ Y) \ E} . \\ & \text{\texttt{\textbackslash end{gathered}}} \end{aligned}$	$\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M + \boldsymbol{a}) \times (N + \boldsymbol{b}))) \\ &= g((M - \boldsymbol{a}) \times (N - \boldsymbol{b})) \\ &= g((M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})) + \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M)) \\ &= \left(\begin{array}{cc} \frac{1}{2} \text{tr}(M \times N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) & \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M) \\ \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M)^* & M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a}) - \frac{1}{2}(\text{tr}(M \times N) - (\boldsymbol{a}, \boldsymbol{b}))E \end{array} \right) \\ &= \dots \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \left(\frac{1}{4}(M, N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) \right) E \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \frac{1}{4}(\gamma(M + \boldsymbol{a}), (N + \boldsymbol{b})) E \\ &= gX \circ gY - \frac{1}{4}(\gamma X, Y) E. \end{aligned}$	$\begin{aligned} g(\gamma(X \times Y)) &= g(\gamma((M + \boldsymbol{a}) \times (N + \boldsymbol{b}))) \\ &= g((M - \boldsymbol{a}) \times (N - \boldsymbol{b})) \\ &= g((M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a})) + \frac{1}{2}(\boldsymbol{a} N + \boldsymbol{b} M)) \\ &= \left(\begin{array}{cc} \frac{1}{2} \text{tr}(M \times N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) & \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M) \\ \frac{i}{2}(\boldsymbol{a} N + \boldsymbol{b} M)^* & M \times N - \frac{1}{2}(\boldsymbol{a}^* \boldsymbol{b} + \boldsymbol{b}^* \boldsymbol{a}) - \frac{1}{2}(\text{tr}(M \times N) - (\boldsymbol{a}, \boldsymbol{b}))E \end{array} \right) \\ &= \dots \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \left(\frac{1}{4}(M, N) - \frac{1}{2}(\boldsymbol{a}, \boldsymbol{b}) \right) E \\ &= g(M + \boldsymbol{a}) \circ g(N + \boldsymbol{b}) - \frac{1}{4}(\gamma(M + \boldsymbol{a}), (N + \boldsymbol{b})) E \\ &= gX \circ gY - \frac{1}{4}(\gamma X, Y) E. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00383	63	0.688	$l=\left(E_{\{1\}},\,E_{\{1\}}\right)=\left(\alpha\,E_{\{1\}},\,\alpha\,E_{\{1\}}\right)=1+\xi_{\{2\}}^2+\xi_{\{3\}}^2+2\left x_{\{1\}}\right ^2,$	$1=\left(E_1,E_1\right)=\left(\alpha E_1,\alpha E_1\right)=1+\xi_2^2+\xi_3^2+2\left x_1\right ^2,$	$1=\left(E_1,E_1\right)=\left(\alpha E_1,\alpha E_1\right)=1+\xi_2^2+\xi_3^2+2\left x_1\right ^2,$
0902.0431_E01181	199	0.697	$\begin{aligned} & (x\wedge a,y\wedge b)=(x,y)(a,b)-\left(a_{\{1\}},y\right)\left(x\wedge a_{\{2\}},b\right)+\left(a_{\{2\}},y\right)\left(x\wedge a_{\{1\}},b\right)\\ & (*b\wedge x,*a\wedge y)=(a,y\wedge (*b\wedge x))=(a_1,y)(x\wedge a_2,b)-(a_2,y)(x\wedge a_1,b), \end{aligned}$	$\begin{aligned} (x\wedge a,y\wedge b) &= (x,y)(a,b)-(a_1,y)(x\wedge a_2,b)+(a_2,y)(x\wedge a_1,b) \\ (*b\wedge x,*a\wedge y) &= (a,y\wedge (*b\wedge x))=(a_1,y)(x\wedge a_2,b)-(a_2,y)(x\wedge a_1,b), \end{aligned}$	$\begin{aligned} (x\wedge a,y\wedge b) &= (x,y)(a,b)-(a_1,y)(x\wedge a_2,b)+(a_2,y)(x\wedge a_1,b), \\ (*b\wedge x,*a\wedge y) &= (a,y\wedge (*b\wedge x))=(a_1,y)(x\wedge a_2,b)-(a_2,y)(x\wedge a_1,b), \end{aligned}$
0902.0431_E00086	18	0.697	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$	$\mathfrak{a}\supset [\mathfrak{a},\mathfrak{S}]\supset [\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}\text{ (Lemma 1.6.1).}$
0902.0431_E00646	107	0.705	$\begin{aligned} & \varphi((A,B),1)X=h(A,B)Xh(A,B)^*, \\ & \varphi((A,B),\epsilon)X=h(A,B)\overline{X}h(A,B)^*, \end{aligned}\quad X\in(\mathfrak{J}_C)^C.$	$\begin{aligned} \varphi((A,B),1)X &= h(A,B)Xh(A,B)^*, \\ \varphi((A,B),\epsilon)X &= h(A,B)\overline{X}h(A,B)^*, \end{aligned}\quad X\in(\mathfrak{J}_C)^C.$	$\begin{aligned} \varphi((A,B),1)X &= h(A,B)Xh(A,B)^*, \\ \varphi((A,B),\epsilon)X &= h(A,B)\overline{X}h(A,B)^*, \end{aligned}\quad X\in(\mathfrak{J}_C)^C.$
0902.0431_E00976	162	0.710	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots\left(\text{using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{ (Proposition 4.3.2),}\right. \\ & \quad \left.(P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0\right. \\ & \quad \left.(\text{Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.})\cdots\right. \\ & \quad \left.=0\right) \end{aligned}$	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots\left(\text{ using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{ (Proposition 4.3.2)}\right. \\ & \quad \left.(P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0\right. \\ & \quad \left.(\text{ Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.})\cdots\right. \\ & \quad \left.=0\right) \end{aligned}$	$\begin{aligned} & \left[R_1,[R_2,R_3]\right]+\left[R_2,[R_3,R_1]\right]+\left[R_3,[R_1,R_2]\right] \\ & =\cdots\left(\text{using }[\Phi,P\times Q]=\Phi P\times Q+P\times\Phi Q\text{ (Proposition 4.3.2),}\right. \\ & \quad \left.(P\times R)Q-(Q\times R)P+\frac{1}{8}\{Q,R\}P-\frac{1}{8}\{P,R\}-\frac{1}{4}\{P,Q\}R=0\right. \\ & \quad \left.(\text{Lemma 4.1.1.(3), }\{\Phi P,Q\}+\{P,\Phi Q\}=0\text{ (Proposition 4.2.2.(2)) etc.})\cdots\right. \\ & \quad \left.=0.\right) \end{aligned}$
0902.0431_E00575	96	0.718	$\begin{aligned} & \{ \}^t\left(k_{\{J\}}(M)\right) \& =\{ \}^t((k(M))\,J)=-J^t(k(M))=-J^t(k(M_1)+ik(M_2)) \\ & =-(\tau^t(k(M_1))+i\tau^t(k(M_2)))J=-(k(M_1^*)+ik(M_2^*))J \\ & =-(k(M_1)+ik(M_2))J=-(k(M))J=-k_J(M) \end{aligned}$	$\begin{aligned} {}^t(k_J(M)) &= {}^t((k(M))J)=-J^t(k(M))=-J^t(k(M_1)+ik(M_2)) \\ &=-(\tau^t(k(M_1))+i\tau^t(k(M_2)))J=-(k(M_1^*)+ik(M_2^*))J \\ &=-(k(M_1)+ik(M_2))J=-(k(M))J=-k_J(M) \end{aligned}$	$\begin{aligned} {}^t(k_J(M)) &= {}^t((k(M))J)=-J^t(k(M))=-J^t(k(M_1)+ik(M_2)) \\ &=-(\tau^t(k(M_1))+i\tau^t(k(M_2)))J=-(k(M_1^*)+ik(M_2^*))J \\ &=-(k(M_1)+ik(M_2))J=-(k(M))J=-k_J(M). \end{aligned}$
0902.0431_E00704	116	0.718	$2\left(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X)\right)\times(X\times X)$	$2\left(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X)\right)\times(X\times X)$	$2\left(\psi(X\times X)+\frac{1}{3}(2\nu+\mu)(X\times X)\right)\times(X\times X)$
0902.0431_E00082	17	0.719	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$	$[\mathfrak{su}(3),\mathfrak{S}]=\mathfrak{S}.$

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0902.0431_E00095	19	■ 0.719	$\mathfrak{h}=\left\{\left(\begin{array}{ccc}\lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3\end{array}\right)\middle \lambda_i\in\boldsymbol{C},\lambda_1+\lambda_2+\lambda_3=0\right\}$		
0902.0431_E00964	160	■ 0.719	$E_{\{7\}}^{\{ \}^{\{C\}}}\simeq E_7\times\boldsymbol{R}^{133}.$		
0902.0431_E00197	33	■ 0.724	$\begin{aligned} & \widetilde{D}_4=\left\{\left(\alpha_1,\alpha_2,\alpha_3\right)\in SO(8)\times SO(8)\times SO(8)\mid\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3(\overline{xy})},x,y\in\mathfrak{C}\right\} \\ & \widetilde{D}_4\text{ is a compact group.} \end{aligned}$		
0902.0431_E00532	89	■ 0.724	$\begin{aligned} & \left \eta_1(t)\right ^2+\left \eta_2(t)\right ^2+\left \eta_3(t)\right ^2 \\ & =\left \xi_1\right ^2+\left \frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ & +\left -\frac{\xi_2-\xi_3}{2}+\frac{\xi_2+\xi_3}{2} \cos t- q \sin t+i \nu \sin t\right ^2 \\ & =\left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\frac{\xi_2+\xi_3}{2} \cos t- q \sin t\right)^2+2 \nu^2 \sin ^2 t \\ & =\left \xi_1\right ^2+2\left(\frac{\xi_2-\xi_3}{2}\right)^2+2\left(\left(\frac{\xi_2+\xi_3}{2}\right)^2+ q ^2\right) \sin ^2\left(t+t_0\right)+2 \nu^2 \sin ^2 t \\ & \leq\left \xi_1\right ^2+\left \xi_2\right ^2+\left \xi_3\right ^2+2 q ^2+2 \nu^2 \cos ^2 t_0 \quad\left(\text { for some } t_0 \in \boldsymbol{R}\right) \end{aligned}$		

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0902.0431_E01178	199	0.725	$\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) a_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$	$\begin{aligned} &= \sum_{j=1}^5 \sum_{i=1}^6 (-1)^i (\mathbf{a}_1 \wedge \cdots \mathbf{a}_{i-1} \wedge \mathbf{a}_{i+1} \wedge \cdots \wedge \mathbf{a}_6, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_5) a_{ij} \mathbf{e}_j \\ &= - \sum_{j=1}^5 \det \begin{pmatrix} a_{1j} & a_{11} & \cdots & a_{15} \\ a_{2j} & a_{21} & \cdots & a_{25} \\ & & \cdots & \\ a_{6j} & a_{61} & \cdots & a_{65} \end{pmatrix} \mathbf{e}_j = 0. \end{aligned}$	
0902.0431_E01148	194	0.732	$\begin{aligned} &(\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) = (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$	$\begin{aligned} &((\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) = (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$	$\begin{aligned} ((\mathbf{u} \times \mathbf{v}) \mathbf{x}, \mathbf{y}) &= (\mathbf{v} \wedge (\mathbf{u} \wedge \mathbf{x}), \mathbf{y}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= -(\mathbf{x} \wedge \mathbf{u}, \mathbf{y} \wedge \mathbf{v}) + \frac{2}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}) \\ &= (\mathbf{x} \wedge \mathbf{u}_2 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_1, \mathbf{y}) - (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_3, \mathbf{v}) (\mathbf{u}_2, \mathbf{y}) \\ &\quad + (\mathbf{x} \wedge \mathbf{u}_1 \wedge \mathbf{u}_2, \mathbf{v}) (\mathbf{u}_3, \mathbf{y}) - \frac{1}{3} (\mathbf{u}, \mathbf{v}) (\mathbf{x}, \mathbf{y}). \end{aligned}$
0902.0431_E01081	183	0.733	$\begin{aligned} &S_1 \tau S_2 = \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 &= \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$	$\begin{aligned} S_1 \tau S_2 &= \chi P_1 \tau \chi P_2 = \chi(X_1, Y_1, \xi_1, \eta_1) \tau \chi(X_2, Y_2, \xi_2, \eta_2) \\ &= \left(k \left(g X_1 - \frac{\xi_1}{2} E \right) + i k \left(g(\gamma Y_1) - \frac{\eta_1}{2} E \right) \right) J \tau \left(\left(k \left(g X_2 - \frac{\xi_2}{2} E \right) \right. \right. \\ &\quad \left. \left. + i k \left(g(\gamma Y_2) - \frac{\eta_2}{2} E \right) \right) J \right) \end{aligned}$

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0902.0431_E01096	186	0.736	$\begin{aligned} & \left[\zeta \left(\left(S_1 \right) I, \zeta \left(\left(S_2 \right) I \right) \right) \right] \\ & = \left[\Theta \left(0, 2 \lambda \gamma \chi^{-1} S_1, 2 \chi^{-1} S_1, 0, 0, 0 \right), \Theta \left(0, 2 \lambda \gamma \chi^{-1} S_2, 2 \chi^{-1} S_2, 0, 0, 0 \right) \right] \\ & = \Theta \left(2 \lambda \gamma \chi^{-1} S_1 \times 2 \chi^{-1} S_2 - 2 \lambda \gamma \chi^{-1} S_2 \times 2 \chi^{-1} S_1, 0, 0, \right. \\ & \quad \left. \frac{1}{8} \left(- \left\{ 2 \lambda \gamma \chi^{-1} S_1, 2 \chi^{-1} S_2 \right\} + \left\{ 2 \lambda \gamma \chi^{-1} S_2, 2 \chi^{-1} S_1 \right\} \right), \right. \\ & \quad \left. \frac{1}{4} \left\{ 2 \lambda \gamma \chi^{-1} S_1, 2 \lambda \gamma \chi^{-1} S_2 \right\}, - \frac{1}{4} \left\{ 2 \chi^{-1} S_1, 2 \chi^{-1} S_2 \right\} \right), \end{aligned}$	$\begin{aligned} & [\zeta l(S_1)I, \zeta(l(S_2)I)] \\ & = [\Theta(0, 2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_1, 0, 0, 0), \Theta(0, 2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_2, 0, 0, 0)] \\ & = \Theta\left(2\lambda\gamma\chi^{-1}S_1 \times 2\chi^{-1}S_2 - 2\lambda\gamma\chi^{-1}S_2 \times 2\chi^{-1}S_1, 0, 0, \right. \\ & \quad \frac{1}{8}(-\{2\lambda\gamma\chi^{-1}S_1, 2\chi^{-1}S_2\} + \{2\lambda\gamma\chi^{-1}S_2, 2\chi^{-1}S_1\}), \\ & \quad \left. \frac{1}{4}\{2\lambda\gamma\chi^{-1}S_1, 2\lambda\gamma\chi^{-1}S_2\}, -\frac{1}{4}\{2\chi^{-1}S_1, 2\chi^{-1}S_2\}\right), \end{aligned}$	
0902.0431_E00158	29	0.746	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \alpha_3 (xy), \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$	
0902.0431_E00621	103	0.747	$\begin{aligned} & \langle \varphi(A)X, \varphi(A)Y \rangle = \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ & = \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$	$\begin{aligned} \langle \varphi(A)X, \varphi(A)Y \rangle &= \langle g\varphi(A)X, g\varphi(A)Y \rangle \\ &= \langle A(gX)A^*, A(gY)A^* \rangle = \langle gX, gY \rangle = \langle X, Y \rangle. \end{aligned}$	
0902.0431_E01102	187	0.751	$\begin{aligned} & \left(\left(\mathfrak{e}_{\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{\tau\gamma} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)^{\tilde{\lambda}\gamma}, \right. \\ & \left. \left(\left(\mathfrak{e}_{-\tilde{\lambda}\gamma} \right)_{\tau\tilde{\lambda}} = \left(\left(\mathfrak{e}_{\tau\gamma} \right)_{-\tilde{\lambda}\gamma} = \left(\mathfrak{e}_{8(8)} \right)_{-\tilde{\lambda}\gamma}, \right. \right. \end{aligned}$	$\begin{aligned} ((\mathfrak{e}_8^C)_{\tilde{\lambda}\gamma})_{\tau\tilde{\lambda}} &= ((\mathfrak{e}_8^C)_{\tau\gamma})_{\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})_{\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, \\ ((\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma})_{\tau\tilde{\lambda}} &= ((\mathfrak{e}_8^C)_{\tau\gamma})_{-\tilde{\lambda}\gamma} = (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}, \end{aligned}$	

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0902.0431_E00250	41	0.754	$\begin{aligned} & \& (4) \quad \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))} \\ & \quad \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E) \\ & \quad = (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-} \end{aligned}$	$(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$	$(4) \Rightarrow (2) \operatorname{tr}(\alpha X) = (\alpha X, E, E) = (\alpha X, \alpha E, \alpha E) = (X, E, E) \text{ (Lemma 2.2.2.(2))}$ $= \operatorname{tr}(X). \text{ Hence } \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y)) = (X, Y, E) \text{ (Lemma 2.1.1)} = (\alpha X, \alpha Y, \alpha E)$ $= (\alpha X, \alpha Y, E) = \frac{1}{2}(\operatorname{tr}(\alpha X) \operatorname{tr}(\alpha Y) - (\alpha X, \alpha Y)) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (\alpha X, \alpha Y)). \text{ There-}$
0902.0431_E00911	151	0.754	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$	$E_{7,C} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{P}_C)^C \right) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \right\}.$
0902.0431_E01107	187	0.754	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$	$k^2 B_8(R, R') = B_8(\alpha R, \alpha R') = B_8(R, R'), \quad R, R' \in (\mathfrak{e}_8^C)_{-\widetilde{\lambda}_\gamma},$
0902.0431_E00448	73	0.755	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$	$\phi = \delta + \widetilde{T}, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C.$
0902.0431_E00318	51	0.755	$\begin{aligned} & \& ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & \quad = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$	$\begin{aligned} & \& ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & \quad = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D) \end{aligned}$	$\begin{aligned} & \& ([\delta, [\widetilde{A}, \widetilde{B}]], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\delta, [\widetilde{C}, \widetilde{D}]])_4 \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}] + [\widetilde{A}, \widetilde{\delta B}], [\widetilde{C}, \widetilde{D}])_4 + ([\widetilde{A}, \widetilde{B}], [\widetilde{\delta C}, \widetilde{D}] + [\widetilde{C}, \widetilde{\delta D}])_4 \text{ (Lemma 2.4.5.(1))} \\ & \quad = ([\widetilde{\delta A}, \widetilde{B}]C, D) + ([\widetilde{A}, \widetilde{\delta B}]C, D) + ([\widetilde{A}, \widetilde{B}]\delta C, D) + ([\widetilde{A}, \widetilde{B}]C, \delta D) \\ & \quad = (\delta A \circ (B \circ C), D) - (B \circ (\delta A \circ C), D) + (A \circ (\delta B \circ C), D) - (\delta B \circ (A \circ C), D) \\ & \quad \quad + (A \circ (B \circ \delta C), D) - (B \circ (A \circ \delta C), D) + (A \circ (B \circ C), \delta D) - (B \circ (A \circ C), \delta D), \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01153	194	■ 0.759	$\begin{aligned} & \& ((u \times w) v - (v \times w) u, a) \setminus \\ & \& = \left(\left((u \times w) v_1 \right) \wedge v_2 \wedge v_3, a \right) \wedge v_2 \wedge v_3, a \right) - \left(\left((u \times w) v_2 \right) \wedge v_1 \wedge v_3, a \right) \\ & \quad + \left(\left((u \times w) v_3 \right) \wedge v_1 \wedge v_2, a \right) - \left(\left((v \times w) u_1 \right) \wedge u_2 \wedge u_3, a \right) \\ & \quad + \left(\left((v \times w) u_2 \right) \wedge u_1 \wedge u_3, a \right) - \left(\left((v \times w) u_3 \right) \wedge u_1 \wedge u_2, a \right) \\ & = - (u, w) (v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$	$\begin{aligned} & ((u \times w) v - (v \times w) u, a) \\ & = ((u \times w) v_1) \wedge v_2 \wedge v_3, a) - ((u \times w) v_2) \wedge v_1 \wedge v_3, a) \\ & \quad + ((u \times w) v_3) \wedge v_1 \wedge v_2, a) - ((v \times w) u_1) \wedge u_2 \wedge u_3, a) \\ & \quad + ((v \times w) u_2) \wedge u_1 \wedge u_3, a) - ((v \times w) u_3) \wedge u_1 \wedge u_2, a) \\ & = - (u, w) (v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$	$\begin{aligned} & ((u \times w) v - (v \times w) u, a) \\ & = ((u \times w) v_1) \wedge v_2 \wedge v_3, a) - ((u \times w) v_2) \wedge v_1 \wedge v_3, a) \\ & \quad + ((u \times w) v_3) \wedge v_1 \wedge v_2, a) - ((v \times w) u_1) \wedge u_2 \wedge u_3, a) \\ & \quad + ((v \times w) u_2) \wedge u_1 \wedge u_3, a) - ((v \times w) u_3) \wedge u_1 \wedge u_2, a) \\ & = - (u, w) (v, a) + \sum_{i=1}^3 \sum_{j=1}^3 (u_i \wedge u_{i+1} \wedge v_j, w) (u_{i+2} \wedge v_{i+1} \wedge v_{j+2}, a) \end{aligned}$
0902.0431_E00134	25	■ 0.760	$\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1)$	$\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1),$	$\alpha = \varphi(1, q) \beta = \varphi(1, q) \varphi(p, 1) = \varphi(p, q), \quad (p, q) \in \operatorname{Sp}(1) \times \operatorname{Sp}(1),$
0902.0431_E00224	37	■ 0.761	$\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$	$\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$	$\begin{aligned} & E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ & F_1(x) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & x \\ 0 & \bar{x} & 0 \end{pmatrix}, \quad F_2(x) = \begin{pmatrix} 0 & 0 & \bar{x} \\ 0 & 0 & 0 \\ x & 0 & 0 \end{pmatrix}, \quad F_3(x) = \begin{pmatrix} 0 & x & 0 \\ \bar{x} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \end{aligned}$

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0902.0431_E00117	22	■ 0.762	\begin{aligned} & \left(\alpha e_4\right) \left(\alpha e_7\right) \& =a_4 a_7=a_4\left(a_3 a_4\right)=-a_4\left(a_4 a_3\right)=a_3=\alpha e_3=\alpha\left(e_4 e_7\right), \\ & \left(\alpha e_1\right) \left(\alpha e_6\right) \& =a_1 a_6=a_1\left(a_4 a_2\right)=-a_4\left(a_1 a_2\right)=-a_4 a_3=a_3 a_4=a_7 \\ & \& =\alpha e_7=\alpha\left(e_1 e_6\right), \\ & \left(\alpha e_3\right) \left(\alpha e_6\right) \& =a_3 a_6=\left(a_1 a_2\right)\left(a_4 a_2\right)=-\left(a_2 a_1\right)\left(a_4 a_2\right)=-a_2\left(a_1 a_4\right) a_2 \\ & \& =-a_2 a_5 a_2=a_2 a_2 a_5=-a_5=-\alpha e_5=\alpha\left(e_3 e_6\right), \text { etc. } \end{aligned}	$ \begin{aligned} (\alpha e_4)(\alpha e_7) &= a_4 a_7 = a_4(a_3 a_4) = -a_4(a_4 a_3) = a_3 = \alpha e_3 = \alpha(e_4 e_7), \\ (\alpha e_1)(\alpha e_6) &= a_1 a_6 = a_1(a_4 a_2) = -a_4(a_1 a_2) = -a_4 a_3 = a_3 a_4 = a_7 \\ &= \alpha e_7 = \alpha(e_1 e_6), \\ (\alpha e_3)(\alpha e_6) &= a_3 a_6 = (a_1 a_2)(a_4 a_2) = -(a_2 a_1)(a_4 a_2) = -a_2(a_1 a_4) a_2 \\ &= -a_2 a_5 a_2 = a_2 a_2 a_5 = -a_5 = -\alpha e_5 = \alpha(e_3 e_6), \text{ etc.} \end{aligned} $	$ \begin{aligned} (\alpha e_4)(\alpha e_7) &= a_4 a_7 = a_4(a_3 a_4) = -a_4(a_4 a_3) = a_3 = \alpha e_3 = \alpha(e_4 e_7), \\ (\alpha e_1)(\alpha e_6) &= a_1 a_6 = a_1(a_4 a_2) = -a_4(a_1 a_2) = -a_4 a_3 = a_3 a_4 = a_7 \\ &= \alpha e_7 = \alpha(e_1 e_6), \\ (\alpha e_3)(\alpha e_6) &= a_3 a_6 = (a_1 a_2)(a_4 a_2) = -(a_2 a_1)(a_4 a_2) = -a_2(a_1 a_4) a_2 \\ &= -a_2 a_5 a_2 = a_2 a_2 a_5 = -a_5 = -\alpha e_5 = \alpha(e_3 e_6), \text{ etc.} \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00466	77	0.763	$ \begin{aligned} & \frac{1}{2}(B, X)A + \frac{1}{6}(A, B)X - 2B \times (A \times X) + \frac{1}{3}(A, B)X \\ & X \times B \times (A \times X) + \frac{1}{3}(A, B)X \\ & X \times = \frac{1}{2}(B, X)A + \frac{1}{2}(A, B)X - 2B \circ (A \times X) + \text{tr}(B)A \times X + \text{tr}(A \times X)B \\ & X \times B \circ (A \times X) + \text{tr}(B)A \times X + \text{tr}(A \times X)B \\ & \times X + \text{tr}(B)A \times X - (B, A \times X))E \\ & \times X + \text{tr}(B)A \times X - 2B \circ (A \times X) + \text{tr}(A)B \circ X + \text{tr}(X)B \circ A \\ & - \text{tr}(A) \text{tr}(X)B + (A, X)B + \text{tr}(B)A \circ X - \frac{1}{2} \text{tr}(B) \text{tr}(A)X \\ & - \frac{1}{2} \text{tr}(B) \text{tr}(X)A + \frac{1}{2} \text{tr}(B) \text{tr}(A) \text{tr}(X)E - \frac{1}{2} \text{tr}(B)(A, X)E \\ & + \frac{1}{2}(\text{tr}(A) \text{tr}(X) - (A, X))B - \frac{1}{2} \text{tr}(B)(\text{tr}(A) \text{tr}(X) - (A, X))E + (A, B, X)E \\ & = \text{tr}(A)B \circ X + \text{tr}(B)X \circ A + \text{tr}(X)A \circ B \\ & - \frac{1}{2}(\text{tr}(A) \text{tr}(B)X + \text{tr}(B) \text{tr}(X)A + \text{tr}(A) \text{tr}(X)B) \\ & + \frac{1}{2}((A, B)X + (B, X)A + (A, X)B) + (A, B, X)E - 2B \circ (A \times X) \\ & = A \circ (B \times X) + B \circ (A \times X) + X \circ (A \times B) - 2B \circ (A \times X) \\ & = A \circ (B \times X) - B \circ (A \times X) + (A \times B) \circ X \\ & = [\tilde{A}, \tilde{B}]X + (A \times B) \sim X. \end{aligned} $	$ \begin{aligned} & \frac{1}{2}(B, X)A + \frac{1}{6}(A, B)X - 2B \times (A \times X) + \frac{1}{3}(A, B)X \\ & = \frac{1}{2}(B, X)A + \frac{1}{2}(A, B)X - 2B \circ (A \times X) + \text{tr}(B)A \times X + \text{tr}(A \times X)B \\ & \quad - (\text{tr}(B) \text{tr}(A \times X) - (B, A \times X))E \\ & = \frac{1}{2}(B, X)A + \frac{1}{2}(A, B)X - 2B \circ (A \times X) + \text{tr}(A)B \circ X + \text{tr}(X)B \circ A \\ & \quad - \text{tr}(A) \text{tr}(X)B + (A, X)B + \text{tr}(B)A \circ X - \frac{1}{2} \text{tr}(B) \text{tr}(A)X \\ & \quad - \frac{1}{2} \text{tr}(B) \text{tr}(X)A + \frac{1}{2} \text{tr}(B) \text{tr}(A) \text{tr}(X)E - \frac{1}{2} \text{tr}(B)(A, X)E \\ & \quad + \frac{1}{2}(\text{tr}(A) \text{tr}(X) - (A, X))B - \frac{1}{2} \text{tr}(B)(\text{tr}(A) \text{tr}(X) - (A, X))E + (A, B, X)E \\ & = \text{tr}(A)B \circ X + \text{tr}(B)X \circ A + \text{tr}(X)A \circ B \\ & \quad - \frac{1}{2}(\text{tr}(A) \text{tr}(B)X + \text{tr}(B) \text{tr}(X)A + \text{tr}(A) \text{tr}(X)B) \\ & \quad + \frac{1}{2}((A, B)X + (B, X)A + (A, X)B) + (A, B, X)E - 2B \circ (A \times X) \\ & = A \circ (B \times X) + B \circ (A \times X) + X \circ (A \times B) - 2B \circ (A \times X) \\ & = A \circ (B \times X) - B \circ (A \times X) + (A \times B) \circ X \\ & = [\tilde{A}, \tilde{B}]X + (A \times B) \sim X. \end{aligned} $	$ \begin{aligned} & \frac{1}{2}(B, X)A + \frac{1}{6}(A, B)X - 2B \times (A \times X) + \frac{1}{3}(A, B)X \\ & = \frac{1}{2}(B, X)A + \frac{1}{2}(A, B)X - 2B \circ (A \times X) + \text{tr}(B)A \times X + \text{tr}(A \times X)B \\ & \quad - (\text{tr}(B) \text{tr}(A \times X) - (B, A \times X))E \\ & = \frac{1}{2}(B, X)A + \frac{1}{2}(A, B)X - 2B \circ (A \times X) + \text{tr}(A)B \circ X + \text{tr}(X)B \circ A \\ & \quad - \text{tr}(A) \text{tr}(X)B + (A, X)B + \text{tr}(B)A \circ X - \frac{1}{2} \text{tr}(B) \text{tr}(A)X \\ & \quad - \frac{1}{2} \text{tr}(B) \text{tr}(X)A + \frac{1}{2} \text{tr}(B) \text{tr}(A) \text{tr}(X)E - \frac{1}{2} \text{tr}(B)(A, X)E \\ & \quad + \frac{1}{2}(\text{tr}(A) \text{tr}(X) - (A, X))B - \frac{1}{2} \text{tr}(B)(\text{tr}(A) \text{tr}(X) - (A, X))E + (A, B, X)E \\ & = \text{tr}(A)B \circ X + \text{tr}(B)X \circ A + \text{tr}(X)A \circ B \\ & \quad - \frac{1}{2}(\text{tr}(A) \text{tr}(B)X + \text{tr}(B) \text{tr}(X)A + \text{tr}(A) \text{tr}(X)B) \\ & \quad + \frac{1}{2}((A, B)X + (B, X)A + (A, X)B) + (A, B, X)E - 2B \circ (A \times X) \\ & = A \circ (B \times X) + B \circ (A \times X) + X \circ (A \times B) - 2B \circ (A \times X) \\ & = A \circ (B \times X) - B \circ (A \times X) + (A \times B) \circ X \\ & = [\tilde{A}, \tilde{B}]X + (A \times B) \sim X. \end{aligned} $

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0902.0431_E00368	60	■ 0.766	$\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{\overline{x_3 a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{\overline{ax_2}}{ a } \sin a \end{array} \right. \end{aligned}$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{\overline{x_3 a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{\overline{ax_2}}{ a } \sin a \end{array} \right.$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 + \xi_3}{2} + \frac{\xi_2 - \xi_3}{2} \cos 2 a + \frac{(a, x_1)}{ a } \sin 2 a \\ \eta_3 = \frac{\xi_2 + \xi_3}{2} - \frac{\xi_2 - \xi_3}{2} \cos 2 a - \frac{(a, x_1)}{ a } \sin 2 a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 - \frac{(\xi_2 - \xi_3)a}{2 a } \sin 2 a - \frac{2(a, x_1)a}{ a ^2} \sin^2 a \\ y_2 = x_2 \cos a - \frac{\overline{x_3 a}}{ a } \sin a \\ y_3 = x_3 \cos a + \frac{\overline{ax_2}}{ a } \sin a \end{array} \right.$
0902.0431_E01100	187	■ 0.766	$\mathfrak{e}_{8(8)} = \left(\mathfrak{e}_8 \right)^{\tau\gamma} = \left\{ R \in \mathfrak{e}_8 \mid \tau\gamma R = R \right\}$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$	$\mathfrak{e}_{8(8)} = (\mathfrak{e}_8^C)^{\tau\gamma} = \{R \in \mathfrak{e}_8^C \mid \tau\gamma R = R\}$
0902.0431_E00816	136	■ 0.770	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$	$\alpha = \left(\begin{array}{cccc} \beta & 0 & M & 0 \\ 0 & \delta & 0 & N \\ a & 0 & \mu & 0 \\ 0 & b & 0 & \nu \end{array} \right), \quad \begin{array}{l} \beta, \delta \in \text{Hom}_C(\mathfrak{J}^C), \\ a, b \in \text{Hom}_C(\mathfrak{J}^C, C), \\ M, N \in \mathfrak{J}^C, \\ \mu, \nu \in C. \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00746	123	0.770	$\begin{aligned} & \pm \left(\lambda_k - \lambda_l \right), \quad \pm \left(\lambda_k + \lambda_l \right), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} \left(\mu_2 - \mu_3 \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \left(\mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left(\frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$	$\begin{aligned} & \pm \left(\lambda_k - \lambda_l \right), \quad \pm \left(\lambda_k + \lambda_l \right), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} \left(\mu_2 - \mu_3 \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_3 - \mu_1 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3 \right) \pm \frac{1}{2} \left(\mu_1 - \mu_2 \right), \\ & \pm \left(\mu_j + \frac{2}{3} \nu \right), \quad 0 \leq j \leq 3, \\ & \pm \lambda_k \pm \left(\frac{1}{2} \mu_1 - \frac{2}{3} \nu \right), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} \left(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \\ & \pm \frac{1}{2} \left(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3 \right) \pm \left(\frac{1}{2} \mu_2 - \frac{2}{3} \nu \right), \end{aligned}$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01119	190	■ 0.778	$\mathfrak{e}_{\{8\}^{\{C\}}}=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_{\{6\}^{\{C\}}}\oplus\left(\mathfrak{J}^{\{C\}}\right)^3\oplus\left(\mathfrak{J}^{\{C\}}\right)^3,$	$\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$	$\mathfrak{e}_8^C=\mathfrak{sl}(3,C)\oplus\mathfrak{e}_6^C\oplus(\mathfrak{J}^C)^3\oplus(\mathfrak{J}^C)^3,$
0902.0431_E00856	142	■ 0.779	$\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$	$\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$	$\begin{aligned} &2\sin^2a\left(E_2,E_3\times X\right)+\tau\xi\sin a\cos a-\left(E_3,Y\right)\sin a\cos a-\eta\cos^2a \\ &=\eta\sin^2a+\left(\tau\xi-\xi\right)\sin a\cos a-\eta\cos^2a \\ &=\frac{1}{2}\left(\tau\xi-\xi\right)\sin 2a-\eta\cos 2a=0. \end{aligned}$
0902.0431_E01077	183	■ 0.782	$\begin{aligned} &g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &=-4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=\left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$	$\begin{aligned} &g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &=-4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=\left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$	$\begin{aligned} &g\left(2\left(X_1\vee\gamma X_2-X_2\vee\gamma X_1\right)X\right) \\ &=-4gX_2\circ\left(gX_1\circ gX\right)+4gX_1\circ\left(gX_2\circ gX\right) \\ &=-gX_2gX_1gX-gX_2gXgX_1-gX_1gXgX_2-gXgX_1gX_2 \\ &\quad +gX_1gX_2gX+gX_1gXgX_2+gX_2gXgX_1+gXgX_2gX_1 \\ &=\left[gX_1,gX_2\right]gX-gX\left[gX_1,gX_2\right]=g\left(\left(\varphi_*\left[gX_1,gX_2\right]\right)X\right). \end{aligned}$
0902.0431_E01137	192	■ 0.785	$\begin{aligned} &\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right)^w=\left\{R\in\mathfrak{e}_{\{8\}^{\{C\}}}\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &=\left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$	$\begin{aligned} &\left(\mathfrak{e}_8\right)^w=\left\{R\in\mathfrak{e}_8^C\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &=\left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$	$\begin{aligned} &\left(\mathfrak{e}_8\right)^w=\left\{R\in\mathfrak{e}_8^C\mid wR=R,\tau\tilde{\lambda}R=R\right\} \\ &=\left\{\left(D,\phi,0,0\right)\in\mathfrak{e}_8^C\mid D\in\mathfrak{su}(3),\phi\in\mathfrak{e}_6\right\}\cong\mathfrak{su}(3)\oplus\mathfrak{e}_6, \end{aligned}$
0902.0431_E00299	48	■ 0.788	$\mathfrak{f}_{\{4\}^{\{C\}}}=\mathfrak{d}_{\{4\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{1\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{2\}^{\{C\}}}\oplus\widetilde{\mathfrak{a}}_{\{3\}^{\{C\}}}=\mathfrak{d}_{\{4\}^{\{C\}}}\oplus\widetilde{\mathfrak{A}}^C.$	$\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$	$\mathfrak{f}_4^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}_1^C\oplus\widetilde{\mathfrak{A}}_2^C\oplus\widetilde{\mathfrak{A}}_3^C=\mathfrak{d}_4^C\oplus\widetilde{\mathfrak{A}}^C.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01034	176	0.789	$\begin{array}{r} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu - r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{1}{3}\nu - r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu - r = 1 \end{array}$	$\begin{array}{r} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu - r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{1}{3}\nu + r = 1 \end{array}$	$\begin{array}{r} \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu - r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}\mu_3 + \frac{1}{3}\nu - r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu + r = 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}\mu_3 - \frac{1}{3}\nu - r = 1 \end{array}$
0902.0431_E00899	149	0.790	$\begin{aligned} & \varphi_*(D)X \\ & \gamma\varphi_*(D)\gamma Y \\ & 0 \\ & 0 \end{aligned}$	$\begin{aligned} & \chi \left(\begin{array}{c} \varphi_*(D)X \\ \gamma\varphi_*(D)\gamma Y \\ 0 \\ 0 \end{array} \right) = \chi \left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ 0 & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix} \right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$	$\begin{aligned} & = \chi \left(\begin{pmatrix} \varphi_*(D)X & 0 & 0 & 0 \\ \gamma\varphi_*(D)\gamma Y & \tau\varphi_*(D)\tau & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ \xi \\ \eta \end{pmatrix} \right) \\ & = \chi(\Phi(\varphi_*(D), 0, 0, 0)P). \end{aligned}$
0902.0431_E00631	104	0.794	$\begin{aligned} & \varphi(B)(M + \mathbf{a}) = g^{-1}(Bg(M + \mathbf{a})B^*) \\ & = g^{-1} \left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix} \right) \right) \\ & = g^{-1} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ iD\mathbf{a}^*\bar{p} & DMD^* - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \right) \\ & = DMD^* + p\mathbf{a}D^* = \beta(M + \mathbf{a}). \end{aligned}$	$\begin{aligned} & \varphi(B)(M + \mathbf{a}) = g^{-1}(Bg(M + \mathbf{a})B^*) \\ & = g^{-1} \left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix} \right) \right) \\ & = g^{-1} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ iD\mathbf{a}^*\bar{p} & DMD^* - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \right) \\ & = DMD^* + p\mathbf{a}D^* = \beta(M + \mathbf{a}). \end{aligned}$	$\begin{aligned} & \varphi(B)(M + \mathbf{a}) = g^{-1}(Bg(M + \mathbf{a})B^*) \\ & = g^{-1} \left(\begin{pmatrix} p & 0 \\ 0 & D \end{pmatrix} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ i\mathbf{a}^* & M - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \begin{pmatrix} \bar{p} & 0 \\ 0 & D^* \end{pmatrix} \right) \right) \\ & = g^{-1} \left(\begin{pmatrix} \frac{1}{2}\text{tr}(M) & i\mathbf{a} \\ iD\mathbf{a}^*\bar{p} & DMD^* - \frac{1}{2}\text{tr}(M)E \end{pmatrix} \right) \\ & = DMD^* + p\mathbf{a}D^* = \beta(M + \mathbf{a}). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_EQ0108	21	0.795	$\begin{aligned} (\alpha x)(\alpha y) &= (a+A\boldsymbol{m})(b+A\boldsymbol{n}) \quad \&= (a\boldsymbol{b}-\langle A\boldsymbol{m}, A\boldsymbol{n} \rangle + (a\boldsymbol{A}\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^* A \boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^* A \boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy) \end{aligned}$	$\begin{aligned} (\alpha x)(\alpha y) &= (a + A\boldsymbol{m})(b + A\boldsymbol{n}) \\ &= (ab - \langle A\boldsymbol{m}, A\boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{A\boldsymbol{m} \times A\boldsymbol{n}}) \\ &= (ab - \langle \boldsymbol{m}, A^* A \boldsymbol{n} \rangle) + (aA\boldsymbol{n} + \bar{b}A\boldsymbol{m} - \overline{\tilde{A}(\boldsymbol{m} \times \boldsymbol{n})}) \\ &= (ab - \langle \boldsymbol{m}, \boldsymbol{n} \rangle) + A(a\boldsymbol{n} + \bar{b}\boldsymbol{m} - \overline{\boldsymbol{m} \times \boldsymbol{n}}) \\ &= \varphi(A)((a + \boldsymbol{m})(b + \boldsymbol{n})) = \alpha(xy). \end{aligned}$
0902.0431_EQ0882	146	0.798	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$	$\alpha P = (X, Y, \xi, \eta), \quad X, Y \quad \text{are diagonal}, \quad \xi > 0.$
0902.0431_EQ0695	115	0.803	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C$	$\Phi = \Phi_{-3} + \Phi_{-2} + \Phi_{-1} + \Phi_0 + \Phi_1 + \Phi_2 + \Phi_3, \quad \Phi_i \in \mathfrak{e}_7^C,$
0902.0431_EQ0931	154	0.803	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$	$\epsilon A = \overline{(\text{Ad } J_3) A}, \quad J_3 = \begin{pmatrix} 0 & E \\ -E & 0 \end{pmatrix},$
0902.0431_EQ0666	110	0.804	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$	$\begin{aligned} E_{6(6)} &\simeq \text{Sp}(4)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{42}, \\ E_{6(2)} &\simeq (\text{Sp}(1) \times \text{SU}(6))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{40}, \\ E_{6(-14)} &\simeq (U(1) \times \text{Spin}(10))/\boldsymbol{Z}_4 \times \boldsymbol{R}^{32}, \\ E_{6(-26)} &\simeq F_4 \times \boldsymbol{R}^{26}. \end{aligned}$
0902.0431_EQ1192	200	0.804	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$	$\begin{aligned} (a \times b)c &= -(a_1 \wedge c_1, b)a_1 \wedge c_2 + (a_1 \wedge c_2, b)a_2 \wedge c_1 \\ &\quad + (a_2 \wedge c_1, b)a_1 \wedge c_2 - (a_2 \wedge c_2, b)a_1 \wedge c_1 + \frac{4}{5}(a, b)c. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00548	92	■ 0.835	$\begin{aligned} 0 &= \left(\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \times \left(\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1) \right) \\ &= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1). \end{aligned}$	$0 = (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1)$	$0 = (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1)) \times (\xi E_1 + \xi_2 E_2 + \xi_3 E_3 + F_1(x_1))$ $= (\xi_2 \xi_3 - (x_1, x_1)) E_1 + \xi \xi_3 E_2 + \xi \xi_2 E_3 - \xi F_1(x_1).$
0902.0431_E00379	63	■ 0.836	$\begin{aligned} \mathfrak{J}_{\sigma} &= \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\} \\ &= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\}) \\ &= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1, \end{aligned}$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$	$\mathfrak{J}_{\sigma} = \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\})$ $= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1,$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$	$\mathfrak{J}_{\sigma} = \{X \in \mathfrak{J} \mid \sigma X = X\} = \left\{ \begin{pmatrix} \xi_1 & 0 & 0 \\ 0 & \xi_2 & x_1 \\ 0 & \bar{x}_1 & \xi_3 \end{pmatrix} \mid \xi_i \in \mathbf{R}, x_1 \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1 \text{ (where } \mathfrak{C}_1 = \{\xi E_1 \mid \xi \in \mathbf{R}\})$ $= \mathfrak{J}(2, \mathfrak{C}) \oplus \mathfrak{C}_1,$ $\mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid \sigma X = -X\} = \left\{ \begin{pmatrix} 0 & x_3 & \bar{x}_2 \\ \bar{x}_3 & 0 & 0 \\ x_2 & 0 & 0 \end{pmatrix} \mid x_i \in \mathfrak{C} \right\}$ $= \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}.$
0902.0431_E00189	32	■ 0.837	$\begin{aligned} \alpha e_1((\alpha e_3)b_0) &= a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0 \\ &= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0, \\ \alpha e_2((\alpha e_5)b_0) &= a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0) \\ &= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0. \end{aligned}$	$(\alpha e_1)((\alpha e_3)b_0) = a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0$ $= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0,$ $(\alpha e_2)((\alpha e_5)b_0) = a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0)$ $= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0.$	$(\alpha e_1)((\alpha e_3)b_0) = a_1(a_3b_0) = a_1(a_2(a_1b_0)) = -a_1(a_1(a_2b_0)) = a_2b_0$ $= \alpha(e_2)b_0 = \alpha(e_1e_3)b_0,$ $(\alpha e_2)((\alpha e_5)b_0) = a_2(a_5b_0) = a_2(a_1(a_4b_0)) = -a_1(a_2(a_4b_0)) = -a_1(a_6b_0)$ $= a_6(a_1b_0) = a_7b_0 = \alpha(e_7)b_0 = \alpha(e_2e_5)b_0.$
0902.0431_E00643	106	■ 0.840	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$	$\tau h(A, B) = h(B, A), \quad \overline{h(A, B)} = h(\bar{A}, \bar{B}), \quad h(A, B)^* = h(B^*, A^*).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
9902.0431_EQ06603	101	0.841	$\begin{aligned} & \begin{aligned} & \left(M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3 \right) \\ & = \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ & = \left\{ i \begin{pmatrix} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3e_4 & -a_2e_4 \\ -a_3e_4 & 0 & a_1e_4 \\ a_2e_4 & -a_1e_4 & 0 \end{pmatrix} \middle \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ & = \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned} \end{aligned}$	$\begin{aligned} & = \{M + iF(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = \mathfrak{J}_{\boldsymbol{H}} \oplus i\boldsymbol{H}^3, \\ & (\mathfrak{J}^C)_{-\tau\gamma} = \{X \in \mathfrak{J}^C \mid \tau\gamma X = -X\} \\ & = \left\{ i \begin{pmatrix} \mu_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \mu_2 & m_1 \\ m_2 & \bar{m}_1 & \mu_3 \end{pmatrix} + \begin{pmatrix} 0 & a_3e_4 & -a_2e_4 \\ -a_3e_4 & 0 & a_1e_4 \\ a_2e_4 & -a_1e_4 & 0 \end{pmatrix} \middle \begin{matrix} \mu_i \in \boldsymbol{R} \\ m_i, a_i \in \boldsymbol{H} \end{matrix} \right\} \\ & = \{iM + F(\boldsymbol{ae}_4) \mid M \in \mathfrak{J}(3, \boldsymbol{H}), \boldsymbol{a} \in \boldsymbol{H}^3\} \\ & = i\mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3, \end{aligned}$	
9902.0431_EQ06348	56	0.842	$\begin{aligned} & \left(\alpha_1, \alpha_1 \right) = \left(\alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left(\alpha_3, \alpha_3 \right) = \left(\alpha_4, \alpha_4 \right) = \frac{1}{18}, \\ & \left(\alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left(\alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left(\alpha_3, \alpha_4 \right) = -\frac{1}{36}, \\ & \left(\alpha_1, \alpha_3 \right) = \left(\alpha_1, \alpha_4 \right) = \left(\alpha_2, \alpha_4 \right) = 0, \\ & \left(-\mu, -\mu \right) = \frac{1}{9}, \quad \left(-\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4, \end{aligned}$	$\begin{aligned} & \left(\alpha_1, \alpha_1 \right) = \left(\alpha_2, \alpha_2 \right) = \frac{1}{9}, \quad \left(\alpha_3, \alpha_3 \right) = \left(\alpha_4, \alpha_4 \right) = \frac{1}{18} \\ & \left(\alpha_1, \alpha_2 \right) = -\frac{1}{18}, \quad \left(\alpha_2, \alpha_3 \right) = -\frac{1}{18}, \quad \left(\alpha_3, \alpha_4 \right) = -\frac{1}{36} \\ & \left(\alpha_1, \alpha_3 \right) = \left(\alpha_1, \alpha_4 \right) = \left(\alpha_2, \alpha_4 \right) = 0, \\ & \left(-\mu, -\mu \right) = \frac{1}{9}, \quad \left(-\mu, \alpha_1 \right) = -\frac{1}{18}, \quad \left(-\mu, \alpha_i \right) = 0, \quad i = 2, 3, 4 \end{aligned}$	
9902.0431_EQ0637	105	0.844	$\begin{aligned} & \left(\phi X, X, X \right) = 0, \left(\phi X, Y \right) + \left(X, \phi Y \right) = 0 \\ & = \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \boldsymbol{C}}, T \in (\mathfrak{J}_{\boldsymbol{C}})_0 \right\} \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6, \boldsymbol{C}} &= \left\{ \phi \in \text{Hom}_{\boldsymbol{C}} \left((\mathfrak{J}_{\boldsymbol{C}})^C \right) \mid (\phi X, X, X) = 0, \langle \phi X, Y \rangle + \langle X, \phi Y \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \boldsymbol{C}}, T \in (\mathfrak{J}_{\boldsymbol{C}})_0 \right\} \end{aligned}$	

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0902.0431_E00686	114	■ 0.844	$\begin{aligned} \phi &= \left[\phi_1, \phi_2 \right] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A &= \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B &= - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu &= (A_1, B_2) - (B_1, A_2). \end{aligned}$	$\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$	$\begin{cases} \phi = [\phi_1, \phi_2] + 2A_1 \vee B_2 - 2A_2 \vee B_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ B = - \left({}^t\phi_1 + \frac{2}{3}\nu_1 \right) B_2 + \left({}^t\phi_2 + \frac{2}{3}\nu_2 \right) B_1 \\ \nu = (A_1, B_2) - (B_1, A_2). \end{cases}$
0902.0431_E00509	83	■ 0.847	$\begin{aligned} &\left[h, \widetilde{F}_{23}(a) \right] = \left[h_\delta, \widetilde{F}_{23}(a) \right] + \left[\widetilde{H}, \widetilde{F}_{23}(a) \right] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$	$\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$	$\begin{aligned} [h, \widetilde{F}_{23}(a)] &= [h_\delta, \widetilde{F}_{23}(a)] + [\widetilde{H}, \widetilde{F}_{23}(a)] \\ &= (h_\delta F_{23}(a))^\sim + \frac{1}{2} [H, F_{23}(a)]^\sim \quad (\text{Lemma 3.6.3}) \\ &= \widetilde{F}_{23}(h_\delta a) + \frac{1}{2} [H, F_{23}(a)]^\sim \\ &= \left(\lambda_k + \frac{1}{2}(\mu_2 - \mu_3) \right) \widetilde{F}_{23}(a), \end{aligned}$
0902.0431_E00091	18	■ 0.848	$\begin{aligned} &\left[H_1, [H_1, S_1] \right] = [H_1, S_2] = -S_1, \quad [H_1, [H_1, S_2]] = [H_1, -S_1] = -S_2, \\ &\left[H_1, [H_1, S_3] \right] = [H_1, -S_4] = -S_3, \quad [H_1, [H_1, S_4]] = [H_1, S_3] = -S_4, \\ &\left[H_1, [H_1, S_5] \right] = 0, \quad [H_1, [H_1, S_6]] = 0 \end{aligned}$	$\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$	$\begin{aligned} [H_1, [H_1, S_1]] &= [H_1, S_2] = -S_1, & [H_1, [H_1, S_2]] &= [H_1, -S_1] = -S_2, \\ [H_1, [H_1, S_3]] &= [H_1, -S_4] = -S_3, & [H_1, [H_1, S_4]] &= [H_1, S_3] = -S_4, \\ [H_1, [H_1, S_5]] &= 0, & [H_1, [H_1, S_6]] &= 0 \end{aligned}$
0902.0431_E00683 (2)	113	■ 0.848	$\begin{aligned} &\left[\alpha P, \alpha Q \right] \alpha P = \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$	$\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$	$\begin{aligned} \{\alpha P, \alpha Q\} \alpha P &= \frac{8}{3} ((\alpha P \times \alpha P) \alpha Q - (\alpha P \times \alpha Q) \alpha P) \quad (\text{Lemma 4.1.1.(2)}) \\ &= \frac{8}{3} (\alpha(P \times P) Q - \alpha(P \times Q) P) = \{P, Q\} \alpha P. \end{aligned}$
0902.0431_E00423	69	■ 0.848	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$	$\beta X = B X B^* \quad \text{or} \quad \beta X = B \bar{X} B^*, \quad X \in \mathfrak{J}_C.$
0902.0431_E00995	165	■ 0.852	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$	$E_8^C = \{ \alpha \in \text{Iso}_C(\mathfrak{e}_8^C) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$

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0902.0431_E00521	86	■ 0.873	$\begin{aligned} B_6(h, h') &= \frac{4}{3} B_4 \left(\sum_{k=0}^3 \lambda_k H_k, \sum_{k=0}^3 \lambda_k' H_k \right) + 12 \left(\sum_{j=1}^3 \mu_j E_j, \sum_{j=1}^3 \mu_j' E_j \right) \\ &= \frac{4}{3} 18 \sum_{k=0}^3 \lambda_k \lambda_k' + 12 \sum_{j=1}^3 \mu_j \mu_j' \quad (\text{Theorem 2.6.2}) \\ &= 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right). \end{aligned}$	$B_6(h, h') = \frac{4}{3} B_4 \left(\sum_{k=0}^3 \lambda_k H_k, \sum_{k=0}^3 \lambda_k' H_k \right) + 12 \left(\sum_{j=1}^3 \mu_j E_j, \sum_{j=1}^3 \mu_j' E_j \right)$ $= \frac{4}{3} 18 \sum_{k=0}^3 \lambda_k \lambda_k' + 12 \sum_{j=1}^3 \mu_j \mu_j' \quad (\text{Theorem 2.6.2})$ $= 12 \left(2 \sum_{k=0}^3 \lambda_k \lambda_k' + \sum_{j=1}^3 \mu_j \mu_j' \right).$	
0902.0431_E00505	83	■ 0.876	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \tilde{\mathfrak{J}}_0^C.$
0902.0431_E01105	187	■ 0.877	$z \left((E_8)^{\tilde{\lambda}\gamma} \right) = \{1, \tilde{\lambda}\gamma\}.$	$z \left((E_8)^{\tilde{\lambda}\gamma} \right) = \{1, \tilde{\lambda}\gamma\}.$	$z((E_8)^{\tilde{\lambda}\gamma}) = \{1, \tilde{\lambda}\gamma\}.$
0902.0431_E00279	45	■ 0.879	$\begin{aligned} &\left(D_1 x_1, \overline{x_2 x_3} \right) = - \left(x_1, D_1 (\overline{x_2 x_3}) \right) = - \left(x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)} \right) \\ &= - \left(x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) x_2} \right) = - \left(x_3 x_1, \overline{D_2 x_2} \right) - \left(x_1 x_2, \overline{D_3 x_3} \right). \end{aligned}$	$\begin{aligned} (D_1 x_1, \overline{x_2 x_3}) &= - (x_1, D_1 (\overline{x_2 x_3})) = - \left(x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)} \right) \\ &= - \left(x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) x_2} \right) = - (x_3 x_1, \overline{D_2 x_2}) - (x_1 x_2, \overline{D_3 x_3}). \end{aligned}$	$\begin{aligned} (D_1 x_1, \overline{x_2 x_3}) &= - (x_1, D_1 (\overline{x_2 x_3})) = - (x_1, \overline{(D_2 x_2) x_3 + x_2 (D_3 x_3)}) \\ &= - (x_1, \overline{x_3 (D_2 x_2) + (D_3 x_3) x_2}) = - (x_3 x_1, \overline{D_2 x_2}) - (x_1 x_2, \overline{D_3 x_3}). \end{aligned}$
0902.0431_E00073	15	■ 0.881	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ e_1(E_{12} + E_{21}), \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ E_{13} - E_{31}, \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$	$\begin{array}{l} e_1(E_{11} - E_{22}), \quad e_1(E_{22} - E_{33}), \quad E_{12} - E_{21}, \quad e_1(E_{12} + E_{21}), \\ E_{13} - E_{31}, \quad e_1(E_{13} + E_{31}), \quad E_{23} - E_{32}, \quad e_1(E_{23} + E_{32}) \end{array}$
0902.0431_E01126	191	■ 0.881	$\langle R_1, R_2 \rangle = 60 \text{tr} (D_1 (\tau^t D_2)) + \frac{5}{2} B_6 (\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$	$\langle R_1, R_2 \rangle = 60 \text{tr} (D_1 (\tau^t D_2)) + \frac{5}{2} B_6 (\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$	$\langle R_1, R_2 \rangle = 60 \text{tr} (D_1 (\tau^t D_2)) + \frac{5}{2} B_6 (\phi_1, \tau^t \phi_2 \tau) + 15 \langle \mathbf{X}_1, \mathbf{X}_2 \rangle + 15 \langle \mathbf{Y}_1, \mathbf{Y}_2 \rangle.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00708	117	■ 0.882	$\begin{aligned} & \{[\] \& \Phi, P \times P\} \setminus = \& \{ \left[\right. \\ & \Phi(\phi, A, B, \nu), \Phi \left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta) \right) \} \\ & = \dots \text{(using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B] \\ & = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \dots \\ & = 2\Phi P \times P. \end{aligned}$	$[\Phi, P \times P]$ $= \left[\Phi(\phi, A, B, \nu), \Phi \left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta) \right) \right]$ $= \dots \text{(using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B]$ $= \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \dots$ $= 2\Phi P \times P.$	$[\Phi, P \times P]$ $= \left[\Phi(\phi, A, B, \nu), \Phi \left(-X \vee Y, -\frac{1}{2}(Y \times Y - \xi X), \frac{1}{2}(X \times X - \eta Y), \frac{1}{4}((X, Y) - 3\xi\eta) \right) \right]$ $= \dots \text{(using } \phi(X \times Y) = \phi'X \times Y + X \times \phi'Y \text{ (Lemma 3.4.3.(1)), } [\phi, A \vee B]$ $= \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4.(1)), the formula about } A \vee B \text{ (Lemma 3.4.1) etc.)} \dots$ $= 2\Phi P \times P.$
0902.0431_EQ1121	190	■ 0.884	$\left\{ \begin{array}{l} D = \left[D_1, D_2 \right] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$	$\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2 \end{array} \right.$	$\left\{ \begin{array}{l} D = [D_1, D_2] + \frac{1}{4} \boldsymbol{X}_1 \cdot \boldsymbol{Y}_2 - \frac{1}{4} \boldsymbol{X}_2 \cdot \boldsymbol{Y}_1 \\ \phi = [\phi_1, \phi_2] + \frac{1}{2} \boldsymbol{X}_1 \vee \boldsymbol{Y}_2 - \frac{1}{2} \boldsymbol{X}_2 \vee \boldsymbol{Y}_1 \\ \boldsymbol{X} = \phi_1 \boldsymbol{X}_2 - \phi_2 \boldsymbol{X}_1 + D_1 \boldsymbol{X}_2 - D_2 \boldsymbol{X}_1 - \boldsymbol{Y}_1 \times \boldsymbol{Y}_2 \\ \boldsymbol{Y} = -{}^t \phi_1 \boldsymbol{Y}_2 + {}^t \phi_2 \boldsymbol{Y}_1 - {}^t D_1 \boldsymbol{Y}_2 + {}^t D_2 \boldsymbol{Y}_1 + \boldsymbol{X}_1 \times \boldsymbol{X}_2, \end{array} \right.$
0902.0431_EQ0821 (iii)	137	■ 0.887	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}) \end{array} \right.$	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}) \end{array} \right.$	$\left\{ \begin{array}{l} 2\delta(X \times X) \times N = a(X)\beta X \\ \delta(X \times X) \times \delta(X \times X) = a(X)(\det X)M \\ (\beta X, \delta(X \times X)) + \det X = 3a(X)b(X \times X) \quad (\text{use (i)}). \end{array} \right.$

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0902.0431_E00172 (ii)	30	■ 0.890	$\overline{\bar{x}}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$	$\bar{x}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$	$\bar{x}(\bar{y}p) = (\overline{xy})p, \quad x, y \in \mathfrak{C}.$
0902.0431_E00235	39	■ 0.891	$\begin{aligned} & \alpha^{-1}(Y \times Y) \times \alpha^{-1}(Y \times Y) = (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ & = (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ & = \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$	$\begin{aligned} & {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) = (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ & = (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ & = \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$	$\begin{aligned} {}^t\alpha^{-1}(Y \times Y) \times {}^t\alpha^{-1}(Y \times Y) &= (\alpha Y \times \alpha Y) \times (\alpha Y \times \alpha Y) \quad (\text{Lemma 2.2.2.(4)}) \\ &= (\det \alpha Y) \alpha Y \quad (\text{Lemma 2.1.1.(3)}) = (\det Y) \alpha Y = \alpha((\det Y) Y) \\ &= \alpha((Y \times Y) \times (Y \times Y)), \quad Y \in \mathfrak{J}. \end{aligned}$
0902.0431_E00087	18	■ 0.892	$\begin{array}{l} \left[\begin{array}{l} H_1, L_{12} \end{array} \right] = 2L_{21}, \quad \left[\begin{array}{l} H_1, L_{21} \end{array} \right] = -2L_{12}, \quad \left[\begin{array}{l} H_1, L_{13} \end{array} \right] = L_{31}, \\ \left[\begin{array}{l} H_1, L_{31} \end{array} \right] = -L_{13}, \quad \left[\begin{array}{l} H_1, L_{23} \end{array} \right] = -L_{32}, \quad \left[\begin{array}{l} H_1, L_{32} \end{array} \right] = L_{23}. \end{array}$	$\begin{array}{l} \left[\begin{array}{l} H_1, L_{12} \end{array} \right] = 2L_{21}, \quad \left[\begin{array}{l} H_1, L_{21} \end{array} \right] = -2L_{12}, \quad \left[\begin{array}{l} H_1, L_{13} \end{array} \right] = L_{31}, \\ \left[\begin{array}{l} H_1, L_{31} \end{array} \right] = -L_{13}, \quad \left[\begin{array}{l} H_1, L_{23} \end{array} \right] = -L_{32}, \quad \left[\begin{array}{l} H_1, L_{32} \end{array} \right] = L_{23}. \end{array}$	$\begin{array}{l} [H_1, L_{12}] = 2L_{21}, \quad [H_1, L_{21}] = -2L_{12}, \quad [H_1, L_{13}] = L_{31}, \\ [H_1, L_{31}] = -L_{13}, \quad [H_1, L_{23}] = -L_{32}, \quad [H_1, L_{32}] = L_{23}. \end{array}$
0902.0431_E00417	68	■ 0.894	$\begin{aligned} & (X+M) \times (Y+N) \quad \&= \left(X \times Y - \frac{1}{2}(M^*N + N^*M) \right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ & (X+M, Y+N) \quad \&= (X, Y) + 2(M, N), \\ & w(X+M) \quad \&= X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \right), \end{aligned}$	$\begin{aligned} & (X+M) \times (Y+N) = \left(X \times Y - \frac{1}{2}(M^*N + N^*M) \right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ & (X+M, Y+N) = (X, Y) + 2(M, N), \\ & w(X+M) = X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \right), \end{aligned}$	$\begin{aligned} (X+M) \times (Y+N) &= \left(X \times Y - \frac{1}{2}(M^*N + N^*M) \right) - \frac{1}{2}(MY + NX + \overline{M \times N}), \\ (X+M, Y+N) &= (X, Y) + 2(M, N), \\ w(X+M) &= X + \omega_1 M, \quad \left(\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \right), \end{aligned}$
0902.0431_E00690 (i)	114	■ 0.894	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$	$\begin{cases} \Phi P \times P = 0, & P \in \mathfrak{M}^C \\ \{\Phi P, Q\} + \{P, \Phi Q\} = 0, & P, Q \in \mathfrak{P}^C. \end{cases}$
0902.0431_E00084	17	■ 0.895	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S} \quad (\text{Theorem 1.5.1}).$

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0902.0431_E00738	122	0.896	$\begin{aligned} & \left(\Phi, P \times Q \right)_7 = \left(\Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \end{pmatrix} \right)_7 \\ &= (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ &\quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \\ &= (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ &\quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi \omega + \zeta \eta) \\ &= \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$	$\begin{aligned} (\Phi, P \times Q)_7 &= \left(\Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \end{pmatrix} \right)_7 \\ &= (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ &\quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \\ &= (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ &\quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi \omega + \zeta \eta) \\ &= \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$	$\begin{aligned} (\Phi, P \times Q)_7 &= \left(\Phi \begin{pmatrix} \phi \\ A \\ B \\ \nu \end{pmatrix}, \Phi \begin{pmatrix} -\frac{1}{2}(X \vee W + Z \vee Y) \\ -\frac{1}{4}(2Y \times W - \xi Z - \zeta X) \\ \frac{1}{4}(2X \times Z - \eta W - \omega Y) \\ \frac{1}{8}((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \end{pmatrix} \right)_7 \\ &= (\phi, X \vee W + Z \vee Y)_6 - (A, 2X \times Z - \eta W - \omega Y) + (2Y \times W - \xi Z - \zeta X, B) \\ &\quad - \frac{1}{3}\nu((X, W) + (Z, Y) - 3(\xi \omega + \zeta \eta)) \\ &= (\phi X, W) + (\phi Z, Y) - 2(A, X, Z) + \eta(A, W) + \omega(A, Y) + 2(Y, W, B) \\ &\quad - \xi(Z, B) - \zeta(X, B) - \frac{1}{3}\nu(X, W) - \frac{1}{3}\nu(Z, Y) + \nu(\xi \omega + \zeta \eta) \\ &= \left\{ \begin{pmatrix} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{pmatrix}, \begin{pmatrix} Z \\ W \\ \zeta \\ \omega \end{pmatrix} \right\} = \{\Phi P, Q\}. \end{aligned}$
0902.0431_E00122	23	0.896	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$	$\alpha e_1 = e_1 \quad \text{or} \quad \alpha e_1 = -e_1.$
0902.0431_E00587	98	0.896	$\begin{aligned} & \left(\phi \in \text{Hom}_C((\mathfrak{H})^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right) \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \right\} \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6, \mathbf{H}} &= \left\{ \phi \in \text{Hom}_C((\mathfrak{H})^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \right\} \\ &= \left\{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \right\} \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6, \mathbf{H}} &= \{ \phi \in \text{Hom}_C((\mathfrak{H})^C) \mid (\phi M, M, M) = 0, \langle \phi M, N \rangle + \langle M, \phi N \rangle = 0 \} \\ &= \{ \delta + i\tilde{T} \mid \delta \in \mathfrak{f}_{4, \mathbf{H}}, T \in (\mathfrak{H})_0 \}, \end{aligned}$
0902.0431_E00615	102	0.896	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$	$\langle gX, gY \rangle = \langle X, Y \rangle, \quad X, Y \in \mathfrak{J}^C.$

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0902.0431_E00307	49	■ 0.899	$\begin{aligned} & \backslash begin{aligned} \& =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & \backslash \& =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & \backslash \& -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & \backslash \& =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & \backslash \& =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$	$\begin{aligned} & =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$	$\begin{aligned} & =\delta(A \circ(B \circ X)-B \circ(A \circ X))-(A \circ(B \circ \delta X)-B \circ(A \circ \delta X)) \\ & =\delta A \circ(B \circ X)+A \circ(\delta B \circ X)+A \circ(B \circ \delta X)-\delta B \circ(A \circ X) \\ & -B \circ(\delta A \circ X)-B \circ(A \circ \delta X)-A \circ(B \circ \delta X)+B \circ(A \circ \delta X) \\ & =\delta A \circ(B \circ X)-B \circ(\delta A \circ X)+A \circ(\delta B \circ X)-\delta B \circ(A \circ X) \\ & =[\widetilde{\delta A}, \widetilde{B}] X+[\widetilde{A}, \widetilde{\delta B}] X, \quad X \in \mathfrak{J}^C . \end{aligned}$
0902.0431_E01108	188	■ 0.902	$\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right)^{\tilde{\lambda}} \gamma=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} \mid R_{\{k\}}, R_{\{l\}} \in \mathfrak{e}_{\{8\}^{\{C\}}}\right)_{-\tilde{\lambda}} \gamma \right\} .$	$\left(\mathfrak{e}_8^C\right)^{\tilde{\lambda} \gamma}=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} .$	$\left(\mathfrak{e}_8^C\right)^{\tilde{\lambda} \gamma}=\left\{\sum_{k, l}\left[R_k, R_l\right] \mid R_k, R_l \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma}\right\} .$
0902.0431_E00372	62	■ 0.903	$\begin{aligned} & \backslash left{\backslash begin{array}{l} \xi_1+\xi_2+\xi_3= \\ & \backslash operatorname{tr}(X) \backslash \backslash \xi_1\{ \}^2+\xi_2\{ \}^2+ \\ & \backslash \xi_3\{ \}^2=(X, X) \backslash \backslash \xi_1\{1\} \{ \}^3+\xi_2\{ \}^3+ \\ & \backslash \xi_3\{ \}^3=\operatorname{tr}(X, X, X) \end{array} \backslash right} .$	$\left\{\begin{aligned} & \xi_1+\xi_2+\xi_3=\operatorname{tr}(X) \\ & \xi_1^2+\xi_2^2+\xi_3^2=(X, X) \\ & \xi_1^3+\xi_2^3+\xi_3^3=\operatorname{tr}(X, X, X) \end{aligned}\right.$	$\left\{\begin{aligned} & \xi_1+\xi_2+\xi_3=\operatorname{tr}(X) \\ & \xi_1^2+\xi_2^2+\xi_3^2=(X, X) \\ & \xi_1^3+\xi_2^3+\xi_3^3=\operatorname{tr}(X, X, X) . \end{aligned}\right.$
0902.0431_E00023	9	■ 0.904	$\begin{aligned} & \backslash begin{array}{ll} \{ \backslash left[F_{\{i\} j}, F_{\{j\} k}\right]=F_{\{i\} k},\} \& i, j, k \text { are distinct } \backslash \backslash \{ \backslash left[F_{\{i\} j}, \\ & F_{\{k\} l}\right]=0,\} \& i, j, k, l \text { are distinct. } \backslash \end{array}$	$\begin{aligned} & \left[F_{i j}, F_{j k}\right]=F_{i k}, \quad i, j, k \text { are distinct} \\ & \left[F_{i j}, F_{k l}\right]=0, \quad i, j, k, l \text { are distinct.} \end{aligned}$	$\begin{aligned} & \left[F_{i j}, F_{j k}\right]=F_{i k}, \quad i, j, k \text { are distinct,} \\ & \left[F_{i j}, F_{k l}\right]=0, \quad i, j, k, l \text { are distinct.} \end{aligned}$
0902.0431_E01106	187	■ 0.904	$\alpha R=k R, \quad R \in\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right)_{-\tilde{\lambda}} \gamma .$	$\alpha R=k R, \quad R \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma} .$	$\alpha R=k R, \quad R \in\left(\mathfrak{e}_8^C\right)_{-\tilde{\lambda} \gamma} .$
0902.0431_E00947	157	■ 0.904	$\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \boldsymbol{C}^6$	$\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \boldsymbol{C}^6$	$\sum(A \boldsymbol{a} \wedge A \boldsymbol{b} \wedge A \boldsymbol{c})=\overline{\boldsymbol{a}} \wedge \overline{\boldsymbol{b}} \wedge \overline{\boldsymbol{c}}, \quad \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \boldsymbol{C}^6$
0902.0431_E01009	168	■ 0.905	$U(248)=U\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right)=\left\{\alpha \in \operatorname{Iso}_{\{C\}}\left(\mathfrak{e}_{\{8\}^{\{C\}}}\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$	$U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{e}_8^C\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$	$U(248)=U\left(\mathfrak{e}_8^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{e}_8^C\right) \mid\left\langle\alpha R_1, \alpha R_2\right\rangle=\left\langle R_1, R_2\right\rangle\right\} .$
0902.0431_E00981	163	■ 0.905	$\begin{aligned} & \backslash begin{aligned} \& \mathfrak{a} \ni\left[1, \backslash left[1_{\{-\}},[1, R]\right] \backslash right]=\left[1, \backslash left[1_{\{-\}},(\theta, P,-Q, 0,2 s,-2 t)\right] \backslash right] \backslash \& \quad=\left[1,(\theta, 0, P,-2 s, 0,0)\right]=-(\theta, 0, P, 0,0,0)=-P_{-} \end{aligned}$	$\begin{aligned} & \mathfrak{a} \ni\left[1, \left[1_{-},\left[1, R\right]\right]\right]=\left[1,\left[1_{-},\left(0, P,-Q, 0,2 s,-2 t\right)\right]\right] \\ & =\left[1,\left(0,0, P,-2 s, 0,0\right)\right]=-(0,0, P, 0,0,0)=-P_{-} \end{aligned}$	$\begin{aligned} & \mathfrak{a} \ni\left[1,\left[1_{-},\left[1, R\right]\right]\right]=\left[1,\left[1_{-},\left(0, P,-Q, 0,2 s,-2 t\right)\right]\right] \\ & =\left[1,\left(0,0, P,-2 s, 0,0\right)\right]=-(0,0, P, 0,0,0)=-P_{-} . \end{aligned}$
0902.0431_E00459	75	■ 0.908	$\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$	$\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$	$\mathfrak{a} \ni\left[\delta_1, \delta+\widetilde{T}\right]=\left[\delta_1, \delta\right]+\widetilde{\delta_1 T} \text { (Theorem 3.2.2), }$

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0902.0431_E00508	83	■ 0.928	$\begin{aligned} \{[h, S]\} &= \left[h_{\delta} + \widetilde{H}, S\right] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$	$\begin{aligned} [h, S] &= \left[h_{\delta} + \widetilde{H}, S\right] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S \end{aligned}$	$\begin{aligned} [h, S] &= [h_{\delta} + \widetilde{H}, S] = [h_{\delta}, S] - [S, \widetilde{H}] \\ &= \alpha(h_{\delta})S - \widetilde{S\widetilde{H}} \text{ (Lemma 3.6.3.(1))} \\ &= \alpha(h_{\delta})S = (\pm\lambda_k \pm \lambda_l)S. \end{aligned}$
0902.0431_E00665	110	■ 0.935	$E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$	$E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$	$E_{6(6)} \cong \left(E_6^C\right)^{\tau\gamma}, E_{6(2)} \cong \left(E_6^C\right)^{\tau\lambda\gamma}, E_{6(-14)} \cong \left(E_6^C\right)^{\tau\lambda\sigma}, E_{6(-26)} \cong \left(E_6^C\right)^{\tau}.$
0902.0431_E01052	179	■ 0.936	$\left\{\beta_2 P, \beta_3 Q\right\} = \left\{\beta_2 P, \beta_2 Q\right\}, \quad P, Q \in \mathfrak{P}^C,$	$\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$	$\{\beta_2 P, \beta_3 Q\} = \{\beta_2 P, \beta_2 Q\}, \quad P, Q \in \mathfrak{P}^C,$
0902.0431_E00512	84	■ 0.938	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$	$\begin{aligned} \nu h_{\delta} &= \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3)H_0 + \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3, \\ \kappa\pi h_{\delta} &= \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3)H_0 + \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3)H_1 \\ &+ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3)H_2 + \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3)H_3 \end{aligned}$

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0902.0431_E00582	97	■ 0.938	$\begin{aligned} & \operatorname{det}(k_J(M)) \\ &= \operatorname{det} \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix} \\ &= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ &\quad - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ &\quad + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2 \end{aligned}$	$\det(k_J(M)) = \det \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix}$ $= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2$	$\det(k_J(M)) = \det \begin{pmatrix} 0 & \xi_1 & -n_3 & m_3 & n_2 & \tau(m_2) \\ -\xi_1 & 0 & -\tau(m_3) & -\tau(n_3) & -m_2 & \tau(n_2) \\ n_3 & \tau(m_3) & 0 & \xi_2 & -n_1 & m_1 \\ -m_3 & \tau(n_3) & -\xi_2 & 0 & -\tau(m_1) & -\tau(n_1) \\ -n_2 & m_2 & n_1 & \tau(m_1) & 0 & \xi_3 \\ -\tau(m_2) & -\tau(n_2) & -m_1 & \tau(n_1) & -\xi_3 & 0 \end{pmatrix}$ $= (\xi_1 \xi_2 \xi_3 - \xi_1 n_1 \tau(n_1) - \xi_1 m_1 \tau(m_1) - n_3 \tau(n_3) \xi_3 - n_3 m_2 \tau(n_1) - n_3 \tau(n_2) \tau(m_1) \\ - m_3 \tau(m_3) \xi_3 + m_3 m_2 m_1 - m_3 \tau(n_2) n_1 - n_2 \tau(m_3) \tau(n_1) - n_2 \tau(n_3) m_1 - n_2 \tau(n_2) \xi_2 \\ + \tau(m_2) \tau(m_3) \tau(m_1) - \tau(m_2) \tau(n_3) n_1 - \tau(m_2) m_2 \xi_2)^2.$
0902.0431_E00892	148	■ 0.939	$\left[\Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0) \right] = \Phi(\phi, A, -\gamma A, 0),$	$[\Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0)] = \Phi(\phi, A, -\gamma A, 0),$	$[\Phi(\phi_1, A_1, -\gamma A_1, 0), \Phi(\phi_2, A_2, -\gamma A_2, 0)] = \Phi(\phi, A, -\gamma A, 0),$
0902.0431_E00737	122	■ 0.939	$\begin{aligned} & \dots \left(\text{using } [\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.} \right) \dots \\ &= -(\Phi_1, [\Phi, \Phi_2])_7. \end{aligned}$	$= \dots \left(\text{using } [\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.} \right) \dots$ $= -(\Phi_1, [\Phi, \Phi_2])_7$	$= \dots \left(\text{using } [\phi, A \vee B] = \phi A \vee B + A \vee \phi' B \text{ (Lemma 3.4.4) etc.} \right) \dots$ $= -(\Phi_1, [\Phi, \Phi_2])_7.$
0902.0431_E01073	182	■ 0.940	$Jl(S)I = l(S)JI = -l(S)IJ$	$Jl(S)I = l(S)JI = -l(S)IJ$ ${}^t(l(S)I) = {}^tI^t(l(S)) = Il(\tau^t S) = -Il(\tau S) = -l(S)I$	$Jl(S)I = l(S)JI = -l(S)IJ,$ ${}^t(l(S)I) = {}^tI^t(l(S)) = Il(\tau^t S) = -Il(\tau S) = -l(S)I.$

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0902.0431_E00413	68	■0.941	$\left(\begin{array}{lll} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$	$\left(\begin{array}{ccc} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$	$\left(\begin{array}{ccc} \xi_1 & a_3 & \bar{a}_2 \\ \bar{a}_3 & \xi_2 & a_1 \\ a_2 & \bar{a}_1 & \xi_3 \end{array}\right) + (\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3)$
0902.0431_E00017	8	■0.942	$\left\{ \begin{array}{l} 2F_{01} = G_{01} + G_{23} + G_{45} + G_{67} \\ 2F_{23} = G_{01} + G_{23} - G_{45} - G_{67} \\ 2F_{45} = G_{01} - G_{23} + G_{45} - G_{67} \\ 2F_{67} = G_{01} - G_{23} - G_{45} + G_{67}, \end{array} \right. \left\{ \begin{array}{l} 2F_{02} = G_{02} - G_{13} - G_{46} + G_{57} \\ 2F_{13} = -G_{02} + G_{13} - G_{46} + G_{57} \\ 2F_{46} = -G_{02} - G_{13} + G_{46} + G_{57} \\ 2F_{57} = G_{02} + G_{13} + G_{46} + G_{57}, \end{array} \right.$	$\left\{ \begin{array}{l} 2F_{01} = G_{01} + G_{23} + G_{45} + G_{67} \\ 2F_{23} = G_{01} + G_{23} - G_{45} - G_{67} \\ 2F_{45} = G_{01} - G_{23} + G_{45} - G_{67} \\ 2F_{67} = G_{01} - G_{23} - G_{45} + G_{67}, \end{array} \right. \left\{ \begin{array}{l} 2F_{02} = G_{02} - G_{13} - G_{46} + G_{57} \\ 2F_{13} = -G_{02} + G_{13} - G_{46} + G_{57} \\ 2F_{46} = -G_{02} - G_{13} + G_{46} + G_{57} \\ 2F_{57} = G_{02} + G_{13} + G_{46} + G_{57}, \end{array} \right.$	$\left\{ \begin{array}{l} 2F_{01} = G_{01} + G_{23} + G_{45} + G_{67} \\ 2F_{23} = G_{01} + G_{23} - G_{45} - G_{67} \\ 2F_{45} = G_{01} - G_{23} + G_{45} - G_{67} \\ 2F_{67} = G_{01} - G_{23} - G_{45} + G_{67}, \end{array} \right. \left\{ \begin{array}{l} 2F_{02} = G_{02} - G_{13} - G_{46} + G_{57} \\ 2F_{13} = -G_{02} + G_{13} - G_{46} + G_{57} \\ 2F_{46} = -G_{02} - G_{13} + G_{46} + G_{57} \\ 2F_{57} = G_{02} + G_{13} + G_{46} + G_{57}, \end{array} \right.$
0902.0431_E00927	153	■0.942	$D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$	$D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$	$D = \left(\begin{array}{cc} B & L \\ -L^* & C \end{array} \right) + \frac{\nu}{3} \left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array} \right), \quad B, C \in \mathfrak{su}(3), L \in M(3, C), \nu \in e_1 \boldsymbol{R}.$
0902.0431_E00421	69	■0.942	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$	$\varphi(A, 1)X = AXA^*, \quad \varphi(A, \epsilon)X = A\bar{X}A^*, \quad X \in \mathfrak{J}_C.$
0902.0431_E00257	42	■0.943	$\begin{aligned} \mathfrak{f}_4 &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \begin{array}{l} \operatorname{tr}(\delta X, Y, Z) + \operatorname{tr}(X, \delta Y, Z) + \operatorname{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array} \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, (\delta X, Y) + (X, \delta Y) = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, \delta E = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \right\}. \end{aligned}$	$\begin{aligned} \mathfrak{f}_4 &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \begin{array}{l} \operatorname{tr}(\delta X, Y, Z) + \operatorname{tr}(X, \delta Y, Z) + \operatorname{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array} \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, (\delta X, Y) + (X, \delta Y) = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, \delta E = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \right\}. \end{aligned}$	$\begin{aligned} \mathfrak{f}_4 &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \begin{array}{l} \operatorname{tr}(\delta X, Y, Z) + \operatorname{tr}(X, \delta Y, Z) + \operatorname{tr}(X, Y, \delta Z) = 0 \\ (\delta X, Y) + (X, \delta Y) = 0 \end{array} \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, (\delta X, Y) + (X, \delta Y) = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid (\delta X, X, X) = 0, \delta E = 0 \right\} \\ &= \left\{ \delta \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{J}) \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y \right\}. \end{aligned}$

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0902.0431_E00431	71	■ 0.943	$\begin{aligned} F_4^C &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\} \\ &= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}. \end{aligned}$	$F_4^C = \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$	$F_4^C = \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, (\alpha X, \alpha Y) = (X, Y)\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \alpha E = E\}$ $= \{\alpha \in \text{Iso}_C(\mathfrak{J}^C) \mid \alpha(X \times Y) = \alpha X \times \alpha Y\}.$
0902.0431_E00645	106	■ 0.946	$\epsilon(A, B) = (\bar{B}, \bar{A}),$	$\epsilon(A, B) = (\bar{B}, \bar{A}),$	$\epsilon(A, B) = (\bar{B}, \bar{A}),$
0902.0431_E01049 (i)	179	■ 0.948	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$ $\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$	$\beta_1(P \times Q) = \beta_2 P \times \beta_3 Q, \quad \beta_2 P, \beta_3 Q = \{P, Q\},$
0902.0431_E00435	72	■ 0.948	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$	$F_4^C \simeq (F_4^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = F_4 \times \mathbf{R}^d,$
0902.0431_E00105	20	■ 0.950	$\begin{aligned} & \left(\frac{1}{8} \right) \left(\frac{1}{8} \right) (1+1) = \frac{1}{4} \\ & \left(\frac{1}{24} \right) \left(\frac{1}{24} \right) (1+4+1) = \frac{1}{12} \\ & \left(\frac{1}{8} \right) \left(\frac{1}{24} \right) (-1-2) = -\frac{1}{8} \\ & (-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$	$(\alpha_1, \alpha_1) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1+1) = \frac{1}{4}$ $(\alpha_2, \alpha_2) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1+4+1) = \frac{1}{12}$ $(\alpha_1, \alpha_2) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1-2) = -\frac{1}{8}$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0$	$(\alpha_1, \alpha_1) = B_2(H_{\alpha_1}, H_{\alpha_1}) = 8 \frac{1}{8} \frac{1}{8} (1+1) = \frac{1}{4}$ $(\alpha_2, \alpha_2) = B_2(H_{\alpha_2}, H_{\alpha_2}) = 8 \frac{1}{24} \frac{1}{24} (1+4+1) = \frac{1}{12}$ $(\alpha_1, \alpha_2) = B_2(H_{\alpha_1}, H_{\alpha_2}) = 8 \frac{1}{8} \frac{1}{24} (-1-2) = -\frac{1}{8}$ $(-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0.$
0902.0431_E00272	44	■ 0.950	$\begin{aligned} & \left(\frac{1}{8} \right) \left(\frac{1}{8} \right) (1+1) = \frac{1}{4} \\ & \left(\frac{1}{24} \right) \left(\frac{1}{24} \right) (1+4+1) = \frac{1}{12} \\ & \left(\frac{1}{8} \right) \left(\frac{1}{24} \right) (-1-2) = -\frac{1}{8} \\ & (-\mu, -\mu) = \frac{1}{4}, \quad (-\mu, \alpha_1) = -\frac{1}{8}, \quad (-\mu, \alpha_2) = 0. \end{aligned}$	$\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{xa}). \end{cases}$	$\begin{cases} \tilde{A}_i(a)E_i = 0 \\ \tilde{A}_i(a)E_{i+1} = -\frac{1}{2}F_i(a) \\ \tilde{A}_i(a)E_{i+2} = \frac{1}{2}F_i(a), \end{cases} \quad \begin{cases} \tilde{A}_i(a)F_i(x) = (a, x)(E_{i+1} - E_{i+2}) \\ \tilde{A}_i(a)F_{i+1}(x) = \frac{1}{2}F_{i+2}(\overline{ax}) \\ \tilde{A}_i(a)F_{i+2}(x) = -\frac{1}{2}F_{i+1}(\overline{xa}). \end{cases}$
0902.0431_E00473	78	■ 0.951	$\left(\phi_1, \phi_2 \right)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$	$(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$	$(\phi_1, \phi_2)_6 = (\delta_1, \delta_2)_4 + (T_1, T_2),$

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0902.0431_E00726	120	■ 0.951	$\begin{aligned} & \& \mathfrak{a} \ni \left(\Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) \right) = \Phi(0, A, -B, 0), \\ & \& \Phi(0, A, -B, 0), \& \mathfrak{a} \ni \left(\Phi(0, A, -B, 0), \Phi \left(0, 0, B_1, 0 \right) \right) = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& \Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0)] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$	$\begin{aligned} & \& \Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) \Big] = \Phi(0, A, -B, 0), \\ & \& [\Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0)] = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& [\Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0)] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$	$\begin{aligned} & \& \left[\Phi(0, A, B, \nu), \Phi \left(0, 0, 0, -\frac{3}{2} \right) \right] = \Phi(0, A, -B, 0), \\ & \& [\Phi(0, A, -B, 0), \Phi(0, 0, B_1, 0)] = \Phi(2A \vee B_1, 0, 0, (A, B_1)), \\ & \& [\Phi(2A \vee B_1, 0, 0, (A, B_1)), \Phi(\phi, 0, 0, 0)] = \Phi(2[A \vee B_1 \cdot \phi], 0, 0, 0), \end{aligned}$
0902.0431_E00323	51	■ 0.953	$B_4(\delta, \delta) = \text{tr} \left(\left(\text{ad} \tilde{A}_1(1) \right)^2 \right) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$	$B_4(\delta, \delta) = \text{tr} \left(\left(\text{ad} \tilde{A}_1(1) \right)^2 \right) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$	$B_4(\delta, \delta) = \text{tr}((\text{ad} \tilde{A}_1(1))^2) = (-1) \times 7 \times 2 + \left(-\frac{1}{4} \right) \times 8 \times 2 = -18.$
0902.0431_E00161	29	■ 0.953	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = xy, \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = xy, \quad x, y \in \mathfrak{C}.$
0902.0431_E00489	80	■ 0.954	$\begin{aligned} & \& = 2 \left(\phi' A \times A, A \right) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$	$\begin{aligned} & \& = 2(\phi' A \times A, A) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$	$\begin{aligned} & \& = 2(\phi' A \times A, A) \text{ (Lemma 3.4.3.(1))} = 2(\phi' A, A \times A) = -2(A, \phi(A \times A)) \\ & \& = -2(\phi, (A \times A) \vee A)_6. \end{aligned}$
0902.0431_E00178	31	■ 0.954	$(a \times \bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$	$(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$	$(ax\bar{a})(ay\bar{a}) = a(xy)\bar{a}, \quad x, y \in \mathfrak{C}.$
0902.0431_E00343	55	■ 0.954	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$	$B_4(H, H') = 18 \sum_{k=0}^3 \lambda_k \lambda_k', \quad H = \sum_{k=0}^3 \lambda_k H_k, H' = \sum_{k=0}^3 \lambda_k' H_k \in \mathfrak{h}_R.$
0902.0431_E00495	81	■ 0.956	$\begin{aligned} & \& R \left(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} x_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik}) \right) \\ & \& = 2R \left(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki})) \right) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$	$\begin{aligned} & \& R \left(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} x_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik}) \right) \\ & \& = 2R(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki}))) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$	$\begin{aligned} & \& R(\sum_{k=1}^3 ((\sigma_{ik} x_{ik}) y_{ki} + \bar{y}_{ki} (\bar{\sigma}_{ik} x_{ik})) + \sum_{k=1}^3 (x_{ik} (\sigma_{ki} y_{ki}) + (\bar{\sigma}_{ki} \bar{y}_{ki}) \bar{x}_{ik})) \\ & \& = 2R(\sum_k ((\sigma_{ik} x_{ik}) y_{ki} + x_{ik} (\sigma_{ki} y_{ki}))) \\ & \& = 2 \sum_k ((\sigma_{ik} x_{ik}, \bar{y}_{ki}) + (x_{ik}, \bar{\sigma}_{ki} \bar{y}_{ki})) \\ & \& = 2 \sum_k ((x_{ik}, -\sigma_{ik} \bar{y}_{ki}) + (x_{ik}, \sigma_{ik} \bar{y}_{ki})) = 0. \end{aligned}$

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0902.0431_E00226	38	■ 0.957	$\left\{ \begin{array}{l} E_{-i} \times E_i = 0 \\ E_i \times F_i(x) = -\frac{1}{2}F_i(x) \\ F_i(x) \times F_i(y) = -(x,y)E_i, \end{array} \right. \quad \left\{ \begin{array}{l} E_i \times E_{i+1} = \frac{1}{2}E_{i+2} \\ E_i \times F_j(x) = 0, \quad i \neq j \\ F_i(x) \times F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy}), \end{array} \right.$		
0902.0431_E00862	143	■ 0.957	$\begin{aligned} & \alpha(t) \left(\begin{pmatrix} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} \eta_1 & y_3 & \bar{y}_2 \\ \bar{y}_3 & \eta_2 & y_1 \\ y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, \xi, \eta \right) \\ & = \left(\begin{pmatrix} e^{2it}\xi_1 & e^{it}x_3 & e^{it}\bar{x}_2 \\ e^{it}\bar{x}_3 & \xi_2 & x_1 \\ e^{it}x_2 & \bar{x}_1 & \xi_3 \end{pmatrix}, \begin{pmatrix} e^{-2it}\eta_1 & e^{-it}y_3 & e^{-it}\bar{y}_2 \\ e^{-it}\bar{y}_3 & \eta_2 & y_1 \\ e^{-it}y_2 & \bar{y}_1 & \eta_3 \end{pmatrix}, e^{-2it}\xi, e^{2it}\eta \right), \end{aligned}$		
0902.0431_E00216	36	■ 0.960	$X=X(\xi, x)=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right), \quad \xi_i \in \boldsymbol{R}, x_i \in \mathfrak{C}.$		
0902.0431_E00164	29	■ 0.960	$(\alpha_1 x)(\alpha_2 y) = \overline{\alpha_3(\overline{xy})}, \quad x, y \in \mathfrak{C}$		
0902.0431_E00656	108	■ 0.960	$\langle \alpha X, \alpha M \rangle = 0 = \langle X, M \rangle, \quad \langle \alpha M, \alpha N \rangle = (\tau \alpha M, \alpha N) = (P \tau M h(A, B)^*, P N \tau h(A, B)^*) = (\tau M, N) = \langle M, N \rangle,$		
0902.0431_E01197	201	■ 0.960	$[(C, 0), (C, D, g_1, g_2, g_{-2}, g_{-1})] = (0, 0, [C, g_1], [C, g_2], [C, g_{-2}], [C, g_{-1}]) \in \mathfrak{q} \cap \mathfrak{a} = \{0\},$		

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00494	81	■ 0.964	$\begin{array}{l} \sigma_{23}=\delta_1, \text{ \& \sigma_{31}=} \\ \delta_2, \text{ \& \sigma_{12}=\delta_3} \\ \sigma_{32}=\kappa\delta_1, \text{ \& \sigma_{13}=\kappa\delta_2,} \\ \sigma_{21}=\kappa\delta_3. \end{array}$	$\begin{array}{l} \sigma_{23} = \delta_1, \quad \sigma_{31} = \delta_2, \quad \sigma_{12} = \delta_3 \\ \sigma_{32} = \kappa\delta_1, \quad \sigma_{13} = \kappa\delta_2, \quad \sigma_{21} = \kappa\delta_3. \end{array}$	$\begin{array}{l} \sigma_{23} = \delta_1, \quad \sigma_{31} = \delta_2, \quad \sigma_{12} = \delta_3, \\ \sigma_{32} = \kappa\delta_1, \quad \sigma_{13} = \kappa\delta_2, \quad \sigma_{21} = \kappa\delta_3. \end{array}$
0902.0431_E00760	127	■ 0.964	$\begin{array}{l} \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$	$\begin{array}{l} \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$	$\begin{array}{l} \lambda_0 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_1 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_2 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_3 - \frac{1}{2}\mu_1 + \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \\ \lambda_0 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_1 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_2 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \\ \lambda_3 + \frac{1}{2}\mu_1 - \frac{2}{3}\nu = 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}$
0902.0431_E00433	71	■ 0.965	$F_4 = \left\{ \alpha \in F_4 \mid \tau \alpha = \alpha \tau \right\}.$	$F_4 = \left\{ \alpha \in F_4^C \mid \tau \alpha = \alpha \tau \right\}.$	$F_4 = \left\{ \alpha \in F_4^C \mid \tau \alpha = \alpha \tau \right\}.$
0902.0431_E00713	118	■ 0.965	$\alpha \in E_7 \quad \text{if and only if} \quad \tau \lambda \alpha = \alpha \tau \lambda.$	$\alpha \in E_7 \quad \text{if and only if} \quad \tau \lambda \alpha = \alpha \tau \lambda.$	$\alpha \in E_7 \quad \text{if and only if} \quad \tau \lambda \alpha = \alpha \tau \lambda.$
0902.0431_E00511	84	■ 0.965	$\begin{array}{l} \left[h, \widetilde{F}_{31}(a) \right] = (h_\delta F_{31}(a))^\sim + \frac{1}{2} \left[\widetilde{H}, \widetilde{F}_{31}(a) \right] = \widetilde{F}_{31}((\nu h_\delta) a) + \frac{1}{2} (\mu_3 - \mu_1) \widetilde{F}_{31}(a), \\ \left[h, \widetilde{F}_{12}(a) \right] = (h_\delta F_{12}(a))^\sim + \frac{1}{2} \left[\widetilde{H}, \widetilde{F}_{12}(a) \right] = \widetilde{F}_{12}((\kappa \pi h_\delta) a) + \frac{1}{2} (\mu_1 - \mu_2) \widetilde{F}_{12}(a), \end{array}$	$\begin{array}{l} \left[h, \widetilde{F}_{31}(a) \right] = (h_\delta F_{31}(a))^\sim + \frac{1}{2} \left[\widetilde{H}, \widetilde{F}_{31}(a) \right] = \widetilde{F}_{31}((\nu h_\delta) a) + \frac{1}{2} (\mu_3 - \mu_1) \widetilde{F}_{31}(a), \\ \left[h, \widetilde{F}_{12}(a) \right] = (h_\delta F_{12}(a))^\sim + \frac{1}{2} \left[\widetilde{H}, \widetilde{F}_{12}(a) \right] = \widetilde{F}_{12}((\kappa \pi h_\delta) a) + \frac{1}{2} (\mu_1 - \mu_2) \widetilde{F}_{12}(a), \end{array}$	$\begin{array}{l} [h, \widetilde{F}_{31}(a)] = (h_\delta F_{31}(a))^\sim + \frac{1}{2} [\widetilde{H}, \widetilde{F}_{31}(a)] = \widetilde{F}_{31}((\nu h_\delta) a) + \frac{1}{2} (\mu_3 - \mu_1) \widetilde{F}_{31}(a), \\ [h, \widetilde{F}_{12}(a)] = (h_\delta F_{12}(a))^\sim + \frac{1}{2} [\widetilde{H}, \widetilde{F}_{12}(a)] = \widetilde{F}_{12}((\kappa \pi h_\delta) a) + \frac{1}{2} (\mu_1 - \mu_2) \widetilde{F}_{12}(a), \end{array}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00906	151	■ 0.966	$\begin{gathered} \gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,} \\ \varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.} \end{gathered}$	$\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$	$\gamma \varphi(D) \gamma Y = g^{-1}(D(g(\gamma Y))D^*) \text{ is real diagonal,}$ $\varphi(D)X = \varphi(D) \left(\frac{1}{\xi} Y \times Y \right) = \frac{1}{\xi} (\gamma \varphi(D) \gamma Y \times \gamma \varphi(D) \gamma Y) \text{ is real diagonal.}$
0902.0431_E00191	33	■ 0.966	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$	$\widetilde{\alpha} e_i = b_i, \quad i = 0, 1, \dots, 7$
0902.0431_E00614	102	■ 0.967	$\begin{aligned} gX \circ gY &= g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C \\ (gX, gY) &= (\gamma X, Y) \end{aligned}$	$gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C$ $(gX, gY) = (\gamma X, Y)$	$gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E, \quad X, Y \in \mathfrak{J}^C.$ $(gX, gY) = (\gamma X, Y),$
0902.0431_E00730	120	■ 0.969	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$	$W \ni \Phi(0, 0, X_1, 0)(X, 0, 0, 3\eta) = (0, 0, 0, (X_1, X)),$
0902.0431_E01004	167	■ 0.969	$\begin{aligned} &\langle R_1, R_2 \rangle \\ &= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2. \end{aligned}$	$\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$	$\langle R_1, R_2 \rangle$ $= (\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7 + \langle P_1, P_2 \rangle + \langle Q_1, Q_2 \rangle + 8(\tau r_1) r_2 + 4(\tau s_1) s_2 + 4(\tau t_1) t_2.$

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0902.0431_E00595	99	■ 0.969	$\begin{aligned} & 2\left\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \right\rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\left\langle M + \mathbf{a}, k_J^{-1} \left(A(k_J(N))^t A \right) + p \mathbf{b} k^{-1}(\tau^t A) \right\rangle \\ &= 2\left\langle M, k_J^{-1} \left(A(k_J(N))^t A \right) \right\rangle + 4\langle \mathbf{a}, p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= \left\langle k_J M, A(k_J(N))^t A \right\rangle + 2\langle k(\overline{p} \mathbf{a}), (k \mathbf{b}) \tau^t A \rangle \\ &= \langle \tau^t A(k_J(M)) \tau A, k_J(N) \rangle + 2\langle k(\overline{p} \mathbf{a}) A, k \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A), N \rangle + 4\langle \overline{p} \mathbf{a} k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A + \overline{p} \mathbf{a} k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\overline{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$	$\begin{aligned} & 2\left\langle \tau^t \varphi(p, A) \tau(M + \mathbf{a}), N + \mathbf{b} \right\rangle \quad M + \mathbf{a}, N + \mathbf{b} \in \mathfrak{J}^C \\ &= 2\langle M + \mathbf{a}, \varphi(p, A)(N + \mathbf{b}) \rangle \\ &= 2\left\langle M + \mathbf{a}, k_J^{-1} \left(A(k_J(N))^t A \right) + p \mathbf{b} k^{-1}(\tau^t A) \right\rangle \\ &= 2\left\langle M, k_J^{-1} \left(A(k_J(N))^t A \right) \right\rangle + 4\langle \mathbf{a}, p \mathbf{b} k^{-1}(\tau^t A) \rangle \\ &= \left\langle k_J M, A(k_J(N))^t A \right\rangle + 2\langle k(\overline{p} \mathbf{a}), (k \mathbf{b}) \tau^t A \rangle \\ &= \langle \tau^t A(k_J(M)) \tau A, k_J(N) \rangle + 2\langle k(\overline{p} \mathbf{a}) A, k \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A), N \rangle + 4\langle \overline{p} \mathbf{a} k^{-1}(A), \mathbf{b} \rangle \\ &= 2\langle k_J^{-1}(\tau^t A(k_J(M)) \tau A + \overline{p} \mathbf{a} k^{-1}(A)), N + \mathbf{b} \rangle \\ &= 2\langle \varphi(\overline{p}, \tau^t A)(M + \mathbf{a}), N + \mathbf{b} \rangle, \end{aligned}$	
0902.0431_E00253	41	■ 0.969	$0(27)=0(\frac{J}{\alpha})=\left\{\alpha\in\operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})\mid(\alpha X,\alpha Y)=(X,Y)\right\}\mid(\alpha X,\alpha Y)=(X,Y)\right\}\quad\text{(Lemma 2.2.4.(2)). }$	$O(27) = O(\mathfrak{J}) = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid (\alpha X, \alpha Y) = (X, Y)\} \quad (\text{Lemma 2.2.4.(2)}).$	$O(27) = O(\mathfrak{J}) = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid (\alpha X, \alpha Y) = (X, Y)\} \quad (\text{Lemma 2.2.4.(2)}).$
0902.0431_E00997	166	■ 0.970	$(R \times R) R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$	$(R \times R) R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$	$(R \times R) R_1 = \Theta(R)^2 R_1 + \frac{1}{30} B_8(R, R_1) R, \quad R_1 \in \mathfrak{e}_8^C,$

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0902.0431_E01016	170	■ 0.971	$\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$	$\begin{gathered} \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_2 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r \\ \pm \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{1}{3}\nu\right) \pm r, \\ \pm \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) \pm \left(\frac{1}{2}\mu_3 + \frac{2}{3}\nu\right) \pm r, \\ \pm 2r, \\ \pm \nu \pm r \end{gathered}$	
0902.0431_E00230	38	■ 0.971	$\begin{aligned} & (\operatorname{tr}(\alpha X) \& - \operatorname{tr}(X)) X^2 + \frac{1}{2} (\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2)) X \\ & = (\det X - \det(\alpha X))E \end{aligned}$	$(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2} (\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2)) X = (\det X - \det(\alpha X))E.$	$(\operatorname{tr}(\alpha X) - \operatorname{tr}(X))X^2 + \frac{1}{2} (\operatorname{tr}(X)^2 - \operatorname{tr}(\alpha X)^2 + \operatorname{tr}((\alpha X)^2) - \operatorname{tr}(X^2)) X = (\det X - \det(\alpha X))E.$
0902.0431_E00944	157	■ 0.971	$\overline{f^{-1}(A(fP))}=f^{-1}\left(\left(\overline{\left(\operatorname{Ad} J_3\right) A f \bar{P}}\right)\right), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$	$\overline{f^{-1}(A(fP))}=f^{-1}\left(\left(\overline{\left(\operatorname{Ad} J_3\right) A f \bar{P}}\right)\right), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$	$\overline{f^{-1}(A(fP))}=f^{-1}(\overline{((\operatorname{Ad} J_3) A f \bar{P})}), \quad A \in S U(6), P \in \left(\mathfrak{P}_C\right)^C .$
0902.0431_E00979	163	■ 0.974	$\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t) \right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0, 0, 0\right),$	$\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t) \right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0, 0, 0\right),$	$\mathfrak{a} \ni \left[\Phi_1,(\Phi, P, Q, r, s, t) \right]=\left(\left[\Phi_1, \Phi\right], \Phi_1 P, \Phi_1 Q, 0, 0, 0\right),$

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0902.0431_E08751	124	■ 0.975	$\begin{aligned} & \left[\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0) \right] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[\Phi\left((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)\right) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$	$\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0)] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[\Phi\left((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)\right) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$	$\begin{aligned} & [\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, E_j, 0, 0)] \\ &= \Phi\left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu\right)E_j, 0, 0\right) = \left(\mu_j + \frac{2}{3}\nu\right)\Phi(0, E_j, 0, 0), \\ & \left[\Phi((h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, 0, E_j, 0)) = \Phi\left(0, 0, \left((h_\delta + \tilde{H})' - \frac{2}{3}\nu\right)E_j, 0\right) \right. \\ & \left. = \Phi\left(0, 0, \left(h_\delta - \tilde{H} - \frac{2}{3}\nu\right)E_j, 0\right) = \left(-\mu_j - \frac{2}{3}\nu\right)\Phi(0, 0, E_j, 0). \right. \end{aligned}$
0902.0431_E08894	148	■ 0.975	$\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$	$\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$	$\mathfrak{S}(8, \boldsymbol{C}) = \{S \in M(8, \boldsymbol{C}) \mid {}^t S = -S\},$
0902.0431_E08608	101	■ 0.976	$\begin{aligned} & \boldsymbol{P}_3 \text{ \& } = \left\{ P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1 \right\} \\ & = \{AE_1A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$	$\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \{AE_1A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H})\}. \end{aligned}$	$\begin{aligned} \boldsymbol{HP}_3 &= \{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid P^2 = P, \operatorname{tr}(P) = 1\} \\ &= \left\{ AE_1A^* \mid A \in \operatorname{Sp}(4), E_1 = \operatorname{diag}(1, 0, 0, 0) \in M(4, \boldsymbol{H}) \right\}. \end{aligned}$
0902.0431_E08813	136	■ 0.976	$\widetilde{\iota}(\alpha) = \iota\alpha\iota^{-1}, \quad \alpha \in E_7.$	$\widetilde{\iota}(\alpha) = \iota\alpha\iota^{-1}, \quad \alpha \in E_7.$	$\widetilde{\iota}(\alpha) = \iota\alpha\iota^{-1}, \quad \alpha \in E_7.$
0902.0431_E08414	68	■ 0.976	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$	$M \times N = \left(\begin{array}{ccc} \boldsymbol{m}_2 \times \boldsymbol{n}_3 & \boldsymbol{m}_3 \times \boldsymbol{n}_1 & \boldsymbol{m}_1 \times \boldsymbol{n}_2 \\ + & + & + \\ \boldsymbol{n}_2 \times \boldsymbol{m}_3 & \boldsymbol{n}_3 \times \boldsymbol{m}_1 & \boldsymbol{n}_1 \times \boldsymbol{m}_2 \end{array} \right) \in M(3, \boldsymbol{C}),$
0902.0431_E01088	167	■ 0.976	$\begin{aligned} & E_8 = \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ & = \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$	$\begin{aligned} E_8 &= \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ &= \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$	$\begin{aligned} E_8 &= \left\{ \alpha \in E_8^C \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \right\} \\ &= \left\{ \alpha \in E_8^C \mid \tau \tilde{\lambda} \alpha = \alpha \tau \tilde{\lambda} \right\}. \end{aligned}$
0902.0431_E08088	18	■ 0.976	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$	$B_2(D_1, D_2) = 4 \operatorname{tr}(D_1 D_2), \quad D_i \in \mathfrak{g}_2^C.$

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0902.0431_E00669	111	■ 0.977	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$	$\begin{aligned} (P, Q) &= (X, Z) + (Y, W) + \xi \zeta + \eta \omega, \\ \langle P, Q \rangle &= \langle X, Z \rangle + \langle Y, W \rangle + (\tau \xi) \zeta + (\tau \eta) \omega, \\ \{P, Q\} &= (X, W) - (Z, Y) + \xi \omega - \zeta \eta, \end{aligned}$
0902.0431_E00684	113	■ 0.977	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$	$\mathfrak{e}_7^C = \left\{ \Phi(\phi, A, B, \nu) \in \text{Hom}_C(\mathfrak{P}^C) \mid \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C \right\}.$
0902.0431_E00948	157	■ 0.979	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$	$((X, Y, \xi, \eta), (M, N)) = (X + M, Y + N, \xi, \eta).$
0902.0431_E00994	19	■ 0.979	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C \end{array}$	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1) \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C \end{array}$	$\begin{array}{l} f_*(A) = \varepsilon A - \bar{\varepsilon}^t A, \quad \varepsilon = \frac{1}{2}(1 + ie_1), \\ \varphi_*(D)(a + \mathbf{m}) = D\mathbf{m}, \quad a + \mathbf{m} \in \mathbf{C}^C \oplus (\mathbf{C}^3)^C = \mathfrak{C}^C, \end{array}$
0902.0431_E00381	63	■ 0.979	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}) \end{aligned}$	$\begin{aligned} \alpha E_2 &= \alpha(-F_2(1) \times F_2(1)) = -\alpha F_2(1) \times \alpha F_2(1) \\ &= -(F_2(x_2) + F_3(x_3)) \times (F_2(x_2) + F_3(x_3)) \quad (\text{for some } x_2, x_3 \in \mathfrak{C}) \\ &= (x_2, x_2)E_2 + (x_3, x_3)E_3 - F_1(\overline{x_2 x_3}) \in \mathfrak{J}(2, \mathfrak{C}). \end{aligned}$
0902.0431_E01171	197	■ 0.979	$\begin{aligned} &\exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) = (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$	$\begin{aligned} &\exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) = (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$	$\begin{aligned} \exp(\text{ad}(X, 0, 0))(D, \mathbf{u}, \mathbf{v}) &= (\exp(\text{ad}(X)D, (\exp X)\mathbf{u}, (\exp(-^t X))\mathbf{v})) \\ &= (\text{Ad}(\exp X)D, (\exp X)\mathbf{u}, {}^t(\exp X)^{-1}\mathbf{v}) \\ &= \varphi(\exp X)(D, \mathbf{u}, \mathbf{v}). \end{aligned}$
0902.0431_E00632	105	■ 0.979	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$	$\alpha = \varphi(A)\beta = \varphi(A)\varphi(B) = \varphi(AB), \quad AB \in \text{Sp}(4),$

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0902.0431_E00599	100	■ 0.980	$\begin{aligned} & (\alpha \boldsymbol{a})^*(\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A))^*(\boldsymbol{p} \boldsymbol{b} k^{-1}(\tau^t A)) \\ & = k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ & \tau \alpha \tau(\boldsymbol{a}^* \boldsymbol{b}) = \tau \left(k_J^{-1} \left(A(k_J \tau(\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A(k(\tau(\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A J \tau k(\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau A J k(\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left(A \tau k(\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ & (\alpha \boldsymbol{a})(\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A)) k_J^{-1} \left(A(k_J(N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A) k^{-1} \left(A(k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} \left(J^t A J \right) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A), \\ & \tau \alpha \tau(\boldsymbol{a} N) = \tau \left(\boldsymbol{p} \tau(\boldsymbol{a} N) k^{-1}(\tau^t A) \right) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A). \end{aligned}$	$\begin{aligned} & (\alpha \boldsymbol{a})^*(\alpha \boldsymbol{b}) = (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A))^*(\boldsymbol{p} \boldsymbol{b} k^{-1}(\tau^t A)) \\ & = k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ & \tau \alpha \tau(\boldsymbol{a}^* \boldsymbol{b}) = \tau \left(k_J^{-1} \left(A(k_J \tau(\boldsymbol{a}^* \boldsymbol{b}))^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A(k(\tau(\boldsymbol{a}^* \boldsymbol{b}))) J^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau \left(A J \tau k(\boldsymbol{a}^* \boldsymbol{b})^t A \right) \right) \\ & = -\tau k^{-1} \left(J \tau A J k(\boldsymbol{a}^* \boldsymbol{b}) \tau^t A \right) \\ & = -k^{-1} \left(A \tau k(\boldsymbol{a}^* \boldsymbol{b}) J^t A J \right) = k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ & (\alpha \boldsymbol{a})(\alpha N) = (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A)) k_J^{-1} \left(A(k_J(N))^t A \right) \\ & = -\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A) k^{-1} \left(A(k(N)) J^t A J \right) \\ & = -\boldsymbol{p} \boldsymbol{a} N k^{-1} \left(J^t A J \right) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A), \\ & \tau \alpha \tau(\boldsymbol{a} N) = \tau \left(\boldsymbol{p} \tau(\boldsymbol{a} N) k^{-1}(\tau^t A) \right) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A). \end{aligned}$	$\begin{aligned} (\alpha \boldsymbol{a})^*(\alpha \boldsymbol{b}) &= (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A))^*(\boldsymbol{p} \boldsymbol{b} k^{-1}(\tau^t A)) \\ &= k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ \tau \alpha \tau(\boldsymbol{a}^* \boldsymbol{b}) &= \tau(k_J^{-1}(A(k_J \tau(\boldsymbol{a}^* \boldsymbol{b}))^t A)) \\ &= -\tau k^{-1}(J \tau(A(k(\tau(\boldsymbol{a}^* \boldsymbol{b}))) J^t A)) \\ &= -\tau k^{-1}(J \tau(A J \tau k(\boldsymbol{a}^* \boldsymbol{b})^t A)) \\ &= -\tau k^{-1}(J \tau A J k(\boldsymbol{a}^* \boldsymbol{b}) \tau^t A) \\ &= -k^{-1}(A \tau k(\boldsymbol{a}^* \boldsymbol{b}) J^t A J) = k^{-1}(A) \boldsymbol{a}^* \boldsymbol{b} k^{-1}(\tau^t A), \\ (\alpha \boldsymbol{a})(\alpha N) &= (\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A)) k_J^{-1}(A(k_J(N))^t A) \\ &= -\boldsymbol{p} \boldsymbol{a} k^{-1}(\tau^t A) k^{-1}(A(k(N)) J^t A J) \\ &= -\boldsymbol{p} \boldsymbol{a} N k^{-1}(J^t A J) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A), \\ \tau \alpha \tau(\boldsymbol{a} N) &= \tau(\boldsymbol{p} \tau(\boldsymbol{a} N) k^{-1}(\tau^t A)) = \boldsymbol{p} \boldsymbol{a} N \tau k^{-1}(\tau^t A). \end{aligned}$
0902.0431_E01089	185	■ 0.980	$\begin{aligned} & \left(\widetilde{\mathfrak{e}}_8 \right)^{\widetilde{\lambda} \gamma} = \left\{ \Theta \in \Theta \left(\mathfrak{e}_8^C \right) \mid \widetilde{\lambda} \gamma \Theta = \Theta \widetilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta(\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$	$\begin{aligned} (\mathfrak{e}_8)^{\widetilde{\lambda} \gamma} &= \left\{ \Theta \in \Theta \left(\mathfrak{e}_8^C \right) \mid \widetilde{\lambda} \gamma \Theta = \Theta \widetilde{\lambda} \gamma \right\} \\ & = \left\{ \Theta(\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \right\} \end{aligned}$	$\begin{aligned} (\mathfrak{e}_8)^{\widetilde{\lambda} \gamma} &= \{ \Theta \in \Theta(\mathfrak{e}_8^C) \mid \widetilde{\lambda} \gamma \Theta = \Theta \widetilde{\lambda} \gamma \} \\ &= \{ \Theta(\Phi, P, -\lambda \gamma P, 0, s, -s) \mid \Phi \in (\mathfrak{e}_7)^{\lambda \gamma}, P \in (\mathfrak{P}^C)_{\lambda \gamma}, s \in \boldsymbol{R} \} \end{aligned}$

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0902.0431_E00506	83	■ 0.980	$\mathfrak{h}=\left\{\begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}} \in \mathfrak{e}_6^C \end{array}\right\} \quad \left\{ \begin{array}{l} h_{\delta}=\sum_{k=0}^3 \lambda_k H_k=-\sum_{k=0}^3 \lambda_k i G_{k4+k}, \lambda_k \in C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$	$\mathfrak{h}=\left\{ \begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$	$\mathfrak{h}=\left\{ \begin{array}{l} h=h_{\delta}+\widetilde{H} \in \mathfrak{e}_6^C \\ \text{\textit{H}}=\sum_{j=1}^3 \mu_j E_j, \mu_j \in C, \mu_1+\mu_2+\mu_3=0 \end{array} \right\},$
0902.0431_E00831	139	■ 0.980	$\begin{aligned} &\left(E_7\right)^{\kappa,\mu}=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ &\left(\left(E_7\right)^{\kappa,\mu}\right)_{(0, E_1, 0, 1)}=\left\{\alpha \in\left(E_7\right)^{\kappa,\mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} . \end{aligned}$	$\left(E_7\right)^{\kappa,\mu}=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa,\mu}\right)_{(0, E_1, 0, 1)}=\left\{\alpha \in\left(E_7\right)^{\kappa,\mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} .$	$\left(E_7\right)^{\kappa,\mu}=\left\{\alpha \in E_7 \mid \kappa \alpha=\alpha \kappa, \mu \alpha=\alpha \mu\right\}, \\ \left(\left(E_7\right)^{\kappa,\mu}\right)_{(0, E_1, 0, 1)}=\left\{\alpha \in\left(E_7\right)^{\kappa,\mu} \mid \alpha\left(0, E_1, 0, 1\right)=\left(0, E_1, 0, 1\right)\right\} .$
0902.0431_E01132	191	■ 0.981	$\begin{aligned} &\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right)=\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) \\ &\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right)=\operatorname{tr}\left(A D_{\{1\}} \tau^t A\right) \end{aligned}$	$\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right), \\ \langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$	$\operatorname{tr}\left(A D_1 \tau^t A\left(\tau^t\left(A D_2 \tau^t A\right)\right)\right)=\operatorname{tr}\left(A D_1\left(\tau^t D_2\right) A^{-1}\right)=\operatorname{tr}\left(D_1\left(\tau^t D_2\right)\right), \\ \langle A \boldsymbol{X}, A \boldsymbol{Y}\rangle=\langle \boldsymbol{X}, \boldsymbol{Y}\rangle,$
0902.0431_E00384	64	■ 0.981	$\begin{aligned} &\alpha \sigma X=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ &=\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ &=\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} . \end{aligned}$	$\alpha \sigma X=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ =\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ =\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J}$	$\alpha \sigma X=\alpha \sigma\left(X_1+X_2\right) \quad X_1 \in \mathfrak{J}_{\sigma}, \quad X_2 \in \mathfrak{J}_{-\sigma} \\ =\alpha\left(X_1-X_2\right)=\alpha X_1-\alpha X_2=\sigma\left(\alpha X_1\right)+\sigma\left(\alpha X_2\right) \\ =\sigma \alpha\left(X_1+X_2\right)=\sigma \alpha X, \quad X \in \mathfrak{J} .$
0902.0431_E00641	106	■ 0.981	$\begin{aligned} &h(a, b)=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \\ &h(A, B)=\frac{A+B}{2}+i \frac{A-B}{2} e_1=\iota A+\bar{\iota} B, \end{aligned}$	$h(a, b)=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \quad \iota=\frac{1+i e_1}{2} .$	$h(a, b)=\frac{a+b}{2}+i \frac{a-b}{2} e_1=\iota a+\bar{\iota} b, \quad \iota=\frac{1+i e_1}{2} .$
0902.0431_E00474	78	■ 0.981	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$	$\left(\left[\phi, \phi_1\right], \phi_2\right)_6+\left(\phi_1,\left[\phi, \phi_2\right]\right)_6=0, \quad \phi, \phi_i \in \mathfrak{e}_6^C .$
0902.0431_E01173	197	■ 0.982	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$	$\mathfrak{e}_8^C=\mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1},$

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0902.0431_E00123	24	■ 0.985	$\overline{\omega_{\{1\}} \overline{\boldsymbol{m}}}=\overline{\gamma_{\{1\}}(\boldsymbol{w} \boldsymbol{m})}=\overline{\gamma_{\{1\}}(\boldsymbol{w} \boldsymbol{m})}=\overline{\gamma_{\{1\}}(\boldsymbol{w} \boldsymbol{m})}$ $\overline{\omega_{\{1\}} \overline{\boldsymbol{m}}}=\overline{\gamma_{\{1\}}(\boldsymbol{w} \boldsymbol{m})}=\overline{\gamma_{\{1\}}(\boldsymbol{w} \boldsymbol{m})}$ for all $\boldsymbol{m} \in \boldsymbol{C}^3$,	$\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}} \quad \text{for all } \boldsymbol{m} \in \boldsymbol{C}^3,$	$\omega_1 \overline{\boldsymbol{m}} = w(\gamma_1 \boldsymbol{m}) = \gamma_1(w \boldsymbol{m}) = \overline{\omega_1 \boldsymbol{m}} = \overline{\omega_1 \boldsymbol{m}} \quad \text{for all } \boldsymbol{m} \in \boldsymbol{C}^3,$
0902.0431_E00740	122	■ 0.985	$B_{\{7\}}(\Phi_1, \Phi_2)=k(\Phi_1, \Phi_2)_7=k' \operatorname{tr}(\Phi_1 \Phi_2)$	$B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \operatorname{tr}(\Phi_1 \Phi_2).$	$B_7(\Phi_1, \Phi_2) = k(\Phi_1, \Phi_2)_7 = k' \operatorname{tr}(\Phi_1 \Phi_2).$
0902.0431_E01007	167	■ 0.985	$\begin{gathered} \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7 \end{gathered}$	$\sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_2(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_1, \widetilde{F}_3(e_i) \right], \quad \sqrt{2} \left[\widetilde{E}_3, \widetilde{F}_1(e_i) \right], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}} \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j) \right], \quad 0 \leq i < j \leq 7$	$\sqrt{2}[\widetilde{E}_1, \widetilde{F}_2(e_i)], \quad \sqrt{2}[\widetilde{E}_1, \widetilde{F}_3(e_i)], \quad \sqrt{2}[\widetilde{E}_3, \widetilde{F}_1(e_i)], \quad 0 \leq i \leq 7, \\ \frac{1}{\sqrt{2}}[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)], \quad 0 \leq i < j \leq 7$
0902.0431_E01214	204	■ 0.985	$\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5.$	$\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5.$	$\operatorname{Ker} \varphi = \{(E, E), (\zeta E, \zeta^2 E), (\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\} = \boldsymbol{Z}_5.$
0902.0431_E00780	131	■ 0.985	$\beta_{\{1\}}(X \times X)=\beta X \times \beta X=\tau \beta \tau(X \times X),$	$\beta_1(X \times X)=\beta X \times \beta X=\tau \beta \tau(X \times X),$	$\beta_1(X \times X)=\beta X \times \beta X=\tau \beta \tau(X \times X),$
0902.0431_E00152	28	■ 0.985	$\begin{aligned} G_{\{2\}} \cap O(8) &\simeq \left(G_{\{2\}} \cap O(8) \right) \times \boldsymbol{R}^d \\ &= \left((G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left((G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \quad (\text{Theorem 1.10.1}), \quad d = 8. \end{aligned}$	$\begin{aligned} G_{2(2)} &\simeq \left(G_{2(2)} \cap O(8) \right) \times \boldsymbol{R}^d \\ &= \left((G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left((G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \quad (\text{Theorem 1.10.1}), \quad d = 8. \end{aligned}$	$\begin{aligned} G_{2(2)} &\simeq \left(G_{2(2)} \cap O(8) \right) \times \boldsymbol{R}^d \\ &= \left((G_2^C)^{\tau \gamma} \right)^\gamma \times \boldsymbol{R}^d = \left((G_2^C)^\tau \right)^\gamma \times \boldsymbol{R}^d = (G_2)^\gamma \times \boldsymbol{R}^d \\ &= (\operatorname{Sp}(1) \times \operatorname{Sp}(1)) / \boldsymbol{Z}_2 \times \boldsymbol{R}^d \quad (\text{Theorem 1.10.1}), \quad d = 8. \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E08529	88	■ 0.985	$\begin{aligned} & \left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a , \end{array} \right. \\ & \left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right. \end{aligned}$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$	$\left\{ \begin{array}{l} \eta_1 = \xi_1 \\ \eta_2 = \frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a \\ \eta_3 = -\frac{\xi_2 - \xi_3}{2} + \frac{\xi_2 + \xi_3}{2} \cos a + i \frac{(a, x_1)}{ a } \sin a , \end{array} \right.$ $\left\{ \begin{array}{l} y_1 = x_1 + i \frac{(\xi_2 + \xi_3)a}{2 a } \sin a - \frac{2(a, x_1)a}{ a ^2} \sin^2 \frac{ a }{2} \\ y_2 = x_2 \cos \frac{ a }{2} + i \frac{\overline{x_3 a}}{ a } \sin \frac{ a }{2} \\ y_3 = x_3 \cos \frac{ a }{2} + i \frac{\overline{a x_2}}{ a } \sin \frac{ a }{2} \end{array} \right.$
0902.0431_E08314	50	■ 0.986	$\begin{aligned} & \begin{aligned} & \left([\widetilde{A}, \widetilde{B}]C, D \right) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D) \\ & = (B \circ C, A \circ D) - (A \circ C, B \circ D) \\ & = (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B). \end{aligned} \end{aligned}$	$([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$	$([\widetilde{A}, \widetilde{B}]C, D) = (A \circ (B \circ C), D) - (B \circ (A \circ C), D)$ $= (B \circ C, A \circ D) - (A \circ C, B \circ D)$ $= (C \circ (D \circ A), B) - (D \circ (C \circ A), B) = ([\widetilde{C}, \widetilde{D}]A, B).$
0902.0431_E08449	73	■ 0.986	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78.$
0902.0431_E01090	185	■ 0.986	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$	$\zeta(l(D) + l(S)I + l(icE)) = \Theta(\varphi_* D, 2\lambda\gamma\chi^{-1}S, 2\chi^{-1}S, 0, 2c, -2c),$
0902.0431_E08718	118	■ 0.987	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \tau A_2 + 2A_2 \vee \tau A_1 \\ A = \left(\phi_1 + \frac{2}{3}\nu_1 \right) A_2 - \left(\phi_2 + \frac{2}{3}\nu_2 \right) A_1 \\ \nu = \langle A_1, A_2 \rangle - \langle A_2, A_1 \rangle. \end{array} \right.$
0902.0431_E08055	13	■ 0.988	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$	$\mathfrak{g}_2 = \{D \in \mathfrak{D}_4 \mid \nu D = D, \pi D = D\}$ $= \{D \in \mathfrak{D}_4 \mid \lambda D = D, \lambda \in \mathfrak{S}_3\}$ $= \{D \in \mathfrak{b}_3 \mid \pi D = D\}.$

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0902.0431_EQ1152	194	■ 0.988	$\text{ (ii) } \begin{aligned} &x_{\textcolor{teal}{k}} \widetilde{u}_{\textcolor{teal}{j} \textcolor{teal}{k}} \boldsymbol{u}_{\textcolor{teal}{j}} - (\det U) \boldsymbol{x} = \sum_{i, \textcolor{teal}{j}, \textcolor{teal}{k}} x_{\textcolor{teal}{k}} \widetilde{u}_{\textcolor{teal}{j} \textcolor{teal}{k}} u_{\textcolor{teal}{j} i} \boldsymbol{e}_i - (\det U) \boldsymbol{x} \\ &= \sum_{\textcolor{teal}{j}, \textcolor{teal}{k}} x_{\textcolor{teal}{k}} (\det U) \delta_{\textcolor{teal}{k} i} \boldsymbol{e}_i - (\det U) \boldsymbol{x} = 0. \end{aligned}$	$\text{(ii)} \quad \begin{aligned} &= \sum_{j, k} x_k \widetilde{u}_{jk} \boldsymbol{u}_j - (\det U) \boldsymbol{x} = \sum_{i, j, k} x_k \widetilde{u}_{jk} u_{ji} \boldsymbol{e}_i - (\det U) \boldsymbol{x} \\ &= \sum_{j, k} x_k (\det U) \delta_{ki} \boldsymbol{e}_i - (\det U) \boldsymbol{x} = 0. \end{aligned}$	$\begin{aligned} \text{(ii)} &= \sum_{j, k} x_k \widetilde{u}_{jk} \boldsymbol{u}_j - (\det U) \boldsymbol{x} = \sum_{i, j, k} x_k \widetilde{u}_{jk} u_{ji} \boldsymbol{e}_i - (\det U) \boldsymbol{x} \\ &= \sum_{j, k} x_k (\det U) \delta_{ki} \boldsymbol{e}_i - (\det U) \boldsymbol{x} = 0. \end{aligned}$
0902.0431_EQ1022	170	■ 0.988	$[(\Phi(h), \textcolor{teal}{0}, \textcolor{teal}{0}, \textcolor{teal}{r}, \textcolor{teal}{0}, \textcolor{teal}{0}), (\textcolor{teal}{0}, \textcolor{teal}{0}, Q, \textcolor{teal}{0}, \textcolor{teal}{0}, \textcolor{teal}{0})] = (\textcolor{teal}{0}, \textcolor{teal}{0}, (\Phi(h) - r)Q, \textcolor{teal}{0}, \textcolor{teal}{0}, \textcolor{teal}{0}),$	$[(\Phi(h), 0, 0, r, 0, 0), (0, 0, Q, 0, 0, 0)] = (0, 0, (\Phi(h) - r)Q, 0, 0, 0),$	$[(\Phi(h), 0, 0, r, 0, 0), (0, 0, Q, 0, 0, 0)] = (0, 0, (\Phi(h) - r)Q, 0, 0, 0),$
0902.0431_EQ1066	181	■ 0.988	$\widetilde{\lambda} \gamma(\Phi, P, Q, \textcolor{teal}{,}, s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$	$\tilde{\lambda} \gamma(\Phi, P, Q, \textcolor{teal}{,}, s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$	$\tilde{\lambda} \gamma(\Phi, P, Q, \textcolor{teal}{,}, s, t) = (\lambda \gamma \Phi \gamma \lambda^{-1}, \lambda \gamma Q, -\lambda \gamma P, -r, -t, -s).$
0902.0431_EQ0033	10	■ 0.988	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$	$\pi L_{e_i} = \pi(2F_{i0}) = -2\pi F_{0i} = -2G_{0i} \text{ (Lemma 1.3.1)} = 2G_{i0} = T_{e_i},$
0902.0431_EQ0762	128	■ 0.989	$\mathfrak{h}_{\boldsymbol{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \boldsymbol{R} \\ \mu_j \in \boldsymbol{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$	$\mathfrak{h}_{\boldsymbol{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \boldsymbol{R} \\ \mu_j \in \boldsymbol{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$	$\mathfrak{h}_{\boldsymbol{R}} = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in \mathfrak{e}_7^C \mid \begin{array}{l} \lambda_k, \nu \in \boldsymbol{R} \\ \mu_j \in \boldsymbol{R}, \mu_1 + \mu_2 + \mu_3 = 0 \end{array} \right\}$
0902.0431_EQ0269	44	■ 0.989	$\begin{aligned} ([A, X], X X) &\&= (A X, X X) - (X A, X X) \\ &\&= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &\&= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \overline{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$	$\begin{aligned} ([A, X], X X) &= (A X, X X) - (X A, X X) \\ &= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \overline{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$	$\begin{aligned} ([A, X], X X) &= (A X, X X) - (X A, X X) \\ &= (A, X(X X)) - (A, (X X) X) \quad (\text{Lemma 2.3.5}) \\ &= (A, [X, X X]) = (A, a E) = \frac{1}{2} \text{tr}(A a + \overline{a} A^*) \\ &= \frac{1}{2} \text{tr}(A a - a A) = \frac{1}{2} (\text{tr}(A) a - a \text{tr}(A)) = 0. \end{aligned}$
0902.0431_EQ0893	148	■ 0.989	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$	$\left\{ \begin{array}{l} \phi = [\phi_1, \phi_2] - 2A_1 \vee \gamma A_2 + 2A_2 \vee \gamma A_1, \\ A = \phi_1 A_2 - \phi_2 A_1. \end{array} \right.$
0902.0431_EQ0267	43	■ 0.989	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J} \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J} \end{array} \right.$	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J} \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J} \end{array} \right.$	$\left\{ \begin{array}{l} ([A, X], Y) + (X, [A, Y]) = 0, \quad X, Y \in \mathfrak{J}, \\ \text{tr}([A, X], Y, Z) + \text{tr}(X, [A, Y], Z) + \text{tr}(X, Y, [A, Z]) = 0, \quad X, Y, Z \in \mathfrak{J}. \end{array} \right.$
0902.0431_EQ0195	33	■ 0.990	$\alpha(x \textcolor{teal}{y}) = (\widetilde{\alpha} x) \overline{(\widetilde{\alpha} \textcolor{teal}{y})}, \quad x, y \in \mathfrak{C}.$	$\alpha(xy) = (\tilde{\alpha} x) \overline{(\tilde{\alpha} y)}, \quad x, y \in \mathfrak{C}.$	$\alpha(xy) = (\tilde{\alpha} x) \overline{(\tilde{\alpha} y)}, \quad x, y \in \mathfrak{C}.$

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0902.0431_E00409	67	■ 0.990	$\begin{aligned} & G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ & \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3 \end{aligned}$	$G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$	$G_{ij}, \quad 0 \leq i < j \leq 3, 4 \leq i < j \leq 7, \\ \widetilde{A}_k(e_j), \quad 0 \leq j \leq 3, k = 1, 2, 3$
0902.0431_E00508	98	■ 0.990	$\mathfrak{e}_{\{6\}} = \mathfrak{f}_{\{4\}} + i \left(\widetilde{\mathfrak{J}}_{\{H\}} \right)_0.$	$\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left(\widetilde{\mathfrak{J}}_H \right)_0.$	$\mathfrak{e}_{6,H} = \mathfrak{f}_{4,H} \oplus i \left(\widetilde{\mathfrak{J}}_H \right)_0.$
0902.0431_E00748	124	■ 0.990	$\mathfrak{e}_{\{7\}}^{\{C\}} = \mathfrak{e}_{\{6\}}^{\{C\}} \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C.$
0902.0431_E00750	124	■ 0.990	$\begin{aligned} & [h, S] = \left[\Phi(h_\delta + \widetilde{H}, 0, 0, \nu), \Phi(S, 0, 0, 0) \right] \\ & = \Phi([h_\delta + \widetilde{H}, S], 0, 0, 0) = \Phi(\alpha(h_\delta + \widetilde{H})S, 0, 0, 0) = \alpha(h)S. \end{aligned}$	$[h, S] = \left[\Phi(h_\delta + \widetilde{H}, 0, 0, \nu), \Phi(S, 0, 0, 0) \right] \\ = \Phi([h_\delta + \widetilde{H}, S], 0, 0, 0) = \Phi(\alpha(h_\delta + \widetilde{H})S, 0, 0, 0) = \alpha(h)S.$	$[h, S] = [\Phi(h_\delta + \widetilde{H}, 0, 0, \nu), \Phi(S, 0, 0, 0)] \\ = \Phi([h_\delta + \widetilde{H}, S], 0, 0, 0) = \Phi(\alpha(h_\delta + \widetilde{H})S, 0, 0, 0) = \alpha(h)S.$
0902.0431_E00969	161	■ 0.990	$E_{\{7(7)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau \gamma}, \quad \text{quad} \quad E_{\{7(-5)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau \lambda \sigma}, \quad \text{quad} \quad E_{\{7(-25)\}} \cong \left(E_{\{7\}}^{\{C\}} \right)^{\tau}.$	$E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$	$E_{7(7)} \cong (E_7^C)^{\tau \gamma}, \quad E_{7(-5)} \cong (E_7^C)^{\tau \lambda \sigma}, \quad E_{7(-25)} \cong (E_7^C)^{\tau}.$
0902.0431_E00293	47	■ 0.990	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$	$\widetilde{A}X = A \circ X, \quad X \in \mathfrak{J}^C.$

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0902.0431_E00018	9	■ 0.990	$\begin{gathered} \left\{ \begin{array}{l} 2F_{3} = G_{3} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{3} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{3} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{3} - G_{12} - G_{47} + G_{56}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$	$\left\{ \begin{array}{l} 2F_{03} = G_{03} + G_{12} + G_{47} + G_{56} \\ 2F_{12} = G_{03} + G_{12} - G_{47} - G_{56} \\ 2F_{47} = G_{03} - G_{12} + G_{47} - G_{56} \\ 2F_{56} = G_{03} - G_{12} - G_{47} + G_{56}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{04} = G_{04} - G_{15} + G_{26} - G_{37} \\ 2F_{15} = -G_{04} + G_{15} + G_{26} - G_{37} \\ 2F_{26} = G_{04} + G_{15} + G_{26} + G_{37} \\ 2F_{37} = -G_{04} - G_{15} + G_{26} + G_{37}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{05} = G_{05} + G_{14} - G_{27} - G_{36} \\ 2F_{14} = G_{05} + G_{14} + G_{27} + G_{36} \\ 2F_{27} = -G_{05} + G_{14} + G_{27} - G_{36} \\ 2F_{36} = -G_{05} + G_{14} - G_{27} + G_{36}, \end{array} \right\} \left\{ \begin{array}{l} 2F_{06} = G_{06} - G_{17} - G_{24} + G_{35} \\ 2F_{17} = -G_{06} + G_{17} - G_{24} + G_{35} \\ 2F_{24} = -G_{06} - G_{17} + G_{24} + G_{35} \\ 2F_{35} = G_{06} + G_{17} + G_{24} + G_{35}, \end{array} \right\} \\ \left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$	$\left\{ \begin{array}{l} 2F_{07} = G_{07} + G_{16} + G_{25} + G_{34} \\ 2F_{16} = G_{07} + G_{16} - G_{25} - G_{34} \\ 2F_{25} = G_{07} - G_{16} + G_{25} - G_{34} \\ 2F_{34} = G_{07} - G_{16} - G_{25} + G_{34}. \end{array} \right.$
0902.0431_E00722	119	■ 0.991	$\mathfrak{e}_{\{7\}}^{\{ \}^{\{C\}}} = \mathfrak{e}_{\{6\}}^{\{ \}^{\{C\}}} \oplus \mathfrak{e}_{\{N\}}^{\{ \}^{\{C\}}},$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C,$
0902.0431_E01001	166	■ 0.991	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$	$\widetilde{\lambda} = \lambda \lambda' = \lambda' \lambda.$
0902.0431_E00162	29	■ 0.991	$(x\,p)(\overline{p}\,y) = x\,y, \quad x, y \in \mathfrak{C}.$	$(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$	$(xp)(\overline{p}y) = xy, \quad x, y \in \mathfrak{C}.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00296	47	■ 0.991	$\begin{array}{ll} \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{E}_{_j}\right]=0, \} \& \{\left[\widetilde{E}_{_i}, \right. \\ \left. \widetilde{F}_{_i}(a)\right]=0, \} \quad \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_i+1}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_i+1}(a), \} \& \{\left[\right. \\ \left. \widetilde{E}_{_i}, \widetilde{F}_{_i+2}(a)\right]= \\ \frac{1}{2} \widetilde{A}_{_i+2}(a), \} \quad \& \{\left[\right. \\ \left. \widetilde{F}_{_i}(a), \widetilde{F}_{_i}(b)\right] \in \mathfrak{d}_4^C, \\ \left. \widetilde{D}, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), \\ \left. \widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, \\ \left. \widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. \end{array}$	$\begin{array}{ll} \left[\widetilde{E}_i, \widetilde{E}_j\right] = 0, & \left[\widetilde{E}_i, \widetilde{F}_i(a)\right] = 0, \\ \left[\widetilde{E}_i, \widetilde{F}_{i+1}(a)\right] = -\frac{1}{2}\widetilde{A}_{i+1}(a), & \left[\widetilde{E}_i, \widetilde{F}_{i+2}(a)\right] = \frac{1}{2}\widetilde{A}_{i+2}(a), \\ \left[\widetilde{F}_i(a), \widetilde{F}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)\right] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ \left[\widetilde{D}, \widetilde{A}_i(a)\right] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ \left[\widetilde{A}_i(a), \widetilde{A}_i(b)\right] \in \mathfrak{d}_4^C, & \left[\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)\right] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ \left[\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)\right] = G_{ij}. & \left[\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)\right] = \frac{1}{2}\widetilde{A}_{i+1}(\overline{ba}). \end{array}$	$\begin{array}{ll} [\widetilde{E}_i, \widetilde{E}_j] = 0, & [\widetilde{E}_i, \widetilde{F}_i(a)] = 0, \\ [\widetilde{E}_i, \widetilde{F}_{i+1}(a)] = -\frac{1}{2}\widetilde{A}_{i+1}(a), & [\widetilde{E}_i, \widetilde{F}_{i+2}(a)] = \frac{1}{2}\widetilde{A}_{i+2}(a), \\ [\widetilde{F}_i(a), \widetilde{F}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{F}_i(a), \widetilde{F}_{i+1}(b)] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ [D, \widetilde{A}_i(a)] = \widetilde{A}_i(D_i a), & \text{where } D = (D_1, D_2, D_3) \in \mathfrak{d}_4^C, \\ [\widetilde{A}_i(a), \widetilde{A}_i(b)] \in \mathfrak{d}_4^C, & [\widetilde{A}_i(a), \widetilde{A}_{i+1}(b)] = -\frac{1}{2}\widetilde{A}_{i+2}(\overline{ab}), \\ [\widetilde{F}_1(e_i), \widetilde{F}_1(e_j)] = G_{ij}. & [\widetilde{A}_i(a), \widetilde{A}_{i+2}(b)] = \frac{1}{2}\widetilde{A}_{i+1}(\overline{ba}). \end{array}$
0902.0431_E01151	194	■ 0.991	$\begin{array}{l} \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \det U, \end{array}$	$\begin{array}{l} \& \left(\boldsymbol{u}_{_1} \wedge \cdots \wedge \boldsymbol{u}_{_j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{_j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ \& \left(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9\right) = \det U, \end{array}$	$\begin{array}{l} (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_{j-1} \wedge \boldsymbol{x} \wedge \boldsymbol{u}_{j+1} \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \sum_{k=1}^9 \widetilde{u}_{jk} x_k, \\ (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_9, e_1 \wedge \cdots \wedge e_9) = \det U, \end{array}$

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0902.0431_E00936	155	■0.992	=\left(\begin{array}{rrrrr} b_{11} & b_{12} & b_{13} & b_{12} & b_{13} \\ & \bar{b}_{12} & b_{22} & b_{23} & b_{23} \\ & & \bar{b}_{13} & b_{23} & b_{33} \\ & & & \bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} \\ & & & & c_{11} & c_{12} & c_{13} \\ & & & & & -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} \\ & & & & & & -\bar{c}_{12} & c_{22} & c_{23} \\ & & & & & & & -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} \\ & & & & & & & & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\in\mathfrak{su}(6),	$= \begin{pmatrix} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{pmatrix} + \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in \mathfrak{su}(6),$	$= \begin{pmatrix} b_{11} & b_{12} & b_{13} & l_{11} & l_{12} & l_{13} \\ -\bar{b}_{12} & b_{22} & b_{23} & l_{21} & l_{22} & l_{23} \\ -\bar{b}_{13} & b_{23} & b_{33} & l_{31} & l_{32} & l_{33} \\ -\bar{l}_{11} & -\bar{l}_{21} & -\bar{l}_{31} & c_{11} & c_{12} & c_{13} \\ -\bar{l}_{12} & -\bar{l}_{22} & -\bar{l}_{32} & -\bar{c}_{12} & c_{22} & c_{23} \\ -\bar{l}_{13} & -\bar{l}_{23} & -\bar{l}_{33} & -\bar{c}_{13} & -\bar{c}_{23} & c_{33} \end{pmatrix} + \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \in \mathfrak{su}(6),$
0902.0431_E00994	165	■0.992	$B_8(1,1)=56 \times 2+4 \times 2=120$.	$B_8(1,1)=56 \times 2+4 \times 2=120.$	$B_8(1,1)=56 \times 2+4 \times 2=120.$
0902.0431_E00998	166	■0.992	$\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\right\}.$ $\mid R \times R=0, R \neq 0\right\}.$	$\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\right\}.$	$\mathfrak{W}^C=\left\{R\in\mathfrak{e}_8^C\mid R\times R=0,R\neq 0\right\}.$
0902.0431_E00034	11	■0.992	$\begin{aligned} &\pi T_{e_i}=\pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ &\pi R_a=\pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$	$\begin{aligned} \pi T_{e_i} &= \pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ \pi R_a &= \pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$	$\begin{aligned} \pi T_{e_i} &= \pi\left(2G_{i0}\right)=2F_{i0}=L_{e_i}, \\ \pi R_a &= \pi\left(T_a-L_a\right)=L_a-T_a=-R_a. \end{aligned}$
0902.0431_E00065	14	■0.992	$\begin{aligned} x &= x_0+x_1 e_1+x_2 e_2+x_3 e_3+x_4 e_4+x_5 e_5+x_6 e_6+x_7 e_7 \\ &= (x_0+x_1 e_1)+(x_2+x_3 e_1) e_2+(x_4+x_5 e_1) e_4+(x_6+x_7 e_1) e_6, \end{aligned}$	$\begin{aligned} x &= x_0+x_1 e_1+x_2 e_2+x_3 e_3+x_4 e_4+x_5 e_5+x_6 e_6+x_7 e_7 \quad (x_i \in \boldsymbol{R}) \\ &= (x_0+x_1 e_1)+(x_2+x_3 e_1) e_2+(x_4+x_5 e_1) e_4+(x_6+x_7 e_1) e_6, \end{aligned}$	$\begin{aligned} x &= x_0+x_1 e_1+x_2 e_2+x_3 e_3+x_4 e_4+x_5 e_5+x_6 e_6+x_7 e_7 \quad (x_i \in \boldsymbol{R}) \\ &= (x_0+x_1 e_1)+(x_2+x_3 e_1) e_2+(x_4+x_5 e_1) e_4+(x_6+x_7 e_1) e_6, \end{aligned}$
0902.0431_E00012	136	■0.992	$\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$	$\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$	$\iota(X,Y,\xi,\eta)=(-iX,iY,-i\xi,i\eta).$
0902.0431_E01046	178	■0.992	$0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$	$0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$	$0=[(\beta_1\Phi,0,0,a_1\Phi,0,0),(0,0,0,0,1,0)]= (0,0,0,0,2a_1\Phi,0),$
0902.0431_E00835	139	■0.993	$(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$	$(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$	$(A,Y)=-\left(A,\kappa_1Y\right),\quad Y\in\mathfrak{J}^C.$
0902.0431_E01019	170	■0.993	$[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$	$[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$	$[(\Phi(h),0,0,r,0,0),(0,P,0,0,0,0)]=(0,(\Phi(h)+r)P,0,0,0,0),$

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0902.0431_E08443	73	■ 0.993	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \\ &\quad \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$	$\begin{aligned} E_6^C &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle\}, \\ E_6 &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y, \alpha Z \rangle = \langle X, Y, Z \rangle, \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\} \\ &= \{\alpha \in \operatorname{Iso}_C(\mathfrak{J}^C) \mid \alpha X \times \alpha Y = \tau\alpha\tau(X \times Y), \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}. \end{aligned}$
0902.0431_E08709	117	■ 0.993	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$	$\begin{aligned} &(\exp t\Phi)(P \times Q)(\exp t\Phi)^{-1} \\ &= (\exp t(\operatorname{ad} \Phi))(P \times Q) \quad ((\operatorname{ad} \Phi)\Phi_1 = [\Phi, \Phi_1], \Phi_1 \in \mathfrak{e}_7^C) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (t(\operatorname{ad} \Phi))^n (P \times Q) \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \left(\sum_{k+l=n} \frac{n!}{k!l!} \Phi^k P \times \Phi^l Q \right) \quad (\text{Proposition 4.3.2}) \\ &= \sum_{n=0}^{\infty} \left(\sum_{k+l=n} \frac{t^k t^l}{k!l!} \Phi^k P \times \Phi^l Q \right) \\ &= \left(\sum_{k=0}^{\infty} \frac{1}{k!} (t\Phi)^k P \right) \times \left(\sum_{l=0}^{\infty} \frac{1}{l!} (t\Phi)^l Q \right) \\ &= (\exp t\Phi)P \times (\exp t\Phi)Q. \end{aligned}$
0902.0431_E08555	93	■ 0.993	$\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$	$\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$	$\beta\alpha_{23}(t_0)X = E_2 - E_3 \in S^8.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00777	130	■0.993	$\backslash\mathrm{mathfrak{M}}^{\mathrm{C}}\backslash\mathrm{ni}\backslash\alpha(X,X\times X,\backslash\mathrm{operatorname{det}}X,1)=\backslash\mathrm{left}(\backslash\beta X,\backslash\beta_{\mathrm{1}}(X\times X)\backslash\mathrm{operatorname{det}}X,1\mathrm{right}),$	$\mathfrak{M}^C \ni \alpha(X, X \times X, \det X, 1) = (\beta X, \beta_1(X \times X), \det X, 1),$	$\mathfrak{M}^C \ni \alpha(X, X \times X, \det X, 1) = (\beta X, \beta_1(X \times X), \det X, 1),$
0902.0431_E00841 (ii)	12	■0.994	$\backslash\mathrm{left}(\mathrm{L}_{\mathrm{b}}\mathrm{x}\mathrm{right})\mathrm{y}+\mathrm{x}\backslash\mathrm{left}(\mathrm{R}_{\mathrm{b}}\mathrm{y}\mathrm{right})=\mathrm{T}_{\mathrm{b}}(\mathrm{x}\mathrm{y}),\quad\mathrm{x},\mathrm{y}\in\mathrm{mathfrak{C}}.$	$(L_b x)y+x(R_b y)=T_b(xy),\quad x,y\in\mathfrak{C}.$	$(L_b x)y+x(R_b y)=T_b(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00077	16	■0.994	$\mathrm{B}=\mathrm{D}+\sum_{\mathrm{i}=1}^6\mathrm{x}_{\mathrm{i}}\mathrm{S}_{\mathrm{i}},\quad\mathrm{D}\in\mathrm{mathfrak{s u}}(3),\mathrm{x}_{\mathrm{i}}\in\mathrm{boldsymbol{R}}$	$B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}$	$B=D+\sum_{i=1}^6x_iS_i,\quad D\in\mathfrak{su}(3),x_i\in\boldsymbol{R}.$
0902.0431_E00996	166	■0.994	$\backslash\mathrm{operatorname{Aut}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right})=\backslash\mathrm{operatorname{Inn}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\mathrm{8}}^{\mathrm{C}}\backslash\mathrm{right}).$	$\mathrm{Aut}\left(\mathfrak{e}_8^C\right)=\mathrm{Inn}\left(\mathfrak{e}_8^C\right).$	$\mathrm{Aut}(\mathfrak{e}_8^C)=\mathrm{Inn}(\mathfrak{e}_8^C).$
0902.0431_E00999	166	■0.994	$\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}/\backslash\mathrm{left}(\mathrm{E}_{\mathrm{8}}^{\mathrm{C}}\mathrm{right})_{\mathrm{1}_{-}}\backslash\mathrm{simeq}\backslash\mathrm{mathfrak{W}}^{\mathrm{C}},$	$E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$	$E_8^C/(E_8^C)_{1_-}\simeq\mathfrak{W}^C,$
0902.0431_E00562	94	■0.994	$\backslash\mathrm{begin}{gathered}\backslash\mathrm{mathfrak{E}}_{\mathrm{1}}^{\mathrm{C}}=\backslash\mathrm{left}\backslash\{\mathrm{x_i}\mathrm{E}_{\mathrm{1}}\backslash\mathrm{mid}\mathrm{x_i}\in\mathrm{C}\mathrm{right}\},\quad\backslash\mathrm{quad}\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}(2,\backslash\mathrm{mathfrak{C}})^{\mathrm{C}}=\backslash\mathrm{left}\backslash\{X\in\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}4\mathrm{E}_{\mathrm{1}}\times(\mathrm{E}_{\mathrm{1}}\times X)=X\}\backslash\mathrm{mid}4\mathrm{E}_{\mathrm{1}}\times\backslash\mathrm{left}(\mathrm{E}_{\mathrm{1}}\times X\mathrm{right})=X\mathrm{right}\},\quad\backslash\backslash\backslash\mathrm{left}(\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\mathrm{right})_{-\backslash\mathrm{sigma}}=\backslash\mathrm{left}\backslash\{X\in\backslash\mathrm{mathfrak{J}}^{\mathrm{C}}\backslash\mathrm{mid}\mathrm{E}_{\mathrm{1}}\times X=0,\backslash\mathrm{left}\backslash\mathrm{angle}\mathrm{E}_{\mathrm{1}},X\mathrm{right}\mathrm{rangle}=0\mathrm{right}\}.$	$\mathfrak{E}_1^C=\{\xi E_1\mid \xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},$ $(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$	$\mathfrak{E}_1^C=\{\xi E_1\mid \xi\in C\},\quad\mathfrak{J}(2,\mathfrak{C})^C=\{X\in\mathfrak{J}^C\mid 4E_1\times(E_1\times X)=X\},$ $(\mathfrak{J}^C)_{-\sigma}=\{X\in\mathfrak{J}^C\mid E_1\times X=0,\langle E_1,X\rangle=0\}.$
0902.0431_E00810	135	■0.994	$\backslash\alpha=\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{cccc}\backslash\omega^{\backslash\mathrm{prime}}1&\&\&\&\&\&\&\backslash\omega^{\backslash\mathrm{prime}-1}1&\&\&\&\&\&\&\&\&\&\&\backslash\omega^{\backslash\mathrm{prime}}=1,\backslash\omega\mathrm{or}\backslash\omega^2}.$	$\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$	$\alpha=\left(\begin{array}{cccc}\omega'1&0&0&0\\0&\omega'^{-1}1&0&0\\0&0&1&0\\0&0&0&1\end{array}\right),\quad\omega'=1,\omega\text{ or }\omega^2.$
0902.0431_E00012	8	■0.994	$\backslash\mathrm{begin}{aligned}(\backslash\alpha\mathrm{x},\backslash\alpha\mathrm{y})&=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\overline{\backslash\alpha\mathrm{y}})+(\backslash\alpha\mathrm{y})(\overline{\backslash\alpha\mathrm{x}}))=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\backslash\alpha\mathrm{y})+(\backslash\alpha\mathrm{y})(\backslash\alpha\mathrm{x}))\\&=\mathrm{frac}{1}{2}((\backslash\alpha\mathrm{x})(\backslash\alpha\mathrm{y})+(\backslash\alpha\mathrm{y})(\backslash\alpha\mathrm{x}))\backslash\&=\backslash\alpha\backslash\mathrm{left}(\mathrm{frac}{1}{2}(\mathrm{x}\mathrm{y}+\mathrm{y}\mathrm{x})\mathrm{right})=\backslash\alpha((\mathrm{x},\mathrm{y}))=(\mathrm{x},\mathrm{y}).\backslash\mathrm{end}{aligned}$	$(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \overline{y})+(\alpha y)(\alpha \overline{x}))$ $=\alpha\left(\frac{1}{2}(x\overline{y}+y\overline{x})\right)=\alpha((x,y))=(x,y).$	$(\alpha x,\alpha y)=\frac{1}{2}((\alpha x)(\overline{\alpha y})+(\alpha y)(\overline{\alpha x}))=\frac{1}{2}((\alpha x)(\alpha \overline{y})+(\alpha y)(\alpha \overline{x}))$ $=\alpha\left(\frac{1}{2}(x\overline{y}+y\overline{x})\right)=\alpha((x,y))=(x,y).$
0902.0431_E00412	68	■0.994	$\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{ccc}\backslash\mathrm{x_i}_{\mathrm{1}}\&\&\backslash\mathrm{x}_{\mathrm{3}}\&\&\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{2}}\backslash\backslash\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{3}}\&\&\backslash\mathrm{x_i}_{\mathrm{2}}\&\&\backslash\mathrm{x}_{\mathrm{1}}\backslash\backslash\backslash\mathrm{x}_{\mathrm{2}}\&\&\backslash\mathrm{bar}{\mathrm{x}}_{\mathrm{1}}\&\&\backslash\mathrm{x_i}_{\mathrm{3}}\backslash\mathrm{end}{array}\mathrm{right}),\quad\backslash\mathrm{quad}\backslash\mathrm{x_i}_{\mathrm{i}}\in\mathrm{boldsymbol{R}},\mathrm{x}_{\mathrm{i}}=\mathrm{a}_{\mathrm{i}}+\mathrm{boldsymbol{m}_{\mathrm{i}}}\in\mathrm{boldsymbol{C}}\oplus\mathrm{boldsymbol{C}^3}=\mathrm{mathfrak{C}}$	$\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$	$\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right),\quad\xi_i\in\boldsymbol{R},x_i=a_i+\boldsymbol{m}_i\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}$
0902.0431_E00673	112	■0.994	$\backslash\mathrm{begin}{aligned}&=\backslash\Phi\backslash\mathrm{left}(-X\vee Y,-\mathrm{frac}{1}{2}(Y\times Y-\backslash\mathrm{x_i}X),\mathrm{frac}{1}{2}(X\times X-\backslash\mathrm{eta}Y),\mathrm{frac}{1}{4}((X,Y)-3\backslash\mathrm{x_i}\backslash\mathrm{eta})\mathrm{right})(Z,W,\backslash\mathrm{zeta},\backslash\mathrm{omega}),\quad\backslash\&-\mathrm{frac}{3}{8}((X,W)-(Z,Y)+\backslash\mathrm{x_i}\backslash\mathrm{omega}-\backslash\mathrm{zeta}\backslash\mathrm{eta})(X,Y,\backslash\mathrm{x_i},\backslash\mathrm{eta})\backslash\&=(P\times P)Q-\mathrm{frac}{3}{8}\{P,Q\}P.\backslash\mathrm{end}{aligned}$	$=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),$ $-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)$ $=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$	$=\Phi\left(-X\vee Y,-\frac{1}{2}(Y\times Y-\xi X),\frac{1}{2}(X\times X-\eta Y),\frac{1}{4}((X,Y)-3\xi\eta)\right)(Z,W,\zeta,\omega),$ $-\frac{3}{8}((X,W)-(Z,Y)+\xi\omega-\zeta\eta)(X,Y,\xi,\eta)$ $=(P\times P)Q-\frac{3}{8}\{P,Q\}P.$

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0902.0431_E00136	25	■ 0.995	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & \\ -G_{47}+G_{56}, & \backslash \backslash 2 G_{13}-G_{46}-G_{57}, & \\ G_{46}+G_{57}, & \backslash \backslash 2 G_{23}-G_{45}-G_{67}, & \\ -G_{45}+G_{67} \end{array} \quad \backslash \end{array}$	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2 G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2 G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$	$\begin{array}{cc} 2 G_{12}-G_{47}-G_{56}, & -G_{47}+G_{56}, \\ 2 G_{13}-G_{46}-G_{57}, & G_{46}+G_{57}, \\ 2 G_{23}-G_{45}-G_{67}, & -G_{45}+G_{67} \end{array}$
0902.0431_E00282	45	■ 0.995	$\left(D_{\{i\}} x, y\right)+\left(x, D_{\{i\}} y\right)=0, \quad \backslash \quad \quad \quad x, y \in \mathfrak{C} .$	$\left(D_i x, y\right)+\left(x, D_i y\right)=0, \quad x, y \in \mathfrak{C} .$	$\left(D_i x, y\right)+\left(x, D_i y\right)=0, \quad x, y \in \mathfrak{C} .$
0902.0431_E01149 (i)	194	■ 0.995	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = \boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = (\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$	$\begin{aligned} & (\boldsymbol{u} \times \boldsymbol{v}) \boldsymbol{x} = (\boldsymbol{x} \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_1 + (\boldsymbol{u}_1 \wedge \boldsymbol{x} \wedge \boldsymbol{u}_3, \boldsymbol{v}) \boldsymbol{u}_2 \\ & + (\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{x}, \boldsymbol{v}) \boldsymbol{u}_3 - \frac{1}{3}(\boldsymbol{u}, \boldsymbol{v}) \boldsymbol{x}. \end{aligned}$
0902.0431_E00292	46	■ 0.995	$\begin{aligned} & \mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid (\phi X, X, X) = 0\} \\ & = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y\}, \\ & \mathfrak{f}_4^C = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y\} \\ & = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y\}. \end{aligned}$	$\begin{aligned} & \mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid (\phi X, X, X) = 0\} \\ & = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y\}, \\ & \mathfrak{f}_4^C = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y\} \\ & = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y\}. \end{aligned}$	$\begin{aligned} & \mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid (\phi X, X, X) = 0\} \\ & = \{\phi \in \operatorname{Hom}_{\mathfrak{C}} \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y\}, \\ & \mathfrak{f}_4^C = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \circ Y) = \delta X \circ Y + X \circ \delta Y\} \\ & = \{\delta \in \operatorname{Hom}_{\mathfrak{C}} \mid \delta(X \times Y) = \delta X \times Y + X \times \delta Y\}. \end{aligned}$
0902.0431_E00822 (iv)	137	■ 0.995	$\begin{aligned} & a(X) \operatorname{det} X = a(X)(\operatorname{det} X)(M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ & = (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & = \frac{1}{2} a(X)(3 a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$	$\begin{aligned} & a(X) \operatorname{det} X = a(X)(\operatorname{det} X)(M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ & = (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & = \frac{1}{2} a(X)(3 a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$	$\begin{aligned} & a(X) \operatorname{det} X = a(X)(\operatorname{det} X)(M, N) = (\delta(X \times X) \times \delta(X \times X), N) \\ & = (\delta(X \times X), \delta(X \times X) \times N) = \frac{1}{2} a(X)(\delta(X \times X), \beta X) \\ & = \frac{1}{2} a(X)(3 a(X) b(X \times X) - \operatorname{det} X). \end{aligned}$
0902.0431_E01042	178	■ 0.995	$\left(E_8^C\right)_{1,1^-,1_-} = \left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$	$\left(E_8^C\right)_{1,1^-,1_-} = \left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$	$\left(E_8^C\right)_{1,1^-,1_-} = \left\{\alpha \in E_8^C \mid \alpha 1=1, \alpha 1^-=1^-, \alpha 1_-=1_-\right\} .$

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0902.0431_E00387	64	■ 0.995	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$	$\gamma\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \gamma x_3 & \overline{\gamma x_2} \\ \overline{\gamma x_3} & \xi_2 & \gamma x_1 \\ \gamma x_2 & \overline{\gamma x_1} & \xi_3\end{array}\right).$
0902.0431_E00744	123	■ 0.995	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$	$\Phi_0\Phi_0(X,Y,\xi,\eta)=\Phi_0\left(-\frac{X}{3},\frac{Y}{3},\xi,-\eta\right)=\left(\frac{X}{9},\frac{Y}{9},\xi,\eta\right),$
0902.0431_E01109	188	■ 0.995	$S\,0(16),\quad S\,s(16)\,.$	$SO(16),\quad Ss(16).$	$SO(16),\quad Ss(16).$
0902.0431_E01124	190	■ 0.995	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$	$\tau\widetilde{\lambda}(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(-\tau^tD,-\tau^t\phi\tau,-\tau\boldsymbol{Y},-\tau\boldsymbol{X}).$
0902.0431_E01044	178	■ 0.995	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$	$\alpha=\left(\begin{array}{cccccc}\beta_1 & \beta_{12} & \beta_{13} & 0 & 0 & 0 \\ \beta_{21} & \beta_2 & \beta_{23} & 0 & 0 & 0 \\ \beta_{31} & \beta_{32} & \beta_3 & 0 & 0 & 0 \\ a_1 & b_1 & c_1 & 1 & 0 & 0 \\ a_2 & b_2 & c_2 & 0 & 1 & 0 \\ a_3 & b_3 & c_3 & 0 & 0 & 1\end{array}\right),$
0902.0431_E00410	68	■ 0.995	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$	$w\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \omega x_3 & \overline{\omega x_2} \\ \overline{\omega x_3} & \xi_2 & \omega x_1 \\ \omega x_2 & \overline{\omega x_1} & \xi_3\end{array}\right).$
0902.0431_E01134	191	■ 0.995	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$	$(\mathrm{ad}\,\phi')(D,\phi,\boldsymbol{X},\boldsymbol{Y})=(0,(\mathrm{ad}\,\phi')\phi,\phi'\boldsymbol{X},-{}^t\phi'\boldsymbol{Y}),$
0902.0431_E00493	81	■ 0.996	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$	$\delta(X\circ Y)=\delta X\circ Y+X\circ\delta Y,\quad X,Y\in(\mathfrak{M}^r)^C.$

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0902.0431_E0861	143	■ 0.996	$\left(\left(\left(E_7\right)^{\kappa,\mu}\right)_{(0,E_1,0,1)}/\{1,\sigma\}\cong SO(11),\right.\\ \left.E_{\{1\}},\left.0,1\right)\right)\wedge\{1,\sigma\}\cong S O(11) \text{ .}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}/\{1,\sigma\}\cong SO(11).$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}/\{1,\sigma\}\cong SO(11).$
0902.0431_E0882	163	■ 0.996	$\left[\frac{a}{n}\right]\left[\Phi,\left[P_1^{-},P_{-}\right],P_{-}\right]\right]=\left[\Phi,P\times P_1\right] \text{ .}$	$\mathfrak{a}\ni\left[\Phi,\left[P_1^{-},P_{-}\right]\right]=\left[\Phi,P\times P_1\right].$	$\mathfrak{a}\ni\left[\Phi,\left[P_1^{-},P_{-}\right]\right]=\left[\Phi,P\times P_1\right].$
0902.0431_E0886	164	■ 0.996	$\operatorname{Der}\left(\frac{e_8}{C}\right)=\left\{\Theta\in\operatorname{Hom}\left(\frac{e_8}{C}\right)\mid\Theta\left[R_1,R_2\right]=\left[\Theta R_1,R_2\right]+\left[R_2,\Theta R_2\right]\right\}.$	$\operatorname{Der}\left(\mathfrak{e}_8^C\right)=\left\{\Theta\in\operatorname{Hom}_C\left(\mathfrak{e}_8^C\right)\mid\Theta\left[R_1,R_2\right]=\left[\Theta R_1,R_2\right]+\left[R_2,\Theta R_2\right]\right\}.$	$\operatorname{Der}\left(\mathfrak{e}_8^C\right)=\left\{\Theta\in\operatorname{Hom}_C\left(\mathfrak{e}_8^C\right)\mid\Theta\left[R_1,R_2\right]=\left[\Theta R_1,R_2\right]+\left[R_2,\Theta R_2\right]\right\}.$
0902.0431_E01182	199	■ 0.996	$\left(\boldsymbol{x}\wedge\boldsymbol{a},\boldsymbol{y}\wedge\boldsymbol{b}\right)+\left(\boldsymbol{*b}\wedge\boldsymbol{x},\boldsymbol{*a}\wedge\boldsymbol{y}\right)=\left(\boldsymbol{x},\boldsymbol{y}\right)\left(\boldsymbol{a},\boldsymbol{b}\right).$	$(\boldsymbol{x}\wedge\boldsymbol{a},\boldsymbol{y}\wedge\boldsymbol{b})+(\boldsymbol{*b}\wedge\boldsymbol{x},\boldsymbol{*a}\wedge\boldsymbol{y})=(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{a},\boldsymbol{b}).$	$(\boldsymbol{x}\wedge\boldsymbol{a},\boldsymbol{y}\wedge\boldsymbol{b})+(\boldsymbol{*b}\wedge\boldsymbol{x},\boldsymbol{*a}\wedge\boldsymbol{y})=(\boldsymbol{x},\boldsymbol{y})(\boldsymbol{a},\boldsymbol{b}).$
0902.0431_E0883	164	■ 0.996	$\begin{aligned} \left[\frac{a}{n}\right]\left[P^{\wedge{-}},\left[1^{\wedge{-}},\left[1_{-},R\right]\right],R\right]\right]=\left[P^{\wedge{-}},\left[1^{\wedge{-}},\left(0,0,0,-s,0,2r\right)\right]\right] \\ =\left[P^{\wedge{-}},\left(0,0,0,2r,2s,0\right)\right]=\left(0,-2rP,0,0,0,0\right) \end{aligned}$	$\mathfrak{a}\ni\left[P^{-},\left[1^{-},\left[1_{-},R\right]\right]\right]=\left[P^{-},\left[1^{-},\left(0,0,0,-s,0,2r\right)\right]\right] \\ =\left[P^{-},\left(0,0,0,2r,2s,0\right)\right]=\left(0,-2rP,0,0,0,0\right)$	$\mathfrak{a}\ni\left[P^{-},\left[1^{-},\left[1_{-},R\right]\right]\right]=\left[P^{-},\left[1^{-},\left(0,0,0,-s,0,2r\right)\right]\right] \\ =\left[P^{-},\left(0,0,0,2r,2s,0\right)\right]=\left(0,-2rP,0,0,0,0\right).$
0902.0431_E01079	183	■ 0.996	$\begin{aligned} \chi\,\lambda\,\gamma\,\chi^{-1}\,S &= \chi\,\lambda\,\gamma(X,\,Y,\,\xi,\,\eta) = \chi(\gamma Y,\,-\gamma X,\,\eta,\,-\xi) \\ &= \left(k\left(g(\gamma Y) - \frac{\eta}{2}E\right) + ik\left(g\left(\gamma(-\gamma X) - \frac{-\xi}{2}E\right)\right)\right)J \\ &= -i\left(k\left(gX - \frac{\xi}{2}E\right) + ik\left(g(\gamma Y) - \frac{\eta}{2}E\right)\right)J \\ &= -i\chi(X,\,Y,\,\xi,\,\eta) = -i\chi P = -iS. \end{aligned}$	$\chi\lambda\gamma\chi^{-1}S = \chi\lambda\gamma(X,Y,\xi,\eta) = \chi(\gamma Y,-\gamma X,\eta,-\xi) \\ = \left(k\left(g(\gamma Y) - \frac{\eta}{2}E\right) + ik\left(g\left(\gamma(-\gamma X) - \frac{-\xi}{2}E\right)\right)\right)J \\ = -i\left(k\left(gX - \frac{\xi}{2}E\right) + ik\left(g(\gamma Y) - \frac{\eta}{2}E\right)\right)J \\ = -i\chi(X,Y,\xi,\eta) = -i\chi P = -iS.$	$\chi\lambda\gamma\chi^{-1}S = \chi\lambda\gamma(X,Y,\xi,\eta) = \chi(\gamma Y,-\gamma X,\eta,-\xi) \\ = \left(k\left(g(\gamma Y) - \frac{\eta}{2}E\right) + ik\left(g\left(\gamma(-\gamma X) - \frac{-\xi}{2}E\right)\right)\right)J \\ = -i\left(k\left(gX - \frac{\xi}{2}E\right) + ik\left(g(\gamma Y) - \frac{\eta}{2}E\right)\right)J \\ = -i\chi(X,Y,\xi,\eta) = -i\chi P = -iS.$
0902.0431_E0897	151	■ 0.996	$\alpha P=(X,Y,\xi,\eta),\quad X,Y\text{ are real diagonal, }0\neq\xi\in\boldsymbol{R}.$	$\alpha P=(X,Y,\xi,\eta),\quad X,Y\text{ are real diagonal, }0\neq\xi\in\boldsymbol{R}.$	$\alpha P=(X,Y,\xi,\eta),\quad X,Y\text{ are real diagonal, }0\neq\xi\in\boldsymbol{R}.$
0902.0431_E00113	22	■ 0.996	$\begin{aligned} \left(a_4,a_7\right) &= \left(a_4,a_3a_4\right) = \left(1,a_3\right)\left(a_4,a_4\right) = 0, \\ \left(a_1,a_6\right) &= \left(a_1,a_4a_2\right) = -\left(a_1a_2,a_4\right) = -\left(a_3,a_4\right) = 0, \\ \left(a_3,a_6\right) &= \left(a_3,a_4a_2\right) = -\left(a_3a_2,a_4\right) = \left(a_1,a_4\right) = 0, \text{ etc.} \end{aligned}$	$\begin{aligned} \left(a_4,a_7\right) &= \left(a_4,a_3a_4\right) = \left(1,a_3\right)\left(a_4,a_4\right) = 0, \\ \left(a_1,a_6\right) &= \left(a_1,a_4a_2\right) = -\left(a_1a_2,a_4\right) = -\left(a_3,a_4\right) = 0, \\ \left(a_3,a_6\right) &= \left(a_3,a_4a_2\right) = -\left(a_3a_2,a_4\right) = \left(a_1,a_4\right) = 0, \text{ etc.} \end{aligned}$	$\begin{aligned} \left(a_4,a_7\right) &= \left(a_4,a_3a_4\right) = \left(1,a_3\right)\left(a_4,a_4\right) = 0, \\ \left(a_1,a_6\right) &= \left(a_1,a_4a_2\right) = -\left(a_1a_2,a_4\right) = -\left(a_3,a_4\right) = 0, \\ \left(a_3,a_6\right) &= \left(a_3,a_4a_2\right) = -\left(a_3a_2,a_4\right) = \left(a_1,a_4\right) = 0, \text{ etc.} \end{aligned}$
0902.0431_E00166	30	■ 0.996	$\left(\alpha_2y\right)\left(\alpha_3\left(\overline{xy}\right)\right)=\overline{\alpha_1x} y ^2.$	$(\alpha_2y)\left(\alpha_3(\overline{xy})\right)=\overline{\alpha_1x} y ^2.$	$(\alpha_2y)(\alpha_3(\overline{xy}))=\overline{\alpha_1x} y ^2.$
0902.0431_E00300	48	■ 0.996	$\frac{a}{n}\left[D^{\prime},D+\sum_{i=1}^3\widetilde{A}_{\left(D_i^{\prime}a_i\right)}\right]=\left[D^{\prime},D\right]+\sum_{i=1}^3\widetilde{A}_{\left(D_i^{\prime}a_i\right)}\text{ (Lemma 2.4.2),}$	$\mathfrak{a}\ni\left[D',D+\sum_{i=1}^3\widetilde{A}_i\left(a_i\right)\right]=\left[D',D\right]+\sum_{i=1}^3\widetilde{A}_i\left(D_i'a_i\right)\text{ (Lemma 2.4.2),}$	$\mathfrak{a}\ni\left[D',D+\sum_{i=1}^3\widetilde{A}_i(a_i)\right]=\left[D',D\right]+\sum_{i=1}^3\widetilde{A}_i(D_i'a_i)\text{ (Lemma 2.4.2),}$

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0902.0431_E00571	95	■ 0.996	J=\left(\begin{array}{ccc} J & \theta & \theta \\ \theta & J & \theta \\ \theta & \theta & J \end{array}\right) \in M(6, C), \quad J=\left(\begin{array}{cc} \theta & 1 \\ -1 & \theta \end{array}\right)	$J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$	$J = \begin{pmatrix} J & 0 & 0 \\ 0 & J & 0 \\ 0 & 0 & J \end{pmatrix} \in M(6, C), \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$
0902.0431_E00638	105	■ 0.996	\operatorname{dim}\left(\mathfrak{e}_{_6, \boldsymbol{C}}\right)=8+8=16 \quad .	$\dim \left(\mathfrak{e}_{6, C}\right)=8+8=16.$	$\dim \left(\mathfrak{e}_{6, C}\right)=8+8=16.$
0902.0431_E00647	107	■ 0.996	\begin{aligned} \operatorname{det}(\alpha X) &= (\operatorname{det}(h(A, B))) (\operatorname{det} X) \\ \left(\operatorname{det}\left(h(A, B)^*\right)\right) &= \operatorname{det} X, \\ \angle \alpha X, \alpha Y &= \angle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle . \end{aligned}	$\begin{aligned} \det(\alpha X) &= (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X, \\ \langle \alpha X, \alpha Y \rangle &= \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \rangle \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle . \end{aligned}$	$\begin{aligned} \det(\alpha X) &= (\det(h(A, B))) (\det X) (\det(h(A, B)^*)) = \det X, \\ \langle \alpha X, \alpha Y \rangle &= \langle h(A, B) X h(A, B)^*, h(A, B) Y h(A, B)^* \rangle \\ &= (\tau h(A, B) \tau X \tau h(A, B)^*, h(A, B) Y h(A, B)^*) \\ &= (\tau X, Y) = \langle X, Y \rangle . \end{aligned}$
0902.0431_E00687	114	■ 0.996	\operatorname{dim}_{\boldsymbol{C}}\left(\mathfrak{e}_{_7}^C\right)=78+27 \times 2+1=133 \quad .	$\dim_C \left(\mathfrak{e}_7^C\right)=78+27 \times 2+1=133.$	$\dim_C \left(\mathfrak{e}_7^C\right)=78+27 \times 2+1=133.$
0902.0431_E00807	135	■ 0.996	\begin{aligned} &\left(\theta, \theta, \theta, \theta^3\right) \eta=\varphi_1(\theta)(\theta, \theta, \theta, \eta)=\varphi_1(\theta) \alpha(0,0,1,0) \\ &=\alpha \varphi_1(\theta)(0,0,1,0)=\alpha\left(0,0, \theta^3, 0\right)=\left(0,0,0, \theta^3 \eta\right), \end{aligned}	$\begin{aligned} (0,0,0,\theta^{-3}\eta) &= \varphi_1(\theta)(0,0,0,\eta) = \varphi_1(\theta)\alpha(0,0,1,0) \\ &= \alpha\varphi_1(\theta)(0,0,1,0) = \alpha(0,0,\theta^3,0) = (0,0,0,\theta^3\eta), \end{aligned}$	$\begin{aligned} (0,0,0,\theta^{-3}\eta) &= \varphi_1(\theta)(0,0,0,\eta) = \varphi_1(\theta)\alpha(0,0,1,0) \\ &= \alpha\varphi_1(\theta)(0,0,1,0) = \alpha(0,0,\theta^3,0) = (0,0,0,\theta^3\eta), \end{aligned}$
0902.0431_E00933	154	■ 0.996	\psi(A, 1) P=f^{-1}(A(f P)), \quad \psi(A, \epsilon) P=f^{-1}(A(f \bar{P})), \quad P \in\left(\mathfrak{P}_{\boldsymbol{C}}\right)^C .	$\psi(A, 1) P=f^{-1}(A(f P)), \quad \psi(A, \epsilon) P=f^{-1}(A(f \bar{P})), \quad P \in\left(\mathfrak{P}_C\right)^C .$	$\psi(A, 1) P=f^{-1}(A(f P)), \quad \psi(A, \epsilon) P=f^{-1}(A(f \bar{P})), \quad P \in\left(\mathfrak{P}_C\right)^C .$
0902.0431_E00977	162	■ 0.996	\begin{aligned} \Phi &= (\Phi, \theta, \theta, \theta, \theta), \quad P^{\{-\}} &= (\theta, P, \theta, \theta, \theta), \\ Q_{-} &= (0,0, Q, \theta, \theta), \quad r &= (0,0, \theta, r, \theta), \\ s^{-} &= (0,0, \theta, \theta, s), \quad t_{-} &= (0,0, \theta, \theta, t) . \end{aligned}	$\begin{aligned} \Phi &= (\Phi, 0, 0, 0, 0), \quad P^{-} = (0, P, 0, 0, 0), \\ Q_{-} &= (0, 0, Q, 0, 0), \quad r = (0, 0, 0, r, 0), \\ s^{-} &= (0, 0, 0, 0, s), \quad t_{-} = (0, 0, 0, 0, t). \end{aligned}$	$\begin{aligned} \Phi &= (\Phi, 0, 0, 0, 0), \quad P^{-} = (0, P, 0, 0, 0), \\ Q_{-} &= (0, 0, Q, 0, 0), \quad r = (0, 0, 0, r, 0), \\ s^{-} &= (0, 0, 0, 0, s), \quad t_{-} = (0, 0, 0, 0, t). \end{aligned}$
0902.0431_E01098	186	■ 0.996	\begin{aligned} &\zeta\left[l\left(i c_1 E\right), l\left(i c_2 E\right)\right]=\zeta l\left[i c_1 E, i c_2 E\right]=\zeta 0=0 \\ &=\Theta\left(0,0,0,0,2 c_1,-2 c_1\right), \Theta\left(0,0,0,0,2 c_2,-2 c_2\right)) \\ &=\left[\zeta l\left(i c_1 E\right), \zeta l\left(i c_2 E\right)\right] . \end{aligned}	$\begin{aligned} \zeta[l(ic_1E),l(ic_2E)] &= \zeta l[ic_1E,ic_2E] = \zeta 0 = 0 \\ &= \Theta(0,0,0,0,2c_1,-2c_1), \Theta(0,0,0,0,2c_2,-2c_2)) \\ &= [\zeta l(ic_1E), \zeta l(ic_2E)] . \end{aligned}$	$\begin{aligned} \zeta[l(ic_1E),l(ic_2E)] &= \zeta l[ic_1E,ic_2E] = \zeta 0 = 0 \\ &= \Theta(0,0,0,0,2c_1,-2c_1), \Theta(0,0,0,0,2c_2,-2c_2)) \\ &= [\zeta l(ic_1E), \zeta l(ic_2E)] . \end{aligned}$
0902.0431_E01186	199	■ 0.996	\begin{aligned} &((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ &= (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}	$\begin{aligned} ((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ = (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}$	$\begin{aligned} ((\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x})) \boldsymbol{v}, \boldsymbol{w}) \\ = (\boldsymbol{v} \wedge \boldsymbol{a}, \boldsymbol{w} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - \frac{3}{5}(*(\boldsymbol{a} \wedge \boldsymbol{b}), \boldsymbol{x})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}$

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0902.0431_E00724	119	■ 0.996	$\begin{aligned} & \mathfrak{a} \supset \left[\mathfrak{a}, \mathfrak{e}_7^C \right] \supset \left[\Phi \left(\mathfrak{e}_6^C, 0, 0, 0 \right), \Phi \left(0, \mathfrak{J}^C, 0, 0 \right) \right] \\ & = \Phi \left(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0 \right) = \Phi \left(0, \mathfrak{J}^C, 0, 0 \right) \text{ (Proposition 3.3.2.(3)).} \end{aligned}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$	$\mathfrak{a} \supset [\mathfrak{a}, \mathfrak{e}_7^C] \supset [\Phi(\mathfrak{e}_6^C, 0, 0, 0), \Phi(0, \mathfrak{J}^C, 0, 0)]$ $= \Phi(0, \mathfrak{e}_6^C \mathfrak{J}^C, 0, 0) = \Phi(0, \mathfrak{J}^C, 0, 0) \text{ (Proposition 3.3.2.(3)).}$
0902.0431_E00063	14	■ 0.997	$\begin{aligned} & \lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0, \\ & \lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0, \\ & \lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0, \\ & \lambda_{16} + \lambda_{25} + \lambda_{34} = 0, \end{aligned}$	$\lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0,$ $\lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0,$ $\lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0,$ $\lambda_{16} + \lambda_{25} + \lambda_{34} = 0,$	$\lambda_{23} + \lambda_{45} + \lambda_{67} = 0, \quad -\lambda_{13} - \lambda_{46} + \lambda_{57} = 0,$ $\lambda_{12} + \lambda_{47} + \lambda_{56} = 0, \quad -\lambda_{15} + \lambda_{26} - \lambda_{37} = 0,$ $\lambda_{14} - \lambda_{27} - \lambda_{36} = 0, \quad -\lambda_{17} - \lambda_{24} + \lambda_{35} = 0,$ $\lambda_{16} + \lambda_{25} + \lambda_{34} = 0,$
0902.0431_E00264	43	■ 0.997	$(X, Y) = \frac{1}{2} \operatorname{tr} \left(X Y + Y^* X^* \right) .$	$(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$	$(X, Y) = \frac{1}{2} \operatorname{tr} (XY + Y^* X^*) .$
0902.0431_E00335	53	■ 0.997	$\begin{aligned} & H \left(e_k + i e_{4+k} \right) = - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right) \\ & = - i \lambda_k \left(- e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right), \end{aligned}$	$H \left(e_k + i e_{4+k} \right) = - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right)$ $= - i \lambda_k \left(- e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right),$	$H \left(e_k + i e_{4+k} \right) = - \sum_{j=0}^3 \lambda_k i G_{j4+j} \left(e_k + i e_{4+k} \right)$ $= - i \lambda_k \left(- e_{4+k} + i e_k \right) = \lambda_k \left(e_k + i e_{4+k} \right),$
0902.0431_E00881	146	■ 0.997	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$	$\alpha X = \left(\begin{array}{ccc} \xi_1 & 0 & 0 \\ 0 & \xi_2 & 0 \\ 0 & 0 & \xi_3 \end{array} \right), \quad \xi_i \in C.$
0902.0431_E01057	180	■ 0.997	$-21_{-} = \left[\left(\Phi, P, Q, r, s, t \right), 1_{-} \right] = (0, 0, -P, s, 0, -2r),$	$-21_{-} = [(\Phi, P, Q, r, s, t), 1_{-}] = (0, 0, -P, s, 0, -2r),$	$-21_{-} = [(\Phi, P, Q, r, s, t), 1_{-}] = (0, 0, -P, s, 0, -2r),$
0902.0431_E00081	17	■ 0.997	$\begin{aligned} & H_1 = -G_{23} + G_{45}, \quad L_{12} = G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37}, \\ & L_{21} = -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}. \end{aligned}$	$H_1 = -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67}$ $L_{12} = G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37},$ $L_{21} = -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}.$	$H_1 = -G_{23} + G_{45}, \quad H_2 = -G_{45} + G_{67}$ $L_{12} = G_{24} + G_{35}, \quad L_{21} = -G_{25} + G_{34}, \quad L_{13} = G_{26} + G_{37},$ $L_{21} = -G_{27} + G_{36}, \quad L_{23} = G_{46} + G_{57}, \quad L_{32} = -G_{47} + G_{56}.$

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0902.0431_E00624	104	■ 0.997	$\begin{aligned} & (g(\alpha E))^2 = g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ & = g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ & = g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$	$\begin{aligned} (g(\alpha E))^2 &= g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ &= g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ &= g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$	$\begin{aligned} (g(\alpha E))^2 &= g(\alpha E) \circ g(\alpha E) = g(\gamma(\alpha E \times \alpha E)) + \frac{1}{4}(\gamma \alpha E, \alpha E)E \text{ (Lemma 3.12.1)} \\ &= g(\gamma \tau \alpha \tau E) + \frac{1}{4}\langle \tau \gamma \alpha E, \alpha E \rangle E = g(\alpha \tau \gamma E) + \frac{1}{4}\langle \alpha \tau \gamma E, \alpha E \rangle E \\ &= g(\alpha E) + \frac{1}{4}\langle \alpha E, \alpha E \rangle E = g(\alpha E) + \frac{1}{4}\langle E, E \rangle E = g(\alpha E) + \frac{3}{4}E. \end{aligned}$
0902.0431_E00374	62	■ 0.997	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$	$F_4 / \operatorname{Spin}(9) \simeq \mathfrak{C}P_2.$
0902.0431_E00434	71	■ 0.997	$F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$	$F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$	$F_4^{\mathcal{C}} \simeq F_4 \times \boldsymbol{R}^{52}.$
0902.0431_E00717	118	■ 0.997	$\left[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2) \right] = \Phi(\phi, A, -\tau A, \nu),$	$[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$	$[\Phi(\phi_1, A_1, -\tau A_1, \nu_1), \Phi(\phi_2, A_2, -\tau A_2, \nu_2)] = \Phi(\phi, A, -\tau A, \nu),$
0902.0431_E00914	152	■ 0.997	$\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$	$\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$	$\dim \mathfrak{e}_{7,C} = 16 + 18 + 1 = 35.$
0902.0431_E00208	35	■ 0.997	$\begin{aligned} & \operatorname{Spin}(7) \cap \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ & G_2 \cap \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ & = \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$	$\begin{aligned} \operatorname{Spin}(7) &= \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ G_2 &= \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$	$\begin{aligned} \operatorname{Spin}(7) &= \{\alpha \in \operatorname{Spin}(8) \mid \kappa \alpha = \alpha\}, \\ G_2 &= \{\alpha \in \operatorname{Spin}(8) \mid \lambda \alpha = \alpha, \lambda \in \mathfrak{S}_3\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \pi \alpha = \alpha, \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(8) \mid \nu \alpha = \alpha\} \\ &= \{\alpha \in \operatorname{Spin}(7) \mid \pi \alpha = \alpha\}. \end{aligned}$

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0902.0431_E00183	31	■ 0.998	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$	$\widetilde{B}_3=\{\widetilde{\alpha}\in SO(8)\mid (\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),x,y\in \mathfrak{C}\text{ for some }\alpha\in SO(7)\}.$
0902.0431_E00255	41	■ 0.998	$\begin{aligned} E_{6(-26)}&=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\right.\\&\quad\left.\mid\operatorname{det}(\alpha X)=\operatorname{det}X\right\}\\&=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}. \end{aligned}$	$E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\}\\=\left\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\right\}.$	$E_{6(-26)}=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X\}\\=\{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha^{-1}(X\times Y)=\alpha X\times\alpha Y\}.$
0902.0431_E00617 (i)	103	■ 0.998	$(g\,X,\,g\,Y)=(\gamma X,\,Y),\quad X,\,Y\in\mathfrak{J}^C.$	$(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$	$(gX,gY)=(\gamma X,Y),\quad X,Y\in\mathfrak{J}^C.$
0902.0431_E00635	105	■ 0.998	$E_{6,\boldsymbol{C}}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$E_{6,\boldsymbol{C}}=\left\{\alpha\in\operatorname{Iso}_C\left((\mathfrak{J}_C)^C\right)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}.$	$E_{6,\boldsymbol{C}}=\{\alpha\in\operatorname{Iso}_C((\mathfrak{J}_C)^C)\mid\det(\alpha X)=\det X,\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\}.$
0902.0431_E00965	160	■ 0.998	$E_7\{ \}^{\{C\}}\simeq\left(E_7\{ \}^{\{C\}}\cap U\left(\mathfrak{P}^C\right)\right)\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$	$E_7^C\simeq\left(E_7^C\cap U\left(\mathfrak{P}^C\right)\right)\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$	$E_7^C\simeq\left(E_7^C\cap U\left(\mathfrak{P}^C\right)\right)\times\boldsymbol{R}^d=E_7\times\boldsymbol{R}^d,\quad d=133.$
0902.0431_E01113	188	■ 0.998	$\left(E_8\right)^{\sigma}\cong Ss(16).$	$(E_8)^{\sigma}\cong Ss(16).$	$(E_8)^{\sigma}\cong Ss(16).$
0902.0431_E01116	189	■ 0.998	$\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$	$\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$	$\left(\mathfrak{J}^C\right)^3=\left\{\boldsymbol{X}=\begin{pmatrix}X_1\\X_2\\X_3\end{pmatrix}\middle X_i\in\mathfrak{J}^C\right\}.$
0902.0431_E01138	192	■ 0.998	$\begin{gathered} \left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\ \left(a,b\right)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C. \end{gathered}$	$\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\ \left(a,b\right)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C.$	$\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k,\boldsymbol{y}_1\wedge\cdots\wedge\boldsymbol{y}_k\right)=\det\left(\left(\boldsymbol{x}_i,\boldsymbol{y}_j\right)\right),\quad k\geq1,\\ \left(a,b\right)=ab,\quad a,b\in\Lambda^0\left(C^n\right)=C.$
0902.0431_E00148	27	■ 0.998	$(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$	$(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$	$(m+ae_4')(n+be_4')=(mn+\overline{b}a)+(a\overline{n}+bm)e_4'.$
0902.0431_E00691 (ii)	114	■ 0.998	$\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$	$\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$	$\Phi=\left(\begin{array}{cccc}g&l&C&A\\k&h&B&D\\c&a&\nu&\lambda\\b&d&\kappa&\mu\end{array}\right),\quad\begin{array}{l}g,h,k,l\in\operatorname{Hom}_C\left(\mathfrak{J}^C\right),\\a,b,c,d\in\operatorname{Hom}_C\left(\mathfrak{J}^C,C\right),\\A,B,C,D\in\mathfrak{J}^C,\\ \nu,\mu,\kappa,\lambda\in C.\end{array}$

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0902.0431_E00745	123	■ 0.998	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.	$\operatorname{tr}\left(\Phi_0^2\right)=\frac{1}{9}\times 27\times 2+1+1=8$.
0902.0431_E00025	10	■ 0.998	$\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$	$\begin{aligned} & \left[F_{i0}, F_{kl}\right] x = \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ & = -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ & \left[F_{ij}, F_{jk}\right] x = \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ & = \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ & \left[F_{ij}, F_{kl}\right] x = \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ & = \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$	$\begin{aligned} \left[F_{i0}, F_{kl}\right] x &= \left(F_{i0} F_{kl} - F_{kl} F_{i0}\right) x = \frac{1}{4} e_i\left(e_l\left(e_k x\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_i x\right)\right) \\ &= -\frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) + \frac{1}{4} e_l\left(e_i\left(e_k x\right)\right) = 0, \\ \left[F_{ij}, F_{jk}\right] x &= \left(F_{ij} F_{jk} - F_{jk} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_k\left(e_j x\right)\right)\right) - \frac{1}{4} e_k\left(e_j\left(e_j\left(e_i x\right)\right)\right) \\ &= \frac{1}{4} e_i\left(e_k\left(e_j\left(e_j x\right)\right)\right) + \frac{1}{4} e_k\left(e_i x\right) = -\frac{1}{4} e_i\left(e_k x\right) + \frac{1}{4} e_k\left(e_i x\right) \\ &= \frac{1}{2} e_k\left(e_i x\right) = F_{ik} x, \\ \left[F_{ij}, F_{kl}\right] x &= \left(F_{ij} F_{kl} - F_{kl} F_{ij}\right) x = \frac{1}{4} e_j\left(e_i\left(e_l\left(e_k x\right)\right)\right) - \frac{1}{4} e_l\left(e_k\left(e_j\left(e_i x\right)\right)\right) \\ &= \frac{1}{4} e_j\left(e_k\left(e_i\left(e_l x\right)\right)\right) - \frac{1}{4} e_j\left(e_l\left(e_k\left(e_i x\right)\right)\right) = \cdots = 0. \end{aligned}$
0902.0431_E00047 (iv)	12	■ 0.998	$x(D\ y)=D(x\ y),\quad \text{ } x,\ y\in\mathfrak{C}$.	$x(Dy)=D(xy),\quad x,y\in\mathfrak{C}$.	$x(Dy)=D(xy),\quad x,y\in\mathfrak{C}$.
0902.0431_E00138	26	■ 0.998	$\alpha=1\quad\text{or}\quad\alpha=\gamma$.	$\alpha=1\quad\text{or}\quad\alpha=\gamma$.	$\alpha=1\quad\text{or}\quad\alpha=\gamma$.
0902.0431_E00330	53	■ 0.998	$\mathfrak{D}_{4}^C\supset D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^C\subset\mathfrak{f}_4^C$	$\mathfrak{D}_4^C\supset D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^C\subset\mathfrak{f}_4^C$	$\mathfrak{D}_4^C\supset D\rightarrow(D,\nu D,\kappa\pi D)\in\mathfrak{d}_4^C\subset\mathfrak{f}_4^C$
0902.0431_E00457	75	■ 0.998	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6$.	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6$.	$\phi=\frac{\phi-\phi^*}{2}+i\frac{\phi+\phi^*}{2i},\quad\frac{\phi-\phi^*}{2},\frac{\phi+\phi^*}{2i}\in\mathfrak{e}_6$.
0902.0431_E00499	81	■ 0.998	$\widetilde{T}X=\frac{1}{2}(TX+XT^*),\quad\text{where}\quad T^*={}^t\bar{T}$.	$\widetilde{T}X=\frac{1}{2}(TX+XT^*),\quad\text{where}\quad T^*={}^t\bar{T}$.	$\widetilde{T}X=\frac{1}{2}(TX+XT^*),\quad\text{where}\quad T^*={}^t\bar{T}$.

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0902.0431_E00513	84	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ \& \ \backslash\alpha_{\mathrm{1}}=\backslash\mathrm{lambda}_{\mathrm{0}}-\backslash\mathrm{lambda}_{\mathrm{1}}, \\ & \backslash\mathrm{quad} \ \backslash\alpha_{\mathrm{2}}=\backslash\mathrm{lambda}_{\mathrm{1}}-\backslash\mathrm{lambda}_{\mathrm{2}}, \ \backslash\mathrm{quad} \\ & \backslash\alpha_{\mathrm{3}}=\backslash\mathrm{lambda}_{\mathrm{2}}-\backslash\mathrm{lambda}_{\mathrm{3}}, \ \backslash\backslash \ \& \ \backslash\alpha_{\mathrm{4}}= \\ & \backslash\mathrm{lambda}_{\mathrm{3}}+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\backslash\mathrm{left}(\backslash\mathrm{mu}_{\mathrm{2}}-\backslash\mathrm{mu}_{\mathrm{3}}\backslash\mathrm{right}), \ \backslash\backslash \\ & \ \& \ \backslash\alpha_{\mathrm{5}}=\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\backslash\mathrm{left}(-\backslash\mathrm{lambda}_{\mathrm{0}}-\backslash\mathrm{lambda}_{\mathrm{1}}- \\ & \backslash\mathrm{lambda}_{\mathrm{2}}+\backslash\mathrm{lambda}_{\mathrm{3}}\backslash\mathrm{right})+\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\backslash\mathrm{left}(\backslash\mathrm{mu}_{\mathrm{3}}- \\ & \backslash\mathrm{mu}_{\mathrm{1}}\backslash\mathrm{right}), \ \backslash\backslash \ \& \ \backslash\alpha_{\mathrm{6}}=\backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\backslash\mathrm{left} \\ & \backslash\mathrm{lambda}_{\mathrm{0}}+\backslash\mathrm{lambda}_{\mathrm{1}}+\backslash\mathrm{lambda}_{\mathrm{2}}+\backslash\mathrm{lambda}_{\mathrm{3}}\backslash\mathrm{right})+ \\ & \backslash\mathrm{frac}\{\mathrm{1}\}\{\mathrm{2}\}\backslash\mathrm{left}(\backslash\mathrm{mu}_{\mathrm{1}}-\backslash\mathrm{mu}_{\mathrm{2}}\backslash\mathrm{right}) \ \backslash\mathrm{end}\{\mathrm{aligned}\} \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ \alpha_4 &= \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3), \\ \alpha_5 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_6 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) \end{aligned}$
0902.0431_E00679	113	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ E_{\mathrm{7}}^{\mathrm{C}} \ \& \ =\backslash\mathrm{left}\backslash\{\backslash\alpha \ \mathrm{in} \\ & \backslash\mathrm{operatorname}\{\mathrm{Iso}\}_{\mathrm{C}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{\mathrm{P}\}^{\mathrm{C}}\backslash\mathrm{right}) \\ & \backslash\mathrm{mid} \ \backslash\alpha(P \ \backslash\mathrm{times} \ Q) \ \backslash\alpha^{\mathrm{-1}}=\backslash\alpha \ P \ \backslash\mathrm{times} \\ & \backslash\alpha \ Q\backslash\mathrm{right}\backslash\}, \ \backslash\backslash \ E_{\mathrm{7}} \ \& \ =\backslash\mathrm{left}\backslash\{\backslash\alpha \ \mathrm{in} \\ & \backslash\mathrm{operatorname}\{\mathrm{Iso}\}_{\mathrm{C}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{\mathrm{P}\}^{\mathrm{C}}\backslash\mathrm{right}) \\ & \backslash\mathrm{mid} \ \backslash\alpha(P \ \backslash\mathrm{times} \ Q) \ \backslash\alpha^{\mathrm{-1}}=\backslash\alpha \ P \ \backslash\mathrm{times} \\ & \backslash\alpha \ Q,\backslash\mathrm{rangle}\backslash\alpha \ P, \ \backslash\alpha \ Q\backslash\mathrm{rangle}=\backslash\mathrm{rangle} \ P, \ Q \\ & \backslash\mathrm{rangle}\backslash\mathrm{right}\backslash\} \ . \ \backslash\mathrm{end}\{\mathrm{aligned}\} \end{aligned}$	$\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$	$\begin{aligned} E_7^C &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q \}, \\ E_7 &= \{ \alpha \in \mathrm{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q) \alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle \}. \end{aligned}$
0902.0431_E00959	159	■ 0.998	$\backslash\alpha \ P=f^{\mathrm{-1}}(A(f \ P)) \ \backslash\mathrm{quad} \ \backslash\mathrm{text} \ \{ \ \mathrm{or} \ \} \ \backslash\mathrm{quad} \ \backslash\alpha \ P=f^{\mathrm{-1}}(A(f \ \mathrm{bar}\{P\})), \ \backslash\mathrm{quad} \ P \ \mathrm{in}\backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{P\}_{\backslash\mathrm{boldsymbol}\{C\}}\backslash\mathrm{right})^{\mathrm{C}}$	$\alpha P = f^{-1}(A(fP)) \quad \text{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$	$\alpha P = f^{-1}(A(fP)) \quad \text{or} \quad \alpha P = f^{-1}(A(f\bar{P})), \quad P \in (\mathfrak{P}_C)^C$
0902.0431_E00985	164	■ 0.998	$\backslash\mathrm{mathfrak}\{\mathrm{e}\}_{\mathrm{8}}\{\ \}^{\mathrm{C}} \ \backslash\mathrm{cong} \ \backslash\mathrm{Theta}\backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{\mathrm{e}\}_{\mathrm{8}}\{\ \}^{\mathrm{C}}\backslash\mathrm{right})=\backslash\mathrm{left}\backslash\{\backslash\mathrm{Theta}(R) \ \mathrm{mid} \ R \ \mathrm{in} \ \backslash\mathrm{mathfrak}\{\mathrm{e}\}_{\mathrm{8}}\{\ \}^{\mathrm{C}}\backslash\mathrm{right}\backslash\}$	$\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$	$\mathfrak{e}_8^C \cong \Theta(\mathfrak{e}_8^C) = \{ \Theta(R) \mid R \in \mathfrak{e}_8^C \}$
0902.0431_E01128	191	■ 0.998	$\mathrm{w}(D, \ \backslash\mathrm{phi}, \ \backslash\mathrm{boldsymbol}\{X\}, \ \backslash\mathrm{boldsymbol}\{Y\})=\backslash\mathrm{left}(D, \ \backslash\mathrm{phi}, \ \backslash\mathrm{omega} \ \backslash\mathrm{boldsymbol}\{X\}, \ \backslash\mathrm{omega}^{\mathrm{2}} \ \backslash\mathrm{boldsymbol}\{Y\}\backslash\mathrm{right}),$	$w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y}),$	$w(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \phi, \omega \boldsymbol{X}, \omega^2 \boldsymbol{Y}),$
0902.0431_E00007	7	■ 0.998	$\begin{aligned} & \backslash\mathrm{begin}\{\mathrm{aligned}\} \ \& \ (\mathrm{a} \ \mathrm{x}) \ \mathrm{y}+\mathrm{x}(\mathrm{y} \ \mathrm{a})=\mathrm{a}(\mathrm{x} \ \mathrm{y})+(\mathrm{x} \ \mathrm{y}) \ \mathrm{a}, \ \backslash\backslash \ \& \\ & (\mathrm{x} \ \mathrm{a}) \ \mathrm{y}+(\mathrm{x} \ \mathrm{y}) \ \mathrm{a}=\mathrm{x}(\mathrm{a} \ \mathrm{y})+\mathrm{x}(\mathrm{y} \ \mathrm{a}), \ \backslash\backslash \ \& \ (\mathrm{a} \ \mathrm{x}) \ \mathrm{y}+(\mathrm{x} \ \mathrm{a}) \ \mathrm{y}=\mathrm{a}(\mathrm{x} \\ & \mathrm{y})+\mathrm{x}(\mathrm{a} \ \mathrm{y}) \ . \ \backslash\mathrm{end}\{\mathrm{aligned}\} \end{aligned}$	$\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$	$\begin{aligned} (ax)y + x(ya) &= a(xy) + (xy)a, \\ (xa)y + (xy)a &= x(ay) + x(ya), \\ (ax)y + (xa)y &= a(xy) + x(ay). \end{aligned}$
0902.0431_E00147	27	■ 0.998	$\mathrm{G}_{\mathrm{2}}\{\ \}^{\mathrm{C}} \ \backslash\mathrm{simeq}\backslash\mathrm{left}(\mathrm{G}_{\mathrm{2}}\{\ \}^{\mathrm{C}} \ \backslash\mathrm{cap} \ U \ \backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{C\}^{\mathrm{C}}\backslash\mathrm{right})\backslash\mathrm{right}) \ \backslash\mathrm{times} \ \backslash\mathrm{boldsymbol}\{R\}^{\mathrm{d}}=\mathrm{G}_{\mathrm{2}} \ \backslash\mathrm{times} \ \backslash\mathrm{boldsymbol}\{R\}^{\mathrm{d}},$	$G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$	$G_2^C \simeq (G_2^C \cap U(\mathfrak{C}^C)) \times \boldsymbol{R}^d = G_2 \times \boldsymbol{R}^d,$

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0902.0431_E00050	12	■ 0.999	$\begin{array}{ll} \left(D_{\{2\}}x\right)y+x\left(D_{\{3\}}y\right)=\left(\kappa D_{\{1\}}\right)(x\,y), & \& x,\,y\in\mathfrak{C}, \\ \left(D_{\{3\}}x\right)y+x\left(D_{\{1\}}y\right)=\left(\kappa D_{\{2\}}\right)(x\,y), & \& x,\,y\in\mathfrak{C}. \end{array}$	$\begin{aligned} (D_2x)y+x(D_3y) &= (\kappa D_1)(xy), & x,y\in\mathfrak{C}, \\ (D_3x)y+x(D_1y) &= (\kappa D_2)(xy), & x,y\in\mathfrak{C}. \end{aligned}$	$\begin{aligned} (D_2x)y+x(D_3y) &= (\kappa D_1)(xy), & x,y\in\mathfrak{C}, \\ (D_3x)y+x(D_1y) &= (\kappa D_2)(xy), & x,y\in\mathfrak{C}. \end{aligned}$
0902.0431_E00180	31	■ 0.999	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$	$\alpha_{\bar{\omega}_1}(a+\boldsymbol{m})=\bar{\omega}_1(a+\boldsymbol{m})\omega_1=\bar{\omega}_1a\omega_1+\bar{\omega}_1^2\boldsymbol{m}=a+\omega_1\boldsymbol{m}=w(a+\boldsymbol{m}),$
0902.0431_E00345	56	■ 0.999	$\begin{aligned} & \& B_{\{4\}}\left(H,\,H^{\prime}\right)=\sum_{k,\,l=0}^3\lambda_k\lambda_l^{\prime}B_{\{4\}}\left(H_k,\,H_l\right) \\ & \& \left(H_{\{k\}},\,H_{\{l\}}\right)\& \& \quad=3\sum_{k,\,l}\left(\lambda_k\lambda_l^{\prime}\left(\mathrm{tr}\left(H_kH_l\right)+\mathrm{tr}\left(\left(\nu H_k\right)\left(\nu H_l\right)\right)+\mathrm{tr}\left(\left(\kappa\pi H_k\right)\left(\kappa\pi H_l\right)\right)\right)\right) \\ & \& \left(\lambda_{\{k\}}\lambda_{\{l\}}^{\prime}\right)\left(\mathrm{tr}\left(\left(\nu H_k\right)\left(\nu H_l\right)\right)+\mathrm{tr}\left(\left(\kappa\pi H_k\right)\left(\kappa\pi H_l\right)\right)\right) \\ & \& \left(\nu H_{\{k\}}\right)\left(\nu H_{\{l\}}\right)\right)+\left(\kappa\pi H_{\{k\}}\right)\left(\kappa\pi H_{\{l\}}\right)\right) \\ & \& \left(\nu H_{\{k\}}\right)\left(\nu H_{\{l\}}\right)\right)+\left(\kappa\pi H_{\{k\}}\right)\left(\kappa\pi H_{\{l\}}\right)\right) \\ & \& \quad=3\sum_{k,\,l}\lambda_k\lambda_l^{\prime}\left(2+2+2\right)\delta_{kl}=18\sum_{k=0}^3\lambda_k\lambda_k^{\prime} \end{aligned}$	$\begin{aligned} B_4(H,H') &= \sum_{k,l=0}^3 \lambda_k \lambda_l' B_4(H_k, H_l) \\ &= 3 \sum_{k,l} (\lambda_k \lambda_l' (\mathrm{tr}(H_k H_l) + \mathrm{tr}((\nu H_k)(\nu H_l)) + \mathrm{tr}((\kappa \pi H_k)(\kappa \pi H_l)))) \\ &= 3 \sum_{k,l} \lambda_k \lambda_l' (2 + 2 + 2) \delta_{kl} = 18 \sum_{k=0}^3 \lambda_k \lambda_k' \end{aligned}$	$\begin{aligned} B_4(H,H') &= \sum_{k,l=0}^3 \lambda_k \lambda_l' B_4(H_k, H_l) \\ &= 3 \sum_{k,l} (\lambda_k \lambda_l' (\mathrm{tr}(H_k H_l) + \mathrm{tr}((\nu H_k)(\nu H_l)) + \mathrm{tr}((\kappa \pi H_k)(\kappa \pi H_l)))) \\ &= 3 \sum_{k,l} \lambda_k \lambda_l' (2 + 2 + 2) \delta_{kl} = 18 \sum_{k=0}^3 \lambda_k \lambda_k' \end{aligned}$
0902.0431_E00405	67	■ 0.999	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$	$\varphi(p,A)((M+\boldsymbol{a})\times(N+\boldsymbol{b}))=\varphi(p,A)(M+\boldsymbol{a})\times\varphi(p,A)(N+\boldsymbol{b}),$
0902.0431_E00441	72	■ 0.999	$\begin{aligned} & \& F_{\{4(4)\}}\& \& \simeq(\operatorname{Sp}(1)\times\operatorname{Sp}(3))/\boldsymbol{Z}_2\times\boldsymbol{R}^{28}, \\ & \& F_{\{4(-20)\}}\& \& \simeq\operatorname{Spin}(9)\times\boldsymbol{R}^{16}. \end{aligned}$	$\begin{aligned} F_{4(4)} &\simeq (\mathrm{Sp}(1)\times\mathrm{Sp}(3))/\boldsymbol{Z}_2\times\boldsymbol{R}^{28}, \\ F_{4(-20)} &\simeq \mathrm{Spin}(9)\times\boldsymbol{R}^{16}. \end{aligned}$	$\begin{aligned} F_{4(4)} &\simeq (\mathrm{Sp}(1)\times\mathrm{Sp}(3))/\boldsymbol{Z}_2\times\boldsymbol{R}^{28}, \\ F_{4(-20)} &\simeq \mathrm{Spin}(9)\times\boldsymbol{R}^{16}. \end{aligned}$
0902.0431_E00485	80	■ 0.999	$B_{\{6\}}(\phi,\phi)=\operatorname{tr}\left(\left(\operatorname{ad}\phi\right)^2\right)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$	$B_6(\phi,\phi)=\mathrm{tr}\left((\mathrm{ad}\,\phi)^2\right)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$	$B_6(\phi,\phi)=\mathrm{tr}((\mathrm{ad}\phi)^2)=\left(\frac{1}{4}\times4+1\times2\right)\times8=24.$
0902.0431_E00496	81	■ 0.999	$\begin{aligned} & \& \sum_{k=1}^3\left(\left(\sigma_{ik}x_{ik}\right)y_{kj}+\overline{y_{ki}\left(\sigma_{jk}x_{jk}\right)}\right)+\sum_{k=1}^3\left(x_{ik}\left(\sigma_{kj}y_{kj}\right)+\overline{\left(\sigma_{ki}y_{ki}\right)x_{jk}}\right) \\ & \& =\sum_k\left(\left(\sigma_{ik}x_{ik}\right)y_{kj}+y_{ik}\left(\sigma_{kj}x_{kj}\right)\right)+\sum_k\left(\overline{\left(\sigma_{jk}x_{jk}\right)y_{ki}}+x_{jk}\left(\sigma_{ki}y_{ki}\right)\right) \end{aligned}$	$\begin{aligned} & \sum_{k=1}^3((\sigma_{ik}x_{ik})y_{kj}+\overline{y_{ki}(\sigma_{jk}x_{jk})})+\sum_{k=1}^3(x_{ik}(\sigma_{kj}y_{kj})+\overline{(\sigma_{ki}y_{ki})x_{jk}}) \\ & =\sum_k((\sigma_{ik}x_{ik})y_{kj}+y_{ik}(\sigma_{kj}x_{kj}))+\sum_k(\overline{(\sigma_{jk}x_{jk})y_{ki}}+x_{jk}(\sigma_{ki}y_{ki})) \end{aligned}$	$\begin{aligned} & \sum_{k=1}^3((\sigma_{ik}x_{ik})y_{kj}+\overline{y_{ki}(\sigma_{jk}x_{jk})})+\sum_{k=1}^3(x_{ik}(\sigma_{kj}y_{kj})+\overline{(\sigma_{ki}y_{ki})x_{jk}}) \\ & =\sum_k((\sigma_{ik}x_{ik})y_{kj}+y_{ik}(\sigma_{kj}x_{kj}))+\sum_k(\overline{(\sigma_{jk}x_{jk})y_{ki}}+x_{jk}(\sigma_{ki}y_{ki})) \end{aligned}$

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0902.0431_E00560	94	■ 0.999	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$	$\phi(\theta)\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\theta^4\xi_1 & \theta x_3 & \theta\bar{x}_2 \\ \theta\bar{x}_3 & \theta^{-2}\xi_2 & \theta^{-2}x_1 \\ \theta x_2 & \theta^{-2}\bar{x}_1 & \theta^{-2}\xi_3\end{array}\right).$
0902.0431_E00578	96	■ 0.999	$k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$	$k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$	$k\left((a+be_2)+i(c+de_2)\right)=0, \quad a,b,c,d\in\boldsymbol{C},$
0902.0431_E00607	101	■ 0.999	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$	$A(P\circ Q)A^*=APA^*\circ AQA^*, \quad A\in\mathrm{Sp}(4), P,Q\in\mathfrak{J}(4,\boldsymbol{H}).$
0902.0431_E00693	114	■ 0.999	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$	$rf_r(P\times Q)f_r^{-1}=f_rP\times f_rQ, \quad P,Q\in\mathfrak{P}^C.$
0902.0431_E00720	118	■ 0.999	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$	$\Phi\in\mathfrak{e}_7 \quad \text{if and only if} \quad \tau\lambda\Phi\lambda^{-1}\tau=\Phi.$
0902.0431_E00836 (i)	139	■ 0.999	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$	$(E_1,\phi X)-\frac{1}{3}\nu(E_1,X)=-\nu(E_1,X).$
0902.0431_E00883	147	■ 0.999	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$	$\tau\gamma(X,Y,\xi,\eta)=(\tau\gamma X,\tau\gamma Y,\tau\xi,\tau\eta),$
0902.0431_E00962	160	■ 0.999	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$	$\gamma_1:\mathfrak{P}^C\rightarrow\mathfrak{P}^C,\gamma_1(P_C+(M+N))=\overline{P_C}+(\bar{M}+\bar{N}), \quad P_C+(M+N)\in\mathfrak{P}^C.$
0902.0431_E00974	162	■ 0.999	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$	$\left\{\begin{array}{l}\Phi=[\Phi_1,\Phi_2]+P_1\times Q_2-P_2\times Q_1\\P=\Phi_1P_2-\Phi_2P_1+r_1P_2-r_2P_1+s_1Q_2-s_2Q_1\\Q=\Phi_1Q_2-\Phi_2Q_1-r_1Q_2+r_2Q_1+t_1P_2-t_2P_1\\r=-\frac{1}{8}\{P_1,Q_2\}+\frac{1}{8}\{P_2,Q_1\}+s_1t_2-s_2t_1\\s=\frac{1}{4}\{P_1,P_2\}+2r_1s_2-2r_2s_1\\t=-\frac{1}{4}\{Q_1,Q_2\}-2r_1t_2+2r_2t_1\end{array}\right.$
0902.0431_E00987	164	■ 0.999	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$	$(R_1,R_2)_8=(\Phi_1,\Phi_2)_7-\{Q_1,P_2\}+\{P_1,Q_2\}-8r_1r_2-4t_1s_2-s_1t_2,$
0902.0431_E01054	179	■ 0.999	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$	$\beta(P\times Q)\beta^{-1}=\beta P\times\beta Q,$
0902.0431_E01099	186	■ 0.999	$\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$	$\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$	$\mathrm{Spin}(16), \quad \mathrm{SO}(16), \quad \mathrm{Ss}(16), \quad \mathrm{SO}(16)/\boldsymbol{Z}_2.$
0902.0431_E01136	192	■ 0.999	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$	$\varphi(A,\alpha)=\varphi_1(A)\varphi_2(\alpha).$

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0902.0431_E01216	204	■ 0.999	$\begin{aligned} E_{8(8)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}, \\ E_{8(-24)} &= \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}. \end{aligned}$	$E_{8(8)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\},$ $E_{8(-24)} = \left\{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \right\}.$	$E_{8(8)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(8)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \},$ $E_{8(-24)} = \{ \alpha \in \operatorname{Iso}_R(\mathfrak{e}_{8(-24)}) \mid \alpha[R_1, R_2] = [\alpha R_1, \alpha R_2] \}.$
0902.0431_E01218	204	■ 0.999	$\begin{aligned} E_{8(8)} &\simeq Ss(16) \times \boldsymbol{R}^{128}, \\ E_{8(-24)} &\simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}. \end{aligned}$	$E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$	$E_{8(8)} \simeq Ss(16) \times \boldsymbol{R}^{128},$ $E_{8(-24)} \simeq (SU(2) \times E_7) / \boldsymbol{Z}_2 \times \boldsymbol{R}^{112}.$
0902.0431_E00092	19	■ 0.999	$B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$	$B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$	$B_2(H_1, H_1) = \operatorname{tr}((\operatorname{ad} H_1)^2) = (-4) \times 2 + (-1) \times 8 = -16.$
0902.0431_E00165	29	■ 0.999	$\begin{array}{ll} \left(\alpha_2 x \right) \left(\alpha_3 y \right) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ \left(\alpha_3 x \right) \left(\alpha_1 y \right) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$	$\begin{array}{ll} (\alpha_2 x) (\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x) (\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$	$\begin{array}{ll} (\alpha_2 x) (\alpha_3 y) = \overline{\alpha_1(\overline{xy})}, & x, y \in \mathfrak{C}, \\ (\alpha_3 x) (\alpha_1 y) = \overline{\alpha_2(\overline{xy})}, & x, y \in \mathfrak{C}. \end{array}$
0902.0431_E00450	74	■ 0.999	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$	$\delta + \widetilde{T} = 0, \quad \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C \quad \text{implies} \quad \delta = 0, T = 0.$
0902.0431_E00541	91	■ 0.999	$\begin{aligned} \left(\mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left(\mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$	$\begin{aligned} \left(\mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left(\mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$	$\begin{aligned} \left(\mathfrak{J}^C \right)_\sigma &= \{ X \in \mathfrak{J}^C \mid \sigma X = X \} \\ &= \{ X \in \mathfrak{J}^C \mid 4E_1 \times (E_1 \times X) = X \} \oplus \mathfrak{E}_1^C, \\ \left(\mathfrak{J}^C \right)_{-\sigma} &= \{ X \in \mathfrak{J}^C \mid \sigma X = -X \} \\ &= \{ X \in \mathfrak{J}^C \mid E_1 \times X = 0, \langle E_1, X \rangle = 0 \}, \end{aligned}$
0902.0431_E00201	34	■ 0.999	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x) (\alpha_2 y) = \overline{\alpha_3(\overline{xy})}. \quad x, y \in \mathfrak{C}.$
0902.0431_E00002	5	■ 0.999	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$	$\gamma, \quad \sigma, \quad \iota, \quad \kappa, \quad \lambda, \quad \mu, \quad \tau, \quad v, \quad \chi, \quad g, \quad w.$
0902.0431_E00004	6	■ 0.999	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$	$e_i^2 = -1, \quad i \neq 0, \quad e_i e_j = -e_j e_i, \quad i \neq j, i \neq 0, j \neq 0,$

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0902.0431_E00668	15	■ 0.999	$\begin{aligned} & (a+\boldsymbol{m})(b+\boldsymbol{n}) \\ & \& =(a\,b-\langle\boldsymbol{m},\boldsymbol{n}\rangle,\,\boldsymbol{m}\boldsymbol{n}) \\ & \& +(a\,\boldsymbol{b}+\boldsymbol{a}\,\overline{\boldsymbol{b}}-\boldsymbol{m}\boldsymbol{n}), \\ & \& \overline{a+\boldsymbol{m}}=\overline{a}-\boldsymbol{m}, \end{aligned}$	$\begin{aligned} (a+\boldsymbol{m})(b+\boldsymbol{n}) &= (ab-\langle\boldsymbol{m},\boldsymbol{n}\rangle)+(\boldsymbol{a}\boldsymbol{n}+\bar{\boldsymbol{b}}\boldsymbol{m}-\overline{\boldsymbol{m}\times\boldsymbol{n}}), \\ (a+\boldsymbol{m},b+\boldsymbol{n}) &= (a,b)+(\boldsymbol{m},\boldsymbol{n}), \\ \overline{a+\boldsymbol{m}} &= \bar{a}-\boldsymbol{m}, \end{aligned}$	$\begin{aligned} (a+\boldsymbol{m})(b+\boldsymbol{n}) &= (ab-\langle\boldsymbol{m},\boldsymbol{n}\rangle)+(\boldsymbol{a}\boldsymbol{n}+\bar{\boldsymbol{b}}\boldsymbol{m}-\overline{\boldsymbol{m}\times\boldsymbol{n}}), \\ (a+\boldsymbol{m},b+\boldsymbol{n}) &= (a,b)+(\boldsymbol{m},\boldsymbol{n}), \\ \overline{a+\boldsymbol{m}} &= \bar{a}-\boldsymbol{m}, \end{aligned}$
0902.0431_E00139	26	■ 0.999	$G_2^{\mathcal{C}}=\left\{\alpha\in\mathrm{Iso}_C(\mathfrak{C}^C)\mid\alpha(xy)=(\alpha x)(\alpha y)\right\}.$	$G_2^{\mathcal{C}}=\left\{\alpha\in\mathrm{Iso}_C(\mathfrak{C}^C)\mid\alpha(xy)=(\alpha x)(\alpha y)\right\}.$	$G_2^{\mathcal{C}}=\left\{\alpha\in\mathrm{Iso}_C(\mathfrak{C}^C)\mid\alpha(xy)=(\alpha x)(\alpha y)\right\}.$
0902.0431_E00182 (ii)	31	■ 0.999	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C},$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00187 (i)	32	■ 0.999	$(\alpha x)\left((\alpha y)b_0\right)=(\alpha(xy))b_0,\quad x,y\in\mathfrak{C}.$	$(\alpha x)\left((\alpha y)b_0\right)=(\alpha(xy))b_0,\quad x,y\in\mathfrak{C}.$	$(\alpha x)\left((\alpha y)b_0\right)=(\alpha(xy))b_0,\quad x,y\in\mathfrak{C}.$
0902.0431_E00209	35	■ 0.999	$\mathrm{Spin}(8)/\{(1,1,1),(-1,-1,1)\}\cong\mathrm{Spin}(8)/\{(1,1,1),(1,-1,-1)\},$	$\mathrm{Spin}(8)/\{(1,1,1),(-1,-1,1)\}\cong\mathrm{Spin}(8)/\{(1,1,1),(1,-1,-1)\},$	$Spin(8)/\{(1,1,1),(-1,-1,1)\}\cong Spin(8)/\{(1,1,1),(1,-1,-1)\},$
0902.0431_E00211	35	■ 0.999	$G_2=\{\alpha\in\mathrm{Spin}(8)\mid\nu\alpha=\alpha\}.$	$G_2=\{\alpha\in\mathrm{Spin}(8)\mid\nu\alpha=\alpha\}.$	$G_2=\{\alpha\in Spin(8)\mid\nu\alpha=\alpha\}.$
0902.0431_E00225	37	■ 0.999	$\begin{aligned} & \left\{\begin{array}{l} E_{-i}\circ E_{-i}=E_{-i} \\ E_{-i}\circ F_i(x)=0 \\ F_i(x)\circ F_i(y)=(x,y)(E_{i+1}+E_{i+2}), \end{array}\right. \\ & \left\{\begin{array}{l} E_i\circ E_j=0,\quad i\neq j \\ E_i\circ F_j(x)=\frac{1}{2}F_j(x),\quad i\neq j \\ F_i(x)\circ F_{i+1}(y)=\frac{1}{2}F_{i+2}(\overline{xy}), \end{array}\right. \end{aligned}$	$\begin{aligned} & \left\{\begin{array}{l} E_{-i}\circ E_{-i}=E_{-i} \\ E_{-i}\circ F_i(x)=0 \\ F_i(x)\circ F_i(y)=(x,y)(E_{i+1}+E_{i+2}), \end{array}\right. \\ & \left\{\begin{array}{l} E_i\circ E_j=0,\quad i\neq j \\ E_i\circ F_j(x)=\frac{1}{2}F_j(x),\quad i\neq j \\ F_i(x)\circ F_{i+1}(y)=\frac{1}{2}F_{i+2}(\overline{xy}), \end{array}\right. \end{aligned}$	$\begin{cases} E_i\circ E_i=E_i \\ E_i\circ F_i(x)=0 \\ F_i(x)\circ F_i(y)=(x,y)(E_{i+1}+E_{i+2}), \end{cases} \quad \begin{cases} E_i\circ E_j=0,\quad i\neq j \\ E_i\circ F_j(x)=\frac{1}{2}F_j(x),\quad i\neq j \\ F_i(x)\circ F_{i+1}(y)=\frac{1}{2}F_{i+2}(\overline{xy}), \end{cases}$
0902.0431_E00240 (3)	40	■ 0.999	$\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y).$	$\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y).$	$\det(\alpha X)=\det X,(\alpha X,\alpha Y)=(X,Y).$
0902.0431_E00259	42	■ 0.999	$[X,Y]=XY-YX.$	$[X,Y]=XY-YX.$	$[X,Y]=XY-YX.$
0902.0431_E00280	45	■ 0.999	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\}.$	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\}.$	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\}.$
0902.0431_E00286	45	■ 0.999	$2\delta E_i\circ E_i=\delta E_i,\quad \delta E_i\circ E_j+E_i\circ\delta E_j=0.$	$2\delta E_i\circ E_i=\delta E_i,\quad \delta E_i\circ E_j+E_i\circ\delta E_j=0.$	$2\delta E_i\circ E_i=\delta E_i,\quad \delta E_i\circ E_j+E_i\circ\delta E_j=0.$
0902.0431_E00328	52	■ 0.999	$\pm(\lambda_k-\lambda_l),\quad \pm(\lambda_k+\lambda_l),\quad 0\leq k<l\leq 3.$	$\pm(\lambda_k-\lambda_l),\quad \pm(\lambda_k+\lambda_l),\quad 0\leq k<l\leq 3.$	$\pm(\lambda_k-\lambda_l),\quad \pm(\lambda_k+\lambda_l),\quad 0\leq k<l\leq 3.$
0902.0431_E00346	56	■ 0.999	$\begin{aligned} & H_{\alpha_1}=\frac{1}{18}(H_0-H_1),\quad H_{\alpha_2}=\frac{1}{18}(H_1-H_2), \\ & H_{\alpha_3}=\frac{1}{18}H_2,\quad H_{\alpha_4}=\frac{1}{36}(-H_0-H_1-H_2+H_3). \end{aligned}$	$\begin{aligned} & H_{\alpha_1}=\frac{1}{18}(H_0-H_1),\quad H_{\alpha_2}=\frac{1}{18}(H_1-H_2), \\ & H_{\alpha_3}=\frac{1}{18}H_2,\quad H_{\alpha_4}=\frac{1}{36}(-H_0-H_1-H_2+H_3). \end{aligned}$	$\begin{aligned} & H_{\alpha_1}=\frac{1}{18}(H_0-H_1),\quad H_{\alpha_2}=\frac{1}{18}(H_1-H_2), \\ & H_{\alpha_3}=\frac{1}{18}H_2,\quad H_{\alpha_4}=\frac{1}{36}(-H_0-H_1-H_2+H_3). \end{aligned}$
0902.0431_E00351	57	■ 0.999	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\},\quad i=1,2,3.$	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\},\quad i=1,2,3.$	$\mathfrak{J}_i=\{F_i(x)\mid x\in\mathfrak{C}\}=\{X\in\mathfrak{J}\mid 2E_{i+1}\circ X=2E_{i+2}\circ X=X\},\quad i=1,2,3.$

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0902.0431_E09362	58	■ 0.999	$\begin{array}{l} \backslash begin{array}{l} \backslash alpha E=E, \& \backslash alpha E_{\{1\}}=E_{\{1\}}, \\ \& \backslash alpha \backslash left(E_{\{2\}}-E_{\{3\}}\backslash right)=A, \backslash \backslash \backslash alpha F_{\{1\}} \\ \backslash left(e_{\{i\}}\backslash right)=X_{\{i\}}, \& \backslash alpha F_{\{2\}}\backslash left(e_{\{i\}} \\ \backslash right)=Y_{\{i\}}, \& \backslash alpha F_{\{3\}}\backslash left(e_{\{i\}}\backslash right)=Z_{\{i\}}, \\ \quad i=0,1, \backslash cdots, 7 \backslash end{array} \end{array}$	$\begin{array}{l} \alpha E = E, \quad \quad \alpha E_1 = E_1, \quad \quad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$	$\begin{array}{l} \alpha E = E, \quad \quad \alpha E_1 = E_1, \quad \quad \alpha (E_2 - E_3) = A, \\ \alpha F_1(e_i) = X_i, \quad \alpha F_2(e_i) = Y_i, \quad \alpha F_3(e_i) = Z_i, \quad i = 0, 1, \cdots, 7 \end{array}$
0902.0431_E09426	70	■ 0.999	$\backslash varphi(P, A)((X+M) \backslash times(Y+N))=\backslash varphi(P, A)(X+M) \backslash times \backslash varphi(P, A)(Y+N),$	$\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$	$\varphi(P,A)((X+M)\times(Y+N))=\varphi(P,A)(X+M)\times\varphi(P,A)(Y+N),$
0902.0431_E09438	72	■ 0.999	$\begin{array}{l} \backslash begin{aligned} \backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\backslash prime\backslash right) \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \\ \backslash mathfrak{C}\wedge\backslash C\backslash right) \backslash mid \backslash tau \backslash gamma X=X\backslash right \\ \backslash \}=\backslash left(\backslash mathfrak{J}\backslash left(3, \backslash mathfrak{C}\wedge\backslash C\backslash right) \\ \backslash right)_{\backslash tau \backslash gamma}, \backslash \backslash \backslash mathfrak{J}\{1,2, \backslash mathfrak{C}\} \\ \& \backslash cong\backslash left\{X \in \backslash mathfrak{J}\}\backslash left(3, \backslash mathfrak{C}\wedge\backslash C \\ \backslash right) \backslash mid \backslash tau \backslash sigma X=X\backslash right\backslash \}=\backslash left(\backslash mathfrak{J} \\ \backslash left(3, \backslash mathfrak{C}\wedge\backslash C\backslash right)\backslash right)_{\backslash tau \backslash sigma} \\ \backslash end{aligned} \end{array}$	$\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$	$\begin{array}{l} \mathfrak{J}(3, \mathfrak{C}') \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\gamma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\gamma}, \\ \mathfrak{J}(1, 2, \mathfrak{C}) \cong \{X \in \mathfrak{J}(3, \mathfrak{C}^C) \mid \tau\sigma X = X\} = (\mathfrak{J}(3, \mathfrak{C}^C))_{\tau\sigma} \end{array}$
0902.0431_E09483	79	■ 0.999	$\backslash left[\backslash phi,\backslash left[\backslash phi, \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}} \\ \backslash right)\backslash right]\backslash right]=\backslash left[\backslash phi,-\backslash frac{1}{2} \\ \backslash widetilde{F}_{\{1\}}\backslash left(e_{\{i\}}\backslash right)\backslash right]=\backslash frac{1}{4} \\ \backslash widetilde{A}_{\{1\}}\backslash left(e_{\{i\}}\backslash right),$	$\left[\phi,\left[\phi,\widetilde{A}_1(e_i)\right]\right]=\left[\phi,-\frac{1}{2}\widetilde{F}_1(e_i)\right]=\frac{1}{4}\widetilde{A}_1(e_i),$	$[\phi,[\phi,\widetilde{A}_1(e_i)]]=\left[\phi,-\frac{1}{2}\widetilde{F}_1(e_i)\right]=\frac{1}{4}\widetilde{A}_1(e_i),$
0902.0431_E09487	80	■ 0.999	$\backslash operatorname{tr}(\backslash phi \backslash phi)=1 \backslash times 2+\backslash frac{1}{4} \\ \backslash times 8 \backslash times 2=6 \; .$	$\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$	$\mathrm{tr}(\phi\phi)=1\times2+\frac{1}{4}\times8\times2=6.$
0902.0431_E09597	99	■ 0.999	$(M+\backslash boldsymbol{a}) \backslash times (N+\backslash boldsymbol{b})=\backslash left(M \\ \backslash times N-\backslash frac{1}{2})\backslash left(\backslash boldsymbol{a}\wedge\backslash * \\ \backslash boldsymbol{b}+\backslash boldsymbol{b}\wedge\backslash * \backslash boldsymbol{a}\backslash right) \\ \backslash right)-\backslash frac{1}{2}(\backslash boldsymbol{a} \backslash N+\backslash boldsymbol{b} \backslash M) \; .$	$(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$	$(M+\boldsymbol{a})\times(N+\boldsymbol{b})=\left(M\times N-\frac{1}{2}(\boldsymbol{a}^*\boldsymbol{b}+\boldsymbol{b}^*\boldsymbol{a})\right)-\frac{1}{2}(\boldsymbol{a}N+\boldsymbol{b}M).$
0902.0431_E09604	101	■ 0.999	$\backslash mathfrak{J}\wedge\backslash C=\backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \\ \backslash gamma} \backslash oplus \backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash -\backslash \tau \\ \backslash gamma}=\backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \backslash gamma} \\ \backslash oplus i\backslash left(\backslash mathfrak{J}\wedge\backslash C\backslash right)_{\backslash \tau \backslash gamma} \; .$	$\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$	$\mathfrak{J}^C=(\mathfrak{J}^C)_{\tau\gamma}\oplus(\mathfrak{J}^C)_{-\tau\gamma}=(\mathfrak{J}^C)_{\tau\gamma}\oplus i(\mathfrak{J}^C)_{\tau\gamma}.$
0902.0431_E09668	111	■ 0.999	$\backslash mathfrak{P}\wedge\backslash C=\backslash mathfrak{J}\wedge\backslash C \backslash oplus \backslash mathfrak{J}\wedge\backslash C \\ \backslash oplus C \backslash oplus C \; .$	$\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$	$\mathfrak{P}^C=\mathfrak{J}^C\oplus\mathfrak{J}^C\oplus C\oplus C.$
0902.0431_E09672	112	■ 0.999	$\begin{array}{l} \backslash begin{aligned} \& (P \backslash times Q) P \backslash \backslash = \& \backslash Phi \backslash left(- \\ \backslash frac{1}{2}(X \vee W+Z \vee Y),-\backslash frac{1}{4}(2 Y \backslash times W- \\ \backslash xi Z-\backslash zeta X), \backslash frac{1}{4}(2 X \backslash times Z-\backslash eta W-\backslash omega \\ Y),\backslash right. \backslash \backslash \& \backslash left.\backslash frac{1}{8}((X, W)+(Z, Y)-3(\xi\omega+\zeta\eta)) \\ \backslash omega+\backslash zeta \backslash eta))\backslash right)(X, Y, \backslash xi, \backslash eta) \backslash end{aligned} \end{array}$	$\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$	$\begin{array}{l} (P\times Q)P \\ =\Phi\left(-\frac{1}{2}(X\vee W+Z\vee Y),-\frac{1}{4}(2Y\times W-\xi Z-\zeta X),\frac{1}{4}(2X\times Z-\eta W-\omega Y),\right. \\ \left.\frac{1}{8}((X,W)+(Z,Y)-3(\xi\omega+\zeta\eta))\right)(X,Y,\xi,\eta) \end{array}$
0902.0431_E09715	118	■ 0.999	$\backslash mathfrak{e}_{\{7\}}=\backslash left\{\backslash Phi(\backslash phi, A,-\backslash tau A, \backslash nu) \backslash mid \\ \backslash phi \in \backslash mathfrak{e}_{\{6\}}, A \in \backslash mathfrak{J}\wedge\backslash C, \backslash nu \\ \in i \backslash boldsymbol{R}\backslash right\},$	$\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$	$\mathfrak{e}_7=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_6,A\in\mathfrak{J}^C,\nu\in i\boldsymbol{R}\right\},$

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0902.0431_E00773	130	■ 0.999	$\alpha=\left(\begin{array}{cccc} \beta & \epsilon & 0 & 0 \\ \delta & \beta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right), \quad \beta, \beta_1, \delta, \epsilon \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right).$	$\alpha=\left(\begin{array}{cccc} \beta & \epsilon & 0 & 0 \\ \delta & \beta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right), \quad \beta, \beta_1, \delta, \epsilon \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right).$	$\alpha=\left(\begin{array}{cccc} \beta & \epsilon & 0 & 0 \\ \delta & \beta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right), \quad \beta, \beta_1, \delta, \epsilon \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right).$
0902.0431_E00832	139	■ 0.999	$\begin{aligned} & \left(\frac{e_7}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \\ & \left(\frac{e_7}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \&= \left(\frac{\Phi}{\sigma}\right)^{\sigma} \end{aligned}$	$(\mathfrak{e}_7)^{\sigma}=\left\{\Phi \in \mathfrak{e}_7 \mid \sigma \Phi=\Phi \sigma\right\}$ $=\left\{\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7 \mid \phi \in\left(\mathfrak{e}_6\right)^{\sigma}, A \in\left(\mathfrak{J}^C\right)_{\sigma}\right\}.$	$(\mathfrak{e}_7)^{\sigma}=\left\{\Phi \in \mathfrak{e}_7 \mid \sigma \Phi=\Phi \sigma\right\}$ $=\left\{\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7 \mid \phi \in\left(\mathfrak{e}_6\right)^{\sigma}, A \in\left(\mathfrak{J}^C\right)_{\sigma}\right\}.$
0902.0431_E00855	142	■ 0.999	$\tan 2 a=\frac{2 \eta}{\tau \xi-\xi}.$	$\tan 2 a=\frac{2 \eta}{\tau \xi-\xi}.$	$\tan 2 a=\frac{2 \eta}{\tau \xi-\xi}.$
0902.0431_E00864	143	■ 0.999	$(P, P)_{\mu}=\frac{1}{2}\{\mu P, P\}=\bar{x} x+(\tau \xi) \xi+(\tau \eta) \eta.$	$(P, P)_{\mu}=\frac{1}{2}\{\mu P, P\}=\bar{x} x+(\tau \xi) \xi+(\tau \eta) \eta.$	$(P, P)_{\mu}=\frac{1}{2}\{\mu P, P\}=\bar{x} x+(\tau \xi) \xi+(\tau \eta) \eta.$
0902.0431_E00905	150	■ 0.999	$D(g(\gamma Y)) D^{*} \quad \text { is real diagonal. }$	$D(g(\gamma Y)) D^{*} \quad \text { is real diagonal. }$	$D(g(\gamma Y)) D^{*} \quad \text { is real diagonal. }$
0902.0431_E00916	152	■ 0.999	$E_{7, C} / E_{6, C} \simeq\left(\mathfrak{M}_C\right)_1.$	$E_{7, C} / E_{6, C} \simeq\left(\mathfrak{M}_C\right)_1.$	$E_{7, C} / E_{6, C} \simeq\left(\mathfrak{M}_C\right)_1.$
0902.0431_E00973	162	■ 0.999	$\left[\left(\Phi_1, P_1, Q_1, r_1, s_1, t_1\right),\left(\Phi_2, P_2, Q_2, r_2, s_2, t_2\right)\right]=\left(\Phi, P, Q, r, s, t\right),$	$\left[\left(\Phi_1, P_1, Q_1, r_1, s_1, t_1\right),\left(\Phi_2, P_2, Q_2, r_2, s_2, t_2\right)\right]=\left(\Phi, P, Q, r, s, t\right),$	$\left[\left(\Phi_1, P_1, Q_1, r_1, s_1, t_1\right),\left(\Phi_2, P_2, Q_2, r_2, s_2, t_2\right)\right]=\left(\Phi, P, Q, r, s, t\right),$
0902.0431_E01118	190	■ 0.999	$\phi \boldsymbol{X}=\left(\begin{array}{c} \phi X_1 \\ \phi X_2 \\ \phi X_3 \end{array}\right), \quad D \boldsymbol{X}=\left(\begin{array}{c} d_{11} X_1+d_{12} X_2+d_{13} X_3 \\ d_{21} X_1+d_{22} X_2+d_{23} X_3 \\ d_{31} X_1+d_{32} X_2+d_{33} X_3 \end{array}\right).$	$\phi \boldsymbol{X}=\left(\begin{array}{c} \phi X_1 \\ \phi X_2 \\ \phi X_3 \end{array}\right), \quad D \boldsymbol{X}=\left(\begin{array}{c} d_{11} X_1+d_{12} X_2+d_{13} X_3 \\ d_{21} X_1+d_{22} X_2+d_{23} X_3 \\ d_{31} X_1+d_{32} X_2+d_{33} X_3 \end{array}\right).$	$\phi \boldsymbol{X}=\left(\begin{array}{c} \phi X_1 \\ \phi X_2 \\ \phi X_3 \end{array}\right), \quad D \boldsymbol{X}=\left(\begin{array}{c} d_{11} X_1+d_{12} X_2+d_{13} X_3 \\ d_{21} X_1+d_{22} X_2+d_{23} X_3 \\ d_{31} X_1+d_{32} X_2+d_{33} X_3 \end{array}\right).$
0902.0431_E01131	191	■ 0.999	$\left(\operatorname{ad} D_1\right)\left(D, \phi, \boldsymbol{X}, \boldsymbol{Y}\right)=\left(\left(\operatorname{ad} D_1\right) D, 0, D_1 \boldsymbol{X},-{}^t D_1 \boldsymbol{Y}\right),$	$\left(\operatorname{ad} D_1\right)\left(D, \phi, \boldsymbol{X}, \boldsymbol{Y}\right)=\left(\left(\operatorname{ad} D_1\right) D, 0, D_1 \boldsymbol{X},-{}^t D_1 \boldsymbol{Y}\right),$	$\left(\operatorname{ad} D_1\right)\left(D, \phi, \boldsymbol{X}, \boldsymbol{Y}\right)=\left(\left(\operatorname{ad} D_1\right) D, 0, D_1 \boldsymbol{X},-{}^t D_1 \boldsymbol{Y}\right),$
0902.0431_E00223	37	■ 0.999	$X \times X=\left(\begin{array}{ccc} \xi_2 \xi_3-x_1 \bar{x}_1 & \overline{x_1 x_2}-\xi_3 x_3 & x_3 x_1-\xi_2 \bar{x}_2 \\ x_1 x_2-\xi_3 \bar{x}_3 & \xi_3 \xi_1-x_2 \bar{x}_2 & \overline{x_2 x_3}-\xi_1 x_1 \\ \overline{x_3 x_1}-\xi_2 x_2 & x_2 x_3-\xi_1 \bar{x}_1 & \xi_1 \xi_2-x_3 \bar{x}_3 \end{array}\right), \quad X=X(\xi, x),$	$X \times X=\left(\begin{array}{ccc} \xi_2 \xi_3-x_1 \bar{x}_1 & \overline{x_1 x_2}-\xi_3 x_3 & x_3 x_1-\xi_2 \bar{x}_2 \\ x_1 x_2-\xi_3 \bar{x}_3 & \xi_3 \xi_1-x_2 \bar{x}_2 & \overline{x_2 x_3}-\xi_1 x_1 \\ \overline{x_3 x_1}-\xi_2 x_2 & x_2 x_3-\xi_1 \bar{x}_1 & \xi_1 \xi_2-x_3 \bar{x}_3 \end{array}\right), \quad X=X(\xi, x),$	$X \times X=\left(\begin{array}{ccc} \xi_2 \xi_3-x_1 \bar{x}_1 & \overline{x_1 x_2}-\xi_3 x_3 & x_3 x_1-\xi_2 \bar{x}_2 \\ x_1 x_2-\xi_3 \bar{x}_3 & \xi_3 \xi_1-x_2 \bar{x}_2 & \overline{x_2 x_3}-\xi_1 x_1 \\ \overline{x_3 x_1}-\xi_2 x_2 & x_2 x_3-\xi_1 \bar{x}_1 & \xi_1 \xi_2-x_3 \bar{x}_3 \end{array}\right), \quad X=X(\xi, x),$

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0902.0431_E00243	40	■ 0.999	$\begin{aligned} & 3 \operatorname{det}(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3 \\ & = \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X. \end{aligned}$	$3 \det(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X.$	$3 \det(\alpha X) = \operatorname{tr}(\alpha X, \alpha X, \alpha X) - \frac{3}{2} \operatorname{tr}(\alpha X)(\alpha X, \alpha X) + \frac{1}{2} \operatorname{tr}(\alpha X)^3$ $= \operatorname{tr}(X, X, X) - \frac{3}{2} \operatorname{tr}(X)(X, X) + \frac{1}{2} \operatorname{tr}(X)^3 = 3 \det X.$
0902.0431_E00706	116	■ 0.999	$\begin{aligned} \Phi &= \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \\ &= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C. \end{aligned}$	$\Phi = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right)$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$	$\Phi = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & \phi' + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right)$ $= \Phi(\phi, A, B, \nu), \quad \phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C.$
0902.0431_E01071	182	■ 0.999	$\begin{aligned} B &= l(D') + l(S)I, & D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C) \\ &= l(D) + l(S)I + l(icE), & D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \boldsymbol{R}. \end{aligned}$	$B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \boldsymbol{R}.$	$B = l(D') + l(S)I, \quad D' \in \mathfrak{u}(8), S \in \mathfrak{S}(8, C)$ $= l(D) + l(S)I + l(icE), \quad D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \boldsymbol{R}.$
0902.0431_E01110	188	■ 0.999	$\begin{array}{l} \omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots \\ 16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots \end{array}$	$\omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots$ $16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots$	$\omega_1 \quad 2\omega_1 \quad \omega_2 \quad 2\omega_2 \quad \omega_3 \quad \omega_4 \quad \omega_5 \quad \omega_6 \quad \omega_7 \quad \omega_8 \quad \cdots$ $16 \quad 135 \quad 120 \quad 5304 \quad 560 \quad 1820 \quad 4368 \quad 8008 \quad 128 \quad 128 \quad \cdots$
0902.0431_E00006	7	■ 1.000	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$	$\{x, y, a\} = \{y, a, x\} = \{a, x, y\} = -\{y, x, a\} = -\{x, a, y\} = -\{a, y, x\}.$
0902.0431_E00011	8	■ 1.000	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$	$\alpha 1 = 1, \quad \overline{\alpha e_i} = -\alpha e_i, \quad i = 1, 2, \dots, 7.$
0902.0431_E00019	9	■ 1.000	$\begin{aligned} (\kappa D)x &= \overline{Dx}, \quad x \in \mathfrak{C}, \\ \pi(G_{ij}) &= F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j, \\ \nu &= \pi\kappa. \end{aligned}$	$(\kappa D)x = \overline{Dx}, \quad x \in \mathfrak{C},$ $\pi(G_{ij}) = F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j,$ $\nu = \pi\kappa.$	$(\kappa D)x = \overline{Dx}, \quad x \in \mathfrak{C},$ $\pi(G_{ij}) = F_{ij}, \quad i, j = 0, 1, \dots, 7, i \neq j,$ $\nu = \pi\kappa.$
0902.0431_E00026	10	■ 1.000	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$	$L_a x = ax, \quad R_a x = xa, \quad T_a x = ax + xa = (L_a + R_a)x, \quad x \in \mathfrak{C}.$
0902.0431_E00032	10	■ 1.000	$\begin{aligned} L_{e_i} x &= e_i x = 2F_{i0}x, \quad x \in \mathfrak{C}, \\ T_{e_i} x &= e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases} \end{aligned}$	$L_{e_i} x = e_i x = 2F_{i0}x, \quad x \in \mathfrak{C},$ $T_{e_i} x = e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases}$	$L_{e_i} x = e_i x = 2F_{i0}x, \quad x \in \mathfrak{C},$ $T_{e_i} x = e_i x + x e_i = \begin{cases} 2e_i, & x = 1 \\ -2, & x = e_i \\ 0, & x = e_j, \quad j \neq 0, i. \end{cases}$

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0902.0431_E00072	15	■ 1.000	$\varphi_*(D)(a+\boldsymbol{m})=D\,\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C},$ $\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$\varphi_*(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$\varphi_*(D)(a+\boldsymbol{m})=D\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$
0902.0431_E00089	18	■ 1.000	$B_2\bigl(D_1,D_2\bigr)=k\,\operatorname{tr}\bigl(D_1D_2\bigr).$	$B_2(D_1,D_2)=k\,\mathrm{tr}\,(D_1D_2).$	$B_2(D_1,D_2)=k\mathrm{tr}(D_1D_2).$
0902.0431_E00090	18	■ 1.000	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$	$\begin{array}{ll} \left[H_1,\left[H_1,L_{12}\right]\right]=\left[H_1,2L_{21}\right]=-4L_{12}, & \left[H_1,\left[H_1,L_{21}\right]\right]=\left[H_1,-2L_{12}\right]=-4L_{21}, \\ \left[H_1,\left[H_1,L_{13}\right]\right]=\left[H_1,L_{31}\right]=-L_{13}, & \left[H_1,\left[H_1,L_{31}\right]\right]=\left[H_1,-L_{13}\right]=-L_{31}, \\ \left[H_1,\left[H_1,L_{23}\right]\right]=\left[H_1,-L_{32}\right]=-L_{23}, & \left[H_1,\left[H_1,L_{32}\right]\right]=\left[H_1,L_{23}\right]=-L_{32}, \end{array}$
0902.0431_E00119	23	■ 1.000	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$	$w(a+\boldsymbol{m})=a+\omega_1\boldsymbol{m},\quad a+\boldsymbol{m}\in C\oplus C^3=\mathfrak{C}.$
0902.0431_E00146	27	■ 1.000	$G_2\{ \}^{\mathfrak{C}}\simeq G_2\,\times\,\boldsymbol{R}^{14}.$	$G_2^C\simeq G_2\times\boldsymbol{R}^{14}.$	$G_2^C\simeq G_2\times\boldsymbol{R}^{14}.$
0902.0431_E00176	31	■ 1.000	$(x\,\overline{a})(a\,y\,a)=(x\,y)\,a,\quad x,\,y\in\mathfrak{C}.$	$(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$	$(x\overline{a})(aya)=(xy)a,\quad x,y\in\mathfrak{C}.$
0902.0431_E00181 (i)	31	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\alpha^{\prime}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\alpha'(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00185	32	■ 1.000	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$	$\begin{array}{l} \left(a_2,a_6\right)=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0, \\ \left(a_3,a_7\right)=\left(a_3b_0,a_7b_0\right)=\left(a_2\left(a_1b_0\right),a_6\left(a_1b_0\right)\right)=\left(a_1\left(a_2b_0\right),a_1\left(a_6b_0\right)\right) \\ \qquad\qquad\qquad=\left(a_2b_0,a_6b_0\right)=\left(a_2b_0,a_2\left(a_4b_0\right)\right)=\left(b_0,a_4b_0\right)=\left(1,a_4\right)=0. \end{array}$
0902.0431_E00192 (ii)	33	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00194	33	■ 1.000	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(x\,y),\quad x,\,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$	$(\alpha x)(\widetilde{\alpha}y)=\widetilde{\alpha}(xy),\quad x,y\in\mathfrak{C}.$
0902.0431_E00200	34	■ 1.000	$\left(\alpha_1,\alpha_2,\alpha_3\right)a=\alpha_1a,\quad a\in S^7.$	$(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$	$(\alpha_1,\alpha_2,\alpha_3)a=\alpha_1a,\quad a\in S^7.$

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0902.0431_E00222	37	■ 1.000	$\begin{aligned} (X, Y, Z) &= \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ &\quad \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right), \\ \det X &= \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3. \end{aligned}$	$(X, Y, Z) = \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right) \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3$	$(X, Y, Z) = \sum_{i=1}^3 \left(\frac{1}{2} (\xi_i \eta_{i+1} \zeta_{i+2} + \xi_i \eta_{i+2} \zeta_{i+1}) + R(x_i y_{i+1} z_{i+2} + x_i z_{i+1} y_{i+2}) \right. \\ \left. - (\xi_i (y_i, z_i) + \eta_i (z_i, x_i) + \zeta_i (x_i, y_i)) \right), \\ \det X = \xi_1 \xi_2 \xi_3 + 2R(x_1 x_2 x_3) - \xi_1 x_1 \bar{x}_1 - \xi_2 x_2 \bar{x}_2 - \xi_3 x_3 \bar{x}_3.$
0902.0431_E00237	39	■ 1.000	$\begin{aligned} &3 \operatorname{det} \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ &= \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X, \end{aligned}$	$3 \det \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ = \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X,$	$3 \det \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \\ = \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) \times \left(\begin{matrix} \alpha^{-1} X \\ \alpha^{-1} X \\ \alpha^{-1} X \end{matrix} \right) = (X, X \times X) = 3 \det X,$
0902.0431_E00252	41	■ 1.000	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z).$
0902.0431_E00256	42	■ 1.000	$\begin{aligned} &\frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) = \frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) \\ &= \frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) = \frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) \\ &= \frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) = \frac{1}{6} \operatorname{tr}(\phi X, \phi X, \phi X) \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6(-26)} &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}. \end{aligned}$	$\begin{aligned} \mathfrak{e}_{6(-26)} &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, X, X) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid (\phi X, Y, Z) + (X, \phi Y, Z) + (X, Y, \phi Z) = 0 \} \\ &= \{ \phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J}) \mid -{}^t \phi(X \times Y) = \phi X \times Y + X \times \phi Y \}. \end{aligned}$
0902.0431_E00260	42	■ 1.000	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$	$\widetilde{A}X = \frac{1}{2}[A, X], \quad X \in \mathfrak{J}.$
0902.0431_E00265	43	■ 1.000	$(X Y, Z) = (Y Z, X) = (Z X, Y) = (X, Y Z) = (Y, Z X) = (Z, X Y).$	$(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$	$(XY, Z) = (YZ, X) = (ZX, Y) = (X, YZ) = (Y, ZX) = (Z, XY).$
0902.0431_E00266	43	■ 1.000	$\begin{aligned} (XY, Z) &= R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ &= \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X). \end{aligned}$	$\begin{aligned} (XY, Z) &= R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ &= \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X). \end{aligned}$	$\begin{aligned} (XY, Z) &= R(XY, Z) = \frac{1}{2} R(\operatorname{tr}((XY)Z + Z^*(Y^* X^*))) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((x_{ij} y_{jk}) z_{ki} + \bar{z}_{ji} (\bar{y}_{kj} \bar{x}_{ik})) \right) \\ &= \frac{1}{2} R \left(\sum_{i,j,k} ((y_{jk} z_{ki}) x_{ij} + \bar{x}_{ik} (\bar{z}_{ji} \bar{y}_{kj})) \right) \\ &= \frac{1}{2} R(\operatorname{tr}((YZ)X + X^*(Z^* Y^*))) = (YZ, X). \end{aligned}$

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0902.0431_E00283	45	■ 1.000	$\left(D_1x\right)y+x\left(D_2y\right)=\overline{\overline{\left(D_3\right)\left(\overline{xy}\right)}},\quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$
0902.0431_E00295 (2)	47	■ 1.000	$\begin{aligned} & \{[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,\\ & [\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0. \end{aligned}$	$[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$	$[\widetilde{A},\widetilde{B}]=\left[\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim},\left(B-\frac{1}{3}\operatorname{tr}(B)E\right)^{\sim}\right]\in e_6^C,$ $[\widetilde{A},\widetilde{B}]E=\widetilde{A}(\widetilde{B}E)-\widetilde{B}(\widetilde{A}E)=A\circ B-B\circ A=0.$
0902.0431_E00304	49	■ 1.000	$\emptyset\not\in\left[\widetilde{A}_{\{1\}}(1),\widetilde{A}_{\{1\}}\left(e_1\right)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$	$0\neq\left[\widetilde{A}_1(1),\widetilde{A}_1(e_1)\right]\in\mathfrak{d}_4^C\cap\mathfrak{a}$	$0\neq[\widetilde{A}_1(1),\widetilde{A}_1(e_1)]\in\mathfrak{d}_4^C\cap\mathfrak{a}$
0902.0431_E00311	50	■ 1.000	$F_{\{3\}}(1)+F_{\{1\}}\left(z_1\right)\in W.$	$F_3(1)+F_1(z_1)\in W.$	$F_3(1)+F_1(z_1)\in W.$
0902.0431_E00320	51	■ 1.000	$B_{\{4\}}\left(\delta_1,\delta_2\right)=k\left(\delta_1,\delta_2\right)_4=k^{\prime}\operatorname{tr}\left(\delta_1\delta_2\right).$	$B_4(\delta_1,\delta_2)=k(\delta_1,\delta_2)_4=k'\operatorname{tr}(\delta_1\delta_2).$	$B_4(\delta_1,\delta_2)=k(\delta_1,\delta_2)_4=k'\operatorname{tr}(\delta_1\delta_2).$
0902.0431_E00321	51	■ 1.000	$\begin{aligned} & (\delta_1,\delta_2)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4\\ & \qquad\qquad=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2. \end{aligned}$	$(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$	$(\delta,\delta)_4=\left(\widetilde{A}_1(1),\widetilde{A}_1(1)\right)_4=-2\left(\widetilde{A}_1(1),\left[\widetilde{E}_3,\widetilde{F}_1(1)\right]\right)_4$ $=-2\left(\widetilde{A}_1(1)E_3,F_1(1)\right)=-(F_1(1),F_1(1))=-2.$
0902.0431_E00325	52	■ 1.000	$\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$	$\operatorname{tr}(\delta\delta)=\operatorname{tr}\left(\left(\widetilde{A}_1(1)\right)^2\right)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$	$\operatorname{tr}(\delta\delta)=\operatorname{tr}((\widetilde{A}_1(1))^2)=\left(-\frac{1}{2}\right)\times2-1+\left(-\frac{1}{4}\right)\times8\times2=-6.$
0902.0431_E00349	57	■ 1.000	$\varphi\left(\alpha_1,\alpha_2,\alpha_3\right)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$	$\varphi(\alpha_1,\alpha_2,\alpha_3)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$	$\varphi(\alpha_1,\alpha_2,\alpha_3)\left(\begin{array}{ccc}\xi_1&x_3&\overline{x_2}\\\overline{x_3}&\xi_2&x_1\\x_2&\overline{x_1}&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&\alpha_3x_3&\overline{\alpha_2x_2}\\\overline{\alpha_3x_3}&\xi_2&\alpha_1x_1\\\alpha_2x_2&\overline{\alpha_1x_1}&\xi_3\end{array}\right).$
0902.0431_E00354	57	■ 1.000	$\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3\left(\overline{xy}\right)},\quad x,y\in\mathfrak{C}.$	$(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$	$(\alpha_1x)(\alpha_2y)=\overline{\alpha_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$
0902.0431_E00361	58	■ 1.000	$\begin{array}{l} Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7,\\ Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7. \end{array}$	$Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7.$	$Y_i=-2Z_0\circ X_i,\quad i=4,5,6,7,$ $Z_i=-2X_i\circ Y_0,\quad i=1,3,5,6,7.$
0902.0431_E00364	59	■ 1.000	$p:\left(F_4\right)_{E_1}\rightarrow SO(9).$	$p:(F_4)_{E_1}\rightarrow SO(9).$	$p:(F_4)_{E_1}\rightarrow SO(9).$

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0902.0431_E00393	65	■ 1.000	$\begin{aligned} F_{4, \boldsymbol{H}} &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ &= \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}. \end{aligned}$	$F_{4, \boldsymbol{H}} = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$	$F_{4, \boldsymbol{H}} = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \circ N) = \alpha M \circ \alpha N\} \\ = \{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{H}}) \mid \alpha(M \times N) = \alpha M \times \alpha N\}.$
0902.0431_E00402	66	■ 1.000	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$	$\alpha = \varphi(A) \beta = \varphi(A) \varphi(B) = \varphi(AB), \quad AB \in \operatorname{Sp}(3).$
0902.0431_E00447	73	■ 1.000	$\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$	$\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$	$\mathfrak{e}_6^C = \{\phi \in \operatorname{Hom}_C(\mathfrak{J}^C) \mid (\phi X, X, X) = 0\}.$
0902.0431_E00468	77	■ 1.000	$\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$	$\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$	$\left(\phi^{\prime} X, Y\right)=-\left(X, \phi Y\right), \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00469	77	■ 1.000	$\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$	$\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$	$\phi(X \times Y) = \phi^{\prime} X \times Y + X \times \phi^{\prime} Y, \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00478	79	■ 1.000	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$	$\begin{aligned} (\phi, A \vee B)_6 &= \left(\delta + \widetilde{T}, [\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E\right)^{\sim}\right)_6 \\ &= (\delta, [\widetilde{A}, \widetilde{B}])_4 + \left(T, A \circ B - \frac{1}{3}(A, B)E\right) \\ &= (\delta A, B) + (\widetilde{T}A, B) = ((\delta + \widetilde{T})A, B) = (\phi A, B). \end{aligned}$

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0902.0431_E00504	82	1.000	$\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \end{aligned}$	$\begin{aligned} & \pm (\lambda_k - \lambda_l), \quad \pm (\lambda_k + \lambda_l), \quad 0 \leq k < l \leq 3, \\ & \pm \lambda_k \pm \frac{1}{2} (\mu_2 - \mu_3), \quad 0 \leq k \leq 3, \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_3 - \mu_1), \\ & \pm \frac{1}{2} (\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \\ & \pm \frac{1}{2} (-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) \pm \frac{1}{2} (\mu_1 - \mu_2), \end{aligned}$	
0902.0431_E00530	89	1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&\theta&\theta\\\xi_2&\theta&\theta\\\xi_3&\theta&\theta\end{array}\right),\quad \xi_i\in C\;.$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in C\;.$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_i\in C\;.$
0902.0431_E00531	89	1.000	$\theta\neq x_1=p+i\,q,\quad p,\,q\in\mathfrak{C}\;.$	$0\neq x_1=p+iq,\quad p,q\in\mathfrak{C}\;.$	$0\neq x_1=p+iq,\quad p,q\in\mathfrak{C}\;.$
0902.0431_E00569	95	1.000	$k\left(a+b\,e_2\right)=\left(\begin{array}{cc}a^{\prime}&b^{\prime}\\\tau a^{\prime}&\tau b^{\prime}\end{array}\right),\quad a,\,b\in\boldsymbol{C}\;.$	$k\left(a+b\,e_2\right)=\left(\begin{array}{cc}a^{\prime}&b^{\prime}\\\tau a^{\prime}&\tau b^{\prime}\end{array}\right),\quad a,b\in C\;.$	$k\left(a+b\,e_2\right)=\left(\begin{array}{cc}a^{\prime}&b^{\prime}\\\tau a^{\prime}&\tau b^{\prime}\end{array}\right),\quad a,b\in C\;.$

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0902.0431_E00572	96	■ 1.000	$\begin{aligned} & \kappa(M_1 + iM_2) = \kappa(M_1) + i\kappa(M_2), \quad M_1, M_2 \in M(3, \boldsymbol{H}), \\ & \kappa(\boldsymbol{a}_1 + i\boldsymbol{a}_2) = \kappa(\boldsymbol{a}_1) + i\kappa(\boldsymbol{a}_2), \quad \boldsymbol{a}_1, \boldsymbol{a}_2 \in \boldsymbol{H}^3. \end{aligned}$	$\begin{aligned} & \varphi(p, A)(M + \boldsymbol{a}) = k_J^{-1}(A(k_J(M))^t A) + p\boldsymbol{a}k_J^{-1}(\tau^t A), \\ & M + \boldsymbol{a} \in (\mathfrak{J}_H)^C \oplus (\boldsymbol{H}^3)^C = \mathfrak{J}^C. \end{aligned}$	$\begin{aligned} & \varphi(p, A)(M + \boldsymbol{a}) = k_J^{-1}(A(k_J(M))^t A) + p\boldsymbol{a}k_J^{-1}(\tau^t A), \\ & M + \boldsymbol{a} \in (\mathfrak{J}_H)^C \oplus (\boldsymbol{H}^3)^C = \mathfrak{J}^C. \end{aligned}$
0902.0431_E00594	99	■ 1.000	$\begin{aligned} & E_6 / F_4 \simeq EIV_C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\begin{aligned} & E_{6(6)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \det(\alpha X) = \det X\}, \\ & E_{6(2)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\gamma = \langle X, Y \rangle_\gamma\}, \\ & E_{6(-14)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\sigma = \langle X, Y \rangle_\sigma\}, \\ & E_{6(-26)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C})) \mid \det(\alpha X) = \det X\}. \end{aligned}$
0902.0431_E00640	106	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$
0902.0431_E00652	108	■ 1.000	$\begin{aligned} & E_{6(6)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \det(\alpha X) = \det X\}, \\ & E_{6(2)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\gamma = \langle X, Y \rangle_\gamma\}, \\ & E_{6(-14)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\sigma = \langle X, Y \rangle_\sigma\}, \\ & E_{6(-26)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C})) \mid \det(\alpha X) = \det X\}. \end{aligned}$	$\begin{aligned} & E_{6(6)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \det(\alpha X) = \det X\}, \\ & E_{6(2)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\gamma = \langle X, Y \rangle_\gamma\}, \\ & E_{6(-14)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\sigma = \langle X, Y \rangle_\sigma\}, \\ & E_{6(-26)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C})) \mid \det(\alpha X) = \det X\}. \end{aligned}$	$\begin{aligned} & E_{6(6)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \det(\alpha X) = \det X\}, \\ & E_{6(2)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\gamma = \langle X, Y \rangle_\gamma\}, \\ & E_{6(-14)} = \{\alpha \in \text{Iso}_C(\mathfrak{J}(3, \mathfrak{C}^C)) \mid \det(\alpha X) = \det X, \langle \alpha X, \alpha Y \rangle_\sigma = \langle X, Y \rangle_\sigma\}, \\ & E_{6(-26)} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C})) \mid \det(\alpha X) = \det X\}. \end{aligned}$
0902.0431_E00658	108	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$
0902.0431_E00664	110	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$
0902.0431_E00682 (2)	113	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$
0902.0431_E00692	114	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$
0902.0431_E00712	118	■ 1.000	$\begin{aligned} & \beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C. \end{aligned}$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$	$\beta(X + M) = X + PM = \varphi(P, E, E)(X + M), \quad X + M \in (\mathfrak{J}_C)^C \oplus M(3, \boldsymbol{C})^C = \mathfrak{J}^C.$

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0902.0431_E00852	141	■ 1.000	$\begin{aligned} V^{\{1\}} &= \left\{ P \in \mathfrak{P} \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0 \right\} \\ &= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\} \end{aligned}$	$V^{11} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0\}$ $= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$	$V^{11} = \{P \in \mathfrak{P}^C \mid \kappa P = P, \mu \tau \lambda P = P, P \times (0, E_1, 0, 1, 0) = 0\}$ $= \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \mid x \in \mathfrak{C}, \xi \in C, \eta \in i\mathbf{R} \right\}$
0902.0431_E00872	145	■ 1.000	$z(\operatorname{Spin}(12)) = \{1, -1, \sigma, -\sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$	$z(\operatorname{Spin}(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$	$z(\operatorname{Spin}(12)) = \{1, -1, \sigma - \sigma\} \cong \{1, -1\} \times \{1, \sigma\} \cong \mathbf{Z}_2 \times \mathbf{Z}_2.$
0902.0431_E00909	151	■ 1.000	$w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$	$w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$	$w(X, Y, \xi, \eta) = (wX, wY, \xi, \eta).$
0902.0431_E00923	153	■ 1.000	$\begin{aligned} h(a, b) &= \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C}, \\ h(a + be_1) &= a + bi, \quad a, b \in \mathbf{R}. \end{aligned}$	$h(a, b) = \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C},$ $h(a + be_1) = a + bi, \quad a, b \in \mathbf{R}.$	$h(a, b) = \frac{a+b}{2} + i \frac{(b-a)e_1}{2}, \quad a, b \in \mathbf{C},$ $h(a + be_1) = a + bi, \quad a, b \in \mathbf{R}.$
0902.0431_E00957	159	■ 1.000	$\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$	$\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$	$\psi(Q, A)P = f^{-1}((Q, A)(fP)), \quad P \in \mathfrak{P}^C.$
0902.0431_E00988	164	■ 1.000	$\left(\left([R, R_1], R_2 \right)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C. \right.$	$([R, R_1], R_2)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C.$	$([R, R_1], R_2)_8 + (R_1, [R, R_2])_8 = 0, \quad R, R_i \in \mathfrak{e}_8^C.$
0902.0431_E01003	167	■ 1.000	$\left\langle R_1, R_2 \right\rangle = -\frac{1}{15} B_8 \left(\tau \tilde{\lambda} R_1, R_2 \right).$	$\langle R_1, R_2 \rangle = -\frac{1}{15} B_8 \left(\tau \tilde{\lambda} R_1, R_2 \right).$	$\langle R_1, R_2 \rangle = -\frac{1}{15} B_8 (\tau \tilde{\lambda} R_1, R_2).$
0902.0431_E01013	168	■ 1.000	$E_{\{8\}} \simeq E_8 \times \mathbf{R}^{248}.$	$E_8^C \simeq E_8 \times \mathbf{R}^{248}.$	$E_8^C \simeq E_8 \times \mathbf{R}^{248}.$
0902.0431_E01050 (ii)	179	■ 1.000	$\left\{ \beta_2 P, \beta_2 Q \right\} = \{P, Q\}.$	$\{\beta_2 P, \beta_2 Q\} = \{P, Q\}.$	$\{\beta_2 P, \beta_2 Q\} = \{P, Q\}.$
0902.0431_E01053	179	■ 1.000	$\beta_1 \Phi = \beta \Phi \beta^{-1}.$	$\beta_1 \Phi = \beta \Phi \beta^{-1}.$	$\beta_1 \Phi = \beta \Phi \beta^{-1}.$
0902.0431_E01144	193	■ 1.000	$(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$	$(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$	$(\mathbf{u} \times \mathbf{v})\mathbf{x} = *(\mathbf{v} \wedge *(\mathbf{u} \wedge \mathbf{x})) + (-1)^{n-k} \frac{n-k}{n} (\mathbf{u}, \mathbf{v})\mathbf{x}, \quad \mathbf{x} \in C^n.$
0902.0431_E01145	193	■ 1.000	$\mathfrak{e}_{\{8\}} \simeq \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$	$\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$	$\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus \Lambda^3(C^9) \oplus \Lambda^3(C^9),$
0902.0431_E01147	193	■ 1.000	$\begin{aligned} D &= [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} &= D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} &= -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{aligned}$	$\begin{cases} D = [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} = D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} = -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{cases}$	$\begin{cases} D = [D_1, D_2] + \mathbf{u}_1 \times \mathbf{v}_2 - \mathbf{u}_2 \times \mathbf{v}_1 \\ \mathbf{u} = D_1 \mathbf{u}_2 - D_2 \mathbf{u}_1 + *(\mathbf{v}_1 \wedge \mathbf{v}_2) \\ \mathbf{v} = -{}^t D_1 \mathbf{v}_2 + {}^t D_2 \mathbf{v}_1 - *(\mathbf{u}_1 \wedge \mathbf{u}_2), \end{cases}$

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0902.0431_E01160	196	■ 1.000	$\begin{aligned} &0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ &= (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}. \end{aligned}$	$0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ = (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}.$	$0 \neq \operatorname{ad}(T) \operatorname{ad}(S_{37}) \operatorname{ad}(S_{27}) \operatorname{ad}(S_{17}) \operatorname{ad}(S_{36}) \operatorname{ad}(S_{25}) \operatorname{ad}(S_{14}) R \\ = (-E_{34}, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}.$
0902.0431_E01165	196	■ 1.000	$\left\langle R_1, R_2 \right\rangle = -B_8 \left(R_1, \tau \widetilde{\lambda} R_2 \right).$	$\langle R_1, R_2 \rangle = -B_8 \left(R_1, \tau \widetilde{\lambda} R_2 \right).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$
0902.0431_E01168	197	■ 1.000	$w_3(D, \boldsymbol{u}, \boldsymbol{v}) = \left(D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v} \right),$	$w_3(D, \boldsymbol{u}, \boldsymbol{v}) = \left(D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v} \right),$	$w_3(D, \boldsymbol{u}, \boldsymbol{v}) = (D, \omega \boldsymbol{u}, \omega^2 \boldsymbol{v}),$
0902.0431_E01180	199	■ 1.000	$(*(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$	$(*(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$	$(*(*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z}), \boldsymbol{v}) = (\boldsymbol{x} \wedge \boldsymbol{y}, \boldsymbol{z} \wedge \boldsymbol{v}) = (\boldsymbol{x}, \boldsymbol{z})(\boldsymbol{y}, \boldsymbol{v}) - (\boldsymbol{y}, \boldsymbol{z})(\boldsymbol{x}, \boldsymbol{v}).$
0902.0431_E01201	202	■ 1.000	$\begin{aligned} &\& B_8 \left(R_1, R_2 \right) = 60 \left(\operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \right. \\ &\quad \left. - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1) \right), \end{aligned}$	$B_8(R_1, R_2) = 60 \left(\operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \right. \\ \left. - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1) \right),$	$B_8(R_1, R_2) = 60(\operatorname{tr}(C_1 C_2) + \operatorname{tr}(D_1 D_2) - (\boldsymbol{x}_1, \boldsymbol{w}_2)(\boldsymbol{a}_1, \boldsymbol{d}_2) - (\boldsymbol{x}_2, \boldsymbol{w}_1)(\boldsymbol{a}_2, \boldsymbol{d}_1) \\ - (\boldsymbol{y}_1, \boldsymbol{z}_2)(\boldsymbol{b}_1, \boldsymbol{c}_2) - (\boldsymbol{y}_2, \boldsymbol{z}_1)(\boldsymbol{b}_2, \boldsymbol{c}_1)),$
0902.0431_E00001	5	■ 1.000	$\tau(u + i v) = u - i v.$	$\tau(u + i v) = u - i v.$	$\tau(u + i v) = u - i v.$
0902.0431_E00008	7	■ 1.000	$G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$	$G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$	$G_2 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$
0902.0431_E00009	7	■ 1.000	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00010	7	■ 1.000	$\overline{\alpha x} = \alpha \bar{x}.$	$\overline{\alpha x} = \alpha \bar{x}.$	$\overline{\alpha x} = \alpha \bar{x}.$
0902.0431_E00013	8	■ 1.000	$O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$	$O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$	$O(8) = O(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y) \}.$
0902.0431_E00014	8	■ 1.000	$\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$	$\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$	$\mathfrak{D}_4 = \mathfrak{so}(8) = \mathfrak{so}(\mathfrak{C}) = \{ D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid (Dx, y) + (x, Dy) = 0 \}$
0902.0431_E00015	8	■ 1.000	$G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$	$G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$	$G_{ij} e_j = e_i, \quad G_{ij} e_i = -e_j, \quad G_{ij} e_k = 0, \quad k \neq i, j.$
0902.0431_E00016	8	■ 1.000	$F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$	$F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$	$F_{ij} x = \frac{1}{2} e_i (\bar{e}_j x), \quad x \in \mathfrak{C}.$
0902.0431_E00020	9	■ 1.000	$\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$	$\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$	$\kappa, \pi, \nu \in \operatorname{Aut}(\mathfrak{D}_4).$
0902.0431_E00022	9	■ 1.000	$[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$	$[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$	$[\pi G_{ij}, \pi G_{kl}] = \pi [G_{ij}, G_{kl}],$

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0902.0431_E00027 (1)	10	■ 1.000	$L_{\{a\}}, R_{\{a\}}, T_{\{a\}} \in \mathfrak{D}_{\{4\}}.$	$L_a, R_a, T_a \in \mathfrak{D}_4.$	$L_a, R_a, T_a \in \mathfrak{D}_4.$
0902.0431_E00028 (2)	10	■ 1.000	$\kappa L_{\{a\}}=-R_{\{a\}}, \quad \quad \quad \kappa R_{\{a\}}=-L_{\{a\}}, \quad \quad \quad \kappa T_{\{a\}}=-T_{\{a\}}.$	$\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$	$\kappa L_a = -R_a, \quad \kappa R_a = -L_a, \quad \kappa T_a = -T_a.$
0902.0431_E00029 (3)	10	■ 1.000	$\pi L_{\{a\}}=T_{\{a\}}, \quad \quad \quad \pi R_{\{a\}}=-R_{\{a\}}, \quad \quad \quad \pi T_{\{a\}}=L_{\{a\}}.$	$\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$	$\pi L_a = T_a, \quad \pi R_a = -R_a, \quad \pi T_a = L_a.$
0902.0431_E00030 (4)	10	■ 1.000	$\nu L_{\{a\}}=R_{\{a\}}, \quad \quad \quad \nu R_{\{a\}}=-T_{\{a\}}, \quad \quad \quad \nu T_{\{a\}}=-L_{\{a\}}.$	$\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$	$\nu L_a = R_a, \quad \nu R_a = -T_a, \quad \nu T_a = -L_a.$
0902.0431_E00031	10	■ 1.000	$L_{\{e_{\{i\}}\}}=2 F_{\{i \ 0\}}, \quad \quad \quad T_{\{e_{\{i\}}\}}=2 G_{\{i \ 0\}}.$	$L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$	$L_{e_i} = 2F_{i0}, \quad T_{e_i} = 2G_{i0}.$
0902.0431_E00035	11	■ 1.000	$D=L_{\{a\}}+\sum_{\{i\}}\left[L_{\{b_{\{i\}}\}}, L_{\{c_{\{i\}}\}}\right], \quad \quad \quad a, b_{\{i\}}, c_{\{i\}} \in \mathfrak{C}_{\{0\}}.$	$D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$	$D = L_a + \sum_i [L_{b_i}, L_{c_i}], \quad a, b_i, c_i \in \mathfrak{C}_0.$
0902.0431_E00036	11	■ 1.000	$\kappa^{\{2\}}=1, \quad \quad \quad \pi^{\{2\}}=1, \quad \quad \quad \nu^{\{3\}}=1, \quad \quad \quad \nu=\pi \kappa.$	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$
0902.0431_E00038	11	■ 1.000	$\left(D_{\{1\}} x\right) y+x\left(D_{\{2\}} y\right)=D_{\{3\}}(x y), \quad \quad \quad x, y \in \mathfrak{C}.$	$(D_1 x) y+x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$	$(D_1 x) y+x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00039	11	■ 1.000	$D_{\{2\}}=\nu D_{\{1\}}, \quad \quad \quad D_{\{3\}}=\pi D_{\{1\}}.$	$D_2 = \nu D_1, \quad D_3 = \pi D_1.$	$D_2 = \nu D_1, \quad D_3 = \pi D_1.$
0902.0431_E00040 (i)	11	■ 1.000	$\left(L_{\{a\}} x\right) y+x\left(R_{\{a\}} y\right)=T_{\{a\}}(x y), \quad \quad \quad x, y \in \mathfrak{C}.$	$(L_a x) y+x\left(R_a y\right)=T_a(x y), \quad x, y \in \mathfrak{C}.$	$(L_a x) y+x\left(R_a y\right)=T_a(x y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00042	12	■ 1.000	$\left(L_{\{a\}} L_{\{b\}} x\right) y+\left(L_{\{b\}} x\right) y+\left(L_{\{a\}} x\right) y+\left(R_{\{b\}} y\right) x+\left(R_{\{a\}} R_{\{b\}} y\right)=T_{\{a\}} T_{\{b\}}(x y).$	$(L_a L_b x) y + (L_b x) (R_a y) + (L_a x) (R_b y) + x (R_a R_b y) = T_a T_b (x y).$	$(L_a L_b x) y + (L_b x) (R_a y) + (L_a x) (R_b y) + x (R_a R_b y) = T_a T_b (x y).$
0902.0431_E00043 (iii)	12	■ 1.000	$\left(\left(L_{\{a\}}, L_{\{b\}}\right) x\right) y+x\left(\left(R_{\{a\}}, R_{\{b\}}\right) y\right)=\left[T_{\{a\}}, T_{\{b\}}\right](x y).$	$([L_a, L_b] x) y + x ([R_a, R_b] y) = [T_a, T_b] (x y).$	$([L_a, L_b] x) y + x ([R_a, R_b] y) = [T_a, T_b] (x y).$
0902.0431_E00044	12	■ 1.000	$D_{\{1\}}=L_{\{a\}}+\sum\left[L_{\{b\}}, L_{\{c\}}\right], \quad \quad \quad a, b, c \in \mathfrak{C}_{\{0\}}$	$D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$	$D_1 = L_a + \sum [L_b, L_c], \quad a, b, c \in \mathfrak{C}_0$
0902.0431_E00045	12	■ 1.000	$\left(D_{\{1\}} x\right) y+x\left(D_{\{2\}} y\right)=D_{\{3\}}(x y), \quad \quad \quad x, y \in \mathfrak{C}.$	$(D_1 x) y+x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$	$(D_1 x) y+x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00046	12	■ 1.000	$x\left(D_{\{2\}} y\right)=D_{\{3\}}(x y), \quad \quad \quad x, y \in \mathfrak{C}.$	$x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$	$x\left(D_2 y\right)=D_3(x y), \quad x, y \in \mathfrak{C}.$
0902.0431_E00048	12	■ 1.000	$x(y p)=(x y) p, \quad \quad \quad \text { for all } x, y \in \mathfrak{C}.$	$x(y p) = (x y) p, \quad \text { for all } x, y \in \mathfrak{C}.$	$x(y p) = (x y) p, \quad \text { for all } x, y \in \mathfrak{C}.$
0902.0431_E00049	12	■ 1.000	$\left(D_{\{1\}} x\right) y+x\left(D_{\{2\}} y\right)=\left(\kappa D_{\{3\}}\right)(x y), \quad \quad \quad x, y \in \mathfrak{C}$	$(D_1 x) y+x\left(D_2 y\right)=\left(\kappa D_3\right)(x y), \quad x, y \in \mathfrak{C}$	$(D_1 x) y+x\left(D_2 y\right)=\left(\kappa D_3\right)(x y), \quad x, y \in \mathfrak{C}$
0902.0431_E00052	13	■ 1.000	$\begin{aligned} & \& D_{\{3\}}^{\prime}=\nu D_{\{2\}}=\nu \nu D_{\{1\}}=\nu^{-1} D_{\{1\}}=\kappa \pi D_{\{1\}}=\kappa \kappa D_{\{3\}}=D_{\{3\}} \\ & D_{\{1\}}^{\prime}=\kappa \pi D_{\{2\}}=\nu^{-1} D_{\{2\}}=D_{\{1\}} \end{aligned}$	$\begin{aligned} D_3' &= \nu D_2 = \nu \nu D_1 = \nu^{-1} D_1 = \kappa \pi D_1 = \kappa \kappa D_3 = D_3 \\ D_1' &= \kappa \pi D_2 = \nu^{-1} D_2 = D_1 \end{aligned}$	$\begin{aligned} D_3' &= \nu D_2 = \nu \nu D_1 = \nu^{-1} D_1 = \kappa \pi D_1 = \kappa \kappa D_3 = D_3, \\ D_1' &= \kappa \pi D_2 = \nu^{-1} D_2 = D_1. \end{aligned}$
0902.0431_E00053	13	■ 1.000	$\mathfrak{g}_{\{2\}}=\left\{D \in \operatorname{Hom}_{\left\{\boldsymbol{R}\right\}}\left(\mathfrak{C}\right) \mid D(x y)=(D x) y+x(D y)\right\}.$	$\mathfrak{g}_2 = \{D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid D(x y) = (D x) y + x(D y)\}.$	$\mathfrak{g}_2 = \{D \in \operatorname{Hom}_{\boldsymbol{R}}(\mathfrak{C}) \mid D(x y) = (D x) y + x(D y)\}.$
0902.0431_E00054	13	■ 1.000	$\begin{aligned} & \mathfrak{b}_{\{3\}}=\left\{D \in \mathfrak{D}_{\{4\}} \mid D 1=0\right\} \\ & =\left\{D \in \mathfrak{D}_{\{4\}} \mid \kappa D=D\right\}=\left\{D \in \mathfrak{D}_{\{4\}} \mid \nu D=\pi D\right\} . \end{aligned}$	$\begin{aligned} \mathfrak{b}_3 &= \{D \in \mathfrak{D}_4 \mid D 1 = 0\} \\ &= \{D \in \mathfrak{D}_4 \mid \kappa D = D\} = \{D \in \mathfrak{D}_4 \mid \nu D = \pi D\}. \end{aligned}$	$\begin{aligned} \mathfrak{b}_3 &= \{D \in \mathfrak{D}_4 \mid D 1 = 0\} \\ &= \{D \in \mathfrak{D}_4 \mid \kappa D = D\} = \{D \in \mathfrak{D}_4 \mid \nu D = \pi D\}. \end{aligned}$
0902.0431_E00056	13	■ 1.000	$D x \in \mathfrak{C}_{\{0\}}, \quad \quad \quad x \in \mathfrak{C}.$	$D x \in \mathfrak{C}_0, \quad x \in \mathfrak{C}.$	$D x \in \mathfrak{C}_0, \quad x \in \mathfrak{C}.$
0902.0431_E00057	13	■ 1.000	$x y+y x=-\left(x \bar{y}+y \bar{x}\right)=-2(x, y), \quad \quad \quad x, y \in \mathfrak{C}_{\{0\}}.$	$x y+y x = -\left(x \bar{y}+y \bar{x}\right) = -2(x, y), \quad x, y \in \mathfrak{C}_0.$	$x y+y x = -\left(x \bar{y}+y \bar{x}\right) = -2(x, y), \quad x, y \in \mathfrak{C}_0.$
0902.0431_E00058	13	■ 1.000	$(D x, y)+(x, D y)=0, \quad \quad \quad x, y \in \mathfrak{C}.$	$(D x, y) + (x, D y) = 0, \quad x, y \in \mathfrak{C}.$	$(D x, y) + (x, D y) = 0, \quad x, y \in \mathfrak{C}.$
0902.0431_E00060	14	■ 1.000	$\operatorname{dim} \mathfrak{g}_{\{2\}}=14.$	$\dim \mathfrak{g}_2 = 14.$	$\dim \mathfrak{g}_2 = 14.$

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0902.0431_E00061	14	■ 1.000	$\sum_{0<i<j}\lambda_{ij}\text{F}_{ij}=\sum_{0<i<j}\lambda_{ij}\text{G}_{ij}.$	$\sum_{0<i<j}\lambda_{ij}F_{ij}=\sum_{0<i<j}\lambda_{ij}G_{ij}.$	$\sum_{0<i<j}\lambda_{ij}F_{ij}=\sum_{0<i<j}\lambda_{ij}G_{ij}.$
0902.0431_E00062	14	■ 1.000	$\sum_{0<i<j}\lambda_{ij}\text{e}_{ij}\text{e}_i=0.$	$\sum_{0<i<j}\lambda_{ij}e_je_i=0.$	$\sum_{0<i<j}\lambda_{ij}e_je_i=0.$
0902.0431_E00064	14	■ 1.000	$\boldsymbol{C}=\left\{x_0+x_1\text{e}_1\mid x_i\in\boldsymbol{R}\right\}.$	$\boldsymbol{C}=\{x_0+x_1e_1\mid x_i\in\boldsymbol{R}\}.$	$\boldsymbol{C}=\{x_0+x_1e_1\mid x_i\in\boldsymbol{R}\}.$
0902.0431_E00066	15	■ 1.000	$x=\text{a+m}_1\text{e}_2+\text{m}_2\text{e}_4+\text{m}_3\text{e}_6, \quad \text{a},\text{m}_i\in\boldsymbol{C}.$	$x=a+m_1e_2+m_2e_4+m_3e_6, \quad a,m_i\in\boldsymbol{C}.$	$x=a+m_1e_2+m_2e_4+m_3e_6, \quad a,m_i\in\boldsymbol{C}.$
0902.0431_E00070	15	■ 1.000	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$
0902.0431_E00071	15	■ 1.000	$\left(\mathfrak{g}_2\right)_{e_1}=\left\{D\in\mathfrak{g}_2\mid De_1=0\right\}.$	$\left(\mathfrak{g}_2\right)_{e_1}=\{D\in\mathfrak{g}_2\mid De_1=0\}.$	$\left(\mathfrak{g}_2\right)_{e_1}=\{D\in\mathfrak{g}_2\mid De_1=0\}.$
0902.0431_E00074	16	■ 1.000	$\begin{array}{ll} \varphi_*\left(\text{e}_{11}-\text{E}_{22}\right)\right)= -G_{23}+G_{45}, & \varphi_*\left(\text{e}_{11}-\text{E}_{22}\right)\right)= -G_{23}+G_{45}, \\ \varphi_*\left(\text{E}_{12}-\text{E}_{21}\right)=G_{24}+G_{35}, & \varphi_*\left(\text{E}_{12}-\text{E}_{21}\right)=G_{24}+G_{35}, \\ \varphi_*\left(\text{E}_{13}-\text{E}_{31}\right)=G_{26}+G_{37}, & \varphi_*\left(\text{E}_{13}-\text{E}_{31}\right)=G_{26}+G_{37}, \\ \varphi_*\left(\text{E}_{23}-\text{E}_{32}\right)=G_{46}+G_{57}, & \varphi_*\left(\text{E}_{23}-\text{E}_{32}\right)=G_{46}+G_{57}, \end{array}$	$\varphi_*\left(e_1\left(E_{11}-E_{22}\right)\right)=-G_{23}+G_{45}, \quad \varphi_*\left(e_1\left(E_{22}-E_{33}\right)\right)=-G_{45}+G_{67},$ $\varphi_*\left(E_{12}-E_{21}\right)=G_{24}+G_{35}, \quad \varphi_*\left(e_1\left(E_{12}+E_{21}\right)\right)=-G_{25}+G_{34},$ $\varphi_*\left(E_{13}-E_{31}\right)=G_{26}+G_{37}, \quad \varphi_*\left(e_1\left(E_{13}+E_{31}\right)\right)=-G_{27}+G_{36},$ $\varphi_*\left(E_{23}-E_{32}\right)=G_{46}+G_{57}, \quad \varphi_*\left(e_1\left(E_{23}+E_{32}\right)\right)=-G_{47}+G_{56}.$	$\varphi_*\left(e_1\left(E_{11}-E_{22}\right)\right)=-G_{23}+G_{45}, \quad \varphi_*\left(e_1\left(E_{22}-E_{33}\right)\right)=-G_{45}+G_{67},$ $\varphi_*\left(E_{12}-E_{21}\right)=G_{24}+G_{35}, \quad \varphi_*\left(e_1\left(E_{12}+E_{21}\right)\right)=-G_{25}+G_{34},$ $\varphi_*\left(E_{13}-E_{31}\right)=G_{26}+G_{37}, \quad \varphi_*\left(e_1\left(E_{13}+E_{31}\right)\right)=-G_{27}+G_{36},$ $\varphi_*\left(E_{23}-E_{32}\right)=G_{46}+G_{57}, \quad \varphi_*\left(e_1\left(E_{23}+E_{32}\right)\right)=-G_{47}+G_{56}.$
0902.0431_E00078	16	■ 1.000	$-2\text{x}_1\text{e}_2-2\text{x}_2\text{e}_3-2\text{x}_3\text{e}_4-2\text{x}_4\text{e}_5-2\text{x}_5\text{e}_6-2\text{x}_6\text{e}_7=0.$	$-2x_1e_2-2x_2e_3-2x_3e_4-2x_4e_5-2x_5e_6-2x_6e_7=0.$	$-2x_1e_2-2x_2e_3-2x_3e_4-2x_4e_5-2x_5e_6-2x_6e_7=0.$
0902.0431_E00079	16	■ 1.000	$\overline{x_1+ix_2}=\bar{x}_1+i\bar{x}_2, \quad \tau\left(x_1+ix_2\right)=x_1-ix_2, \quad x_i\in\mathfrak{C}.$	$\overline{x_1+ix_2}=\bar{x}_1+i\bar{x}_2, \quad \tau\left(x_1+ix_2\right)=x_1-ix_2, \quad x_i\in\mathfrak{C}.$	$\overline{x_1+ix_2}=\bar{x}_1+i\bar{x}_2, \quad \tau\left(x_1+ix_2\right)=x_1-ix_2, \quad x_i\in\mathfrak{C}.$
0902.0431_E00080	16	■ 1.000	$\tau(x\,y)=(\tau x)(\tau y), \quad x,y\in\mathfrak{C}^C.$	$\tau(xy)=(\tau x)(\tau y), \quad x,y\in\mathfrak{C}^C.$	$\tau(xy)=(\tau x)(\tau y), \quad x,y\in\mathfrak{C}^C.$
0902.0431_E00083 (cdots ii)	17	■ 1.000	$-x_1S_6+x_2S_5\in W \quad \cdots\text{(i)} \quad \text{and} \quad -x_1S_5-x_2S_6\in W$	$-x_1S_6+x_2S_5\in W \quad \cdots\text{(i)} \quad \text{and} \quad -x_1S_5-x_2S_6\in W$	$-x_1S_6+x_2S_5\in W \quad \cdots\text{(i)} \quad \text{and} \quad -x_1S_5-x_2S_6\in W$
0902.0431_E00085	17	■ 1.000	$\mathfrak{a}\ni\left[D',D+S\right]=\left[D',D\right]+\left[D',S\right], \quad \left[D',S\right]\in\mathfrak{S}$	$\mathfrak{a}\ni\left[D',D+S\right]=\left[D',D\right]+\left[D',S\right], \quad \left[D',S\right]\in\mathfrak{S}$	$\mathfrak{a}\ni\left[D',D+S\right]=\left[D',D\right]+\left[D',S\right], \quad \left[D',S\right]\in\mathfrak{S}$
0902.0431_E00096	19	■ 1.000	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$	$\pm\left(\lambda_1-\lambda_2\right), \quad \pm\left(\lambda_1-\lambda_3\right), \quad \pm\left(\lambda_2-\lambda_3\right), \quad \pm\lambda_1, \quad \pm\lambda_2 \quad \pm\lambda_3$
0902.0431_E00099	20	■ 1.000	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$	$\alpha_1=\lambda_1-\lambda_2, \quad \alpha_2=\lambda_2$
0902.0431_E00100	20	■ 1.000	$\mu=2\alpha_1+3\alpha_3$	$\mu=2\alpha_1+3\alpha_3$	$\mu=2\alpha_1+3\alpha_3$
0902.0431_E00102	20	■ 1.000	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67}\mid\lambda_i\in\boldsymbol{R},\lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67}\mid\lambda_i\in\boldsymbol{R},\lambda_1+\lambda_2+\lambda_3=0\right\}$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{-i\lambda_1G_{23}-i\lambda_2G_{45}-i\lambda_3G_{67}\mid\lambda_i\in\boldsymbol{R},\lambda_1+\lambda_2+\lambda_3=0\right\}$
0902.0431_E00103	20	■ 1.000	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$	$B_2\left(H,H'\right)=8\sum_{k=1}^3\lambda_k\lambda_k',$

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0902.0431_E00104	20	■ 1.000	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$	$H_{\alpha_1}=-\frac{1}{8}iG_{23}+\frac{1}{8}iG_{45},\quad H_{\alpha_2}=\frac{1}{24}iG_{23}-\frac{1}{12}iG_{45}+\frac{1}{24}iG_{67}.$
0902.0431_E00106	21	■ 1.000	$\left(G_2\right)_{e_1}=\left\{\alpha\in G_2\mid\alpha e_1=e_1\right\}.$	$(G_2)_{e_1}=\{\alpha\in G_2\mid\alpha e_1=e_1\}.$	$(G_2)_{e_1}=\{\alpha\in G_2\mid\alpha e_1=e_1\}.$
0902.0431_E00107	21	■ 1.000	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$	$\varphi(A)(a+\boldsymbol{m})=a+A\boldsymbol{m},\quad a+\boldsymbol{m}\in\boldsymbol{C}\oplus\boldsymbol{C}^3=\mathfrak{C}.$
0902.0431_E00109	21	■ 1.000	$\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$	$\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$	$\alpha e_2=\boldsymbol{a}_1,\quad \alpha e_4=\boldsymbol{a}_2,\quad \alpha e_6=\boldsymbol{a}_3$
0902.0431_E00110	21	■ 1.000	$\left\langle\boldsymbol{a}_1,\boldsymbol{a}_2\right\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$	$\langle\boldsymbol{a}_1,\boldsymbol{a}_2\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$	$\langle\boldsymbol{a}_1,\boldsymbol{a}_2\rangle=0,\quad \boldsymbol{a}_3=\overline{\boldsymbol{a}_1\times\boldsymbol{a}_2}.$
0902.0431_E00111	22	■ 1.000	$a_3=a_1a_2.$	$a_3=a_1a_2.$	$a_3=a_1a_2.$
0902.0431_E00112	22	■ 1.000	$a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$	$a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$	$a_5=a_1a_4,\quad a_6=a_4a_2,\quad a_7=a_3a_4.$
0902.0431_E00114	22	■ 1.000	$\alpha e_i=a_i,\quad i=0,1,\cdots,7$	$\alpha e_i=a_i,\quad i=0,1,\cdots,7$	$\alpha e_i=a_i,\quad i=0,1,\cdots,7$
0902.0431_E00115	22	■ 1.000	$\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$	$\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$	$\alpha(xy)=(\alpha x)(\alpha y),\quad x,y\in\mathfrak{C}.$
0902.0431_E00116	22	■ 1.000	$\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$	$\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$	$\alpha(e_ie_j)=(\alpha e_i)(\alpha e_j),\quad i,j=0,1,\cdots,7$
0902.0431_E00118	23	■ 1.000	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$	$w=\varphi(\operatorname{diag}(\omega_1,\omega_1,\omega_1))$
0902.0431_E00120	23	■ 1.000	$(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$	$(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$	$(G_2)^w=\{\alpha\in G_2\mid w\alpha=\alpha w\}.$
0902.0431_E00121	23	■ 1.000	$w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$	$w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$	$w\varphi(A)(a+\boldsymbol{m})=w(a+A\boldsymbol{m})=a+\omega_1A\boldsymbol{m}=a+A\omega_1\boldsymbol{m}=\varphi(A)w(a+\boldsymbol{m}).$
0902.0431_E00124	24	■ 1.000	$\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$	$\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$	$\boldsymbol{H}=\{x_0+x_1e_1+x_2e_2+x_3e_3\mid x_i\in\boldsymbol{R}\}.$
0902.0431_E00126	24	■ 1.000	$x=m+ae_4,\quad m,a\in\boldsymbol{H}.$	$x=m+ae_4,\quad m,a\in\boldsymbol{H}.$	$x=m+ae_4,\quad m,a\in\boldsymbol{H}.$
0902.0431_E00127	24	■ 1.000	$(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\bar{a}n+bm)e_4,$ $(m+ae_4,n+be_4)=(m,n)+(a,b),$ $\overline{m+ae_4}=\bar{m}-ae_4,$ $\gamma(m+ae_4)=m-ae_4.$	$(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\bar{a}n+bm)e_4,$ $(m+ae_4,n+be_4)=(m,n)+(a,b),$ $\overline{m+ae_4}=\bar{m}-ae_4,$ $\gamma(m+ae_4)=m-ae_4.$	$(m+ae_4)(n+be_4)=(mn-\bar{b}a)+(\bar{a}n+bm)e_4,$ $(m+ae_4,n+be_4)=(m,n)+(a,b),$ $\overline{m+ae_4}=\bar{m}-ae_4,$ $\gamma(m+ae_4)=m-ae_4.$
0902.0431_E00128	24	■ 1.000	$\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$	$\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$	$\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$
0902.0431_E00129	24	■ 1.000	$(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$	$(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$	$(G_2)^\gamma=\{\alpha\in G_2\mid\gamma\alpha=\alpha\gamma\}.$
0902.0431_E00130	24	■ 1.000	$\varphi(p,q)(m+ae_4)=qm\bar{q}+(p\bar{a}q)e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$	$\varphi(p,q)(m+ae_4)=qm\bar{q}+(p\bar{a}q)e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$	$\varphi(p,q)(m+ae_4)=qm\bar{q}+(p\bar{a}q)e_4,\quad m+ae_4\in\boldsymbol{H}\oplus\boldsymbol{H}e_4=\mathfrak{C}.$
0902.0431_E00132	25	■ 1.000	$\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$	$\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$	$\alpha m=qm\bar{q},\quad m\in\boldsymbol{H}$
0902.0431_E00135	25	■ 1.000	$f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$	$f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$	$f(p,q)x=px\bar{q},\quad x\in\boldsymbol{H}$
0902.0431_E00137	26	■ 1.000	$z(G_2)=\{1\}.$	$z(G_2)=\{1\}.$	$z(G_2)=\{1\}.$

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0902.0431_E00140	26	■ 1.000	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}$.	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}^C.$	$(\alpha x, \alpha y) = (x, y), \quad x, y \in \mathfrak{C}^C.$
0902.0431_E00141	26	■ 1.000	$\overline{\alpha e_k} = -\alpha e_k, \quad k=1,2, \dots, 7.$	$\overline{\alpha e_k} = -\alpha e_k, \quad k = 1, 2, \dots, 7.$	$\overline{\alpha e_k} = -\alpha e_k, \quad k = 1, 2, \dots, 7.$
0902.0431_E00144	26	■ 1.000	$\langle x, y \rangle = (\tau x, y),$	$\langle x, y \rangle = (\tau x, y),$	$\langle x, y \rangle = (\tau x, y),$
0902.0431_E00149	27	■ 1.000	$G_{2(2)} = \left\{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \right\}.$	$G_{2(2)} = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$	$G_{2(2)} = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}') \mid \alpha(xy) = (\alpha x)(\alpha y) \}.$
0902.0431_E00150	28	■ 1.000	$G_{2(2)} = \left(G_{2\{ \}^{\{ \}} \right)^{\tau \gamma} = \left\{ \alpha \in G_{2^C} \mid \tau \gamma \alpha = \alpha \gamma \tau \right\}.$	$G_{2(2)} = \left(G_{2^C} \right)^{\tau \gamma} = \{ \alpha \in G_{2^C} \mid \tau \gamma \alpha = \alpha \gamma \tau \}.$	$G_{2(2)} = (G_{2^C})^{\tau \gamma} = \{ \alpha \in G_{2^C} \mid \tau \gamma \alpha = \alpha \gamma \tau \}.$
0902.0431_E00153	28	■ 1.000	$z \left(G_{2(2)} \right) = \{ 1 \}.$	$z \left(G_{2(2)} \right) = \{ 1 \}.$	$z(G_{2(2)}) = \{ 1 \}.$
0902.0431_E00155	28	■ 1.000	$A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A = D_{a_m} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$
0902.0431_E00156	28	■ 1.000	$A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}$	$A = D_{a_{2m}} \cdots D_{a_2} D_{a_1}, \quad a_i \in S^{n-1}.$
0902.0431_E00157	28	■ 1.000	$SO(\mathfrak{C}) = \left\{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \right\}.$	$SO(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \}.$	$SO(\mathfrak{C}) = \{ \alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid (\alpha x, \alpha y) = (x, y), \det \alpha = 1 \}.$
0902.0431_E00159	29	■ 1.000	$D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$	$D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$	$D_a x = x - 2(x, a)a = x - (x \bar{a} + a \bar{x})a = -a \bar{x}a, \quad x \in \mathfrak{C},$
0902.0431_E00160	29	■ 1.000	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$	$(\alpha_1 x)(\alpha_2 y) = (b(\bar{a}x))((y \bar{a})b) = b(\bar{a}(xy) \bar{a})b = \alpha_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00163	29	■ 1.000	$(x p) y = x(p y), \quad x, y \in \mathfrak{C}.$	$(xp)y = x(py), \quad x, y \in \mathfrak{C}.$	$(xp)y = x(py), \quad x, y \in \mathfrak{C}.$
0902.0431_E00167	30	■ 1.000	$\left(\alpha_2 y \right) \left(\alpha_3 z \right) = \overline{\alpha_1(\overline{y z})}.$	$(\alpha_2 y)(\alpha_3 z) = \overline{\alpha_1(\overline{y z})}.$	$(\alpha_2 y)(\alpha_3 z) = \overline{\alpha_1(\overline{y z})}.$
0902.0431_E00168	30	■ 1.000	$\left(\alpha_1 x \right) \left(\alpha_2 y \right) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$	$(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$	$(\alpha_1 x)(\alpha_2 y) = \alpha_3(xy), \quad x, y \in \mathfrak{C},$
0902.0431_E00169	30	■ 1.000	$\left(\beta_1(\alpha_1 x) \right) \left(\beta_2(\alpha_2 y) \right) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$	$(\beta_1(\alpha_1 x))(\beta_2(\alpha_2 y)) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$	$(\beta_1(\alpha_1 x))(\beta_2(\alpha_2 y)) = \beta_3((\alpha_1 x)(\alpha_2 y)) = \beta_3(\alpha_3(xy)), \quad x, y \in \mathfrak{C}.$
0902.0431_E00170	30	■ 1.000	$\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$	$\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$	$\bar{x}(\gamma_2 y) = \gamma_3(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00171 (i)	30	■ 1.000	$\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$	$\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$	$\bar{x}(\gamma_2 y) = \gamma_2(xy), \quad x, y \in \mathfrak{C}.$
0902.0431_E00173	30	■ 1.000	$p x = x p \quad \text{for all } x \in \mathfrak{C}.$	$p x = x p \quad \text{for all } x \in \mathfrak{C}.$	$p x = x p \quad \text{for all } x \in \mathfrak{C}.$
0902.0431_E00174	30	■ 1.000	$x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$	$x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$	$x y = y x \quad \text{for all } x, y \in \mathfrak{C}.$
0902.0431_E00175	31	■ 1.000	$\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$	$\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$	$\alpha_a x = a x a^{-1}, \quad x \in \mathfrak{C}$
0902.0431_E00177	31	■ 1.000	$(a x \bar{a}) \left(a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$	$(a x \bar{a}) \left(a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$	$(a x \bar{a}) \left(a y a^2 \right) = a(x y) a^2, \quad x, y \in \mathfrak{C}.$
0902.0431_E00179	31	■ 1.000	$a y a^2 = \pm a y \bar{a}.$	$a y a^2 = \pm a y \bar{a}.$	$a y a^2 = \pm a y \bar{a}.$
0902.0431_E00184	32	■ 1.000	$a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$	$a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$	$a_5 b_0 = a_1(a_4 b_0), \quad a_6 b_0 = a_2(a_4 b_0), \quad a_7 b_0 = a_6(a_1 b_0).$
0902.0431_E00186	32	■ 1.000	$\alpha e_i = a_i, \quad i = 0, 1, \dots, 7$	$\alpha e_i = a_i, \quad i = 0, 1, \dots, 7$	$\alpha e_i = a_i, \quad i = 0, 1, \dots, 7$
0902.0431_E00188	32	■ 1.000	$\left(\alpha e_i \right) \left(\left(\alpha e_j \right) b_0 \right) = \left(\alpha \left(e_i e_j \right) \right) b_0, \quad i, j = 0, 1, \dots, 7.$	$(\alpha e_i)((\alpha e_j) b_0) = (\alpha(e_i e_j)) b_0, \quad i, j = 0, 1, \dots, 7.$	$(\alpha e_i)((\alpha e_j) b_0) = (\alpha(e_i e_j)) b_0, \quad i, j = 0, 1, \dots, 7.$

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0902.0431_E00190	33	■ 1.000	$b_{\{i\}=a_{\{i\}}\,b_{\{0\}}}$, $\quad i=0,1,\,\cdots,\,7$,	$b_i = a_i b_0, \quad i = 0, 1, \cdots, 7,$	$b_i = a_i b_0, \quad i = 0, 1, \cdots, 7,$
0902.0431_E00193	33	■ 1.000	$\widetilde{B}_3 \cong \operatorname{Spin}(7)$.	$\widetilde{B}_3 \cong \operatorname{Spin}(7).$	$\widetilde{B}_3 \cong Spin(7).$
0902.0431_E00196	33	■ 1.000	$\widetilde{B}_3/\{1,-1\} \cong S\,0(7)$.	$\widetilde{B}_3/\{1,-1\} \cong SO(7).$	$\widetilde{B}_3/\{1,-1\} \cong SO(7).$
0902.0431_E00198	34	■ 1.000	$\operatorname{Spin}(7) \ni \widetilde{\alpha}$ $\xrightarrow{\quad \alpha, \, \widetilde{\alpha}, \, \kappa \widetilde{\alpha} \quad}$ $\in \widetilde{D}_4$.	$\operatorname{Spin}(7) \ni \widetilde{\alpha} \longleftrightarrow (\alpha, \widetilde{\alpha}, \kappa \widetilde{\alpha}) \in \widetilde{D}_4.$	$Spin(7) \ni \widetilde{\alpha} \longleftrightarrow (\alpha, \widetilde{\alpha}, \kappa \widetilde{\alpha}) \in \widetilde{D}_4.$
0902.0431_E00199	34	■ 1.000	$\widetilde{D}_4/\operatorname{Spin}(7) \simeq S^7$.	$\widetilde{D}_4/\operatorname{Spin}(7) \simeq S^7.$	$\widetilde{D}_4/Spin(7) \simeq S^7.$
0902.0431_E00202	34	■ 1.000	$\widetilde{D}_4 \cong \operatorname{Spin}(8)$.	$\widetilde{D}_4 \cong \operatorname{Spin}(8).$	$\widetilde{D}_4 \cong Spin(8).$
0902.0431_E00203	34	■ 1.000	$p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1$.	$p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1.$	$p(\alpha_1, \alpha_2, \alpha_3) = \alpha_1.$
0902.0431_E00204	34	■ 1.000	$\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong S\,0(8)$.	$\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong SO(8).$	$\widetilde{D}_4/\{(1,1,1),(1,-1,-1)\} \cong SO(8).$
0902.0431_E00205	34	■ 1.000	$\begin{aligned} z(\operatorname{Spin}(8)) \, \&= \\ &\{(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \, \&= \\ &\{(1,1,1),(1,-1,-1)\} \, \times \{(1,1,1),(-1,-1,1)\} \\ &\cong \boldsymbol{\mathbb{Z}}_2 \, \times \boldsymbol{\mathbb{Z}}_2 \, . \end{aligned}$	$\begin{aligned} z(\operatorname{Spin}(8)) = \{ &(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \\ &= \{(1,1,1),(1,-1,-1)\} \times \{(1,1,1),(-1,-1,1)\} \cong \boldsymbol{Z}_2 \times \boldsymbol{Z}_2. \end{aligned}$	$\begin{aligned} z(Spin(8)) \, = \, &\{(1,1,1),(1,-1,-1),(-1,-1,1),(-1,1,-1)\} \\ &= \{(1,1,1),(1,-1,-1)\} \times \{(1,1,1),(-1,-1,1)\} \cong \boldsymbol{Z}_2 \times \boldsymbol{Z}_2. \end{aligned}$
0902.0431_E00207	35	■ 1.000	$\kappa^2=1$, $\quad \pi^2=1$, $\quad \nu^3=1$, $\quad \nu=\pi\kappa$.	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$	$\kappa^2 = 1, \quad \pi^2 = 1, \quad \nu^3 = 1, \quad \nu = \pi \kappa.$
0902.0431_E00210	35	■ 1.000	$S\,0(8) \cong S\,s(8)$	$SO(8) \cong Ss(8)$	$SO(8) \cong Ss(8).$
0902.0431_E00212 (i)	35	■ 1.000	$(\alpha x)(\alpha y) = \kappa \alpha(x y)$, $\quad x, y \in \mathfrak{C}$,	$(\alpha x)(\alpha y) = \kappa \alpha(xy), \quad x, y \in \mathfrak{C},$	$(\alpha x)(\alpha y) = \kappa \alpha(xy), \quad x, y \in \mathfrak{C},$
0902.0431_E00213	35	■ 1.000	$a(\alpha x) = (\alpha x) a$ $\quad$ for all $\quad x \in \mathfrak{C}$.	$a(\alpha x) = (\alpha x) a \quad$ for all $\quad x \in \mathfrak{C}.$	$a(\alpha x) = (\alpha x) a \quad$ for all $\quad x \in \mathfrak{C}.$
0902.0431_E00214	35	■ 1.000	$\nu^2(-1,-1,1) = (1,-1,-1)$,	$\nu^2(-1,-1,1) = (1,-1,-1),$	$\nu^2(-1,-1,1) = (1,-1,-1),$
0902.0431_E00215	36	■ 1.000	$\mathfrak{J} = \left\{ X \in M(3, \mathfrak{C}) \mid X^* = X \right.$ $\left. \right\}$,	$\mathfrak{J} = \{ X \in M(3, \mathfrak{C}) \mid X^* = X \},$	$\mathfrak{J} = \{ X \in M(3, \mathfrak{C}) \mid X^* = X \},$
0902.0431_E00217	36	■ 1.000	$X \circ Y = \frac{1}{2}(XY + YX)$	$X \circ Y = \frac{1}{2}(XY + YX)$	$X \circ Y = \frac{1}{2}(XY + YX).$
0902.0431_E00218	36	■ 1.000	$\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z) . \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z). \end{aligned}$	$\begin{aligned} \operatorname{tr}(X) &= \xi_1 + \xi_2 + \xi_3, \quad X = X(\xi, x), \\ (X, Y) &= \operatorname{tr}(X \circ Y), \quad \operatorname{tr}(X, Y, Z) = (X, Y \circ Z). \end{aligned}$
0902.0431_E00219	36	■ 1.000	$X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E)$,	$X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E),$	$X \times Y = \frac{1}{2}(2X \circ Y - \operatorname{tr}(X)Y - \operatorname{tr}(Y)X + (\operatorname{tr}(X)\operatorname{tr}(Y) - (X, Y))E),$
0902.0431_E00220	36	■ 1.000	$(X, Y, Z) = (X, Y \times Z)$, $\quad \operatorname{det} X = \frac{1}{3}(X, X, X)$.	$(X, Y, Z) = (X, Y \times Z), \quad \det X = \frac{1}{3}(X, X, X).$	$(X, Y, Z) = (X, Y \times Z), \quad \det X = \frac{1}{3}(X, X, X).$
0902.0431_E00227	38	■ 1.000	$F_4 = \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right.$ $\left. \right\}$.	$F_4 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$	$F_4 = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$
0902.0431_E00228 (i)	38	■ 1.000	$X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\operatorname{det} X)E$.	$X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\det X)E.$	$X \circ (X \circ X) - \operatorname{tr}(X)X^2 + \frac{1}{2}(\operatorname{tr}(X)^2 - \operatorname{tr}(X^2))X = (\det X)E.$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E00229 (ii)	38	■ 1.000	$X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$	$X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$	$X \circ (X \circ X) - \operatorname{tr}(\alpha X) X^2 + \frac{1}{2} (\operatorname{tr}(\alpha X)^2 - \operatorname{tr}((\alpha X^2)) X = (\det \alpha X) E.$
0902.0431_E00231	38	■ 1.000	$\begin{aligned} & \operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ & = -\det(\alpha F_i(e_j)) E. \end{aligned}$	$\operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ = -\det(\alpha F_i(e_j)) E.$	$\operatorname{tr}(\alpha F_i(e_j)) (E_{i+1} + E_{i+2}) + \frac{1}{2} (-\operatorname{tr}(\alpha F_i(e_j))^2 + \operatorname{tr}((\alpha F_i(e_j))^2) - 2) F_i(e_j) \\ = -\det(\alpha F_i(e_j)) E.$
0902.0431_E00232	38	■ 1.000	$\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$	$\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$	$\operatorname{tr}(\alpha F_i(e_j)) = 0 (= \operatorname{tr}(F_i(e_j)))$
0902.0431_E00233	38	■ 1.000	$\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$	$\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$	$\operatorname{tr}(\alpha E_i) = \operatorname{tr}(\alpha (E - F_i(1)^2)) = \operatorname{tr}(E) - \operatorname{tr}((\alpha F_i(1))^2) = 3 - 2 = 1 = \operatorname{tr}(E_i),$
0902.0431_E00236	39	■ 1.000	$\alpha(X \circ Y) = \alpha X \circ \alpha Y.$	$\alpha(X \circ Y) = \alpha X \circ \alpha Y.$	$\alpha(X \circ Y) = \alpha X \circ \alpha Y.$
0902.0431_E00238 (1)	40	■ 1.000	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z), (\alpha X, \alpha Y) = (X, Y).$
0902.0431_E00239 (2)	40	■ 1.000	$\det(\alpha X) = \det X, \alpha E = E.$	$\det(\alpha X) = \det X, \alpha E = E.$	$\det(\alpha X) = \det X, \alpha E = E.$
0902.0431_E00241 (4)	40	■ 1.000	$\alpha(X \times Y) = \alpha X \times \alpha Y.$	$\alpha(X \times Y) = \alpha X \times \alpha Y.$	$\alpha(X \times Y) = \alpha X \times \alpha Y.$
0902.0431_E00242 (5)	40	■ 1.000	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$	$\alpha X \circ \alpha E = \alpha(X \times E) = \frac{1}{2} \alpha(\operatorname{tr}(X) E - X),$
0902.0431_E00245	40	■ 1.000	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$	$\alpha X \times P = \frac{1}{2} \operatorname{tr}(X) P - \frac{1}{2} \alpha X, \quad X \in \mathfrak{J}.$
0902.0431_E00246	40	■ 1.000	$\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1)$	$\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1)$	$\frac{1}{2} (\rho_2 E_3 + \rho_3 E_2 - F_1(p_1)) = \frac{1}{2} (\lambda(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)) - E_1),$
0902.0431_E00247	40	■ 1.000	$0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$	$0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$	$0 = \lambda \rho_1 - 1, \quad \rho_3 = \lambda \rho_2, \quad \rho_2 = \lambda \rho_3, \quad -p_1 = \lambda p_1, \quad 0 = \lambda p_2, \quad 0 = \lambda p_3.$
0902.0431_E00248	41	■ 1.000	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$	$-\frac{1}{2} \rho_1 F_1(1) = \frac{1}{2} (\mu(\rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3) - F_1(1)),$
0902.0431_E00249	41	■ 1.000	$(\alpha X, \alpha Y) = (X, Y).$	$(\alpha X, \alpha Y) = (X, Y).$	$(\alpha X, \alpha Y) = (X, Y).$
0902.0431_E00251	41	■ 1.000			

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0902.0431_E00254	41	■ 1.000	$\widetilde{\alpha}\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1 & \alpha x_3 & \overline{\alpha x_2} \\ \overline{\alpha x_3} & \xi_2 & \alpha x_1 \\ \alpha x_2 & \overline{\alpha x_1} & \xi_3\end{array}\right).$		
0902.0431_E00258	42	■ 1.000	$\mathfrak{M}^{\scriptscriptstyle -}=\left\{A\in M(3,\mathfrak{C})\mid A^{\scriptscriptstyle *}=-A\right\}.$	$\mathfrak{M}^{-}=\{A\in M(3,\mathfrak{C})\mid A^*=-A\}.$	$\mathfrak{M}^{-}=\{A\in M(3,\mathfrak{C})\mid A^*=-A\}.$
0902.0431_E00261	42	■ 1.000	$[X,\,X\,X]=a\,E\;.$	$[X,XX]=aE.$	$[X,XX]=aE.$
0902.0431_E00262	42	■ 1.000	$a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$	$a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$	$a_{ij}=\sum_{k,l}\left(x_{ik}\left(x_{kl}x_{lj}\right)-\left(x_{ik}x_{kl}\right)x_{lj}\right)=-\sum_{k,l}\left\{x_{ik},x_{kl},x_{lj}\right\},\quad i,j=1,2,3.$
0902.0431_E00268	43	■ 1.000	$([A,\,X],\,X\,X)=0,\quad X\in\mathfrak{J}\;.$	$([A,X],XX)=0,\quad X\in\mathfrak{J}.$	$([A,X],XX)=0,\quad X\in\mathfrak{J}.$
0902.0431_E00270	44	■ 1.000	$([A,\,X],\,Y\,Z+Z\,Y)+([A,\,Y],\,X\,Z+Z\,X)+([A,\,Z],\,X\,Y+Y\,X)=0,$	$([A,X],YZ+ZY)+([A,Y],XZ+ZX)+([A,Z],XY+YX)=0,$	$([A,X],YZ+ZY)+([A,Y],XZ+ZX)+([A,Z],XY+YX)=0,$
0902.0431_E00271	44	■ 1.000	$A_{\scriptscriptstyle 1}(a)=\left(\begin{array}{ccc}\theta&\theta&\theta\\ \theta&-\bar{a}&\theta\end{array}\right),\quad A_{\scriptscriptstyle 2}(a)=\left(\begin{array}{ccc}\theta&\theta&-\bar{a}\\ \theta&\theta&\theta\end{array}\right),\quad A_{\scriptscriptstyle 3}(a)=\left(\begin{array}{ccc}\theta&a&\theta\\ \theta&-\bar{a}&\theta\end{array}\right).$	$A_1(a)=\left(\begin{array}{ccc}0&0&0\\ 0&0&a\\ 0&-\bar{a}&0\end{array}\right),\quad A_2(a)=\left(\begin{array}{ccc}0&0&-\bar{a}\\ 0&0&0\\ a&0&0\end{array}\right),\quad A_3(a)=\left(\begin{array}{ccc}0&a&0\\ -\bar{a}&0&0\\ 0&0&0\end{array}\right).$	$A_1(a)=\left(\begin{array}{ccc}0&0&0\\ 0&0&a\\ 0&-\bar{a}&0\end{array}\right),\quad A_2(a)=\left(\begin{array}{ccc}0&0&-\bar{a}\\ 0&0&0\\ a&0&0\end{array}\right),\quad A_3(a)=\left(\begin{array}{ccc}0&a&0\\ -\bar{a}&0&0\\ 0&0&0\end{array}\right).$
0902.0431_E00273	44	■ 1.000	$\mathfrak{d}_{\scriptscriptstyle 4}=\left\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\right\}$	$\mathfrak{d}_4=\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\}$	$\mathfrak{d}_4=\{\delta\in\mathfrak{f}_4\mid\delta E_i=0,i=1,2,3\}$
0902.0431_E00274	44	■ 1.000	$\mathfrak{D}_{\scriptscriptstyle 4}=\left\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}\left(\mathfrak{C}\right)\mid\left(Dx,y\right)+\left(x,Dy\right)=0\right\}$	$\mathfrak{D}_4=\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}(\mathfrak{C})\mid(Dx,y)+(x,Dy)=0\}$	$\mathfrak{D}_4=\{D\in\operatorname{Hom}_{\boldsymbol{\mathfrak{R}}}(\mathfrak{C})\mid(Dx,y)+(x,Dy)=0\}$
0902.0431_E00275	44	■ 1.000	$\mathfrak{D}_{\scriptscriptstyle 4}\ni D_{\scriptscriptstyle 1}\longrightarrow\delta\in\mathfrak{d}_{\scriptscriptstyle 4}$	$\mathfrak{D}_4\ni D_1\longrightarrow\delta\in\mathfrak{d}_4$	$\mathfrak{D}_4\ni D_1\longrightarrow\delta\in\mathfrak{d}_4$
0902.0431_E00276	44	■ 1.000	$\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$	$\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$	$\delta\left(\begin{array}{ccc}\xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3\end{array}\right)=\left(\begin{array}{ccc}0 & D_3x_3 & \overline{D_2x_2} \\ \overline{D_3x_3} & 0 & D_1x_1 \\ D_2x_2 & \overline{D_1x_1} & 0\end{array}\right),$
0902.0431_E00277	44	■ 1.000	$\left(D_{\scriptscriptstyle 1}x\right)y+x\left(D_{\scriptscriptstyle 2}y\right)=\overline{\left(D_{\scriptscriptstyle 3}\left(\overline{xy}\right)\right)},\quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$	$(D_1x)y+x(D_2y)=\overline{D_3(\overline{xy})},\quad x,y\in\mathfrak{C}.$
0902.0431_E00281	45	■ 1.000	$\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$	$\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$	$\delta F_i(x)=F_i(D_ix),\quad x\in\mathfrak{C},$
0902.0431_E00284	45	■ 1.000	$\delta=D+\widetilde{A},\quad D\in\mathfrak{d},A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$	$\delta=D+\widetilde{A},\quad D\in\mathfrak{d}_4,A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$	$\delta=D+\widetilde{A},\quad D\in\mathfrak{d}_4,A\in\mathfrak{M}^{-},\operatorname{diag}A=0,$
0902.0431_E00285	45	■ 1.000	$\operatorname{dim}\mathfrak{f}_{\scriptscriptstyle 4}=28+24=52\;.$	$\dim\mathfrak{f}_4=28+24=52.$	$\dim\mathfrak{f}_4=28+24=52.$

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0902.0431_E00287	46	■ 1.000	$\backslash\mathrm{delta}\ E_{\{1\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}\ 0\ \&\ -a_{\{3\}}\ \&\ \backslash\mathrm{bar}\{a\}_{\{2\}}\ \backslash\backslash\ -\backslash\mathrm{bar}\{a\}_{\{3\}}\ \&\ 0\ \&\ 0\ \backslash\backslash\ a_{\{2\}}\ \&\ 0\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{delta}\ E_{\{2\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}\ 0\ \&\ a_{\{3\}}\ \&\ 0\ \backslash\backslash\ \backslash\mathrm{bar}\{a\}_{\{3\}}\ \&\ 0\ \&\ -a_{\{1\}}\ \backslash\backslash\ 0\ \&\ -\backslash\mathrm{bar}\{a\}_{\{1\}}\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),\ \backslash\mathrm{delta}\ E_{\{3\}}=\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}\ 0\ \&\ 0\ \&\ -\backslash\mathrm{bar}\{a\}_{\{2\}}\ \backslash\backslash\ 0\ \&\ 0\ \&\ a_{\{1\}}\ \backslash\backslash\ -a_{\{2\}}\ \&\ \backslash\mathrm{bar}\{a\}_{\{1\}}\ \&\ 0\ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right}),$	$\delta E_1 = \begin{pmatrix} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{pmatrix}, \delta E_2 = \begin{pmatrix} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{pmatrix}, \delta E_3 = \begin{pmatrix} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{pmatrix},$	$\delta E_1 = \begin{pmatrix} 0 & -a_3 & \bar{a}_2 \\ -\bar{a}_3 & 0 & 0 \\ a_2 & 0 & 0 \end{pmatrix}, \delta E_2 = \begin{pmatrix} 0 & a_3 & 0 \\ \bar{a}_3 & 0 & -a_1 \\ 0 & -\bar{a}_1 & 0 \end{pmatrix}, \delta E_3 = \begin{pmatrix} 0 & 0 & -\bar{a}_2 \\ 0 & 0 & a_1 \\ -a_2 & \bar{a}_1 & 0 \end{pmatrix},$
0902.0431_E00288	46	■ 1.000	$\backslash\mathrm{delta}\ E_{\{i\}}=\backslash\mathrm{widetilde}\{A\}\ E_{\{i\}},\ \backslash\mathrm{quad}\ i=1,2,3\ .$	$\delta E_i = \tilde{A}E_i,\quad i = 1, 2, 3.$	$\delta E_i = \tilde{A}E_i,\quad i = 1, 2, 3.$
0902.0431_E00290	46	■ 1.000	$\overline{\mathrm{X}_{\{1\}}+i\ \mathrm{X}_{\{2\}}}=\mathrm{bar}\{\mathrm{X}_{\{1\}}+i\ \mathrm{bar}\{\mathrm{X}_{\{2\}}\},\ \backslash\mathrm{quad}\ \backslash\mathrm{tau}\backslash\mathrm{left}(\mathrm{X}_{\{1\}}+i\ \mathrm{X}_{\{2\}}\backslash\mathrm{right})=\mathrm{X}_{\{1\}}-i\ \mathrm{X}_{\{2\}},\ \backslash\mathrm{quad}\ \mathrm{X}_{\{i\}}\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{\mathrm{J}\}\ .$	$\overline{X_1+iX_2}=\bar{X}_1+i\bar{X}_2,\quad \tau(X_1+iX_2)=X_1-iX_2,\quad X_i\in\mathfrak{J}.$	$\overline{X_1+iX_2}=\bar{X}_1+i\bar{X}_2,\quad \tau(X_1+iX_2)=X_1-iX_2,\quad X_i\in\mathfrak{J}.$
0902.0431_E00291	46	■ 1.000	$\backslash\mathrm{tau}(\mathrm{X}\ \backslash\mathrm{circ}\ \mathrm{Y})=\backslash\mathrm{tau}\ \mathrm{X}\ \backslash\mathrm{circ}\ \backslash\mathrm{tau}\ \mathrm{Y},\ \backslash\mathrm{quad}\ \backslash\mathrm{tau}(\mathrm{X}\ \backslash\mathrm{times}\ \mathrm{Y})=\backslash\mathrm{tau}\ \mathrm{X}\ \backslash\mathrm{times}\ \backslash\mathrm{tau}\ \mathrm{Y},\ \backslash\mathrm{quad}\ \mathrm{X},\ \mathrm{Y}\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{\mathrm{J}\}^{\mathrm{C}}\ .$	$\tau(X\circ Y)=\tau X\circ\tau Y,\quad \tau(X\times Y)=\tau X\times\tau Y,\quad X,Y\in\mathfrak{J}^C.$	$\tau(X\circ Y)=\tau X\circ\tau Y,\quad \tau(X\times Y)=\tau X\times\tau Y,\quad X,Y\in\mathfrak{J}^C.$
0902.0431_E00294	47	■ 1.000	$=(\mathrm{A},(\backslash\mathrm{operatorname}\{\mathrm{det}\}\ \mathrm{X})\ \mathrm{E})=(\backslash\mathrm{operatorname}\{\mathrm{det}\}\ \mathrm{X})(\mathrm{A},\ \mathrm{E})=(\backslash\mathrm{operatorname}\{\mathrm{det}\}\ \mathrm{X})\ \backslash\mathrm{operatorname}\{\mathrm{tr}\}(\mathrm{A})=0,\ \backslash\mathrm{quad}\ \mathrm{X}\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{\mathrm{J}\}^{\mathrm{C}}\ .$	$=(\mathrm{A},(\det X)E)=(\det X)(\mathrm{A},E)=(\det X)\mathrm{tr}(A)=0,\quad X\in\mathfrak{J}^C.$	$=(\mathrm{A},(\det X)E)=(\det X)(\mathrm{A},E)=(\det X)\mathrm{tr}(A)=0,\quad X\in\mathfrak{J}^C.$
0902.0431_E00303	48	■ 1.000	$\backslash\mathrm{widetilde}\{A\}_{\{1\}}(1)+\backslash\mathrm{widetilde}\{A\}_{\{2\}}(b)+\backslash\mathrm{widetilde}\{A\}_{\{3\}}(c)\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{a\}\ .$	$\tilde{A}_1(1)+\tilde{A}_2(b)+\tilde{A}_3(c)\in\mathfrak{a}.$	$\tilde{A}_1(1)+\tilde{A}_2(b)+\tilde{A}_3(c)\in\mathfrak{a}.$
0902.0431_E00305	49	■ 1.000	$-\backslash\mathrm{widetilde}\{A\}_{\{3\}}(\backslash\mathrm{bar}\{b\})+\backslash\mathrm{widetilde}\{A\}_{\{2\}}(\backslash\mathrm{bar}\{c\})\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{a\}\ \backslash\mathrm{quad}\ \backslash\mathrm{text}\ \{\ \mathrm{and\ then}\ \}\ \backslash\mathrm{quad}\ \backslash\mathrm{widetilde}\{A\}_{\{3\}}(1)+\backslash\mathrm{widetilde}\{A\}_{\{2\}}\backslash\mathrm{left}(c^{\wedge}\backslash\mathrm{prime}\}\ \backslash\mathrm{right})\ \backslash\mathrm{in}\ \backslash\mathrm{mathfrak}\{a\}$	$-\tilde{A}_3(\bar{b})+\tilde{A}_2(\bar{c})\in\mathfrak{a}\quad\text{and then}\quad\tilde{A}_3(1)+\tilde{A}_2(c')\in\mathfrak{a}$	$-\tilde{A}_3(\bar{b})+\tilde{A}_2(\bar{c})\in\mathfrak{a}\quad\text{and then}\quad\tilde{A}_3(1)+\tilde{A}_2(c')\in\mathfrak{a}$
0902.0431_E00306	49	■ 1.000	$[\backslash\mathrm{delta},[\backslash\mathrm{widetilde}\{A\},\ \backslash\mathrm{widetilde}\{B\}]]=[\backslash\mathrm{widetilde}\{\backslash\mathrm{delta}\ A\},\ \backslash\mathrm{widetilde}\{B\}]+[\backslash\mathrm{widetilde}\{A\},\ \backslash\mathrm{widetilde}\{\backslash\mathrm{delta}\ B\}]\ .$	$[\delta, [\tilde{A}, \tilde{B}]] = [\widetilde{\delta A}, \tilde{B}] + [\tilde{A}, \widetilde{\delta B}].$	$[\delta, [\tilde{A}, \tilde{B}]] = [\widetilde{\delta A}, \tilde{B}] + [\tilde{A}, \widetilde{\delta B}].$
0902.0431_E00308	49	■ 1.000	$\mathrm{F}_{\{i\}}(1)\ \backslash\mathrm{xrightarrow}\backslash\mathrm{widetilde}\{A\}_{\{i\}}(1)\}\ \mathrm{E}_{\{i+1\}}-\mathrm{E}_{\{i+2\}}\ \backslash\mathrm{xrightarrow}\backslash\mathrm{widetilde}\{A\}_{\{i+2\}}(1)\}\ \mathrm{F}_{\{i+2\}}(1)\ \backslash\mathrm{longrightarrow}\backslash\mathrm{cdots}\ \backslash\mathrm{longrightarrow}\ \mathrm{F}_{\{i+1\}}(1),$	$F_i(1)\xrightarrow{\tilde{A}_i(1)}E_{i+1}-E_{i+2}\xrightarrow{\tilde{A}_{i+2}(1)}F_{i+2}(1)\longrightarrow\cdots\longrightarrow F_{i+1}(1),$	$F_i(1)\xrightarrow{\tilde{A}_i(1)}E_{i+1}-E_{i+2}\xrightarrow{\tilde{A}_{i+2}(1)}F_{i+2}(1)\longrightarrow\cdots\longrightarrow F_{i+1}(1),$
0902.0431_E00309	49	■ 1.000	$0\ \backslash\mathrm{neq}\ \mathrm{X}=\backslash\mathrm{x}\backslash\mathrm{left}(\mathrm{E}_{\{1\}}-\mathrm{E}_{\{2\}}\backslash\mathrm{right})+\backslash\mathrm{eta}\ \backslash\mathrm{left}(\mathrm{E}_{\{2\}}-\mathrm{E}_{\{3\}}\backslash\mathrm{right})+\mathrm{F}_{\{1\}}\backslash\mathrm{left}(\mathrm{x}_{\{1\}}\backslash\mathrm{right})+\mathrm{F}_{\{2\}}\ \backslash\mathrm{left}(\mathrm{x}_{\{2\}}\backslash\mathrm{right})+\mathrm{F}_{\{3\}}\backslash\mathrm{left}(\mathrm{x}_{\{3\}}\backslash\mathrm{right})\ \backslash\mathrm{in}\ W\ .$	$0\neq X=\xi(E_1-E_2)+\eta(E_2-E_3)+F_1(x_1)+F_2(x_2)+F_3(x_3)\in W.$	$0\neq X=\xi(E_1-E_2)+\eta(E_2-E_3)+F_1(x_1)+F_2(x_2)+F_3(x_3)\in W.$
0902.0431_E00310	50	■ 1.000	$\mathrm{F}_{\{1\}}(1)+\mathrm{F}_{\{2\}}\backslash\mathrm{left}(\mathrm{y}_{\{2\}}\backslash\mathrm{right})+\mathrm{F}_{\{3\}}\backslash\mathrm{left}(\mathrm{y}_{\{3\}}\backslash\mathrm{right})\ \backslash\mathrm{in}\ W\ .$	$F_1(1)+F_2(y_2)+F_3(y_3)\in W.$	$F_1(1)+F_2(y_2)+F_3(y_3)\in W.$
0902.0431_E00312	50	■ 1.000	$\mathrm{X}=\backslash\mathrm{x}\backslash\mathrm{left}(\mathrm{E}_{\{1\}}-\mathrm{E}_{\{2\}}\backslash\mathrm{right})+\backslash\mathrm{eta}\backslash\mathrm{left}(\mathrm{E}_{\{2\}}-\mathrm{E}_{\{3\}}\ \backslash\mathrm{right})\ \backslash\mathrm{in}\ W,$	$X=\xi(E_1-E_2)+\eta(E_2-E_3)\in W,$	$X=\xi(E_1-E_2)+\eta(E_2-E_3)\in W,$
0902.0431_E00313	50	■ 1.000	$([\backslash\mathrm{widetilde}\{A\},\ \backslash\mathrm{widetilde}\{B\}]\ C,\ D)=([\backslash\mathrm{widetilde}\{C\},\ \backslash\mathrm{widetilde}\{D\}]\ A,\ B)\ .$	$([\tilde{A}, \tilde{B}]C, D) = ([\tilde{C}, \tilde{D}]A, B).$	$([\tilde{A}, \tilde{B}]C, D) = ([\tilde{C}, \tilde{D}]A, B).$
0902.0431_E00316	50	■ 1.000	$\backslash\mathrm{left}(\backslash\mathrm{delta}_{\{1\}},\ \backslash\mathrm{delta}_{\{2\}}\backslash\mathrm{right})_{\{4\}}=\backslash\mathrm{sum}_{\{i,\ j\}}\backslash\mathrm{left}(\backslash\mathrm{left}[\backslash\mathrm{widetilde}\{A\}_{\{i\}},\ \backslash\mathrm{widetilde}\{B\}_{\{j\}}\backslash\mathrm{right}]\ C_{\{j\}},\ \mathrm{D}_{\{j\}}\backslash\mathrm{right})=\backslash\mathrm{sum}_{\{i,\ j\}}\backslash\mathrm{left}(\backslash\mathrm{left}[\backslash\mathrm{widetilde}\{C\}_{\{j\}},\ \backslash\mathrm{widetilde}\{D\}_{\{j\}}\backslash\mathrm{right}]\ A_{\{i\}},\ \mathrm{B}_{\{i\}}\backslash\mathrm{right}),$	$(\delta_1, \delta_2)_4 = \sum_{i,j} \left(\left[\tilde{A}_i, \tilde{B}_i \right] C_j, D_j \right) = \sum_{i,j} \left(\left[\tilde{C}_j, \tilde{D}_j \right] A_i, B_i \right),$	$(\delta_1, \delta_2)_4 = \sum_{i,j} ([\tilde{A}_i, \tilde{B}_i]C_j, D_j) = \sum_{i,j} ([\tilde{C}_j, \tilde{D}_j]A_i, B_i),$
0902.0431_E00326	52	■ 1.000	$\backslash\mathrm{mathfrak}\{\mathrm{D}\}_{\{4\}}\ \backslash\mathrm{^}\{\mathrm{C}\}=\backslash\mathrm{left}\backslash\{\mathrm{D}\ \backslash\mathrm{in}\ \backslash\mathrm{operatorname}\{\mathrm{Hom}\}_{\{C\}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak}\{C\}^{\mathrm{C}}\backslash\mathrm{right})\ \backslash\mathrm{mid}(\mathrm{D}\ \mathrm{x},\ \mathrm{y})+(\mathrm{x},\ \mathrm{D}\ \mathrm{y})=0\backslash\mathrm{right}\backslash\}\ .$	$\mathfrak{D}_4^C = \{D \in \mathrm{Hom}_C(\mathfrak{C}^C) \mid (Dx, y) + (x, Dy) = 0\}.$	$\mathfrak{D}_4^C = \{D \in \mathrm{Hom}_C(\mathfrak{C}^C) \mid (Dx, y) + (x, Dy) = 0\}.$
0902.0431_E00327	52	■ 1.000	$\backslash\mathrm{mathfrak}\{\mathfrak{h}\}=\backslash\mathrm{left}\backslash\{\mathrm{H}=\backslash\mathrm{sum}_{\{k=0\}^{\{3\}}\ \backslash\mathrm{lambda}_{\{i\}}\ \mathrm{H}_{\{k\}}\ \backslash\mathrm{mid}\ \backslash\mathrm{lambda}_{\{k\}}\ \backslash\mathrm{in}\ C\backslash\mathrm{right}\backslash\}$	$\mathfrak{h} = \left\{ H = \sum_{k=0}^3 \lambda_i H_k \mid \lambda_k \in C \right\}$	$\mathfrak{h} = \{ H = \sum_{k=0}^3 \lambda_i H_k \mid \lambda_k \in C \}$

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0902.0431_E00331	53	■ 1.000	$\nu=\pi \kappa=\frac{1}{2}\left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2}\left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$	$\nu=\pi\kappa=\frac{1}{2}\left(\begin{array}{rrrr} -1 & -1 & 1 & -1 \\ 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \end{array}\right), \quad \pi=\frac{1}{2}\left(\begin{array}{rrrr} 1 & -1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{array}\right)$	
0902.0431_E00334	53	■ 1.000	$\pm\lambda_k-\lambda_l, \quad \pm(\lambda_k+\lambda_l), \quad 0\leq k<l\leq 3$	$\pm(\lambda_k-\lambda_l), \quad \pm(\lambda_k+\lambda_l), \quad 0\leq k<l\leq 3$	$\pm(\lambda_k-\lambda_l), \quad \pm(\lambda_k+\lambda_l), \quad 0\leq k<l\leq 3$
0902.0431_E00336	54	■ 1.000	$\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3\lambda_kH_k\right) \\ & =\frac{1}{2}(\lambda_0(-H_0+H_1-H_2+H_3)+\lambda_1(-H_0+H_1+H_2-H_3) \\ & \quad +\lambda_2(H_0+H_1+H_2+H_3)+\lambda_3(-H_0-H_1+H_2+H_3)) \\ & =\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3)H_0+\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1 \\ & \quad +\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3, \end{aligned}$	$\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3\lambda_kH_k\right) \\ & =\frac{1}{2}(\lambda_0(-H_0+H_1-H_2+H_3)+\lambda_1(-H_0+H_1+H_2-H_3) \\ & \quad +\lambda_2(H_0+H_1+H_2+H_3)+\lambda_3(-H_0-H_1+H_2+H_3)) \\ & =\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3)H_0+\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1 \\ & \quad +\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3, \end{aligned}$	$\begin{aligned} \nu H = & \nu\left(\sum_{k=0}^3\lambda_kH_k\right) \\ & =\frac{1}{2}(\lambda_0(-H_0+H_1-H_2+H_3)+\lambda_1(-H_0+H_1+H_2-H_3) \\ & \quad +\lambda_2(H_0+H_1+H_2+H_3)+\lambda_3(-H_0-H_1+H_2+H_3)) \\ & =\frac{1}{2}(-\lambda_0-\lambda_1+\lambda_2-\lambda_3)H_0+\frac{1}{2}(\lambda_0+\lambda_1+\lambda_2-\lambda_3)H_1 \\ & \quad +\frac{1}{2}(-\lambda_0+\lambda_1+\lambda_2+\lambda_3)H_2+\frac{1}{2}(\lambda_0-\lambda_1+\lambda_2+\lambda_3)H_3, \end{aligned}$
0902.0431_E00338	54	■ 1.000	$\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)$	$\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)$	$\alpha_1=\lambda_0-\lambda_1, \quad \alpha_2=\lambda_1-\lambda_2, \quad \alpha_3=\lambda_2, \quad \alpha_4=\frac{1}{2}(-\lambda_0-\lambda_1-\lambda_2+\lambda_3)$
0902.0431_E00339	54	■ 1.000	$\mu=2\alpha_1+3\alpha_2+4\alpha_3+2\alpha_4$	$\mu=2\alpha_1+3\alpha_2+4\alpha_3+2\alpha_4$	$\mu=2\alpha_1+3\alpha_2+4\alpha_3+2\alpha_4$
0902.0431_E00342	55	■ 1.000	$\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3\lambda_kH_k\mid\lambda_k\in\boldsymbol{R}\right\}.$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3\lambda_kH_k\mid\lambda_k\in\boldsymbol{R}\right\}.$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{H=\sum_{k=0}^3\lambda_kH_k\mid\lambda_k\in\boldsymbol{R}\right\}.$
0902.0431_E00347	56	■ 1.000	$\left(\alpha_1,\alpha_1\right)=B_4\left(H_{\alpha_1},H_{\alpha_1}\right)=18\frac{1}{18}\frac{1}{18}(1+1)=\frac{1}{9}$	$\left(\alpha_1,\alpha_1\right)=B_4\left(H_{\alpha_1},H_{\alpha_1}\right)=18\frac{1}{18}\frac{1}{18}(1+1)=\frac{1}{9}$	$\left(\alpha_1,\alpha_1\right)=B_4\left(H_{\alpha_1},H_{\alpha_1}\right)=18\frac{1}{18}\frac{1}{18}(1+1)=\frac{1}{9}$
0902.0431_E00352	57	■ 1.000	$\alpha F_i(x)=F_i(\alpha_i x), \quad x\in\mathfrak{C},$	$\alpha F_i(x)=F_i(\alpha_i x), \quad x\in\mathfrak{C},$	$\alpha F_i(x)=F_i(\alpha_i x), \quad x\in\mathfrak{C},$
0902.0431_E00353	57	■ 1.000	$\left(\alpha_i x,\alpha_i y\right)=\left(x,y\right), \quad x,y\in\mathfrak{C}.$	$\left(\alpha_i x,\alpha_i y\right)=\left(x,y\right), \quad x,y\in\mathfrak{C}.$	$\left(\alpha_i x,\alpha_i y\right)=\left(x,y\right), \quad x,y\in\mathfrak{C}.$
0902.0431_E00355	57	■ 1.000	$\left(F_4\right)_{E_1}=\left\{\alpha\in F_4\mid\alpha E_1=E_1\right\}.$	$\left(F_4\right)_{E_1}=\left\{\alpha\in F_4\mid\alpha E_1=E_1\right\}.$	$\left(F_4\right)_{E_1}=\left\{\alpha\in F_4\mid\alpha E_1=E_1\right\}.$

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0902.0431_E00356	57	■ 1.000	$\mathfrak{J}_{01}=\left\{X\in\mathfrak{J}\mid\mathfrak{J}\circ X=0,\mathrm{tr}(X)=0\right\}$ $\circ X=0$, $\mathrm{tr}(X)=0$ $\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)$ $\xi\in\boldsymbol{R},x\in\mathfrak{C}$	$\mathfrak{J}_{01}=\{X\in\mathfrak{J}\mid E_1\circ X=0,\mathrm{tr}(X)=0\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)\middle \xi\in\boldsymbol{R},x\in\mathfrak{C}\right\},$	$\mathfrak{J}_{01}=\{X\in\mathfrak{J}\mid E_1\circ X=0,\mathrm{tr}(X)=0\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\xi\end{array}\right)\middle \xi\in\boldsymbol{R},x\in\mathfrak{C}\right\},$
0902.0431_E00357	58	■ 1.000	$\mathfrak{J}_{23}=\left\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\right\}$ $Y=Y$ $\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)$ $y_2,y_3\in\mathfrak{C}$	$\mathfrak{J}_{23}=\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\}=\left\{\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)\middle y_2,y_3\in\mathfrak{C}\right\}.$	$\mathfrak{J}_{23}=\{Y\in\mathfrak{J}\mid 2E_1\circ Y=Y\}=\left\{\left(\begin{array}{ccc}0&y_3&\bar{y}_2\\y_3&0&0\\y_2&0&0\end{array}\right)\middle y_2,y_3\in\mathfrak{C}\right\}.$
0902.0431_E00358	58	■ 1.000	$Z_0=2X_0\circ Y_0$	$Z_0=2X_0\circ Y_0$	$Z_0=2X_0\circ Y_0$
0902.0431_E00359	58	■ 1.000	$Y_1=-2Z_0\circ X_1$, $Z_2=-2X_2\circ Y_0$, $X_3=-2Y_1\circ Z_2$	$Y_1=-2Z_0\circ X_1$, $Z_2=-2X_2\circ Y_0$, $X_3=-2Y_1\circ Z_2$	$Y_1=-2Z_0\circ X_1$, $Z_2=-2X_2\circ Y_0$, $X_3=-2Y_1\circ Z_2$
0902.0431_E00360	58	■ 1.000	$Z_4=-2X_4\circ Y_0$, $Y_2=-2Z_0\circ X_2$, $Y_3=-2Z_0\circ X_3$, $X_5=-2Y_1\circ Z_4$, $X_6=2Y_2\circ Z_4$, $X_7=-2Y_3\circ Z_4$	$Z_4=-2X_4\circ Y_0$, $Y_2=-2Z_0\circ X_2$, $Y_3=-2Z_0\circ X_3$, $X_5=-2Y_1\circ Z_4$, $X_6=2Y_2\circ Z_4$, $X_7=-2Y_3\circ Z_4$	$Z_4=-2X_4\circ Y_0$, $Y_2=-2Z_0\circ X_2$, $Y_3=-2Z_0\circ X_3$, $X_5=-2Y_1\circ Z_4$, $X_6=2Y_2\circ Z_4$, $X_7=-2Y_3\circ Z_4$
0902.0431_E00363	58	■ 1.000	$\alpha(X\circ Y)=\alpha X\circ\alpha Y$, $X,Y\in\mathfrak{J}$	$\alpha(X\circ Y)=\alpha X\circ\alpha Y$, $X,Y\in\mathfrak{J}$	$\alpha(X\circ Y)=\alpha X\circ\alpha Y$, $X,Y\in\mathfrak{J}$
0902.0431_E00366	59	■ 1.000	$(F_4)_{E_1}/\{1,\sigma\}\cong SO(9)$	$(F_4)_{E_1}/\{1,\sigma\}\cong SO(9)$	$(F_4)_{E_1}/\{1,\sigma\}\cong SO(9)$
0902.0431_E00367	60	■ 1.000	$f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$, $(\epsilon x=\bar{x},x\in\mathfrak{C})$	$f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$, $(\epsilon x=\bar{x},x\in\mathfrak{C})$	$f(\alpha')=\varepsilon\alpha'\varepsilon^{-1}$, $(\epsilon x=\bar{x},x\in\mathfrak{C})$
0902.0431_E00369	61	■ 1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$
0902.0431_E00373	62	■ 1.000	$\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}$	$\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}$	$\mathfrak{C}P_2=\{X\in\mathfrak{J}\mid X^2=X,\mathrm{tr}(X)=1\}$
0902.0431_E00375	62	■ 1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$, $\xi_i\in\boldsymbol{R}$
0902.0431_E00376	62	■ 1.000	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$	$\left(\begin{array}{ccc}\xi_1^2&0&0\\0&\xi_2^2&0\\0&0&\xi_3^2\end{array}\right)=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right)$
0902.0431_E00377	63	■ 1.000	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right)$	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right)$	$\sigma\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\x_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\left(\begin{array}{ccc}\xi_1&-x_3&-\bar{x}_2\\-x_3&\xi_2&x_1\\-x_2&\bar{x}_1&\xi_3\end{array}\right)$

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0902.0431_E00378	63	■ 1.000	$\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \alpha \sigma=\sigma \alpha\right\} .$	$\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \sigma \alpha=\alpha \sigma\right\} .$	$\left(F_4\right)^{\sigma}=\left\{\alpha \in F_4 \mid \sigma \alpha=\alpha \sigma\right\} .$
0902.0431_E00380	63	■ 1.000	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$	$\alpha E_2, \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}) .$
0902.0431_E00382	63	■ 1.000	$\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$	$\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$	$\alpha E_1=E_1+\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right) .$
0902.0431_E00385	64	■ 1.000	$z\left(F_4\right)=\{1\} .$	$z\left(F_4\right)=\{1\} .$	$z\left(F_4\right)=\{1\} .$
0902.0431_E00386	64	■ 1.000	$\alpha=1 \quad \text { or } \quad \alpha=\sigma .$	$\alpha=1 \quad \text { or } \quad \alpha=\sigma .$	$\alpha=1 \quad \text { or } \quad \alpha=\sigma .$
0902.0431_E00388	65	■ 1.000	$\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$	$\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$	$\left(F_4\right)^{\gamma}=\left\{\alpha \in F_4 \mid \gamma \alpha=\alpha \gamma\right\} .$
0902.0431_E00389	65	■ 1.000	$X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$	$X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$	$X=\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(\begin{array}{ccc} 0 & a_3 e_4 & -a_2 e_4 \\ -a_3 e_4 & 0 & a_1 e_4 \\ a_2 e_4 & -a_1 e_4 & 0 \end{array}\right),$
0902.0431_E00390	65	■ 1.000	$\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$	$\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$	$\left(\begin{array}{ccc} \xi_1 & m_3 & \bar{m}_2 \\ \bar{m}_3 & \xi_2 & m_1 \\ m_2 & \bar{m}_1 & \xi_3 \end{array}\right)+\left(a_1, a_2, a_3\right) .$
0902.0431_E00391	65	■ 1.000	$\begin{aligned} (M+\boldsymbol{a}) \times(N+\boldsymbol{b}) &=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M), \\ (M+\boldsymbol{a}, N+\boldsymbol{b}) &=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}), \\ \gamma(M+\boldsymbol{a}) &=M-\boldsymbol{a} . \end{aligned}$	$\begin{aligned} (M+\boldsymbol{a}) \times(N+\boldsymbol{b}) &=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M), \\ (M+\boldsymbol{a}, N+\boldsymbol{b}) &=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}), \\ \gamma(M+\boldsymbol{a}) &=M-\boldsymbol{a} . \end{aligned}$	$\begin{aligned} (M+\boldsymbol{a}) \times(N+\boldsymbol{b}) &=\left(M \times N-\frac{1}{2}\left(\boldsymbol{a}^* \boldsymbol{b}+\boldsymbol{b}^* \boldsymbol{a}\right)\right)-\frac{1}{2}(\boldsymbol{a} N+\boldsymbol{b} M), \\ (M+\boldsymbol{a}, N+\boldsymbol{b}) &=(M, N)+2(\boldsymbol{a}, \boldsymbol{b}), \\ \gamma(M+\boldsymbol{a}) &=M-\boldsymbol{a} . \end{aligned}$
0902.0431_E00392	65	■ 1.000	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$	$\mathfrak{J}(3, \boldsymbol{H}) \oplus \boldsymbol{H}^3=\mathfrak{J} .$
0902.0431_E00394	65	■ 1.000	$\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$	$\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$	$\varphi(A) M=A M A^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}} .$
0902.0431_E00395	66	■ 1.000	$\alpha E_i=A E_i A^*, \quad i=1,2,3 .$	$\alpha E_i=A E_i A^*, \quad i=1,2,3 .$	$\alpha E_i=A E_i A^*, \quad i=1,2,3 .$
0902.0431_E00396	66	■ 1.000	$\beta E_i=E_i, \quad i=1,2,3 .$	$\beta E_i=E_i, \quad i=1,2,3 .$	$\beta E_i=E_i, \quad i=1,2,3 .$
0902.0431_E00397 (i)	66	■ 1.000	$\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$	$\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$	$\left(\beta_1 m\right)\left(\beta_2 n\right)=\overline{\beta_3(\overline{m n})}, \quad m, n \in \boldsymbol{H} .$
0902.0431_E00398	66	■ 1.000	$\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$	$\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$	$\beta_2 m=\bar{p}\left(\beta_1 m\right) q, \quad \beta_3 m=\overline{\left(\beta_1 \bar{m}\right)} q .$
0902.0431_E00399	66	■ 1.000	$(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$	$(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$	$(\zeta m)(\zeta n)=\zeta(m n), \quad m, n \in \boldsymbol{H},$
0902.0431_E00401	66	■ 1.000	$\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$	$\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$	$\beta M=B M B^*, \quad M \in \mathfrak{J}_{\boldsymbol{H}},$

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0902.0431_E00403	67	■ 1.000	$\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$ $\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$	$\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$	$\varphi(p, A)(M+\boldsymbol{a})=A\,M\,A^*+p\boldsymbol{a}\,A^*, \quad M+\boldsymbol{a}\in\mathfrak{J}_H\oplus\boldsymbol{H}^3=\mathfrak{J}.$
0902.0431_E00406	67	■ 1.000	$\alpha\,M=A\,M\,A^*, \quad M\in\mathfrak{J}_H$	$\alpha M=A\,M\,A^*, \quad M\in\mathfrak{J}_H$	$\alpha M=A\,M\,A^*, \quad M\in\mathfrak{J}_H$
0902.0431_E00407	67	■ 1.000	$\beta(m+a\boldsymbol{e}_4)=m+(p\boldsymbol{a})\boldsymbol{e}_4, \quad m+a\boldsymbol{e}_4\in\boldsymbol{H}\oplus\boldsymbol{H}\boldsymbol{e}_4=\mathfrak{C},$ $\boldsymbol{e}_4\in\boldsymbol{H}\oplus\boldsymbol{H}\boldsymbol{e}_4=\mathfrak{C},$	$\beta(m+a\boldsymbol{e}_4)=m+(p\boldsymbol{a})\boldsymbol{e}_4, \quad m+a\boldsymbol{e}_4\in\boldsymbol{H}\oplus\boldsymbol{H}\boldsymbol{e}_4=\mathfrak{C},$	$\beta(m+a\boldsymbol{e}_4)=m+(p\boldsymbol{a})\boldsymbol{e}_4, \quad m+a\boldsymbol{e}_4\in\boldsymbol{H}\oplus\boldsymbol{H}\boldsymbol{e}_4=\mathfrak{C},$
0902.0431_E00408	67	■ 1.000	$\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$	$\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$	$\beta(M+\boldsymbol{a})=M+p\boldsymbol{a}=\varphi(p, E)(M+\boldsymbol{a}),$
0902.0431_E00411	68	■ 1.000	$(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$ $\alpha\in F_4\mid w\alpha=\alpha w\}.$	$(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$	$(F_4)^w=\{\alpha\in F_4\mid w\alpha=\alpha w\}.$
0902.0431_E00415	68	■ 1.000	$P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)^t\widetilde{A},$	$P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)^t\widetilde{A},$	$P\,M\times P\,N=\widetilde{P}(M\times N), \quad M\,A\times N\,A=(M\times N)^t\widetilde{A},$
0902.0431_E00416	68	■ 1.000	$(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$ $(M, N)=\sum_{i,j}(m_{ij}, n_{ij}),$	$(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$	$(M, N)=\frac{1}{2}\operatorname{tr}(M^*N+N^*M)=\sum_{i,j}(m_{ij}, n_{ij}),$
0902.0431_E00418	69	■ 1.000	$\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}$	$\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}$	$\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$
0902.0431_E00419	69	■ 1.000	$\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\circ Y)=\alpha X\circ\alpha Y\} \\ &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\times Y)=\alpha X\times\alpha Y\}. \end{aligned}$	$\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\circ Y)=\alpha X\circ\alpha Y\} \\ &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\times Y)=\alpha X\times\alpha Y\}. \end{aligned}$	$\begin{aligned} F_{4, \boldsymbol{C}} &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\circ Y)=\alpha X\circ\alpha Y\} \\ &= \{\alpha\in\operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}})\mid \alpha(X\times Y)=\alpha X\times\alpha Y\}. \end{aligned}$
0902.0431_E00420	69	■ 1.000	$\epsilon A=\overline{A}, \quad A\in SU(3),$	$\epsilon A=\overline{A}, \quad A\in SU(3),$	$\epsilon A=\overline{A}, \quad A\in SU(3),$
0902.0431_E00422	69	■ 1.000	$\beta_1x=px, \beta_2x=xq, \beta_3x=\overline{q}x\overline{p} \quad \text{or} \quad \beta_1x=p\overline{x}, \beta_2x=\overline{x}q, \beta_3x=\overline{q}\overline{x}\overline{p}.$	$\beta_1x=px, \beta_2x=xq, \beta_3x=\overline{q}x\overline{p} \quad \text{or} \quad \beta_1x=p\overline{x}, \beta_2x=\overline{x}q, \beta_3x=\overline{q}\overline{x}\overline{p}.$	$\beta_1x=px, \beta_2x=xq, \beta_3x=\overline{q}x\overline{p} \quad \text{or} \quad \beta_1x=p\overline{x}, \beta_2x=\overline{x}q, \beta_3x=\overline{q}\overline{x}\overline{p}.$
0902.0431_E00424	69	■ 1.000	$\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$ $\boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$	$\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$	$\varphi(P, A)(X+M)=A\,X\,A^*+P\,M\,A^*, \quad X+M\in\mathfrak{J}(3, \boldsymbol{C})\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$
0902.0431_E00427	70	■ 1.000	$\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\,\overline{X}\,A^*, \quad X\in\mathfrak{J}_{\boldsymbol{C}}$	$\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\,\overline{X}\,A^*, \quad X\in\mathfrak{J}_{\boldsymbol{C}}$	$\alpha X=A\,X\,A^* \quad \text{or} \quad \alpha X=A\,\overline{X}\,A^*, \quad X\in\mathfrak{J}_{\boldsymbol{C}}$
0902.0431_E00428	70	■ 1.000	$\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_{\boldsymbol{C}}\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$	$\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_{\boldsymbol{C}}\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$	$\beta(X+M)=X+P\,M=\varphi(P, E)(X+M), \quad X+M\in\mathfrak{J}_{\boldsymbol{C}}\oplus M(3, \boldsymbol{C})=\mathfrak{J}.$
0902.0431_E00429	70	■ 1.000	$\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$	$\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$	$\alpha=\varphi(E, A)\beta=\varphi(E, A)\varphi(P, E)=\varphi(P, A).$
0902.0431_E00430	70	■ 1.000	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\widetilde{A}_1(1), \quad \widetilde{A}_2(1), \quad \widetilde{A}_3(1), \quad \widetilde{A}_1(e_1), \quad \widetilde{A}_2(e_1), \quad \widetilde{A}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\widetilde{A}_1(1), \quad \widetilde{A}_2(1), \quad \widetilde{A}_3(1), \quad \widetilde{A}_1(e_1), \quad \widetilde{A}_2(e_1), \quad \widetilde{A}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26}+G_{37}, \quad -G_{27}+G_{36}, \\ &G_{24}+G_{35}, \quad -G_{25}+G_{34}, \quad G_{46}+G_{57}, \quad -G_{47}+G_{56}, \\ &\widetilde{A}_1(1), \quad \widetilde{A}_2(1), \quad \widetilde{A}_3(1), \quad \widetilde{A}_1(e_1), \quad \widetilde{A}_2(e_1), \quad \widetilde{A}_3(e_1) \end{aligned}$

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0902.0431_E00432	71	■ 1.000	$\backslash\langle X, Y\rangle=\backslash\tau X, Y) \ .$	$\langle X, Y\rangle=(\tau X, Y).$	$\langle X, Y\rangle=(\tau X, Y).$
0902.0431_E00437	72	■ 1.000	$X \circ Y=\frac{1}{2}(X\,Y+Y\,X)$	$X\circ Y=\frac{1}{2}(XY+YX)$	$X\circ Y=\frac{1}{2}(XY+YX).$
0902.0431_E00439	72	■ 1.000	$\begin{aligned} F_{4(4)} &= \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right\}, \\ F_{4(-20)} &= \left\{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \right\}. \end{aligned}$	$F_{4(4)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \},$ $F_{4(-20)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$	$F_{4(4)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(3, \mathfrak{C}')) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \},$ $F_{4(-20)} = \{ \alpha \in \operatorname{Iso}_{\boldsymbol{R}}(\mathfrak{J}(1, 2, \mathfrak{C})) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y \}.$
0902.0431_E00442	72	■ 1.000	$z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\} \ .$	$z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\}.$	$z\left(F_{4(4)}\right)=\{1\}, \quad z\left(F_{4(-20)}\right)=\{1\}.$
0902.0431_E00444	73	■ 1.000	$\lambda(\alpha)=\{ \ }^t\alpha^{-1}, \quad \alpha \in E_6^{\,C}$	$\lambda(\alpha)=\,^t\alpha^{-1}, \quad \alpha \in E_6^{\,C}$	$\lambda(\alpha)=\,^t\alpha^{-1}, \quad \alpha \in E_6^{\,C}$
0902.0431_E00445	73	■ 1.000	$E_6=\left\{\alpha \in E_6^{\,C} \mid \tau \lambda(\alpha) \tau=\alpha\right\}=\left(E_6^{\,C}\right)^{\tau \lambda}.$	$E_6=\left\{\alpha \in E_6^{\,C} \mid \tau \lambda(\alpha) \tau=\alpha\right\}=\left(E_6^{\,C}\right)^{\tau \lambda}.$	$E_6=\left\{\alpha \in E_6^{\,C} \mid \tau \lambda(\alpha) \tau=\alpha\right\}=\left(E_6^{\,C}\right)^{\tau \lambda}.$
0902.0431_E00446	73	■ 1.000	$U(27)=U\left(\mathfrak{J}^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid\left\langle\alpha X, \alpha Y\right\rangle=\langle X, Y\rangle\right\}.$	$U(27)=U\left(\mathfrak{J}^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid\left\langle\alpha X, \alpha Y\right\rangle=\langle X, Y\rangle\right\}.$	$U(27)=U\left(\mathfrak{J}^C\right)=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid\left\langle\alpha X, \alpha Y\right\rangle=\langle X, Y\rangle\right\}.$
0902.0431_E00451	74	■ 1.000	$\left[\delta_1+\widetilde{T}_1, \delta_2+\widetilde{T}_2\right]=\left(\left[\delta_1, \delta_2\right]+\left[\widetilde{T}_1, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1 T_2}-\widetilde{\delta_2 T_1}\right),$	$\left[\delta_1+\widetilde{T}_1, \delta_2+\widetilde{T}_2\right]=\left(\left[\delta_1, \delta_2\right]+\left[\widetilde{T}_1, \widetilde{T}_2\right]\right)+\left(\widetilde{\delta_1 T_2}-\widetilde{\delta_2 T_1}\right),$	$[\delta_1+\widetilde{T}_1, \delta_2+\widetilde{T}_2]=([\delta_1, \delta_2]+[\widetilde{T}_1, \widetilde{T}_2])+(\widetilde{\delta_1 T_2}-\widetilde{\delta_2 T_1}),$
0902.0431_E00453	74	■ 1.000	$\lambda(\phi)=-\{ \ }^t\phi=-\{ \ }^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$	$\lambda(\phi)=-\,^t\phi=-\,^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$	$\lambda(\phi)=-\,^t\phi=-\,^t(\delta+\widetilde{T})=\delta-\widetilde{T}.$
0902.0431_E00455	74	■ 1.000	$\mathfrak{e}_6=\left\{\phi \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0,\left\langle\phi X, Y\right\rangle+\langle X, \phi Y\rangle=0\right\}.$	$\mathfrak{e}_6=\left\{\phi \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0,\left\langle\phi X, Y\right\rangle+\langle X, \phi Y\rangle=0\right\}.$	$\mathfrak{e}_6=\left\{\phi \in \operatorname{Hom}_C\left(\mathfrak{J}^C\right) \mid\left(\phi X, X, X\right)=0,\left\langle\phi X, Y\right\rangle+\langle X, \phi Y\rangle=0\right\}.$
0902.0431_E00456	74	■ 1.000	$\phi=\delta+i \widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$	$\phi=\delta+i \widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$	$\phi=\delta+i \widetilde{T}, \quad \delta \in \mathfrak{f}_4, T \in \mathfrak{J}_0.$
0902.0431_E00461	76	■ 1.000	$A \circ X=\left(A-\frac{1}{3} \operatorname{tr}(A) E\right)^{\sim} X+\frac{1}{3} \operatorname{tr}(A) X \in W.$	$A\circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X\in W.$	$A\circ X=\left(A-\frac{1}{3}\operatorname{tr}(A)E\right)^{\sim}X+\frac{1}{3}\operatorname{tr}(A)X\in W.$
0902.0431_E00462	76	■ 1.000	$A \vee B=\left[\widetilde{A}, \widetilde{B}\right]+\left(A \circ B-\frac{1}{3}(A, B) E\right)^{\sim}$	$A\vee B=[\widetilde{A},\widetilde{B}]+\left(A\circ B-\frac{1}{3}(A,B)E\right)^{\sim}$	$A\vee B=[\widetilde{A},\widetilde{B}]+\left(A\circ B-\frac{1}{3}(A,B)E\right)^{\sim}$
0902.0431_E00463	76	■ 1.000	$(A \vee B) X=\frac{1}{2}(B, X) A+\frac{1}{6}(A, B) X-2 B \times(A \times X), \quad X \in \mathfrak{J}^C.$	$(A\vee B)X=\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X), \quad X\in\mathfrak{J}^C.$	$(A\vee B)X=\frac{1}{2}(B,X)A+\frac{1}{6}(A,B)X-2B\times(A\times X), \quad X\in\mathfrak{J}^C.$
0902.0431_E00464	76	■ 1.000	$X \circ(X \circ X)-\operatorname{tr}(X) X \circ X+\frac{1}{2}\left(\operatorname{tr}(X)^2-(X, X)\right) X=\frac{1}{3}(X, X, X) E.$	$X\circ(X\circ X)-\operatorname{tr}(X)X\circ X+\frac{1}{2}\left(\operatorname{tr}(X)^2-(X,X)\right)X=\frac{1}{3}(X,X,X)E.$	$X\circ(X\circ X)-\operatorname{tr}(X)X\circ X+\frac{1}{2}(\operatorname{tr}(X)^2-(X,X))X=\frac{1}{3}(X,X,X)E.$

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0902.0431_E00467	77	■ 1.000	$\left[\left(E_1\vee E_1\right)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).\right]$	$(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$	$(E_1\vee E_1)\left(\begin{array}{ccc}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{array}\right)=\frac{1}{6}\left(\begin{array}{ccc}4\xi_1&x_3&\bar{x}_2\\\bar{x}_3&-2\xi_2&-2x_1\\x_2&-2\bar{x}_1&-2\xi_3\end{array}\right).$
0902.0431_E00470	77	■ 1.000	$\left[(A\vee B)^\prime=-B\vee A.\right]$	$(A\vee B)^\prime=-B\vee A.$	$(A\vee B)^\prime=-B\vee A.$
0902.0431_E00471	78	■ 1.000	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B.$	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B.$	$[\phi,A\vee B]=\phi A\vee B+A\vee\phi^\prime B.$
0902.0431_E00475	78	■ 1.000	$\left[(\phi,A\vee B)_6=(\phi A,B).\right]$	$(\phi,A\vee B)_6=(\phi A,B).$	$(\phi,A\vee B)_6=(\phi A,B).$
0902.0431_E00476	78	■ 1.000	$\begin{aligned}&\left(\left[\phi,\phi_1\right],\phi_2\right)_6\\&=\left(\left[\delta+\tilde{T},\delta_1+\tilde{T}_1\right],\delta_2+\tilde{T}_2\right)_6\end{aligned}$	$\begin{aligned}&([\phi,\phi_1],\phi_2)_6\\&=\left(\left[\delta+\tilde{T},\delta_1+\tilde{T}_1\right],\delta_2+\tilde{T}_2\right)_6\end{aligned}$	$\begin{aligned}&([\phi,\phi_1],\phi_2)_6\\&=([\delta+\tilde{T},\delta_1+\tilde{T}_1],\delta_2+\tilde{T}_2)_6\end{aligned}$
0902.0431_E00481	79	■ 1.000	$B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$	$B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$	$B_6\left(\phi_1,\phi_2\right)=k\left(\phi_1,\phi_2\right)_6=k'\operatorname{tr}\left(\phi_1\phi_2\right).$
0902.0431_E00482	79	■ 1.000	$\left(\phi,\phi\right)_6=\left(\left(E_1-E_2\right)^\sim,\left(E_1-E_2\right)^\sim\right)_6=\left(E_1-E_2,E_1-E_2\right)=2.$	$(\phi,\phi)_6=((E_1-E_2)^\sim,(E_1-E_2)^\sim)_6=(E_1-E_2,E_1-E_2)=2.$	$(\phi,\phi)_6=((E_1-E_2)^\sim,(E_1-E_2)^\sim)_6=(E_1-E_2,E_1-E_2)=2.$
0902.0431_E00486	80	■ 1.000	$\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$	$\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$	$\begin{array}{l} \phi\phi E_1=\phi E_1=E_1, \quad \phi\phi\tilde{F}_1(e_i)=-\frac{1}{2}\phi\tilde{F}_1(e_i)=\frac{1}{4}\tilde{F}_1(e_i), \\ \phi\phi E_2=-\phi E_2=E_2, \quad \phi\phi\tilde{F}_2(e_i)=\frac{1}{2}\phi\tilde{F}_2(e_i)=\frac{1}{4}\tilde{F}_2(e_i), \\ \phi\phi E_3=\phi 0=0, \quad \phi\phi\tilde{F}_3(e_i)=\phi 0=0. \end{array}$
0902.0431_E00488	80	■ 1.000	$A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$	$A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$	$A\vee(B\times C)+B\vee(C\times A)+C\vee(A\times B)=0, \quad A,B,C\in\mathfrak{J}^C.$
0902.0431_E00490	80	■ 1.000	$\mathfrak{M}^r=\{X\in M(3,\mathfrak{C})\mid\text{all diagonal elements of }X\text{ are real}\},$	$\mathfrak{M}^r=\{X\in M(3,\mathfrak{C})\mid\text{all diagonal elements of }X\text{ are real}\},$	$\mathfrak{M}^r=\{X\in M(3,\mathfrak{C})\mid\text{all diagonal elements of }X\text{ are real}\},$
0902.0431_E00491	81	■ 1.000	$X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right)$	$X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right)$	$X\circ Y=\frac{1}{2}\left(XY+Y^*X^*\right),$
0902.0431_E00498	81	■ 1.000	$\begin{aligned}&\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$	$\begin{aligned}&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$	$\begin{aligned}&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\overline{\sigma_{ji}(x_{jk}y_{ki})}\\&=\sum_k\sigma_{ij}(x_{ik}y_{kj})+\sum_k\sigma_{ij}(\overline{x_{jk}y_{ki}})=(i,j)\text{-element of }\delta(X\circ Y).\end{aligned}$
0902.0431_E00500	82	■ 1.000	$[\delta,\widetilde{R}]=\widetilde{\delta R}.$	$[\delta,\widetilde{R}]=\widetilde{\delta R}.$	$[\delta,\widetilde{R}]=\widetilde{\delta R}.$
0902.0431_E00501	82	■ 1.000	$\left[\widetilde{H},\widetilde{T}\right]=\frac{1}{2}[H,T]^\sim.$	$[\widetilde{H},\widetilde{T}]=\frac{1}{2}[H,T]^\sim.$	$[\widetilde{H},\widetilde{T}]=\frac{1}{2}[H,T]^\sim.$

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0902.0431_E00503	82	■ 1.000	$=\frac{1}{4}\left([H,T]X+X[H,T]^*\right)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$	$=\frac{1}{4}([H,T]X+X[H,T]^*)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$	$=\frac{1}{4}([H,T]X+X[H,T]^*)=\frac{1}{2}[H,T]\sim X,\quad X\in\mathfrak{J}^C.$
0902.0431_E00510	84	■ 1.000	$\left[h,\widetilde{F}_{32}(a)\right]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$	$\left[h,\widetilde{F}_{32}(a)\right]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$	$[h,\widetilde{F}_{32}(a)]=\left(-\lambda_k+\frac{1}{2}(\mu_3-\mu_2)\right)\widetilde{F}_{32}(a),\quad a=e_k+ie_{4+k},$
0902.0431_E00514	84	■ 1.000	$\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$	$\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$	$\mu=\alpha_1+2\alpha_2+3\alpha_3+2\alpha_4+2\alpha_5+\alpha_6$
0902.0431_E00519	86	■ 1.000	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\sum_{k=0}^3\lambda_kH_k+\left(\sum_{j=1}^3\mu_jE_j\right)^{\sim}\mid\lambda_k,\mu_j\in\boldsymbol{R},\mu_1+\mu_2+\mu_3=0\right\}.$
0902.0431_E00523	86	■ 1.000	$\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$	$\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$	$\left(\alpha_1,\alpha_1\right)=B_6\left(H_{\alpha_1},H_{\alpha_1}\right)=24\frac{1}{24}\frac{1}{24}2=\frac{1}{12},$
0902.0431_E00525	87	■ 1.000	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = \left(E_6\right)^{\lambda}. \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = \left(E_6\right)^{\lambda}. \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\tau\alpha=\alpha\tau\right\} \\ &= \left\{\alpha\in E_6\mid\lambda(\alpha)=\alpha\right\} = \left(E_6\right)^{\lambda}. \end{aligned}$
0902.0431_E00526	87	■ 1.000	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\left(\alpha X,\alpha Y\right)=\left(X,Y\right),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = \left(E_6\right)_E. \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\left(\alpha X,\alpha Y\right)=\left(X,Y\right),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = \left(E_6\right)_E. \end{aligned}$	$\begin{aligned} \left(E_6\right)^{\tau} &= \left\{\alpha\in E_6\mid\left(\alpha X,\alpha Y\right)=\left(X,Y\right),X,Y\in\mathfrak{J}^C\right\} \\ &= \left\{\alpha\in E_6\mid\alpha E=E\right\} = \left(E_6\right)_E. \end{aligned}$
0902.0431_E00527	88	■ 1.000	$F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,\left(\alpha X,\alpha Y\right)=\left(X,Y\right)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$	$F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,\left(\alpha X,\alpha Y\right)=\left(X,Y\right)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$	$F_4=\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\det(\alpha X)=\det X,\left(\alpha X,\alpha Y\right)=\left(X,Y\right)\right\} \\ =\left\{\alpha\in\mathrm{Iso}_{\boldsymbol{R}}(\mathfrak{J})\mid\alpha(X\times Y)=\alpha X\times\alpha Y\right\}.$
0902.0431_E00528	88	■ 1.000	$\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$	$\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$	$\alpha_{12}(t)\begin{pmatrix}\xi_1&x_3&\bar{x}_2\\\bar{x}_3&\xi_2&x_1\\x_2&\bar{x}_1&\xi_3\end{pmatrix}=\begin{pmatrix}e^{it}\xi_1&x_3&e^{it/2}\bar{x}_2\\\bar{x}_3&e^{-it}\xi_2&e^{-it/2}x_1\\e^{it/2}x_2&e^{-it/2}\bar{x}_1&\xi_3\end{pmatrix},$
0902.0431_E00533	90	■ 1.000	$EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$	$EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$	$EIV=\left\{X\in\mathfrak{J}^C\mid\det X=1,\langle X,X\rangle=3\right\}.$

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0902.0431_E00534	90	■ 1.000	$\alpha X=\left(\begin{array}{ccc}\xi_1&\theta&\theta\\\xi_2&\theta&\theta\\\xi_3&\theta&\theta\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$	$\alpha X=\left(\begin{array}{ccc}\xi_1&0&0\\0&\xi_2&0\\0&0&\xi_3\end{array}\right),\quad \xi_1\in C,\xi_2\geq 0,\xi_3\geq 0$
0902.0431_E00536	90	■ 1.000	$z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$	$z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$	$z\left(E_6\right)=\left\{1,\omega 1,\omega ^21\right\},\quad \omega =-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C.$
0902.0431_E00537	90	■ 1.000	$\beta Y=Y,\quad \text{for all }\beta\in F_4.$	$\beta Y=Y,\quad \text{for all }\beta\in F_4.$	$\beta Y=Y,\quad \text{for all }\beta\in F_4.$
0902.0431_E00538	90	■ 1.000	$\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$	$\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$	$\beta X=TXT^{-1},\quad X\in\mathfrak{J}^C,$
0902.0431_E00539	91	■ 1.000	$\alpha E=Y=\omega E,\quad \omega\in C,$	$\alpha E=Y=\omega E,\quad \omega\in C,$	$\alpha E=Y=\omega E,\quad \omega\in C,$
0902.0431_E00540	91	■ 1.000	$\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$	$\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$	$\left(E_6\right)^{\sigma}=\left\{\alpha\in E_6\mid\sigma\alpha=\alpha\sigma\right\}.$
0902.0431_E00542	91	■ 1.000	$\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$	$\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$	$\alpha E_1=\xi E_1,\quad (\tau\xi)\xi=1.$
0902.0431_E00543	91	■ 1.000	$\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$	$\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$	$\alpha E_2,\alpha E_3\in\mathfrak{J}\left(2,\mathfrak{C}^C\right).$
0902.0431_E00545	92	■ 1.000	$\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$	$\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$	$\alpha E_1\notin\mathfrak{J}\left(2,\mathfrak{C}^C\right).$
0902.0431_E00547	92	■ 1.000	$\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$	$\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$	$\alpha E_1=\xi E_1+\xi_2E_2+\xi_3E_3+F_1\left(x_1\right),\quad \xi\neq 0.$
0902.0431_E00549	92	■ 1.000	$\alpha E_1=\xi E_1,\quad \xi\neq 0.$	$\alpha E_1=\xi E_1,\quad \xi\neq 0.$	$\alpha E_1=\xi E_1,\quad \xi\neq 0.$
0902.0431_E00550	92	■ 1.000	$1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$	$1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$	$1=\left\langle E_1,E_1\right\rangle =\left\langle \alpha E_1,\alpha E_1\right\rangle =\left\langle \xi E_1,\xi E_1\right\rangle =\left(\tau\xi\right)\xi\left\langle E_1,E_1\right\rangle =\left(\tau\xi\right)\xi,$
0902.0431_E00551	92	■ 1.000	$\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$	$\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$	$\left(E_6\right)_{E_1}=\left\{\alpha\in E_6\mid\alpha E_1=E_1\right\}.$
0902.0431_E00552	93	■ 1.000	$V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$	$V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$	$V^{10}=\left\{X\in\mathfrak{J}^C\mid 2E_1\times X=-\tau X\right\}=\left\{\left(\begin{array}{ccc}0&0&0\\0&\xi&x\\0&\bar{x}&-\tau\xi\end{array}\right)\mid\xi\in C,x\in\mathfrak{C}\right\}.$
0902.0431_E00553	93	■ 1.000	$2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$	$2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$	$2E_1\times\alpha X=2\alpha E_1\times\alpha X=2\tau\alpha\tau\left(E_1\times X\right)=\tau\alpha\tau\left(-\tau X\right)=-\tau\left(\alpha X\right),\quad\left\langle\alpha X,\alpha X\right\rangle=\left\langle X,X\right\rangle=2.$
0902.0431_E00554	93	■ 1.000	$\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$	$\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$	$\alpha_{23}\left(t_0\right)X\in S^8=\left\{X\in V^9\mid\left\langle X,X\right\rangle=2\right\}$
0902.0431_E00556	93	■ 1.000	$\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$	$\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$	$\alpha_{23}\left(\pi/2\right)\beta\alpha_{23}\left(t_0\right)X=i\left(E_2+E_3\right).$
0902.0431_E00557	93	■ 1.000	$p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$	$p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$	$p:\left(E_6\right)_{E_1}\rightarrow SO\left(10\right)=SO\left(V^{10}\right)$
0902.0431_E00559	94	■ 1.000	$\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$	$\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$	$\left(E_6\right)_{E_1}/\left\{1,\sigma\right\}\cong SO\left(10\right).$

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0902.0431_E00561	94	■ 1.000	$\begin{aligned} U(1) \, \&= \left\{ \alpha_{12}(t) \right. \\ \alpha_{13}(t) &= \exp i \, t \left(2 \, E_1 - E_2 - E_3 \right) \\ \sim \mid t \, \in \, \boldsymbol{R} \, \Big\} \, \&= \left\{ \phi(\theta) \right. \\ \theta &\in C, (\tau \theta) \theta = 1 \Big\} \end{aligned}$	$U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^\sim \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$	$U(1) = \{ \alpha_{12}(t) \alpha_{13}(t) = \exp i t (2E_1 - E_2 - E_3)^\sim \mid t \in \boldsymbol{R} \}$ $= \{ \phi(\theta) \mid \theta \in C, (\tau \theta) \theta = 1 \}$
0902.0431_E00563	94	■ 1.000	$\phi(\theta) \big \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \big \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \big (\mathfrak{J}^C)_{-\sigma} = \theta 1.$	$\phi(\theta) \big \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \big \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \big (\mathfrak{J}^C)_{-\sigma} = \theta 1.$	$\phi(\theta) \big \mathfrak{E}_1^C = \theta^4 1, \quad \phi(\theta) \big \mathfrak{J}(2, \mathfrak{C})^C = \theta^{-2} 1, \quad \phi(\theta) \big (\mathfrak{J}^C)_{-\sigma} = \theta 1.$
0902.0431_E00564	95	■ 1.000	$\varphi(\theta, \beta) = \phi(\theta) \beta.$	$\varphi(\theta, \beta) = \phi(\theta) \beta.$	$\varphi(\theta, \beta) = \phi(\theta) \beta.$
0902.0431_E00565	95	■ 1.000	$\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$	$\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$	$\alpha E_1 = \theta^4 E_1 = \phi(\theta) E_1$
0902.0431_E00566	95	■ 1.000	$\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$	$\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$	$\operatorname{Ker} \varphi = \{ (\theta, \phi(\theta)^{-1}) \mid \theta \in C, \theta^4 = 1 \} = \{ (\theta, \phi(\theta^{-1})) \mid \theta = \pm 1, \pm i \} = \boldsymbol{Z}_4.$
0902.0431_E00567	95	■ 1.000	$(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$	$(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$	$(E_6)^\gamma = \{ \alpha \in E_6 \mid \gamma \alpha = \alpha \gamma \}.$
0902.0431_E00568	95	■ 1.000	$\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$	$\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$	$\mathfrak{J}^C = \mathfrak{J}(3, \boldsymbol{H})^C \oplus (\boldsymbol{H}^3)^C,$
0902.0431_E00570	95	■ 1.000	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$	$k: M(3, \boldsymbol{H}) \rightarrow M(6, C) \quad \text{and} \quad k: \boldsymbol{H}^3 \rightarrow M(2, 6, C).$
0902.0431_E00574	96	■ 1.000	$k_J(M) = k(M) J.$	$k_J(M) = k(M) J.$	$k_J(M) = k(M) J.$
0902.0431_E00576	96	■ 1.000	$\langle S, T \rangle = \operatorname{tr} \left((\tau^t S) T \right), \quad \langle P, Q \rangle = \operatorname{tr} \left((\tau^t P) Q \right).$	$\langle S, T \rangle = \operatorname{tr} \left((\tau^t S) T \right), \quad \langle P, Q \rangle = \operatorname{tr} \left((\tau^t P) Q \right).$	$\langle S, T \rangle = \operatorname{tr} \left((\tau^t S) T \right), \quad \langle P, Q \rangle = \operatorname{tr} \left((\tau^t P) Q \right).$
0902.0431_E00577	96	■ 1.000	$\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$	$\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$	$\langle k_J(M), k_J(N) \rangle = 2 \langle M, N \rangle, \quad M, N \in \mathfrak{J}(3, \boldsymbol{H})^C,$ $\langle k(\boldsymbol{a}), k(\boldsymbol{b}) \rangle = 2 \langle \boldsymbol{a}, \boldsymbol{b} \rangle, \quad \boldsymbol{a}, \boldsymbol{b} \in (\boldsymbol{H}^3)^C.$ $\det(k_J(M)) = (\det M)^2, \quad M \in \mathfrak{J}(3, \boldsymbol{H})^C.$
0902.0431_E00579	96	■ 1.000	$\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$	$\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$	$\begin{pmatrix} a' & b' \\ -\tau b' & \tau a' \end{pmatrix} + i \begin{pmatrix} c' & d' \\ -\tau d' & \tau c' \end{pmatrix} = 0.$
0902.0431_E00580	97	■ 1.000	$\langle k(m_1 + i n_1), k(m_2 + i n_2) \rangle = 2 \langle m_1 + i n_1, m_2 + i n_2 \rangle.$	$\langle k(m_1 + i n_1), k(m_2 + i n_2) \rangle = 2 \langle m_1 + i n_1, m_2 + i n_2 \rangle.$	$\langle k(m_1 + i n_1), k(m_2 + i n_2) \rangle = 2 \langle m_1 + i n_1, m_2 + i n_2 \rangle.$

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0902.0431_E00583	97	■ 1.000	$\begin{aligned} & \operatorname{det}\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right) \\ & = \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2) \\ & \quad + \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2) \\ & = \text{the contents of in the parenthesis above.} \end{aligned}$	$\det\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right)$ $= \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)$ $+ \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2)$ $= \text{the contents of in the parenthesis above.}$	$\det\left(\begin{array}{ccc} \xi_1 & m_3+n_3e_2 & \tau(m_2+n_2e_2) \\ \tau(m_3+n_3e_2) & \xi_2 & m_1+n_1e_2 \\ m_2+n_2e_2 & \tau(m_1+n_1e_2) & \xi_3 \end{array}\right)$ $= \xi_1\xi_2\xi_3 + (m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)$ $+ \tau((m_1+n_1e_2)(m_2+n_2e_2)(m_3+n_3e_2)) - \sum_{i=1}^3 \xi_i(m_i+n_ie_2)\tau(m_i+n_ie_2)$ $= \text{the contents of in the parenthesis above.}$
0902.0431_E00584	98	■ 1.000	$E_{\{6, \boldsymbol{H}\}} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$	$E_{6,\boldsymbol{H}} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$	$E_{6,\boldsymbol{H}} = \left\{ \alpha \in \operatorname{Iso}_C \left((\mathfrak{H}_{\boldsymbol{H}})^C \right) \mid \det(\alpha M) = \det M, \langle \alpha M, \alpha N \rangle = \langle M, N \rangle \right\}.$
0902.0431_E00585	98	■ 1.000	$F_{\{4, \boldsymbol{H}\}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$	$F_{4,\boldsymbol{H}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$	$F_{4,\boldsymbol{H}} = \left\{ \alpha \in E_{6,\boldsymbol{H}} \mid \alpha E = E \right\},$
0902.0431_E00589	98	■ 1.000	$\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$	$\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$	$\varphi_*(C)M = CM + MC^* = [C, M], \quad M \in \mathfrak{H}_{\boldsymbol{H}}$
0902.0431_E00590	98	■ 1.000	$E_{\{6, \boldsymbol{H}\}} / F_{\{4, \boldsymbol{H}\}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$	$E_{6,\boldsymbol{H}}/F_{4,\boldsymbol{H}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$	$E_{6,\boldsymbol{H}}/F_{4,\boldsymbol{H}} \simeq EIV_{\boldsymbol{H}} = \left\{ X \in (\mathfrak{H}_{\boldsymbol{H}})^C \mid \det M = 1, \langle M, M \rangle = 3 \right\}$
0902.0431_E00592	98	■ 1.000	$\operatorname{det}(\varphi(A)M) = \det M.$	$\det(\varphi(A)M) = \det M.$	$\det(\varphi(A)M) = \det M.$
0902.0431_E00596	99	■ 1.000	$\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$	$\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$	$\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y), \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00598	99	■ 1.000	$k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$	$k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$	$k_J^{-1}(M) = -k^{-1}(MJ) = -k^{-1}(J(\tau M)), \quad \tau k^{-1}(M) = -k^{-1}(JMJ),$
0902.0431_E00600	100	■ 1.000	$\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$	$\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$	$\alpha = \varphi(1, A)\beta = \varphi(1, A)\varphi(p, E) = \varphi(p, A).$
0902.0431_E00601	100	■ 1.000	$(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$	$(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$	$(E_6)^{\tau\gamma} = \left\{ \alpha \in E_6 \mid \tau\gamma\alpha = \alpha\tau\gamma \right\}$ $= \left\{ \alpha \in E_6 \mid \gamma\lambda(\alpha)\gamma = \alpha \right\} = (E_6)^{\lambda\gamma}.$
0902.0431_E00605	101	■ 1.000	$\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$	$\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$	$\mathfrak{J}(4, \boldsymbol{H}) = \left\{ P \in M(4, \boldsymbol{H}) \mid P^* = P \right\},$
0902.0431_E00606	101	■ 1.000	$P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$	$P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$	$P \circ Q = \frac{1}{2}(PQ + QP), \quad (P, Q) = \operatorname{tr}(P \circ Q).$
0902.0431_E00609	101	■ 1.000	$\langle P, Q \rangle = (\tau P, Q).$	$\langle P, Q \rangle = (\tau P, Q).$	$\langle P, Q \rangle = (\tau P, Q).$

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0902.0431_E00610	101	■ 1.000	$\backslash\left(X_{\{1\}}+i X_{\{2\}}\right) A^{*}=A X_{\{1\}} A^{*}+i A X_{\{2\}} A^{*}, \quad \backslash\quad A \in \operatorname{Sp}(4), X_{\{1\}}, X_{\{2\}} \in \mathfrak{J}(4, \boldsymbol{H})$.	$A\left(X_1+i X_2\right) A^{*}=A X_1 A^{*}+i A X_2 A^{*}, \quad A \in \operatorname{Sp}(4), X_1, X_2 \in \mathfrak{J}(4, \boldsymbol{H})$.	$A\left(X_1+i X_2\right) A^{*}=A X_1 A^{*}+i A X_2 A^{*}, \quad A \in \operatorname{Sp}(4), X_1, X_2 \in \mathfrak{J}(4, \boldsymbol{H})$.
0902.0431_E00611	101	■ 1.000	$\backslash\left\langle\right\rangle A P A^{*}, A Q A^{*} \backslash\right\rangle=\left\langle P, Q\right\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$.	$\langle P A^{*}, A Q A^{*}\rangle=\langle P, Q\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$.	$\langle P A^{*}, A Q A^{*}\rangle=\langle P, Q\rangle, \quad P, Q \in \mathfrak{J}(4, \boldsymbol{H})^C$.
0902.0431_E00612	102	■ 1.000	$g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$.	$g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$.	$g(M+\boldsymbol{a})=\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & i \boldsymbol{a} \\ i \boldsymbol{a}^{*} & M-\frac{1}{2} \operatorname{tr}(M) E \end{array}\right), \quad M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$.
0902.0431_E00613	102	■ 1.000	$g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$.	$g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$.	$g(M+i \boldsymbol{a})=\left(\begin{array}{cc} \operatorname{tr}(M) & -\boldsymbol{a} \\ -\boldsymbol{a}^{*} & M \end{array}\right)-\frac{1}{2} \operatorname{tr}(M) E, \quad M+i \boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus i \boldsymbol{H}^3=\left(\mathfrak{J}^C\right)_{\tau \gamma}$.
0902.0431_E00618	103	■ 1.000	$\backslash\left\langle\right\rangle g\left(X_{\{1\}}+i X_{\{2\}}\right), g\left(Y_{\{1\}}+i Y_{\{2\}}\right) \backslash\right\rangle=\left\langle\right\rangle X_{\{1\}}+i X_{\{2\}}, Y_{\{1\}}+i Y_{\{2\}} \backslash\right\rangle, \quad \backslash\quad X_{\{i\}}, Y_{\{i\}} \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$.	$\langle g\left(X_1+i X_2\right), g\left(Y_1+i Y_2\right)\rangle=\left\langle X_1+i X_2, Y_1+i Y_2\right\rangle, \quad X_i, Y_i \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$.	$\langle g\left(X_1+i X_2\right), g\left(Y_1+i Y_2\right)\rangle=\left\langle X_1+i X_2, Y_1+i Y_2\right\rangle, \quad X_i, Y_i \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$.
0902.0431_E00619	103	■ 1.000	$\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$.	$\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$.	$\varphi(A) X=g^{-1}\left(A(g X) A^{*}\right), \quad X \in \mathfrak{J}^C$.
0902.0431_E00622	103	■ 1.000	$\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$,	$\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$,	$\tau \gamma \varphi(A) \tau \gamma X=\varphi(A) X, \quad X \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$,
0902.0431_E00623	103	■ 1.000	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$.	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$.	$(g(\alpha E))^2=g(\alpha E)+\frac{3}{4} E$.
0902.0431_E00625	104	■ 1.000	$P=\frac{1}{4}(2 g(\alpha E)+E),$	$P=\frac{1}{4}(2 g(\alpha E)+E),$	$P=\frac{1}{4}(2 g(\alpha E)+E),$
0902.0431_E00626	104	■ 1.000	$\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$	$\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$	$\begin{aligned} P^2 &=\frac{1}{16}\left(4(g(\alpha E))^2+4 g(\alpha E)+E\right)=\frac{1}{4}(2 g(\alpha E)+E)=P, \\ \operatorname{tr}(P) &=\frac{1}{4} \operatorname{tr}(2 g(\alpha E))+\frac{1}{4} \operatorname{tr}(E)=0+1=1. \end{aligned}$
0902.0431_E00627	104	■ 1.000	$P=A E_1 A^{*}$.	$P=A E_1 A^{*}$.	$P=A E_1 A^{*}$.
0902.0431_E00629	104	■ 1.000	$\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J}$.	$\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J}$.	$\beta(M+\boldsymbol{a})=D M D^{*}+p \boldsymbol{a} D^{*}, \quad M+\boldsymbol{a} \in \mathfrak{J}_{\boldsymbol{H}} \oplus \boldsymbol{H}^3=\mathfrak{J}$.
0902.0431_E00630	104	■ 1.000	$\beta=\varphi(B)$.	$\beta=\varphi(B)$.	$\beta=\varphi(B)$.
0902.0431_E00633	105	■ 1.000	$\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\}$.	$\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\}$.	$\left(E_6\right)^w=\left\{\alpha \in E_6 \mid w \alpha=\alpha w\right\}$.
0902.0431_E00634	105	■ 1.000	$\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C$.	$\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C$.	$\mathfrak{J}(3, \boldsymbol{C})^C \oplus M(3, \boldsymbol{C})^C=\mathfrak{J}^C$.
0902.0431_E00636	105	■ 1.000	$F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$	$F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$	$F_{4, \boldsymbol{C}}=\left\{\alpha \in E_{6, \boldsymbol{C}} \mid \alpha E=E\right\},$

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0902.0431_E00639	105	■ 1.000	$\mathbf{E} \mathbf{I} \mathbf{V}_{\{\boldsymbol{C}\}} = \left\{ X \in \left(\mathfrak{J}_{\{\boldsymbol{C}\}} \mid \det X = 1, \langle X, X \rangle = 3 \right) \right\}$	$EIV_C = \left\{ X \in (\mathfrak{J}_C)^C \mid \det X = 1, \langle X, X \rangle = 3 \right\}$	$EIV_C = \{ X \in (\mathfrak{J}_C)^C \mid \det X = 1, \langle X, X \rangle = 3 \}$
0902.0431_E00644	106	■ 1.000	$\phi_{\boldsymbol{C}}(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_{\boldsymbol{C}})^C$	$\phi_C(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_C)^C$	$\phi_C(C, D)X = h(C, D)X + Xh(C, D)^*, \quad X \in (\mathfrak{J}_C)^C$
0902.0431_E00650	107	■ 1.000	$h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_{\boldsymbol{C}})^C$	$h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_C)^C$	$h(A, B)Xh(A, B)^* = \bar{X}, \quad X \in (\mathfrak{J}_C)^C$
0902.0431_E00653	108	■ 1.000	$\begin{aligned} \tau \alpha \tau((X + M) \times (Y + N)) &= \alpha(X + M) \times \alpha(Y + N), \\ \langle \alpha(X + M), \alpha(Y + N) \rangle &= \langle X + M, Y + N \rangle. \end{aligned}$	$\tau \alpha \tau((X + M) \times (Y + N)) = \alpha(X + M) \times \alpha(Y + N), \\ \langle \alpha(X + M), \alpha(Y + N) \rangle = \langle X + M, Y + N \rangle.$	$\tau \alpha \tau((X + M) \times (Y + N)) = \alpha(X + M) \times \alpha(Y + N), \\ \langle \alpha(X + M), \alpha(Y + N) \rangle = \langle X + M, Y + N \rangle.$
0902.0431_E00654	108	■ 1.000	$\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$	$\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$	$\tau \alpha \tau(X \times Y) = \alpha X \times \alpha Y, \quad \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle$
0902.0431_E00657	108	■ 1.000	$\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$	$\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$	$\alpha X = h(A, B)Xh(A, B)^* \quad \text{or} \quad \alpha X = h(A, B)\bar{X}h(A, B)^*, \quad X \in (\mathfrak{J}_C)^C$
0902.0431_E00659	109	■ 1.000	$\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$	$\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$	$\alpha = \varphi(E, A, B)\beta = \varphi(E, A, B)\varphi(P, E, E) = \varphi(P, A, B).$
0902.0431_E00660	109	■ 1.000	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$	$\begin{aligned} &G_{01}, \quad G_{23}, \quad G_{45}, \quad G_{67}, \quad G_{26} + G_{37}, \quad -G_{27} + G_{36}, \\ &G_{24} + G_{35}, \quad -G_{25} + G_{34}, \quad G_{46} + G_{57}, \quad -G_{47} + G_{56}, \\ &\tilde{A}_1(1), \quad \tilde{A}_2(1), \quad \tilde{A}_3(1), \quad \tilde{A}_1(e_1), \quad \tilde{A}_2(e_1), \quad \tilde{A}_3(e_1), \\ &(E_1 - E_2)^\sim, \quad \tilde{F}_1(1), \quad \tilde{F}_2(1), \quad \tilde{F}_3(1), \\ &(E_2 - E_3)^\sim, \quad \tilde{F}_1(e_1), \quad \tilde{F}_2(e_1), \quad \tilde{F}_3(e_1) \end{aligned}$
0902.0431_E00661	109	■ 1.000	$E_{\{6\}} \simeq E_{\{6\}} \times \mathbf{R}^{78}.$	$E_6^C \simeq E_6 \times \mathbf{R}^{78}.$	$E_6^C \simeq E_6 \times \mathbf{R}^{78}.$
0902.0431_E00662	109	■ 1.000	$E_{\{6\}} \simeq \left(E_{\{6\}} \cap U(\mathfrak{J}^C) \right) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$	$E_6^C \simeq (E_6^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$	$E_6^C \simeq (E_6^C \cap U(\mathfrak{J}^C)) \times \mathbf{R}^d = E_6 \times \mathbf{R}^d, \quad d = 78.$
0902.0431_E00663	110	■ 1.000	$\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$	$\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$	$\langle X, Y \rangle_\gamma = \langle \gamma X, Y \rangle, \quad \langle X, Y \rangle_\sigma = \langle \sigma X, Y \rangle.$
0902.0431_E00667	110	■ 1.000	$z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$	$z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$	$z(E_{6(6)}) = \{1\}, \quad z(E_{6(2)}) = \mathbf{Z}_3, \quad z(E_{6(-14)}) = \mathbf{Z}_3, \quad z(E_{6(-26)}) = \{1\}.$

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0902.0431_E00670	111	■ 1.000	$\begin{aligned} & \Phi(\phi, A, B, \nu) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ &= \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ &= \left(\begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right). \end{aligned}$	$\Phi(\phi, A, B, \nu) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ = \left(\begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right).$	$\Phi(\phi, A, B, \nu) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) = \left(\begin{array}{cccc} \phi - \frac{1}{3}\nu & 2B & 0 & A \\ 2A & -{}^t\phi + \frac{1}{3}\nu & B & 0 \\ 0 & A & \nu & 0 \\ B & 0 & 0 & -\nu \end{array} \right) \left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array} \right) \\ = \left(\begin{array}{l} \phi X - \frac{1}{3}\nu X + 2B \times Y + \eta A \\ 2A \times X - {}^t\phi Y + \frac{1}{3}\nu Y + \xi B \\ (A, Y) + \nu \xi \\ (B, X) - \nu \eta \end{array} \right).$
0902.0431_E00676	112	■ 1.000	$(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$	$(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$	$(P \times R)Q - (Q \times R)P + \frac{1}{8}\{Q, R\}P - \frac{1}{8}\{P, R\}Q - \frac{1}{4}\{P, Q\}R = 0.$
0902.0431_E00680	113	■ 1.000	$U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$	$U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$	$U(56) = U(\mathfrak{P}^C) = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \langle \alpha P, \alpha Q \rangle = \langle P, Q \rangle\}.$
0902.0431_E00681 (1)	113	■ 1.000	$\alpha \mathfrak{M}^C = \mathfrak{M}^C.$	$\alpha \mathfrak{M}^C = \mathfrak{M}^C.$	$\alpha \mathfrak{M}^C = \mathfrak{M}^C.$
0902.0431_E00697	115	■ 1.000	$l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$	$l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$	$l(Y) = 2B \times Y, \quad Y \in \mathfrak{J}^C.$
0902.0431_E00698	115	■ 1.000	$2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$	$2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$	$2(B \times (X \times X), X \times X) + (\det X)(X, B) = 3(\det X)b(X),$
0902.0431_E00699	115	■ 1.000	$b(X) = (B, X), \quad X \in \mathfrak{J}^C.$	$b(X) = (B, X), \quad X \in \mathfrak{J}^C.$	$b(X) = (B, X), \quad X \in \mathfrak{J}^C.$
0902.0431_E00700 (i)	116	■ 1.000	$k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$	$k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$	$k(X) = 2A \times X, \quad a(Y) = (A, Y), \quad X, Y \in \mathfrak{J}^C.$
0902.0431_E00701 (ii)	116	■ 1.000	$\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$	$\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$	$\begin{aligned} & 2g(X) \times X = \mu X \times X + h(X \times X), \\ & 2h(X \times X) \times (X \times X) = (\det X)(\nu X + g(X)), \\ & (g(X), X \times X) + (h(X \times X), X) = 3(\nu + \mu) \det X. \end{aligned}$
0902.0431_E00703	116	■ 1.000	$\phi \in \mathfrak{e}_6^C.$	$\phi \in \mathfrak{e}_6^C.$	$\phi \in \mathfrak{e}_6^C.$
0902.0431_E00705	116	■ 1.000	$\psi' = \phi.$	$\psi' = \phi.$	$\psi' = \phi.$
0902.0431_E00707	117	■ 1.000	$[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$	$[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$	$[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q.$
0902.0431_E00710	117	■ 1.000	$\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$	$\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$	$\lambda(X, Y, \xi, \eta) = (Y, -X, \eta, -\xi).$
0902.0431_E00711	117	■ 1.000	$(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$	$(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$	$(P, Q) = \{\lambda P, Q\} = -\{P, \lambda Q\}, \quad \langle P, Q \rangle = \{\tau \lambda P, Q\}.$

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0902.0431_E00716	118	■ 1.000	$\backslash\mathrm{Phi}(\backslash\mathrm{phi}, A, -\backslash\mathrm{tau} A, \backslash\mathrm{nu})=\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{cccc}\backslash\mathrm{phi}-\backslash\mathrm{frac}{1}{3}\backslash\mathrm{nu} \& -2\backslash\mathrm{tau} A \& \theta \& A \backslash\backslash 2 A \& \backslash\mathrm{tau} \backslash\mathrm{phi} \backslash\mathrm{tau}+\backslash\mathrm{frac}{1}{3}\backslash\mathrm{nu} \& -\backslash\mathrm{tau} A \& \theta \backslash\backslash \theta \& A \& \backslash\mathrm{nu} \& \theta \backslash\backslash -\backslash\mathrm{tau} A \& \theta \& \theta \& -\backslash\mathrm{nu} \backslash\mathrm{end}{array}\backslash\mathrm{right})$.	$\Phi(\phi, A, -\tau A, \nu) = \begin{pmatrix} \phi - \frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau + \frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{pmatrix}.$	$\Phi(\phi, A, -\tau A, \nu) = \begin{pmatrix} \phi - \frac{1}{3}\nu & -2\tau A & 0 & A \\ 2A & \tau\phi\tau + \frac{1}{3}\nu & -\tau A & 0 \\ 0 & A & \nu & 0 \\ -\tau A & 0 & 0 & -\nu \end{pmatrix}.$
0902.0431_E00719	118	■ 1.000	$\backslash\mathrm{operatorname}{dim}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\backslash\mathrm{7}}\backslash\mathrm{right})=78+54+1=133$.	$\dim(\mathfrak{e}_7) = 78 + 54 + 1 = 133.$	$\dim(\mathfrak{e}_7) = 78 + 54 + 1 = 133.$
0902.0431_E00721	119	■ 1.000	$\backslash\mathrm{Phi}=\backslash\mathrm{frac}\{\backslash\mathrm{Phi}-\backslash\mathrm{Phi}^{\backslash\mathrm{*}}\}\{2\}+i\backslash\mathrm{frac}\{\backslash\mathrm{Phi}+\backslash\mathrm{Phi}^{\backslash\mathrm{*}}\}\{2\}i\}, \backslash\mathrm{quad} \backslash\mathrm{frac}\{\backslash\mathrm{Phi}-\backslash\mathrm{Phi}^{\backslash\mathrm{*}}\}\{2\}, \backslash\mathrm{frac}\{\backslash\mathrm{Phi}+\backslash\mathrm{Phi}^{\backslash\mathrm{*}}\}\{2\}i\} \backslash\mathrm{in} \backslash\mathrm{mathfrak{e}}_{\backslash\mathrm{7}}\}.$	$\Phi = \frac{\Phi - \Phi^*}{2} + i\frac{\Phi + \Phi^*}{2i}, \quad \frac{\Phi - \Phi^*}{2}, \frac{\Phi + \Phi^*}{2i} \in \mathfrak{e}_7.$	$\Phi = \frac{\Phi - \Phi^*}{2} + i\frac{\Phi + \Phi^*}{2i}, \quad \frac{\Phi - \Phi^*}{2}, \frac{\Phi + \Phi^*}{2i} \in \mathfrak{e}_7.$
0902.0431_E00723	119	■ 1.000	$\backslash\mathrm{mathfrak{a}} \backslash\mathrm{ni}\backslash\mathrm{left}[\backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{phi}_{\backslash\mathrm{1}}, \theta, \theta, \theta\backslash\mathrm{right}), \backslash\mathrm{Phi}(\backslash\mathrm{phi}, A, B, \backslash\mathrm{nu})\backslash\mathrm{right}]=\backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{left}[\backslash\mathrm{phi}_{\backslash\mathrm{1}}, \backslash\mathrm{phi}\backslash\mathrm{right}], \backslash\mathrm{phi}_{\backslash\mathrm{1}} A, \backslash\mathrm{phi}_{\backslash\mathrm{1}}\{ \}^{\backslash\mathrm{prime}} B, \theta\backslash\mathrm{right})$	$\mathfrak{a} \ni [\Phi(\phi_1, 0, 0, 0), \Phi(\phi, A, B, \nu)] = \Phi([\phi_1, \phi], \phi_1 A, \phi_1' B, 0)$	$\mathfrak{a} \ni [\Phi(\phi_1, 0, 0, 0), \Phi(\phi, A, B, \nu)] = \Phi([\phi_1, \phi], \phi_1 A, \phi_1' B, 0)$
0902.0431_E00725	119	■ 1.000	$\backslash\mathrm{mathfrak{a}} \backslash\mathrm{ni}\backslash\mathrm{left}[\backslash\mathrm{Phi}\backslash\mathrm{left}(\theta, E_{\backslash\mathrm{1}}, \theta, \theta\backslash\mathrm{right}), \backslash\mathrm{Phi}\backslash\mathrm{left}(\theta, \theta, E_{\backslash\mathrm{1}}, \theta\backslash\mathrm{right})\backslash\mathrm{right}]=\backslash\mathrm{Phi}\backslash\mathrm{left}(2 E_{\backslash\mathrm{1}} \vee E_{\backslash\mathrm{1}}, \theta, \theta, 1\backslash\mathrm{right}),$	$\mathfrak{a} \ni [\Phi(0, E_1, 0, 0), \Phi(0, 0, E_1, 0)] = \Phi(2E_1 \vee E_1, 0, 0, 1),$	$\mathfrak{a} \ni [\Phi(0, E_1, 0, 0), \Phi(0, 0, E_1, 0)] = \Phi(2E_1 \vee E_1, 0, 0, 1),$
0902.0431_E00727	120	■ 1.000	$\backslash[\backslash\mathrm{mathfrak{a}} \backslash\mathrm{ni}[\backslash\mathrm{Phi}(\theta, \theta, \theta, \backslash\mathrm{nu}), \backslash\mathrm{Phi}(\theta, A, \theta, \theta)]=\backslash\mathrm{Phi}\backslash\mathrm{left}(\theta, \backslash\mathrm{frac}\{2\}{3}\backslash\mathrm{nu} A, \theta, \theta\backslash\mathrm{right}) . \backslash\backslash\mathrm{end}{itemize}$	$\mathfrak{a} \ni [\Phi(0, 0, 0, \nu), \Phi(0, A, 0, 0)] = \Phi\left(0, \frac{2}{3}\nu A, 0, 0\right).$	$\mathfrak{a} \ni [\Phi(0, 0, 0, \nu), \Phi(0, A, 0, 0)] = \Phi\left(0, \frac{2}{3}\nu A, 0, 0\right).$
0902.0431_E00729 (c)	120	■ 1.000	$\backslash\mathrm{begin}{gathered} W \backslash\mathrm{ni} \backslash\mathrm{Phi}(\theta, \theta, \theta, 3)(X, Y, \backslash\mathrm{xi}, \backslash\mathrm{eta})=(-X, Y, 3\backslash\mathrm{xi}, -3\backslash\mathrm{eta}), \backslash\backslash W \backslash\mathrm{ni} \backslash\mathrm{Phi}(\theta, \theta, \theta, 3)(-X, Y, 3\backslash\mathrm{xi}, -3\backslash\mathrm{eta})=(X, Y, 9\backslash\mathrm{xi}, 9\backslash\mathrm{eta}) . \backslash\mathrm{end}{gathered}$	$W \ni \Phi(0, 0, 0, 3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta),$ $W \ni \Phi(0, 0, 0, 3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$	$W \ni \Phi(0, 0, 0, 3)(X, Y, \xi, \eta) = (-X, Y, 3\xi, -3\eta),$ $W \ni \Phi(0, 0, 0, 3)(-X, Y, 3\xi, -3\eta) = (X, Y, 9\xi, 9\eta).$
0902.0431_E00731	121	■ 1.000	$\backslash[\backslash W \backslash\mathrm{ni} \backslash\mathrm{Phi}(\theta, \theta, B, \theta)(\theta, Y, \backslash\mathrm{xi}, \backslash\mathrm{eta})=(2 B \times Y, \backslash\mathrm{xi} B, \theta, \theta) . \backslash\backslash\mathrm{end}{itemize}$	$W \ni \Phi(0, 0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$	$W \ni \Phi(0, 0, B, 0)(0, Y, \xi, \eta) = (2B \times Y, \xi B, 0, 0).$
0902.0431_E00732	121	■ 1.000	$\backslash[\backslash W \backslash\mathrm{ni} \backslash\mathrm{Phi}(\theta, \theta, B, \theta)(\theta, \theta, \backslash\mathrm{xi}, \backslash\mathrm{eta})=(\theta, \backslash\mathrm{xi} B, \theta, \theta) . \backslash\backslash\mathrm{end}{itemize}$	$W \ni \Phi(0, 0, B, 0)(0, 0, \xi, \eta) = (0, \xi B, 0, 0).$	$W \ni \Phi(0, 0, B, 0)(0, 0, \xi, \eta) = (0, \xi B, 0, 0).$
0902.0431_E00733	121	■ 1.000	$\backslash\mathrm{left}(\backslash\mathrm{Phi}_{\backslash\mathrm{1}}, \backslash\mathrm{Phi}_{\backslash\mathrm{2}}\backslash\mathrm{right})_{\backslash\mathrm{7}}=-2\backslash\mathrm{left}(\backslash\mathrm{phi}_{\backslash\mathrm{1}}, \backslash\mathrm{phi}_{\backslash\mathrm{2}}\backslash\mathrm{right})_{\backslash\mathrm{6}}-4\backslash\mathrm{left}(A_{\backslash\mathrm{1}}, B_{\backslash\mathrm{2}}\backslash\mathrm{right})-4\backslash\mathrm{left}(A_{\backslash\mathrm{2}}, B_{\backslash\mathrm{1}}\backslash\mathrm{right})-\backslash\mathrm{frac}\{8\}{3}\backslash\mathrm{nu}_{\backslash\mathrm{1}}\backslash\mathrm{nu}_{\backslash\mathrm{2}}$	$(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2$	$(\Phi_1, \Phi_2)_7 = -2(\phi_1, \phi_2)_6 - 4(A_1, B_2) - 4(A_2, B_1) - \frac{8}{3}\nu_1\nu_2,$
0902.0431_E00735	121	■ 1.000	$\backslash[\backslash(\backslash\mathrm{Phi}, P \times Q)_{\backslash\mathrm{7}}]=\{\backslash\mathrm{Phi} P, Q\} . \backslash\backslash\mathrm{end}{itemize}$	$(\Phi, P \times Q)_7 = \{\Phi P, Q\}.$	$(\Phi, P \times Q)_7 = \{\Phi P, Q\}.$
0902.0431_E00736	121	■ 1.000	$=\backslash\mathrm{left}(\backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{c}\{ \}\backslash\mathrm{left}[\backslash\mathrm{phi}, \backslash\mathrm{phi}_{\backslash\mathrm{1}}\backslash\mathrm{right}]+2 A \vee B_{\backslash\mathrm{1}}-2 A_{\backslash\mathrm{1}} \vee B \backslash\backslash \backslash\mathrm{left}(\backslash\mathrm{phi}+\backslash\mathrm{frac}\{2\}{3}\backslash\mathrm{nu}\backslash\mathrm{right}) A_{\backslash\mathrm{1}}-\backslash\mathrm{left}(\backslash\mathrm{phi}_{\backslash\mathrm{1}}+\backslash\mathrm{frac}\{2\}{3}\backslash\mathrm{nu}_{\backslash\mathrm{1}}\backslash\mathrm{right}) A \backslash\backslash -\backslash\mathrm{left}(\{ \}^{\backslash\mathrm{t}}\backslash\mathrm{phi}+\backslash\mathrm{frac}\{2\}{3}\backslash\mathrm{nu}\backslash\mathrm{right}) B_{\backslash\mathrm{1}}+\backslash\mathrm{left}(\{ \}^{\backslash\mathrm{t}}\backslash\mathrm{phi}_{\backslash\mathrm{1}}+\backslash\mathrm{frac}\{2\}{3}\backslash\mathrm{nu}_{\backslash\mathrm{1}}\backslash\mathrm{right}) B \backslash\backslash \backslash\mathrm{left}(A, B_{\backslash\mathrm{1}}\backslash\mathrm{right})-\backslash\mathrm{left}(B, A_{\backslash\mathrm{1}}\backslash\mathrm{right})\backslash\mathrm{end}{array}\backslash\mathrm{right}), \backslash\mathrm{Phi}\backslash\mathrm{left}(\backslash\mathrm{begin}{array}{c}\{ \}\backslash\mathrm{phi}_{\backslash\mathrm{2}} \backslash\backslash A_{\backslash\mathrm{2}} \backslash\backslash B_{\backslash\mathrm{2}} \backslash\backslash \backslash\mathrm{nu}_{\backslash\mathrm{2}} \backslash\mathrm{end}{array}\backslash\mathrm{right})\backslash\mathrm{right})_{\backslash\mathrm{7}}$	$= \left(\Phi \left(\begin{pmatrix} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ (\phi + \frac{2}{3}\nu) A_1 - (\phi_1 + \frac{2}{3}\nu_1) A \\ -({}^t\phi + \frac{2}{3}\nu) B_1 + ({}^t\phi_1 + \frac{2}{3}\nu_1) B \\ (A, B_1) - (B, A_1) \end{pmatrix}, \Phi \left(\begin{pmatrix} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{pmatrix} \right) \right)_7$	$= \left(\Phi \left(\begin{pmatrix} [\phi, \phi_1] + 2A \vee B_1 - 2A_1 \vee B \\ (\phi + \frac{2}{3}\nu) A_1 - (\phi_1 + \frac{2}{3}\nu_1) A \\ -({}^t\phi + \frac{2}{3}\nu) B_1 + ({}^t\phi_1 + \frac{2}{3}\nu_1) B \\ (A, B_1) - (B, A_1) \end{pmatrix}, \Phi \left(\begin{pmatrix} \phi_2 \\ A_2 \\ B_2 \\ \nu_2 \end{pmatrix} \right) \right)_7$

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0902.0431_E00739	122	■ 1.000	$\begin{aligned} B_7 \left(\Phi_1, \Phi_2 \right) &= -9 \left(\Phi_1, \Phi_2 \right)_7 \\ &= 18 \left(\phi_1, \phi_2 \right)_6 + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= \frac{3}{2} B_6 \left(\phi_1, \phi_2 \right) + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= 3 \operatorname{tr} \left(\Phi_1 \Phi_2 \right) \end{aligned}$	$\begin{aligned} B_7 \left(\Phi_1, \Phi_2 \right) &= -9 \left(\Phi_1, \Phi_2 \right)_7 \\ &= 18 \left(\phi_1, \phi_2 \right)_6 + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= \frac{3}{2} B_6 \left(\phi_1, \phi_2 \right) + 36 \left(A_1, B_2 \right) + 36 \left(A_2, B_1 \right) + 24 \nu_1 \nu_2 \\ &= 3 \operatorname{tr} \left(\Phi_1 \Phi_2 \right) \end{aligned}$	$\begin{aligned} B_7(\Phi_1, \Phi_2) &= -9(\Phi_1, \Phi_2)_7 \\ &= 18(\phi_1, \phi_2)_6 + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= \frac{3}{2}B_6(\phi_1, \phi_2) + 36(A_1, B_2) + 36(A_2, B_1) + 24\nu_1\nu_2 \\ &= 3\operatorname{tr}(\Phi_1\Phi_2), \end{aligned}$
0902.0431_E00742	122	■ 1.000	$\left[\Phi_0, \left[\Phi_0, \Phi(\phi, A, B, \nu) \right] \right] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$	$[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$	$[\Phi_0, [\Phi_0, \Phi(\phi, A, B, \nu)]] = \left[\Phi_0, \Phi \left(0, \frac{2}{3}A, -\frac{2}{3}B, 0 \right) \right] = \Phi \left(0, \frac{4}{9}A, \frac{4}{9}B, 0 \right).$
0902.0431_E00752	124	■ 1.000	$\begin{aligned} &\left[\Phi \left(h_\delta + \tilde{H}, 0, 0, \nu \right), \Phi \left(0, F_1(a), 0, 0 \right) \right] \quad a = e_k \pm ie_{4+k} \\ &= \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ &= \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right) \end{aligned}$	$\begin{aligned} &\left[\Phi \left(h_\delta + \tilde{H}, 0, 0, \nu \right), \Phi \left(0, F_1(a), 0, 0 \right) \right] \quad a = e_k \pm ie_{4+k} \\ &= \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ &= \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right) \end{aligned}$	$\begin{aligned} &[\Phi(h_\delta + \tilde{H}, 0, 0, \nu), \Phi(0, F_1(a), 0, 0)] \quad a = e_k \pm ie_{4+k} \\ &= \Phi \left(0, \left(h_\delta + \tilde{H} + \frac{2}{3}\nu \right) F_1(a), 0, 0 \right) \\ &= \Phi \left(0, F_1(h_\delta a) + \frac{1}{2}(\mu_2 + \mu_3)F_1(a) + \frac{2}{3}\nu F_1(a), 0, 0 \right) \end{aligned}$
0902.0431_E00753	125	■ 1.000	$= \left(\pm \lambda_k - \frac{1}{2}\mu_1 + \frac{2}{3}\nu \right) \Phi(0, F_1(a), 0, 0).$	$= \left(\pm \lambda_k - \frac{1}{2}\mu_1 + \frac{2}{3}\nu \right) \Phi(0, F_1(a), 0, 0).$	$= \left(\pm \lambda_k - \frac{1}{2}\mu_1 + \frac{2}{3}\nu \right) \Phi(0, F_1(a), 0, 0).$
0902.0431_E00754	125	■ 1.000	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ \alpha_4 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_5 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2), \\ \alpha_6 &= \mu_2 + \frac{3}{2}\nu, \quad \alpha_7 = -\mu_3 - \frac{3}{2}\nu \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ \alpha_4 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_5 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2), \\ \alpha_6 &= \mu_2 + \frac{3}{2}\nu, \quad \alpha_7 = -\mu_3 - \frac{3}{2}\nu \end{aligned}$	$\begin{aligned} \alpha_1 &= \lambda_0 - \lambda_1, \quad \alpha_2 = \lambda_1 - \lambda_2, \quad \alpha_3 = \lambda_2 - \lambda_3, \\ \alpha_4 &= \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1), \\ \alpha_5 &= \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2), \\ \alpha_6 &= \mu_2 + \frac{3}{2}\nu, \quad \alpha_7 = -\mu_3 - \frac{3}{2}\nu \end{aligned}$
0902.0431_E00755	125	■ 1.000	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7$	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7$	$\mu = \alpha_1 + 2\alpha_2 + 3\alpha_3 + 4\alpha_4 + 3\alpha_5 + 2\alpha_6 + 2\alpha_7$
0902.0431_E00763	128	■ 1.000	$B_7(h, h') = 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda'_k + 3 \sum_{j=1}^3 \mu_j \mu'_j + 4\nu\nu' \right)$	$B_7(h, h') = 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda'_k + 3 \sum_{j=1}^3 \mu_j \mu'_j + 4\nu\nu' \right)$	$B_7(h, h') = 6 \left(6 \sum_{k=0}^3 \lambda_k \lambda'_k + 3 \sum_{j=1}^3 \mu_j \mu'_j + 4\nu\nu' \right)$

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0902.0431_E00766	128	■ 1.000	$\left(\alpha_{\{1\}}, \alpha_{\{1\}}\right)=B_{\{7\}}\left(H_{\{\alpha_{\{1\}}\}}, H_{\{\alpha_{\{1\}}\}}\right)=36 \frac{1}{36} \frac{1}{36} 2=\frac{1}{18}$	$(\alpha_1, \alpha_1) = B_7(H_{\alpha_1}, H_{\alpha_1}) = 36 \frac{1}{36} \frac{1}{36} 2 = \frac{1}{18}$	$(\alpha_1, \alpha_1) = B_7(H_{\alpha_1}, H_{\alpha_1}) = 36 \frac{1}{36} \frac{1}{36} 2 = \frac{1}{18}$
0902.0431_E00768	129	■ 1.000	$\left(E_{\{7\}}\right)_{\{(0,0,1,0)\}}=\left\{\alpha \in E_{\{7\}} \mid \alpha(0,0,1,0)=(0,0,1,0)\right\} .$	$(E_7)_{(0,0,1,0)} = \{\alpha \in E_7 \mid \alpha(0,0,1,0) = (0,0,1,0)\} .$	$(E_7)_{(0,0,1,0)} = \{\alpha \in E_7 \mid \alpha(0,0,1,0) = (0,0,1,0)\} .$
0902.0431_E00769	129	■ 1.000	$\alpha(0,0,0,1)=\alpha(-\tau \lambda(0,0,1,0))=-\tau \lambda \alpha(0,0,1,0)=-\tau \lambda(0,0,1,0)=(0,0,0,1) .$	$\alpha(0,0,0,1) = \alpha(-\tau \lambda(0,0,1,0)) = -\tau \lambda \alpha(0,0,1,0) = -\tau \lambda(0,0,1,0) = (0,0,0,1).$	$\alpha(0,0,0,1) = \alpha(-\tau \lambda(0,0,1,0)) = -\tau \lambda \alpha(0,0,1,0) = -\tau \lambda(0,0,1,0) = (0,0,0,1).$
0902.0431_E00770	129	■ 1.000	$\left(E_{\{7\}}\right)_{\{(0,0,1,0)\}} \cong E_{\{6\}} .$	$(E_7)_{(0,0,1,0)} \cong E_6 .$	$(E_7)_{(0,0,1,0)} \cong E_6 .$
0902.0431_E00771	130	■ 1.000	$\widetilde{\alpha}=\left(\begin{array}{cccc} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right) \in\left(E_7\right)_{(0,0,1,0)} \subset E_7 .$	$\tilde{\alpha} = \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (E_7)_{(0,0,1,0)} \subset E_7 .$	$\tilde{\alpha} = \begin{pmatrix} \alpha & 0 & 0 & 0 \\ 0 & \tau \alpha \tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in (E_7)_{(0,0,1,0)} \subset E_7 .$
0902.0431_E00775	130	■ 1.000	$\mathfrak{M}^{\wedge\{C\}} \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array}\right),$	$\mathfrak{M}^C \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right),$	$\mathfrak{M}^C \ni \alpha \left(\begin{array}{c} X \\ \frac{1}{\eta} X \times X \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right) = \left(\begin{array}{c} \beta X + \frac{1}{\eta} \epsilon(X \times X) \\ \delta X + \frac{1}{\eta} \beta_1(X \times X) \\ \frac{1}{\eta^2} \det X \\ \eta \end{array} \right),$
0902.0431_E00776	130	■ 1.000	$\left(\beta X+\frac{1}{\eta} \epsilon(X \times X)\right) \times\left(\beta X+\frac{1}{\eta} \epsilon(X \times X)\right)=\eta\left(\delta X+\frac{1}{\eta} \beta_1(X \times X)\right)$	$\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$	$\left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) \times \left(\beta X + \frac{1}{\eta} \epsilon(X \times X)\right) = \eta \left(\delta X + \frac{1}{\eta} \beta_1(X \times X)\right)$
0902.0431_E00778	130	■ 1.000	$\beta X \times \beta X=\beta_1(X \times X), \quad\left(\beta X, \beta_1(X \times X)\right)=3 \det X .$	$\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X .$	$\beta X \times \beta X = \beta_1(X \times X), \quad (\beta X, \beta_1(X \times X)) = 3 \det X .$
0902.0431_E00779	131	■ 1.000	$\operatorname{det}(\beta X)=\frac{1}{3}(\beta X, \beta X \times \beta X)=\frac{1}{3}(\beta X, \beta_1(X \times X))=\det X,$	$\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$	$\det(\beta X) = \frac{1}{3}(\beta X, \beta X \times \beta X) = \frac{1}{3}(\beta X, \beta_1(X \times X)) = \det X,$
0902.0431_E00781	131	■ 1.000	$(\operatorname{det} X) \beta_1 X=(\operatorname{det} X) \tau \beta \tau X .$	$(\det X) \beta_1 X = (\det X) \tau \beta \tau X .$	$(\det X) \beta_1 X = (\det X) \tau \beta \tau X .$
0902.0431_E00782	131	■ 1.000	$\varphi_1(\theta)(X, Y, \xi, \eta)=\left(\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta\right) .$	$\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta) .$	$\varphi_1(\theta)(X, Y, \xi, \eta) = (\theta^{-1} X, \theta Y, \theta^3 \xi, \theta^{-3} \eta) .$
0902.0431_E00783	131	■ 1.000	$U(1)=\left\{\varphi_1(\theta) \mid \theta \in C,(\tau \theta) \theta=1\right\}$	$U(1) = \{\varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1\}$	$U(1) = \{\varphi_1(\theta) \mid \theta \in C, (\tau \theta) \theta = 1\}$

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0902.0431_E00784	131	■ 1.000	$\alpha_{\{i\}}(a)=\left(\begin{array}{cccc} 1+(\cos a -1)p_i & 2a\frac{\sin a }{ a }E_i & 0 & -\tau a\frac{\sin a }{ a }E_i \\ -2\tau a\frac{\sin a }{ a }E_i & 1+(\cos a -1)p_i & a\frac{\sin a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin a }{ a }E_i & \cos a & 0 \\ a\frac{\sin a }{ a }E_i & 0 & 0 & \cos a \end{array}\right)$	$\alpha_i(a)=\left(\begin{array}{cccc} 1+(\cos a -1)p_i & 2a\frac{\sin a }{ a }E_i & 0 & -\tau a\frac{\sin a }{ a }E_i \\ -2\tau a\frac{\sin a }{ a }E_i & 1+(\cos a -1)p_i & a\frac{\sin a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin a }{ a }E_i & \cos a & 0 \\ a\frac{\sin a }{ a }E_i & 0 & 0 & \cos a \end{array}\right)$	$\alpha_i(a)=\left(\begin{array}{cccc} 1+(\cos a -1)p_i & 2a\frac{\sin a }{ a }E_i & 0 & -\tau a\frac{\sin a }{ a }E_i \\ -2\tau a\frac{\sin a }{ a }E_i & 1+(\cos a -1)p_i & a\frac{\sin a }{ a }E_i & 0 \\ 0 & -\tau a\frac{\sin a }{ a }E_i & \cos a & 0 \\ a\frac{\sin a }{ a }E_i & 0 & 0 & \cos a \end{array}\right)$
0902.0431_E00785	132	■ 1.000	$p_{\{i\}}\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$	$p_i\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$	$p_i\left(\begin{array}{ccc} \xi_1 & x_3 & \bar{x}_2 \\ \bar{x}_3 & \xi_2 & x_1 \\ x_2 & \bar{x}_1 & \xi_3 \end{array}\right)=\left(\begin{array}{ccc} \xi_1 & \delta_{i3}x_3 & \delta_{i2}\bar{x}_2 \\ \delta_{i3}\bar{x}_3 & \xi_2 & \delta_{i1}x_1 \\ \delta_{i2}x_2 & \delta_{i1}\bar{x}_1 & \xi_3 \end{array}\right),$
0902.0431_E00786	132	■ 1.000	$\Phi_{\{i\}}(a)=\Phi\left(0,-\tau a E_{\{i\}},\tau a E_{\{i\}},0\right)=\left(\begin{array}{cccc} 0 & 2a E_{\{i\}} & 0 & -\tau a E_{\{i\}} \\ -2\tau a E_{\{i\}} & 0 & a E_{\{i\}} & 0 \\ 0 & -\tau a E_{\{i\}} & 0 & 0 \\ a E_{\{i\}} & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$	$\Phi_i(a)=\Phi(0,-\tau a E_i,\tau a E_i,0)=\left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$	$\Phi_i(a)=\Phi(0,-\tau a E_i,\tau a E_i,0)=\left(\begin{array}{cccc} 0 & 2a E_i & 0 & -\tau a E_i \\ -2\tau a E_i & 0 & a E_i & 0 \\ 0 & -\tau a E_i & 0 & 0 \\ a E_i & 0 & 0 & 0 \end{array}\right)\in\mathfrak{e}_7$
0902.0431_E00788	132	■ 1.000	$\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$	$\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$	$\beta X=\frac{1}{\xi}\beta(Y\times Y)=\frac{1}{\xi}\tau\beta\tau Y\times\tau\beta\tau Y$
0902.0431_E00789	132	■ 1.000	$\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$	$\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$	$\tau\beta\tau Y=\left(\begin{array}{ccc} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{array}\right),\quad \eta_i\in C$
0902.0431_E00790	132	■ 1.000	$\alpha_{\{i\}}(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$	$\alpha_i(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$	$\alpha_i(-\pi/2)\beta P=\left(\begin{array}{cccc} 1-p_i & -2E_i & 0 & E_i \\ 2E_i & 1-p_i & -E_i & 0 \\ 0 & E_i & 0 & 0 \\ -E_i & 0 & 0 & 0 \end{array}\right)\left(\begin{array}{c} \beta X \\ \tau\beta\tau Y \\ 0 \\ \eta \end{array}\right)=\left(\begin{array}{c} * \\ * \\ \eta_i \\ * \end{array}\right),\quad \eta_i\neq 0,$
0902.0431_E00791	132	■ 1.000	$\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$	$\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$	$\alpha_{i+1}(-\pi/2)\beta P=(*,\xi_i E_{i+2}+\xi_{i+2} E_i,0,*),\quad \xi_i\neq 0,$
0902.0431_E00792	133	■ 1.000	$\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$	$\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$	$\alpha_1(-\pi/2)P=(\eta E_1,0,0,0),\quad \eta\neq 0,$
0902.0431_E00793	133	■ 1.000	$\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$	$\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$	$\mathfrak{M}_1=\{P\in\mathfrak{P}^C\mid P\times P=0,\langle P,P\rangle=1\}.$
0902.0431_E00794	133	■ 1.000	$E_{\{7\}}/E_{\{6\}}\simeq\mathfrak{M}_1.$	$E_7/E_6\simeq\mathfrak{M}_1.$	$E_7/E_6\simeq\mathfrak{M}_1.$

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0902.0431_E00795	133	■ 1.000	$\alpha P = \left(\frac{1}{\xi} \begin{pmatrix} \eta_2 \eta_3 & 0 & 0 \\ 0 & \eta_3 \eta_1 & 0 \\ 0 & 0 & \eta_1 \eta_2 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{pmatrix}, \xi, \frac{1}{\xi^2} \eta_1 \eta_2 \eta_3 \right), \quad \xi > 0$	$\alpha P = \left(\frac{1}{\xi} \begin{pmatrix} \eta_2 \eta_3 & 0 & 0 \\ 0 & \eta_3 \eta_1 & 0 \\ 0 & 0 & \eta_1 \eta_2 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{pmatrix}, \xi, \frac{1}{\xi^2} \eta_1 \eta_2 \eta_3 \right), \quad \xi > 0$	$\alpha P = \left(\frac{1}{\xi} \begin{pmatrix} \eta_2 \eta_3 & 0 & 0 \\ 0 & \eta_3 \eta_1 & 0 \\ 0 & 0 & \eta_1 \eta_2 \end{pmatrix}, \begin{pmatrix} \eta_1 & 0 & 0 \\ 0 & \eta_2 & 0 \\ 0 & 0 & \eta_3 \end{pmatrix}, \xi, \frac{1}{\xi^2} \eta_1 \eta_2 \eta_3 \right), \quad \xi > 0$
0902.0431_E00796	133	■ 1.000	$\frac{1}{\xi^2} (\eta_2 \eta_3 ^2 + \eta_3 \eta_1 ^2 + \eta_1 \eta_2 ^2) + (\eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2) + \xi^2 + \frac{1}{\xi^4} \eta_1 \eta_2 \eta_3 ^2 = 1,$	$\frac{1}{\xi^2} (\eta_2 \eta_3 ^2 + \eta_3 \eta_1 ^2 + \eta_1 \eta_2 ^2) + (\eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2) + \xi^2 + \frac{1}{\xi^4} \eta_1 \eta_2 \eta_3 ^2 = 1,$	$\frac{1}{\xi^2} (\eta_2 \eta_3 ^2 + \eta_3 \eta_1 ^2 + \eta_1 \eta_2 ^2) + (\eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2) + \xi^2 + \frac{1}{\xi^4} \eta_1 \eta_2 \eta_3 ^2 = 1,$
0902.0431_E00797 (i)	133	■ 1.000	$\left(1 + \frac{ \eta_1 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_2 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_3 ^2}{\xi^2} \right) = \frac{1}{\xi^2}.$	$\left(1 + \frac{ \eta_1 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_2 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_3 ^2}{\xi^2} \right) = \frac{1}{\xi^2}.$	$\left(1 + \frac{ \eta_1 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_2 ^2}{\xi^2} \right) \left(1 + \frac{ \eta_3 ^2}{\xi^2} \right) = \frac{1}{\xi^2}.$
0902.0431_E00798	133	■ 1.000	$\tan r_i = \frac{ \eta_i }{\xi}, \quad i = 1, 2, 3,$	$\tan r_i = \frac{ \eta_i }{\xi}, \quad i = 1, 2, 3,$	$\tan r_i = \frac{ \eta_i }{\xi}, \quad i = 1, 2, 3,$
0902.0431_E00800	134	■ 1.000	$a_i = \frac{\eta_i}{ \eta_i } r_i, \quad i = 1, 2, 3$	$a_i = \frac{\eta_i}{ \eta_i } r_i, \quad i = 1, 2, 3$	$a_i = \frac{\eta_i}{ \eta_i } r_i, \quad i = 1, 2, 3$
0902.0431_E00801	134	■ 1.000	$r_i = a_i , \quad \eta_i = \frac{1}{ a_i } \frac{\eta_i}{ \eta_i } r_i \frac{ \eta_i }{\xi} \xi = \frac{a_i}{ a_i } \tan r_i \cos r_1 \cos r_2 \cos r_3.$	$r_i = a_i , \quad \eta_i = \frac{1}{ a_i } \frac{\eta_i}{ \eta_i } r_i \frac{ \eta_i }{\xi} \xi = \frac{a_i}{ a_i } \tan r_i \cos r_1 \cos r_2 \cos r_3.$	$r_i = a_i , \quad \eta_i = \frac{1}{ a_i } \frac{\eta_i}{ \eta_i } r_i \frac{ \eta_i }{\xi} \xi = \frac{a_i}{ a_i } \tan r_i \cos r_1 \cos r_2 \cos r_3.$
0902.0431_E00803	134	■ 1.000	$\alpha_1 (a_1)^{-1} \alpha_2 (a_2)^{-1} \alpha_3 (a_3)^{-1} \alpha P = (0, 0, 1, 0).$	$\alpha_1 (a_1)^{-1} \alpha_2 (a_2)^{-1} \alpha_3 (a_3)^{-1} \alpha P = (0, 0, 1, 0).$	$\alpha_1 (a_1)^{-1} \alpha_2 (a_2)^{-1} \alpha_3 (a_3)^{-1} \alpha P = (0, 0, 1, 0).$
0902.0431_E00804	134	■ 1.000	$z(E_7) = \{1, -1\}.$	$z(E_7) = \{1, -1\}.$	$z(E_7) = \{1, -1\}.$
0902.0431_E00805	134	■ 1.000	$\beta X = X, \quad \tau \beta \tau Y = Y \quad \text{for all } \beta \in E_6.$	$\beta X = X, \quad \tau \beta \tau Y = Y \quad \text{for all } \beta \in E_6.$	$\beta X = X, \quad \tau \beta \tau Y = Y \quad \text{for all } \beta \in E_6.$
0902.0431_E00806	135	■ 1.000	$\alpha(0, 0, 1, 0) = (0, 0, \xi, \eta).$	$\alpha(0, 0, 1, 0) = (0, 0, \xi, \eta).$	$\alpha(0, 0, 1, 0) = (0, 0, \xi, \eta).$
0902.0431_E00808	135	■ 1.000	$\alpha(0, 0, 1, 0) = (0, 0, \xi, 0), \quad \alpha(0, 0, 0, 1) = (0, 0, 0, \xi^{-1}).$	$\alpha(0, 0, 1, 0) = (0, 0, \xi, 0), \quad \alpha(0, 0, 0, 1) = (0, 0, 0, \xi^{-1}).$	$\alpha(0, 0, 1, 0) = (0, 0, \xi, 0), \quad \alpha(0, 0, 0, 1) = (0, 0, 0, \xi^{-1}).$
0902.0431_E00814	136	■ 1.000	$(E_7)^\iota = \{ \alpha \in E_7 \mid \iota \alpha = \alpha \iota \} \\ \cong \{ \alpha \in E_7 \mid \lambda \alpha = \alpha \lambda \} = (E_7)^\lambda.$	$(E_7)^\iota = \{ \alpha \in E_7 \mid \iota \alpha = \alpha \iota \} \\ \cong \{ \alpha \in E_7 \mid \lambda \alpha = \alpha \lambda \} = (E_7)^\lambda.$	$(E_7)^\iota = \{ \alpha \in E_7 \mid \iota \alpha = \alpha \iota \} \\ \cong \{ \alpha \in E_7 \mid \lambda \alpha = \alpha \lambda \} = (E_7)^\lambda.$

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0902.0431_E00815	136	■1.000	$\varphi(\theta,\beta)=\varphi_1(\theta)\beta,$ $\quad \varphi_1(\theta)=\left(\begin{array}{cccc}\theta^{-1} & 0 & 0 & 0 \\ 0 & \theta 1 & 0 & 0 \\ 0 & 0 & \theta^3 & 0 \\ 0 & 0 & 0 & \theta^{-3}\end{array}\right), \beta=\left(\begin{array}{cccc}\beta & 0 & 0 & 0 \\ 0 & \tau\beta\tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1\end{array}\right).$	$\varphi(\theta,\beta)=\varphi_1(\theta)\beta,$ $\quad \varphi_1(\theta)=\left(\begin{array}{cccc}\theta^{-1}1 & 0 & 0 & 0 \\ 0 & \theta 1 & 0 & 0 \\ 0 & 0 & \theta^3 & 0 \\ 0 & 0 & 0 & \theta^{-3}\end{array}\right), \beta=\left(\begin{array}{cccc}\beta & 0 & 0 & 0 \\ 0 & \tau\beta\tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1\end{array}\right).$	$\varphi(\theta,\beta)=\varphi_1(\theta)\beta,$ $\quad \varphi_1(\theta)=\left(\begin{array}{cccc}\theta^{-1}1 & 0 & 0 & 0 \\ 0 & \theta 1 & 0 & 0 \\ 0 & 0 & \theta^3 & 0 \\ 0 & 0 & 0 & \theta^{-3}\end{array}\right), \beta=\left(\begin{array}{cccc}\beta & 0 & 0 & 0 \\ 0 & \tau\beta\tau & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1\end{array}\right).$
0902.0431_E00817	136	■1.000	$\mu M=0,$ $\quad \nu N=0.$	$\mu M=0,$ $\quad \nu N=0.$	$\mu M=0,$ $\quad \nu N=0.$
0902.0431_E00818 (i)	136	■1.000	$(M, N)=1.$	$(M, N)=1.$	$(M, N)=1.$
0902.0431_E00819	137	■1.000	$\mathfrak{M}^C\supset\alpha\left(\begin{array}{c}X\\\frac{1}{\eta}X\times X\\\frac{1}{\eta^2}\det X\\\eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta^2}(\det X)M\\\frac{1}{\eta}\delta(X\times X)+\eta N\\a(X)\\\frac{1}{\eta}b(X\times X)\end{array}\right),$	$\mathfrak{M}^C\supset\alpha\left(\begin{array}{c}X\\\frac{1}{\eta}X\times X\\\frac{1}{\eta^2}\det X\\\eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta^2}(\det X)M\\\frac{1}{\eta}\delta(X\times X)+\eta N\\a(X)\\\frac{1}{\eta}b(X\times X)\end{array}\right),$	$\mathfrak{M}^C\supset\alpha\left(\begin{array}{c}X\\\frac{1}{\eta}X\times X\\\frac{1}{\eta^2}\det X\\\eta\end{array}\right)=\left(\begin{array}{c}\beta X+\frac{1}{\eta^2}(\det X)M\\\frac{1}{\eta}\delta(X\times X)+\eta N\\a(X)\\\frac{1}{\eta}b(X\times X)\end{array}\right),$
0902.0431_E00823 (v)	137	■1.000	$\operatorname{det} X=a(X)b(X\times X).$	$\det X=a(X)b(X\times X).$	$\det X=a(X)b(X\times X).$
0902.0431_E00824	137	■1.000	$\alpha(0,0,1,0)=(0,0,\mu,0),$ $\quad \alpha(0,0,0,1)=(0,0,0,\mu^{-1}),$ $\quad \mu\in C,(\tau\mu)\mu=1.$	$\alpha(0,0,1,0)=(0,0,\mu,0),$ $\quad \alpha(0,0,0,1)=(0,0,0,\mu^{-1}),$ $\quad \mu\in C,(\tau\mu)\mu=1.$	$\alpha(0,0,1,0)=(0,0,\mu,0),$ $\quad \alpha(0,0,0,1)=(0,0,0,\mu^{-1}),$ $\quad \mu\in C,(\tau\mu)\mu=1.$
0902.0431_E00825	137	■1.000	$\alpha=\varphi_1(\theta)\beta,$ $\quad \theta\in U(1),\beta\in E_6.$	$\alpha=\varphi_1(\theta)\beta,$ $\quad \theta\in U(1),\beta\in E_6.$	$\alpha=\varphi_1(\theta)\beta,$ $\quad \theta\in U(1),\beta\in E_6.$
0902.0431_E00826	138	■1.000	$\sigma(X,Y,\xi,\eta)=(\sigma X,\sigma Y,\xi,\eta).$	$\sigma(X,Y,\xi,\eta)=(\sigma X,\sigma Y,\xi,\eta).$	$\sigma(X,Y,\xi,\eta)=(\sigma X,\sigma Y,\xi,\eta).$
0902.0431_E00827	138	■1.000	$(E_7)^\sigma=\{\alpha\in E_7\mid\sigma\alpha=\alpha\sigma\}.$	$(E_7)^\sigma=\{\alpha\in E_7\mid\sigma\alpha=\alpha\sigma\}.$	$(E_7)^\sigma=\{\alpha\in E_7\mid\sigma\alpha=\alpha\sigma\}.$
0902.0431_E00828	138	■1.000	$\kappa=\Phi(-2E_1\vee E_1,0,0,-1),$ $\quad \mu=\Phi(0,E_1,E_1,0).$	$\kappa=\Phi(-2E_1\vee E_1,0,0,-1),$ $\quad \mu=\Phi(0,E_1,E_1,0).$	$\kappa=\Phi(-2E_1\vee E_1,0,0,-1),$ $\quad \mu=\Phi(0,E_1,E_1,0).$
0902.0431_E00837	139	■1.000	$-4\tau A\times(E_1\times X)+(E_1,X)A=4E_1\times(A\times X)-\langle A,X\rangle E_1,$ $\quad X\in\mathfrak{J}^C.$	$-4\tau A\times(E_1\times X)+(E_1,X)A=4E_1\times(A\times X)-\langle A,X\rangle E_1,$ $\quad X\in\mathfrak{J}^C.$	$-4\tau A\times(E_1\times X)+(E_1,X)A=4E_1\times(A\times X)-\langle A,X\rangle E_1,$ $\quad X\in\mathfrak{J}^C.$
0902.0431_E00838	140	■1.000	$\nu=-\left(A,E_1\right)=-\tau\left(A,E_1\right),$	$\nu=-\left(A,E_1\right)=-\tau\left(A,E_1\right),$	$\nu=-\left(A,E_1\right)=-\tau\left(A,E_1\right),$
0902.0431_E00839	140	■1.000	$\phi(\nu)=2\nu E_1\vee E_1,$	$\phi(\nu)=2\nu E_1\vee E_1,$	$\phi(\nu)=2\nu E_1\vee E_1,$
0902.0431_E00842	140	■1.000	$\varphi_*(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right)$	$\varphi_*(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right)$	$\varphi_*(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right)$
0902.0431_E00843	140	■1.000	$\left[\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right),\left(\begin{array}{cc}\rho & b \\ -\tau b & -\rho\end{array}\right)\right]=\left(\begin{array}{cc}b(\tau a)-a(\tau b) & 2(b\nu-a\rho) \\ -2\tau(b\nu-a\rho) & a(\tau b)-b(\tau a)\end{array}\right),$	$\left[\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right),\left(\begin{array}{cc}\rho & b \\ -\tau b & -\rho\end{array}\right)\right]=\left(\begin{array}{cc}b(\tau a)-a(\tau b) & 2(b\nu-a\rho) \\ -2\tau(b\nu-a\rho) & a(\tau b)-b(\tau a)\end{array}\right),$	$\left[\left(\begin{array}{cc}\nu & a \\ -\tau a & -\nu\end{array}\right),\left(\begin{array}{cc}\rho & b \\ -\tau b & -\rho\end{array}\right)\right]=\left(\begin{array}{cc}b(\tau a)-a(\tau b) & 2(b\nu-a\rho) \\ -2\tau(b\nu-a\rho) & a(\tau b)-b(\tau a)\end{array}\right),$

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0902.0431_E00844	140	■1.000	$\begin{aligned} & \left[\Phi(\nu, aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho) \right] \\ & = \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)). \end{aligned}$	$[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho)]$ $= \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$	$\left[\Phi(\phi(\nu), aE_1, -\tau aE_1, \nu), \Phi(\phi(\rho), bE_1, -\tau bE_1, \rho) \right]$ $= \Phi(\phi(b\tau a) - a(\tau b)), 2(b\nu - a\rho)E_1, -2\tau(b\nu - a\rho)E_1, (\tau a)b - a(\tau b)).$
0902.0431_E00845	140	■1.000	$\left(\frac{e_7}{\nu} \right)^{\sigma} \ni \Phi \left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array} \right) = \Phi \left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array} \right) + \Phi \left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array} \right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$	$(\mathfrak{e}_7)^{\sigma} \ni \Phi \left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array} \right) = \Phi \left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array} \right) + \Phi \left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array} \right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$	$(\mathfrak{e}_7)^{\sigma} \ni \Phi \left(\begin{array}{c} \phi \\ A \\ -\tau A \\ \nu \end{array} \right) = \Phi \left(\begin{array}{c} \phi(\nu') \\ aE_1 \\ -\tau aE_1 \\ \nu' \end{array} \right) + \Phi \left(\begin{array}{c} \phi - \phi(\nu') \\ A - aE_1 \\ -\tau A + \tau aE_1 \\ \nu - \nu' \end{array} \right) \in \mathfrak{a}_1 \oplus (\mathfrak{e}_7)^{\kappa, \mu},$
0902.0431_E00846	141	■1.000	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)},$
0902.0431_E00847	141	■1.000	$\begin{aligned} & \alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a) \\ & = \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0). \end{aligned}$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a)$ $= \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$	$\alpha_{23}(a) = \alpha_2(a)\alpha_3(\tau a)$ $= \exp \Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0).$
0902.0431_E00848	141	■1.000	$\text{Spin}(10) = \{ \alpha \in E_6 \mid \alpha E_1 = E_1 \}$ $= \{ \alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1 \} \subset E_7$	$\text{Spin}(10) = \{ \alpha \in E_6 \mid \alpha E_1 = E_1 \}$ $= \{ \alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1 \} \subset E_7$	$\text{Spin}(10) = \{ \alpha \in E_6 \mid \alpha E_1 = E_1 \}$ $= \{ \alpha \in E_6 \mid \sigma \alpha = \alpha \sigma, \alpha E_1 = E_1 \} \subset E_7$
0902.0431_E00849	141	■1.000	$S^9 = \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$	$S^9 = \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$	$S^9 = \left\{ \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, 0, 0, 0 \right) \mid \xi \in C, x \in \mathfrak{C}, \bar{x}x + (\tau \xi)\xi = 1 \right\}.$
0902.0431_E00850	141	■1.000	$\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$	$\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$	$\alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \quad \text{if and only if} \quad \alpha(0, 0, 1, 0) = (0, 0, 1, 0).$
0902.0431_E00851	141	■1.000	$\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$	$\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$	$\left\{ \alpha \in ((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)} \mid \alpha(0, -E_1, 0, 1) = (0, -E_1, 0, 1) \right\} \cong \text{Spin}(10).$
0902.0431_E00854	142	■1.000	$P = \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$	$P = \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$	$P = \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau \xi \end{pmatrix}, \begin{pmatrix} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, 0, \tau \eta \right) \in S^{10},$
0902.0431_E00858	142	■1.000	$\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$	$\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$	$\beta \alpha_{23}(a)P = (i(E_2 + E_3), 0, 0, 0).$
0902.0431_E00859	142	■1.000	$\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$	$\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$	$\alpha_{23}(-\pi/4)\beta \alpha_{23}(a)P = (0, -iE_1, 0, i).$

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0902.0431_E0865	144	■ 1.000	$P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$	$P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$	$P=\left(\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & \xi & x \\ 0 & \bar{x} & -\tau\xi \end{array}\right),\left(\begin{array}{ccc} \eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right),0,\tau\eta\right)\in S^{11},$
0902.0431_E0866	144	■ 1.000	$\alpha(t) \, P \, \in S^{\wedge 10} \, .$	$\alpha(t)P\in S^{10}.$	$\alpha(t)P\in S^{10}.$
0902.0431_E0867	144	■ 1.000	$\beta\alpha(t) \, P=\left(0,-i\,E_1,0,i\right) \, .$	$\beta\alpha(t)P=(0,-iE_1,0,i) \, .$	$\beta\alpha(t)P=(0,-iE_1,0,i).$
0902.0431_E0868	144	■ 1.000	$\alpha(-\pi/4)\beta\alpha(t) \, P=\left(0,E_1,0,1\right) \, .$	$\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1) \, .$	$\alpha(-\pi/4)\beta\alpha(t)P=(0,E_1,0,1).$
0902.0431_E0869	144	■ 1.000	$p:\left(E_7\right)^{\kappa,\mu}\rightarrow SO(12)=SO\left(V^{12}\right)$	$p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$	$p:(E_7)^{\kappa,\mu}\rightarrow SO(12)=SO(V^{12})$
0902.0431_E0871	144	■ 1.000	$\left(E_7\right)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12) \, .$	$(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$	$(E_7)^{\kappa,\mu}/\{1,\sigma\}\cong SO(12).$
0902.0431_E0874	145	■ 1.000	$\varphi_2(SU(2))=\left\{\varphi_2(A)\in E_7\mid A\in SU(2)\right\}$	$\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$	$\varphi_2(SU(2))=\{\varphi_2(A)\in E_7\mid A\in SU(2)\}$
0902.0431_E0877	145	■ 1.000	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)(X,Y,\xi,\eta)=(X',Y',\xi',\eta')$
0902.0431_E0879	145	■ 1.000	$\exp\left(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)\right)=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma \, .$	$\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma \, .$	$\exp(\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu))=\varphi_2(A)\in\varphi_2(SU(2))\subset(E_7)^\sigma \, .$
0902.0431_E0880	146	■ 1.000	$\varphi(A,\beta)=\varphi_2(A)\beta \, .$	$\varphi(A,\beta)=\varphi_2(A)\beta.$	$\varphi(A,\beta)=\varphi_2(A)\beta.$
0902.0431_E0884	147	■ 1.000	$\begin{aligned} \left(E_7\right)^{\tau\gamma} &= \left\{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\right\} \\ &= \left\{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\right\} = \left(E_7\right)^{\lambda\gamma} \, . \end{aligned}$	$\begin{aligned} (E_7)^{\tau\gamma} &= \{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\} \\ &= \{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\} = (E_7)^{\lambda\gamma} \, . \end{aligned}$	$\begin{aligned} (E_7)^{\tau\gamma} &= \{\alpha\in E_7\mid \tau\gamma\alpha=\alpha\tau\gamma\} \\ &= \{\alpha\in E_7\mid \lambda\gamma\alpha=\alpha\lambda\gamma\} = (E_7)^{\lambda\gamma} \, . \end{aligned}$
0902.0431_E0885	147	■ 1.000	$\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \left\{P\in\mathfrak{P}^C\mid \tau\gamma P=P\right\} \\ &= \left\{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\right\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \left\{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\right\} \\ &= \left\{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\right\}, \\ &= i\left(\mathfrak{P}^C\right)_{\tau\gamma} \, . \end{aligned}$	$\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\}, \\ &= i(\mathfrak{P}^C)_{\tau\gamma}. \end{aligned}$	$\begin{aligned} \left(\mathfrak{P}^C\right)_{\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{\tau\gamma},\xi,\eta\in\boldsymbol{R}\}, \\ \left(\mathfrak{P}^C\right)_{-\tau\gamma} &= \{P\in\mathfrak{P}^C\mid \tau\gamma P=-P\} \\ &= \{(X,Y,\xi,\eta)\in\mathfrak{P}^C\mid X,Y\in(\mathfrak{J}^C)_{-\tau\gamma},\xi,\eta\in i\boldsymbol{R}\}, \\ &= i(\mathfrak{P}^C)_{\tau\gamma}. \end{aligned}$
0902.0431_E0887	147	■ 1.000	$k\left(\left(a+be_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$	$k((a+be_2))=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$	$k\left(\left(a+be_2\right)\right)=\left(\left(\begin{array}{cc} a & b \\ -\bar{b} & \bar{a} \end{array}\right)\right),\quad a,b\in\boldsymbol{C}.$
0902.0431_E0888	147	■ 1.000	$B=k(D)+e_1k(T),\quad D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0.$	$B=k(D)+e_1k(T),\quad D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0.$	$B=k(D)+e_1k(T),\quad D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0.$

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0902.0431_E00890	147	■1.000	$D_{\{1\}}+e_{\{1\}}T_{\{1\}}=0, \ D_{\{1\}}\in k(\mathfrak{sp}(4)), \ T_{\{1\}}\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{implies} \quad C_{\{1\}}=T_{\{1\}}=0.$	$D_1+e_1T_1=0, D_1\in k(\mathfrak{sp}(4)), T_1\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{implies} \quad C_1=T_1=0.$	$D_1+e_1T_1=0, D_1\in k(\mathfrak{sp}(4)), T_1\in k(\mathfrak{J}(4,\boldsymbol{H})_0) \quad \text{implies} \quad C_1=T_1=0.$
0902.0431_E00891	148	■1.000	$\begin{aligned} \left(\mathfrak{e}_{\{7\}}\right)^{\tau\gamma}&=\left(\Phi\in\mathfrak{e}_7\mid\tau\gamma\Phi=\Phi\tau\gamma\right)\\&=\left\{\Phi(\phi,A,-\gamma A,0)\in\mathfrak{e}_7\mid\phi\in(\mathfrak{e}_6)^{\tau\gamma},A\in(\mathfrak{J}^C)_{\tau\gamma}\right\}\\&=\left\{\Phi(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\right\}.\end{aligned}$	$\begin{aligned} (\mathfrak{e}_7)^{\tau\gamma}&=\{\Phi\in\mathfrak{e}_7\mid\tau\gamma\Phi=\Phi\tau\gamma\}\\&=\left\{\Phi(\phi,A,-\gamma A,0)\in\mathfrak{e}_7\mid\phi\in(\mathfrak{e}_6)^{\tau\gamma},A\in(\mathfrak{J}^C)_{\tau\gamma}\right\}\\&=\left\{\Phi(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\right\}.\end{aligned}$	$\begin{aligned} (\mathfrak{e}_7)^{\tau\gamma}&=\{\Phi\in\mathfrak{e}_7\mid\tau\gamma\Phi=\Phi\tau\gamma\}\\&=\{\Phi(\phi,A,-\gamma A,0)\in\mathfrak{e}_7\mid\phi\in(\mathfrak{e}_6)^{\tau\gamma},A\in(\mathfrak{J}^C)_{\tau\gamma}\}\\&=\{\Phi(\varphi_*(D),g^{-1}(T),-\gamma g^{-1}(T),0)\in\mathfrak{e}_7\mid D\in\mathfrak{sp}(4),T\in\mathfrak{J}(4,\boldsymbol{H})_0\}.\end{aligned}$
0902.0431_E00895	148	■1.000	$k_{\{J\}}\left(M_{\{1\}}+iM_{\{2\}}\right)=k\left(M_{\{1\}}\right)J+i\,k\left(M_{\{2\}}\right)J, \quad \text{\textnormal{$\quad$}} M_{\{1\}},M_{\{2\}}\in\mathfrak{J}(4,\boldsymbol{H}),$	$k_J(M_1+iM_2)=k(M_1)J+ik(M_2)J, \quad M_1,M_2\in\mathfrak{J}(4,\boldsymbol{H}),$	$k_J(M_1+iM_2)=k(M_1)J+ik(M_2)J, \quad M_1,M_2\in\mathfrak{J}(4,\boldsymbol{H}),$
0902.0431_E00896	148	■1.000	$\chi(X,Y,\xi,\eta)=k_{\{J\}}\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$	$\chi(X,Y,\xi,\eta)=k_J\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$	$\chi(X,Y,\xi,\eta)=k_J\left(gX-\frac{\xi}{2}E\right)+e_1k_J\left(g(\gamma Y)-\frac{\eta}{2}E\right).$
0902.0431_E00897	148	■1.000	$\varphi_*(B)P=\chi^{-1}\left(B(\chi P)+(\chi P)^tB\right), \quad P\in\mathfrak{P}^C.$	$\varphi_*(B)P=\chi^{-1}\left(B(\chi P)+(\chi P)^tB\right), \quad P\in\mathfrak{P}^C.$	$\varphi_*(B)P=\chi^{-1}(B(\chi P)+(\chi P)^tB), \quad P\in\mathfrak{P}^C.$
0902.0431_E00901	150	■1.000	$\varphi(A)P=\chi^{-1}\left(A(\chi P)^tA\right), \quad P\in\mathfrak{P}^C.$	$\varphi(A)P=\chi^{-1}\left(A(\chi P)^tA\right), \quad P\in\mathfrak{P}^C.$	$\varphi(A)P=\chi^{-1}(A(\chi P)^tA), \quad P\in\mathfrak{P}^C.$
0902.0431_E00902	150	■1.000	$\varphi_*(D)P=\chi^{-1}\left(D(\chi P)+(\chi P)^tD\right), \quad P\in\mathfrak{P}^C,$	$\varphi_*(D)P=\chi^{-1}\left(D(\chi P)+(\chi P)^tD\right), \quad P\in\mathfrak{P}^C,$	$\varphi_*(D)P=\chi^{-1}(D(\chi P)+(\chi P)^tD), \quad P\in\mathfrak{P}^C,$
0902.0431_E00904	150	■1.000	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$	$\tau\gamma Y=Y, \quad X=\frac{1}{\xi}(Y\times Y), \quad \tau\xi=\xi, \quad \tau\eta=\eta.$
0902.0431_E00910	151	■1.000	$\left(E_{\{7\}}\right)^w=\left\{\alpha\in E_{\{7\}}\mid w\alpha=\alpha w\right\}.$	$(E_7)^w=\{\alpha\in E_7\mid w\alpha=\alpha w\}.$	$(E_7)^w=\{\alpha\in E_7\mid w\alpha=\alpha w\}.$
0902.0431_E00912	151	■1.000	$E_{\{6,\boldsymbol{C}\}}=\left\{\alpha\in E_{\{7\}}\mid\alpha(0,0,1,0)=(0,0,1,0)\right\},$	$E_{6,C}=\{\alpha\in E_{7,C}\mid\alpha(0,0,1,0)=(0,0,1,0)\},$	$E_{6,C}=\{\alpha\in E_{7,C}\mid\alpha(0,0,1,0)=(0,0,1,0)\},$
0902.0431_E00913	152	■1.000	$\mathfrak{e}_{\{7,\boldsymbol{C}\}}=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\right\}$	$\mathfrak{e}_{7,C}=\left\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\right\}$	$\mathfrak{e}_{7,C}=\{\Phi(\phi,A,-\tau A,\nu)\mid\phi\in\mathfrak{e}_{6,C},A\in(\mathfrak{J}_C)^C,\nu\in i\boldsymbol{R}\}$
0902.0431_E00915	152	■1.000	$\left(\mathfrak{M}_{\{\boldsymbol{C}\}}\right)_1=\left\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\right\}$	$(\mathfrak{M}_C)_1=\left\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\right\}$	$(\mathfrak{M}_C)_1=\{P\in(\mathfrak{M}_C)^C\mid P\times P=0,\langle P,P\rangle=1\}$
0902.0431_E00917	152	■1.000	$h^{\prime}(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$	$h'(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$	$h'(a+bi)=a+be_1, \quad a,b\in\boldsymbol{R}.$
0902.0431_E00918	152	■1.000	$f(a\,v)=h^{\prime}(a)\,f(v), \quad a\in C, \, v\in V.$	$f(av)=h'(a)f(v), \quad a\in C, v\in V.$	$f(av)=h'(a)f(v), \quad a\in C, v\in V.$

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0902.0431_E00921	153	■ 1.000	$\begin{array}{lll} x_{156}=h^{\{\prime\}}\!\left(\xi_1\right), & x_{164}=h^{\{\prime\}}\!\left(x_3\right), & x_{145}=h^{\{\prime\}}\!\left(\overline{x}_2\right), \\ x_{256}=h^{\{\prime\}}\!\left(\overline{x}_3\right), & x_{264}=h^{\{\prime\}}\!\left(\xi_2\right), & x_{245}=h^{\{\prime\}}\!\left(x_1\right), \\ x_{356}=h^{\{\prime\}}\!\left(x_2\right), & x_{364}=h^{\{\prime\}}\!\left(\overline{x}_1\right), & x_{345}=h^{\{\prime\}}\!\left(\xi_3\right), \\ x_{423}=h^{\{\prime\}}\!\left(\eta_1\right), & x_{431}=h^{\{\prime\}}\!\left(y_3\right), & x_{412}=h^{\{\prime\}}\!\left(\overline{y}_2\right), \\ x_{523}=h^{\{\prime\}}\!\left(\overline{y}_3\right), & x_{531}=h^{\{\prime\}}\!\left(\eta_2\right), & x_{512}=h^{\{\prime\}}\!\left(y_1\right), \\ x_{623}=h^{\{\prime\}}\!\left(y_2\right), & x_{631}=h^{\{\prime\}}\!\left(\overline{y}_1\right), & x_{612}=h^{\{\prime\}}\!\left(\eta_3\right), \\ & x_{123}=h^{\{\prime\}}\!\left(\xi\right), & \\ & x_{456}=h^{\{\prime\}}\!\left(\eta\right). & \end{array}$	$\begin{array}{lll} x_{156}=h^{\{\prime\}}\!\left(\xi_1\right), & x_{164}=h^{\{\prime\}}\!\left(x_3\right), & x_{145}=h^{\{\prime\}}\!\left(\overline{x}_2\right), \\ x_{256}=h^{\{\prime\}}\!\left(\overline{x}_3\right), & x_{264}=h^{\{\prime\}}\!\left(\xi_2\right), & x_{245}=h^{\{\prime\}}\!\left(x_1\right), \\ x_{356}=h^{\{\prime\}}\!\left(x_2\right), & x_{364}=h^{\{\prime\}}\!\left(\overline{x}_1\right), & x_{345}=h^{\{\prime\}}\!\left(\xi_3\right), \\ x_{423}=h^{\{\prime\}}\!\left(\eta_1\right), & x_{431}=h^{\{\prime\}}\!\left(y_3\right), & x_{412}=h^{\{\prime\}}\!\left(\overline{y}_2\right), \\ x_{523}=h^{\{\prime\}}\!\left(\overline{y}_3\right), & x_{531}=h^{\{\prime\}}\!\left(\eta_2\right), & x_{512}=h^{\{\prime\}}\!\left(y_1\right), \\ x_{623}=h^{\{\prime\}}\!\left(y_2\right), & x_{631}=h^{\{\prime\}}\!\left(\overline{y}_1\right), & x_{612}=h^{\{\prime\}}\!\left(\eta_3\right), \\ & x_{123}=h^{\{\prime\}}\!\left(\xi\right), & \\ & x_{456}=h^{\{\prime\}}\!\left(\eta\right). & \end{array}$	
0902.0431_E00925	153	■ 1.000	$A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$	$A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$	$A(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=A \boldsymbol{a} \wedge \boldsymbol{A} \boldsymbol{b} \wedge \boldsymbol{A} \boldsymbol{c}.$
0902.0431_E00926	153	■ 1.000	$D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$	$D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$	$D(\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c})=D \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge D \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{a} \wedge \boldsymbol{b} \wedge D \boldsymbol{c}.$
0902.0431_E00929	154	■ 1.000	$\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$	$\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$	$\varphi_{\boldsymbol{C}}\left(\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)\right)=\Phi\left(\phi_{\boldsymbol{C}}(B, C), h(L),-\tau h(L),-i \nu e_1\right)$
0902.0431_E00930	154	■ 1.000	$\nu=\mathrm{tr}\left(B'\right)=-\mathrm{tr}\left(C'\right), \quad B=B'-\frac{\nu}{3} E, \quad C=C'+\frac{\nu}{3} E,$	$\nu=\mathrm{tr}\left(B'\right)=-\mathrm{tr}\left(C'\right), \quad B=B'-\frac{\nu}{3} E, \quad C=C'+\frac{\nu}{3} E,$	$\nu=\mathrm{tr}\left(B'\right)=-\mathrm{tr}\left(C'\right), \quad B=B'-\frac{\nu}{3} E, \quad C=C'+\frac{\nu}{3} E,$
0902.0431_E00934	154	■ 1.000	$\psi_{\ast}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$	$\psi_{\ast}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$	$\psi_{\ast}(D) P=f^{-1}(D(f P)), \quad P \in\left(\mathfrak{P}_C\right)^C$
0902.0431_E00935	154	■ 1.000	$D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$	$D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$	$D=\left(\begin{array}{cc} B & L \\ -L^* & C \end{array}\right)+\frac{\nu}{3}\left(\begin{array}{cc} E & 0 \\ 0 & -E \end{array}\right)$

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0902.0431_E00942	157	■ 1.000	$=\left(\begin{array}{c} \boldsymbol{\phi}(B, C) F_1(1) + \frac{i\nu e_1}{3} F_1(1) \\ 2h(L) \times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array}\right) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} F_1(1) \\ 0 \\ 0 \\ 0 \end{pmatrix}.$	$= \left(\begin{array}{c} \phi_C(B, C) F_1(1) + \frac{i\nu e_1}{3} F_1(1) \\ 2h(L) \times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array} \right) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} F_1(1) \\ 0 \\ 0 \\ 0 \end{pmatrix}.$	$= \left(\begin{array}{c} \phi_C(B, C) F_1(1) + \frac{i\nu e_1}{3} F_1(1) \\ 2h(L) \times F_1(1) \\ 0 \\ -(h(L), F_1(1)) \end{array} \right) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \begin{pmatrix} F_1(1) \\ 0 \\ 0 \\ 0 \end{pmatrix}.$
0902.0431_E00943	157	■ 1.000	$f^{-1}(D(fP)) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) P.$	$f^{-1}(D(fP)) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) P.$	$f^{-1}(D(fP)) = \Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) P.$
0902.0431_E00950	158	■ 1.000	$\mu^{-1}(M_1 + iM_2, N_1 + iN_2) = \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{pmatrix}, \quad M_i, N_i \in M(3, \mathcal{C}).$	$\mu^{-1}(M_1 + iM_2, N_1 + iN_2) = \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{pmatrix}, \quad M_i, N_i \in M(3, \mathcal{C}).$	$\mu^{-1}(M_1 + iM_2, N_1 + iN_2) = \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 - M_1 e_1 & N_2 - N_1 e_1 \end{pmatrix}, \quad M_i, N_i \in M(3, \mathcal{C}).$
0902.0431_E00951	158	■ 1.000	$\mu(\widetilde{MD}^*) = \psi_*(D)(\mu\widetilde{M}).$	$\mu(\widetilde{MD}^*) = \psi_*(D)(\mu\widetilde{M}).$	$\mu(\widetilde{MD}^*) = \psi_*(D)(\mu\widetilde{M}).$
0902.0431_E00954	159	■ 1.000	$= \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 + M_1 e_1 & N_2 + N_1 e_1 \end{pmatrix} \left(\begin{pmatrix} -B & -L \\ L^* & -C \end{pmatrix} - \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \right) = \widetilde{MD}^*.$	$= \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 + M_1 e_1 & N_2 + N_1 e_1 \end{pmatrix} \left(\begin{pmatrix} -B & -L \\ L^* & -C \end{pmatrix} - \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \right) = \widetilde{MD}^*.$	$= \begin{pmatrix} -N_2 - N_1 e_1 & M_2 + M_1 e_1 \\ M_2 + M_1 e_1 & N_2 + N_1 e_1 \end{pmatrix} \left(\begin{pmatrix} -B & -L \\ L^* & -C \end{pmatrix} - \frac{\nu}{3} \begin{pmatrix} E & 0 \\ 0 & -E \end{pmatrix} \right) = \widetilde{MD}^*.$
0902.0431_E00956	159	■ 1.000	$(Q, A) \left(\sum (\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) + \widetilde{M} \right) = \sum (A\boldsymbol{a} \wedge A\boldsymbol{b} \wedge A\boldsymbol{c}) + Q\widetilde{M}A^*,$	$(Q, A) \left(\sum (\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) + \widetilde{M} \right) = \sum (A\boldsymbol{a} \wedge A\boldsymbol{b} \wedge A\boldsymbol{c}) + Q\widetilde{M}A^*,$	$(Q, A) \left(\sum (\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}) + \widetilde{M} \right) = \sum (A\boldsymbol{a} \wedge A\boldsymbol{b} \wedge A\boldsymbol{c}) + Q\widetilde{M}A^*,$
0902.0431_E00958	159	■ 1.000	$\Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \in \mathfrak{e}_7.$	$\Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \in \mathfrak{e}_7.$	$\Phi(\phi_C(B, C), h(L), -\tau h(L), -i\nu e_1) \in \mathfrak{e}_7.$
0902.0431_E00961	160	■ 1.000	$\alpha = \psi(E, A)\beta = \psi(E, A)\psi(Q, E) = \psi(Q, A).$	$\alpha = \psi(E, A)\beta = \psi(E, A)\psi(Q, E) = \psi(Q, A).$	$\alpha = \psi(E, A)\beta = \psi(E, A)\psi(Q, E) = \psi(Q, A).$
0902.0431_E00966	161	■ 1.000	$\mathfrak{P} = \mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}' = \mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$	$\mathfrak{P} = \mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}' = \mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$	$\mathfrak{P} = \mathfrak{J}(3, \mathfrak{C}) \oplus \mathfrak{J}(3, \mathfrak{C}) \oplus \boldsymbol{R} \oplus \boldsymbol{R},$ $\mathfrak{P}' = \mathfrak{J}(3, \mathfrak{C}') \oplus \mathfrak{J}(3, \mathfrak{C}') \oplus \boldsymbol{R} \oplus \boldsymbol{R}.$
0902.0431_E00967	161	■ 1.000	$\langle P, Q \rangle_\sigma = \langle \sigma P, Q \rangle.$	$\langle P, Q \rangle_\sigma = \langle \sigma P, Q \rangle.$	$\langle P, Q \rangle_\sigma = \langle \sigma P, Q \rangle.$
0902.0431_E00970	161	■ 1.000	$E_{7(7)} \simeq SU(8)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{70},$ $E_{7(-5)} \simeq (SU(2) \times Spin(12))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{64},$ $E_{7(-25)} \simeq (U(1) \times E_6)/\boldsymbol{Z}_3 \times \boldsymbol{R}^{54}.$	$E_{7(7)} \simeq SU(8)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{70},$ $E_{7(-5)} \simeq (SU(2) \times Spin(12))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{64},$ $E_{7(-25)} \simeq (U(1) \times E_6)/\boldsymbol{Z}_3 \times \boldsymbol{R}^{54}.$	$E_{7(7)} \simeq SU(8)/\boldsymbol{Z}_2 \times \boldsymbol{R}^{70},$ $E_{7(-5)} \simeq (SU(2) \times Spin(12))/\boldsymbol{Z}_2 \times \boldsymbol{R}^{64},$ $E_{7(-25)} \simeq (U(1) \times E_6)/\boldsymbol{Z}_3 \times \boldsymbol{R}^{54}.$

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0902.0431_E09971	161	■ 1.000	$z\left(E_{7(7)}\right)=\boldsymbol{Z}_{\{2\}}, \quad z\left(E_{7(-5)}\right)=\boldsymbol{Z}_{\{2\}}, \quad z\left(E_{7(-25)}\right)=\boldsymbol{Z}_{\{2\}}.$	$z\left(E_{7(7)}\right)=\boldsymbol{Z}_2, \quad z\left(E_{7(-5)}\right)=\boldsymbol{Z}_2, \quad z\left(E_{7(-25)}\right)=\boldsymbol{Z}_2.$	$z(E_{7(7)})=\boldsymbol{Z}_2, \quad z(E_{7(-5)})=\boldsymbol{Z}_2, \quad z(E_{7(-25)})=\boldsymbol{Z}_2.$
0902.0431_E09972	162	■ 1.000	$\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{e}_{\{7\}}^{\{C\}} \oplus \mathfrak{e}_{\{P\}}^{\{C\}} \oplus \mathfrak{e}_{\{C\}} \oplus \mathfrak{e}_{\{C\}} \oplus \mathfrak{e}_{\{C\}},$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C,$
0902.0431_E09975	162	■ 1.000	$\begin{aligned} & \left[R_1, R_2+R_3\right]=\left[R_1, R_2\right]+\left[R_1, R_3\right], \\ & \left[k R_1, R_2\right]=k\left[R_1, R_2\right], \quad k \in C, \\ & \left[R_1, R_2\right]=-\left[R_2, R_1\right] \end{aligned}$	$\begin{aligned} \left[R_1, R_2+R_3\right] &= \left[R_1, R_2\right] + \left[R_1, R_3\right], \\ \left[k R_1, R_2\right] &= k \left[R_1, R_2\right], \quad k \in C, \\ \left[R_1, R_2\right] &= -\left[R_2, R_1\right] \end{aligned}$	$\begin{aligned} \left[R_1, R_2+R_3\right] &= \left[R_1, R_2\right] + \left[R_1, R_3\right], \\ \left[k R_1, R_2\right] &= k \left[R_1, R_2\right], \quad k \in C, \\ \left[R_1, R_2\right] &= -\left[R_2, R_1\right] \end{aligned}$
0902.0431_E09978	163	■ 1.000	$\mathfrak{e}_{\{8\}}^{\{C\}}=\mathfrak{e}_{\{7\}}^{\{C\}} \oplus \mathfrak{e}_{\{K\}}^{\{C\}},$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C,$
0902.0431_E09980	163	■ 1.000	$\begin{aligned} & \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] = (0,0,1,0)^-, \\ & \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] = (0,0,0,1)_-, \\ & \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] = 1^-, \\ & \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] = 1_-, \\ & \mathfrak{a} \ni \left[1^-, 1_-\right] = 1, \\ & \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] = P^- + Q_-, \end{aligned}$	$\begin{aligned} \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] &= (0,0,1,0)^-, \\ \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] &= (0,0,0,1)_-, \\ \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] &= 1^-, \\ \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] &= 1_-, \\ \mathfrak{a} \ni \left[1^-, 1_-\right] &= 1, \\ \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] &= P^- + Q_-, \end{aligned}$	$\begin{aligned} \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,1,0)^-\right] &= (0,0,1,0)^-, \\ \mathfrak{a} \ni \left[\Phi(0,0,0,1), (0,0,0,-1)_-\right] &= (0,0,0,1)_-, \\ \mathfrak{a} \ni \left[(0,0,1,0)^-, (0,0,0,4)^-\right] &= 1^-, \\ \mathfrak{a} \ni \left[(0,0,0,1)_-, (0,0,4,0)_-\right] &= 1_-, \\ \mathfrak{a} \ni \left[1^-, 1_-\right] &= 1, \\ \mathfrak{a} \ni \left[1^- + 1_-, Q^- + P_-\right] &= P^- + Q_-, \end{aligned}$
0902.0431_E09984	164	■ 1.000	$\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$	$\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$	$\mathfrak{a} \ni \left[1_-, R\right] = (0,0,0,-s,0,0).$
0902.0431_E09990	165	■ 1.000	$\begin{aligned} & B_8\left(R_1, R_2\right) \\ & =-15\left(R_1, R_2\right)_8 \\ & =-15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ & =\frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \end{aligned}$	$\begin{aligned} B_8\left(R_1, R_2\right) &= -15\left(R_1, R_2\right)_8 \\ &= -15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ &= \frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \end{aligned}$	$\begin{aligned} B_8\left(R_1, R_2\right) &= -15\left(R_1, R_2\right)_8 \\ &= -15\left(\Phi_1, \Phi_2\right)_7+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2 \\ &= \frac{5}{3} B_7\left(\Phi_1, \Phi_2\right)+15\left\{Q_1, P_2\right\}-15\left\{P_1, Q_2\right\}+120 r_1 r_2+60 t_1 s_2+60 s_1 t_2, \end{aligned}$
0902.0431_E09991	165	■ 1.000	$B_{\{8\}}\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$	$B_8\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$	$B_8\left(R_1, R_2\right)=k\left(R_1, R_2\right)_8, \quad R_i \in \mathfrak{e}_8^C.$
0902.0431_E09992	165	■ 1.000	$(1,1)_{\{8\}}=-8.$	$(1,1)_8=-8.$	$(1,1)_8=-8.$
0902.0431_E09993	165	■ 1.000	$\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$	$\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$	$\left[1,\left[1,\left(\Phi, P, Q, r, s, t\right)\right]\right]=\left[1,\left(0, P,-Q, 0,2 s,-2 t\right)\right]=\left(0, P, Q, 0,4 s, 4 t\right),$
0902.0431_E01002	167	■ 1.000	$\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$	$\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$	$\tau(\Phi, P, Q, r, s, t)=\left(\tau \Phi \tau, \tau P, \tau Q, \tau r, \tau s, \tau t\right),$

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0902.0431_E01005	167	■1.000	$\left(\tau\lambda\lambda^{-1}\tau,\Phi_2\right)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$	$\left(\tau\lambda\Phi_1\lambda^{-1}\tau,\Phi_2\right)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$	$\left(\tau\lambda\Phi_1\lambda^{-1}\tau,\Phi_2\right)_7=2\left(\tau^t\phi_1\tau,\phi_2\right)_6+4\left\langle A_1,A_2\right\rangle +4\left\langle B_1,B_2\right\rangle +\frac{8}{3}\left(\tau\nu_1\right)\nu_2.$
0902.0431_E01006	167	■1.000	$\left(\tau^t\phi_1\tau,\phi_2\right)_6=-\left(\tau\delta_1\tau,\delta_2\right)_4+\left\langle T_1,T_2\right\rangle .$	$\left(\tau^t\phi_1\tau,\phi_2\right)_6=-\left(\tau\delta_1\tau,\delta_2\right)_4+\left\langle T_1,T_2\right\rangle .$	$\left(\tau^t\phi_1\tau,\phi_2\right)_6=-\left(\tau\delta_1\tau,\delta_2\right)_4+\left\langle T_1,T_2\right\rangle .$
0902.0431_E01011	168	■1.000	$\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$	$\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$	$\tau\widetilde{\lambda}R=\left(\tau\lambda\Phi\lambda^{-1}\tau,\tau\lambda Q,-\tau\lambda P,-\tau r,-\tau t,-\tau s\right).$
0902.0431_E01012	168	■1.000	$R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$	$R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$	$R=\frac{R-R^*}{2}+i\frac{R+R^*}{2i},\quad\frac{R-R^*}{2},\frac{R+R^*}{2i}\in\mathfrak{e}_8.$
0902.0431_E01017	170	■1.000	$\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$	$\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$	$\mathfrak{h}=\left\{\left(\Phi(h),0,0,r,0,0\right)\mid\Phi(h)\in\mathfrak{h}_7,r\in C\right\}$
0902.0431_E01018	170	■1.000	$\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right),0,0,0,0,0\right).$	$\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right),0,0,0,0,0\right).$	$\left[\left(\Phi(h),0,0,r,0,0\right),\left(\Phi,0,0,0,0,0\right)\right]=\left[\left(\Phi(h),\Phi\right),0,0,0,0,0\right).$
0902.0431_E01020	170	■1.000	$\begin{aligned} &\left(\Phi\left(h_\delta+\widetilde{H},0,0,\nu\right)+r1\right)(X,Y,\xi,\eta)\\ &=\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$	$\begin{aligned} &\left(\Phi\left(h_\delta+\widetilde{H},0,0,\nu\right)+r1\right)(X,Y,\xi,\eta)\\ &=\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$	$\begin{aligned} &\left(\Phi\left(h_\delta+\widetilde{H},0,0,\nu\right)+r1\right)(X,Y,\xi,\eta)\\ &=\left(\left(h_\delta+\widetilde{H}-\frac{1}{3}\nu+r\right)X,\left(h_\delta-\widetilde{H}+\frac{1}{3}\nu+r\right)Y,(\nu+r)\xi,(-\nu+r)\eta\right). \end{aligned}$
0902.0431_E01024	171	■1.000	$\begin{aligned} &\alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right),\\ &\alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r,\\ &\alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right),\\ &\alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$	$\begin{aligned} &\alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right),\\ &\alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r,\\ &\alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right),\\ &\alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$	$\begin{aligned} &\alpha_1=\frac{1}{2}\left(\lambda_0-\lambda_1-\lambda_2-\lambda_3\right)+\frac{1}{2}\left(\mu_3-\mu_1\right),\\ &\alpha_2=\mu_1-\frac{1}{3}\nu-r,\quad\alpha_3=2r,\\ &\alpha_4=\mu_2-\frac{1}{3}\nu-r,\quad\alpha_5=\lambda_3-\frac{1}{2}\left(\mu_2-\mu_3\right),\\ &\alpha_6=\lambda_2-\lambda_3,\quad\alpha_7=\lambda_1-\lambda_2,\quad\alpha_8=\nu-r \end{aligned}$
0902.0431_E01025	171	■1.000	$\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$	$\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$	$\mu=2\alpha_1+4\alpha_2+6\alpha_3+5\alpha_4+4\alpha_5+3\alpha_6+2\alpha_7+3\alpha_8$
0902.0431_E01035	176	■1.000	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$	$\mathfrak{h}_{\boldsymbol{R}}=\left\{\left(\Phi(h),0,0,r,0\right)\mid\Phi(h)\in\left(\mathfrak{h}_7\right)_{\boldsymbol{R}},r\in\boldsymbol{R}\right\},$
0902.0431_E01040	177	■1.000	$\left(\alpha_1,\alpha_1\right)=B_7\left(H_{\alpha_1},H_{\alpha_1}\right)=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$	$\left(\alpha_1,\alpha_1\right)=B_7\left(H_{\alpha_1},H_{\alpha_1}\right)=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$	$\left(\alpha_1,\alpha_1\right)=B_7\left(H_{\alpha_1},H_{\alpha_1}\right)=60\frac{1}{120}\frac{1}{120}4+30\frac{1}{120}\frac{1}{120}8=\frac{1}{30},$

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0902.0431_EQ1043	178	■1.000	$\widetilde{\beta}=\left(\begin{array}{cccccc}\operatorname{Ad}\beta&\beta&\beta&\beta&\beta&\beta\\0&0&0&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{array}\right),$	$\widetilde{\beta}=\begin{pmatrix}\operatorname{Ad}\beta&0&0&0&0&0\\0&\beta&0&0&0&0\\0&0&\beta&0&0&0\\0&0&0&1&0&0\\0&0&0&0&1&0\\0&0&0&0&0&1\end{pmatrix},$	
0902.0431_EQ1045	178	■1.000	$\begin{gathered} 0=\left[\left(\begin{array}{c}\beta_1\Phi,\beta_{21}\Phi,\beta_{31}\Phi,a_1\Phi,a_2\Phi,a_3\Phi\end{array}\right),\left(0,0,0,1,0,0\right)\right] \\ =\left(0,-\beta_{21}\Phi,\beta_{31}\Phi,0,-2a_2\Phi,2a_3\Phi\right) \end{gathered}$	$0=[(\beta_1\Phi,\beta_{21}\Phi,\beta_{31}\Phi,a_1\Phi,a_2\Phi,a_3\Phi),(0,0,0,1,0,0)]$ $=(0,-\beta_{21}\Phi,\beta_{31}\Phi,0,-2a_2\Phi,2a_3\Phi)$	$0=[(\beta_1\Phi,\beta_{21}\Phi,\beta_{31}\Phi,a_1\Phi,a_2\Phi,a_3\Phi),(0,0,0,1,0,0)]$ $=(0,-\beta_{21}\Phi,\beta_{31}\Phi,0,-2a_2\Phi,2a_3\Phi),$
0902.0431_EQ1048	179	■1.000	$\alpha=\left(\begin{array}{cccc}\beta_1&0&0&0\\0&\beta_2&0&0\\0&0&\beta_3&0\\0&0&0&E\end{array}\right).$	$\alpha=\begin{pmatrix}\beta_1&0&0&0\\0&\beta_2&0&0\\0&0&\beta_3&0\\0&0&0&E\end{pmatrix}.$	$\alpha=\begin{pmatrix}\beta_1&0&0&0\\0&\beta_2&0&0\\0&0&\beta_3&0\\0&0&0&E\end{pmatrix}.$
0902.0431_EQ1051 (iii)	179	■1.000	$\left(\beta_1\Phi\right)\left(\beta_2P\right)=\beta_2\left(\Phi P\right).$	$(\beta_1\Phi)(\beta_2P)=\beta_2(\Phi P).$	$(\beta_1\Phi)(\beta_2P)=\beta_2(\Phi P).$
0902.0431_EQ1055	179	■1.000	$v(\Phi,P,Q,r,s,t)=(\Phi.-P,-Q,r,s,t).$	$v(\Phi,P,Q,r,s,t)=(\Phi.-P,-Q,r,s,t).$	$v(\Phi,P,Q,r,s,t)=(\Phi.-P,-Q,r,s,t).$
0902.0431_EQ1056	179	■1.000	$(E_8)^v=\{\alpha\in E_8\mid v\alpha=\alpha v\}.$	$(E_8)^v=\{\alpha\in E_8\mid v\alpha=\alpha v\}.$	$(E_8)^v=\{\alpha\in E_8\mid v\alpha=\alpha v\}.$
0902.0431_EQ1058	180	■1.000	$(E_8)_{1-}=\{\alpha\in E_8\mid \alpha 1_-=1_-\}.$	$(E_8)_{1-}=\{\alpha\in E_8\mid \alpha 1_-=1_-\}.$	$(E_8)_{1-}=\{\alpha\in E_8\mid \alpha 1_-=1_-\}.$
0902.0431_EQ1061	180	■1.000	$E_8/E_7\simeq\mathfrak{W}_1.$	$E_8/E_7\simeq\mathfrak{W}_1.$	$E_8/E_7\simeq\mathfrak{W}_1.$
0902.0431_EQ1062	180	■1.000	$\varphi_3(SU(2))=\{\varphi_3(A)\in E_8\mid A\in SU(2)\}$	$\varphi_3(SU(2))=\{\varphi_3(A)\in E_8\mid A\in SU(2)\}$	$\varphi_3(SU(2))=\{\varphi_3(A)\in E_8\mid A\in SU(2)\}$
0902.0431_EQ1063	180	■1.000	$\varphi_3(A)=\left(\begin{array}{cccccc}1&0&0&0&0&0\\0&a1&-\tau b1&0&0&0\\0&b1&\tau a1&0&0&0\\0&0&0&(\tau a)a-(\tau b)b&-(\tau a)b&a(\tau b)\\0&0&0&2a(\tau b)&a^2&-(\tau b)^2\\0&0&0&2(\tau a)b&-b^2&(\tau a)^2\end{array}\right).$	$\varphi_3(A)=\begin{pmatrix}1&0&0&0&0&0\\0&a1&-\tau b1&0&0&0\\0&b1&\tau a1&0&0&0\\0&0&0&(\tau a)a-(\tau b)b&-(\tau a)b&a(\tau b)\\0&0&0&2a(\tau b)&a^2&-(\tau b)^2\\0&0&0&2(\tau a)b&-b^2&(\tau a)^2\end{pmatrix}.$	$\varphi_3(A)=\begin{pmatrix}1&0&0&0&0&0\\0&a1&-\tau b1&0&0&0\\0&b1&\tau a1&0&0&0\\0&0&0&(\tau a)a-(\tau b)b&-(\tau a)b&a(\tau b)\\0&0&0&2a(\tau b)&a^2&-(\tau b)^2\\0&0&0&2(\tau a)b&-b^2&(\tau a)^2\end{pmatrix}.$
0902.0431_EQ1064	181	■1.000	$\varphi(A,\beta)=\varphi_3(A)\beta.$	$\varphi(A,\beta)=\varphi_3(A)\beta.$	$\varphi(A,\beta)=\varphi_3(A)\beta.$
0902.0431_EQ1065	181	■1.000	$\begin{gathered} \left(\mathfrak{e}_8\right)^v=\{\Theta(R)\in\Theta(\mathfrak{e}_8)\mid v\Theta(R)=\Theta(R)v\}\cong\{R\in\mathfrak{e}_8\mid vR=R\} \\ =\{(\Phi,0,0,r,s,-\tau s)\mid\Phi\in\mathfrak{e}_7,r\in i\boldsymbol{R},s\in C\}. \end{gathered}$	$(\mathfrak{e}_8)^v=\{\Theta(R)\in\Theta(\mathfrak{e}_8)\mid v\Theta(R)=\Theta(R)v\}\cong\{R\in\mathfrak{e}_8\mid vR=R\}$ $=\{(\Phi,0,0,r,s,-\tau s)\mid\Phi\in\mathfrak{e}_7,r\in i\boldsymbol{R},s\in C\}.$	$(\mathfrak{e}_8)^v=\{\Theta(R)\in\Theta(\mathfrak{e}_8)\mid v\Theta(R)=\Theta(R)v\}\cong\{R\in\mathfrak{e}_8\mid vR=R\}$ $=\{(\Phi,0,0,r,s,-\tau s)\mid\Phi\in\mathfrak{e}_7,r\in i\boldsymbol{R},s\in C\}.$

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0902.0431_EQ1068	181	■ 1.000	$\left(\left(\left(x_{kl}+iy_{kl}\right)\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$	$l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$	$l\left(\left(x_{kl}+iy_{kl}\right)\right)=\left(\left(\begin{array}{cc} x_{kl} & y_{kl} \\ -y_{kl} & x_{kl} \end{array}\right)\right), \quad x_{kl}, y_{kl} \in \boldsymbol{R}.$
0902.0431_EQ1069	182	■ 1.000	$I=\operatorname{diag}(I,\cdots,I), \quad I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \quad J=\operatorname{diag}(J,\cdots,J), \quad J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$	$I=\operatorname{diag}(I,\cdots,I), \quad I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \quad J=\operatorname{diag}(J,\cdots,J), \quad J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$	$I=\operatorname{diag}(I,\cdots,I), \quad I=\left(\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array}\right), \quad J=\operatorname{diag}(J,\cdots,J), \quad J=\left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right),$
0902.0431_EQ1070	182	■ 1.000	$l(\frac{S}{8},C)) \quad I=\{B \in \mathfrak{so}(16) \mid JB=-BJ\}.$	$l(\mathfrak{S}(8,C))I=\{B \in \mathfrak{so}(16) \mid JB=-BJ\}.$	$l(\mathfrak{S}(8,C))I=\{B \in \mathfrak{so}(16) \mid JB=-BJ\}.$
0902.0431_EQ1072	182	■ 1.000	$l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$	$l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$	$l(-D)=-B=\left\{ \right\}^t B=\left\{ \right\}^t l(D)=l\left(\tau^t D\right)$
0902.0431_EQ1074	182	■ 1.000	$l(-S) \quad I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=I l\left(\tau^t S\right)=l\left({}^t S\right) I$	$l(-S)I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=I l\left(\tau^t S\right)=l\left({}^t S\right) I$	$l(-S)I=-B=\left\{ \right\}^t B=\left\{ \right\}^t I^t(l(S))=I l\left(\tau^t S\right)=l\left({}^t S\right) I$
0902.0431_EQ1075	182	■ 1.000	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$	$\varphi_*([gX_1,gX_2])=2(X_1\vee\gamma X_2-X_2\vee\gamma X_1), \quad X_1,X_2\in\mathfrak{J}^C.$
0902.0431_EQ1084	184	■ 1.000	$\begin{aligned} &\&\left.A=2\,\gamma\,X_1\,\times\,Y_2-2\,\gamma\,X_2\,\times\,Y_1+\eta_1\,X_2-\eta_2\,X_1-\xi_1\,\gamma\,X_2+\xi_2\,\gamma\,Y_1\in\mathfrak{J}^C\right)\\ &=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E. \end{aligned}$	$\begin{aligned} &A=2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1-\xi_1\gamma X_2+\xi_2\gamma Y_1\in\mathfrak{J}^C)\\ &=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E. \end{aligned}$	$\begin{aligned} &A=2\gamma X_1\times Y_2-2\gamma X_2\times Y_1+\eta_1X_2-\eta_2X_1-\xi_1\gamma X_2+\xi_2\gamma Y_1\in\mathfrak{J}^C)\\ &=kD+ik(gA)+\frac{i}{2}\{P_1,P_2\}E. \end{aligned}$
0902.0431_EQ1085	184	■ 1.000	$\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$	$\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$	$\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=4i\left\{P_1,P_2\right\}=4i\left\{\chi^{-1}S_1,\chi^{-1}S_2\right\}.$
0902.0431_EQ1086	184	■ 1.000	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA)$	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA)$	$S_1\tau S_2-S_2\tau S_1-\frac{1}{8}\operatorname{tr}\left(S_1\tau S_2-S_2\tau S_1\right)=kD+ik(gA),$
0902.0431_EQ1087	184	■ 1.000	$\begin{aligned} &\&\lambda\gamma P_1\times P_2=\left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array}\right)\times\left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array}\right)\\ &=\Phi\left(\begin{array}{c} -\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1) \\ \frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1) \\ \frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1)) \end{array}\right). \end{aligned}$	$\begin{aligned} &\lambda\gamma P_1\times P_2=\left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array}\right)\times\left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array}\right)\\ &=\Phi\left(\begin{array}{c} -\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1) \\ \frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1) \\ \frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1)) \end{array}\right). \end{aligned}$	$\begin{aligned} &\lambda\gamma P_1\times P_2=\left(\begin{array}{c} \gamma Y_1 \\ -\gamma X_1 \\ \eta_1 \\ -\xi_1 \end{array}\right)\times\left(\begin{array}{c} X_2 \\ Y_2 \\ \xi_2 \\ \eta_2 \end{array}\right)\\ &=\Phi\left(\begin{array}{c} -\frac{1}{2}(\gamma Y_1\vee Y_2-X_2\vee\gamma X_1) \\ \frac{1}{4}(2\gamma X_1\times Y_2+\eta_1X_2+\xi_2\gamma Y_1) \\ \frac{1}{4}(2\gamma Y_1\times X_2+\xi_1Y_2+\eta_2\gamma X_1) \\ \frac{1}{8}((\gamma Y_1,Y_2)-(X_2,\gamma X_1)-3(\eta_1\eta_2-\xi_2\xi_1)) \end{array}\right). \end{aligned}$

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0902.0431_E01094	186	■ 1.000	=\left[\backslash\Theta\left(\backslash\varphi_{*}\,D,\,0,0,0,0,0\right),\backslash\Theta(0,0,0,0,2\,c,-2\,c)=[\backslash\zeta\,l(D),\,\backslash\zeta\,l(i\,c\,E)]\,\backslash\right].	$= [\Theta(\varphi_*D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)].$	$= [\Theta(\varphi_*D, 0, 0, 0, 0, 0), \Theta(0, 0, 0, 0, 2c, -2c) = [\zeta l(D), \zeta l(icE)].$
0902.0431_E01111	188	■ 1.000	\sigma(\backslash\Phi,\,P,\,Q,\,r,\,s,\,t)=(\backslash\sigma\,\backslash\Phi\,\backslash\sigma\sigma,\,\backslash\sigma\sigma\,P,\,\backslash\sigma\sigma\,Q,\,r,\,s,\,t)\,.	$\sigma(\Phi, P, Q, r, s, t) = (\sigma\Phi\sigma, \sigma P, \sigma Q, r, s, t).$	$\sigma(\Phi, P, Q, r, s, t) = (\sigma\Phi\sigma, \sigma P, \sigma Q, r, s, t).$
0902.0431_E01112	188	■ 1.000	\left(E_8\right)^{\backslash\sigma}=\left\{\backslash\alpha\,\backslash\mathrm{in}\,E_8\,\backslash\mathrm{mid}\,\backslash\sigma\,\backslash\alpha=\alpha\,\backslash\sigma\right\}\,.	$(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma\alpha = \alpha\sigma\}.$	$(E_8)^\sigma = \{\alpha \in E_8 \mid \sigma\alpha = \alpha\sigma\}.$
0902.0431_E01114	189	■ 1.000	z\left(E_8\right)=\{1\}\,.	$z(E_8) = \{1\}.$	$z(E_8) = \{1\}.$
0902.0431_E01115	189	■ 1.000	\alpha=\varphi(E,\,1)=1\,\mathrm{or}\,\,\mathrm{or}\,\,\alpha=\varphi(E,\,-1)=v\,.	$\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$	$\alpha = \varphi(E, 1) = 1 \quad \text{or} \quad \alpha = \varphi(E, -1) = v.$
0902.0431_E01120	190	■ 1.000	\left[\left(D_1,\phi_1,\boldsymbol{X}_1,\boldsymbol{Y}_1\right),\left(D_2,\phi_2,\boldsymbol{X}_2,\boldsymbol{Y}_2\right)\right]=\left(D,\phi,\boldsymbol{X},\boldsymbol{Y}\right),	$[(D_1, \phi_1, \boldsymbol{X}_1, \boldsymbol{Y}_1), (D_2, \phi_2, \boldsymbol{X}_2, \boldsymbol{Y}_2)] = (D, \phi, \boldsymbol{X}, \boldsymbol{Y}),$	$[(D_1, \phi_1, \boldsymbol{X}_1, \boldsymbol{Y}_1), (D_2, \phi_2, \boldsymbol{X}_2, \boldsymbol{Y}_2)] = (D, \phi, \boldsymbol{X}, \boldsymbol{Y}),$
0902.0431_E01123	190	■ 1.000	B_8\left(R_1,R_2\right)=60\,\mathrm{tr}\left(D_1D_2\right)+\frac{5}{2}B_6\left(\phi_1,\phi_2\right)+15\left(\boldsymbol{X}_1,\boldsymbol{Y}_2\right)+15\left(\boldsymbol{X}_2,\boldsymbol{Y}_1\right)	$B_8(R_1, R_2) = 60 \mathrm{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\boldsymbol{X}_1, \boldsymbol{Y}_2) + 15(\boldsymbol{X}_2, \boldsymbol{Y}_1)$	$B_8(R_1, R_2) = 60 \mathrm{tr}(D_1 D_2) + \frac{5}{2} B_6(\phi_1, \phi_2) + 15(\boldsymbol{X}_1, \boldsymbol{Y}_2) + 15(\boldsymbol{X}_2, \boldsymbol{Y}_1)$
0902.0431_E01125	190	■ 1.000	\left\langle R_1,R_2\right\rangle=-B_8\left(R_1,\tau\widetilde{\lambda}R_2\right).	$\langle R_1, R_2 \rangle = -B_8\left(R_1, \tau \widetilde{\lambda} R_2\right).$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$
0902.0431_E01127	191	■ 1.000	E_8=\left\{\backslash\alpha\,\backslash\mathrm{in}\,\mathrm{Aut}\left(\frac{\mathfrak{e}_8^C}{}\right)\,\backslash\mathrm{mid}\,\left\langle\backslash\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}	$E_8 = \{\alpha \in \mathrm{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$	$E_8 = \{\alpha \in \mathrm{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle\}$
0902.0431_E01129	191	■ 1.000	\left(E_8\right)^w=\left\{\backslash\alpha\,\backslash\mathrm{in}\,E_8\,\backslash\mathrm{mid}\,w\,\backslash\alpha=\alpha\,w\right\}\,.	$(E_8)^w = \{\alpha \in E_8 \mid w\alpha = \alpha w\}.$	$(E_8)^w = \{\alpha \in E_8 \mid w\alpha = \alpha w\}.$
0902.0431_E01130	191	■ 1.000	\varphi_1(A)(D,\phi,\boldsymbol{X},\boldsymbol{Y})=\left(ADA^{-1},\phi,A\boldsymbol{X},{}^tA^{-1}\boldsymbol{Y}\right).	$\varphi_1(A)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (ADA^{-1}, \phi, A\boldsymbol{X}, {}^tA^{-1}\boldsymbol{Y}).$	$\varphi_1(A)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (ADA^{-1}, \phi, A\boldsymbol{X}, {}^tA^{-1}\boldsymbol{Y}).$
0902.0431_E01133	191	■ 1.000	\varphi_2(\alpha)(D,\phi,\boldsymbol{X},\boldsymbol{Y})=\left(D,\alpha\phi\alpha^{-1},\alpha\boldsymbol{X},{}^t\alpha^{-1}\boldsymbol{Y}\right).	$\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \alpha\phi\alpha^{-1}, \alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}).$	$\varphi_2(\alpha)(D, \phi, \boldsymbol{X}, \boldsymbol{Y}) = (D, \alpha\phi\alpha^{-1}, \alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}).$
0902.0431_E01135	192	■ 1.000	\begin{aligned} B_6\left(\alpha\phi_1\alpha^{-1},\tau^t\left(\alpha\phi_2\alpha^{-1}\right)\tau\right)&=B_6\left(\alpha\phi_1\alpha^{-1},\alpha\tau^t\phi_2\tau\alpha^{-1}\right)=B_6\left(\phi_1,\tau^t\phi_2\tau\right),\\ \left\langle\alpha\boldsymbol{X},{}^t\alpha^{-1}\boldsymbol{Y}\right\rangle&=\left\langle\boldsymbol{X},\boldsymbol{Y}\right\rangle, \end{aligned}	$B_6\left(\alpha\phi_1\alpha^{-1},\tau^t\left(\alpha\phi_2\alpha^{-1}\right)\tau\right)=B_6\left(\alpha\phi_1\alpha^{-1},\alpha\tau^t\phi_2\tau\alpha^{-1}\right)=B_6\left(\phi_1,\tau^t\phi_2\tau\right),$ $\langle\alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}\rangle = \langle\boldsymbol{X}, \boldsymbol{Y}\rangle,$	$B_6(\alpha\phi_1\alpha^{-1}, \tau^t(\alpha\phi_2\alpha^{-1})\tau) = B_6(\alpha\phi_1\alpha^{-1}, \alpha\tau^t\phi_2\tau\alpha^{-1}) = B_6(\phi_1, \tau^t\phi_2\tau),$ $\langle\alpha\boldsymbol{X}, {}^t\alpha^{-1}\boldsymbol{Y}\rangle = \langle\boldsymbol{X}, \boldsymbol{Y}\rangle,$
0902.0431_E01139	192	■ 1.000	(\boldsymbol{*}\,\boldsymbol{u},\boldsymbol{v})=(\boldsymbol{u}\wedge\boldsymbol{v},\boldsymbol{e}_1\wedge\cdots\wedge\boldsymbol{e}_n),\quad\boldsymbol{v}\in\Lambda^{n-k}(C^m).	$(\boldsymbol{*}\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^m).$	$(\boldsymbol{*}\boldsymbol{u}, \boldsymbol{v}) = (\boldsymbol{u} \wedge \boldsymbol{v}, \boldsymbol{e}_1 \wedge \cdots \wedge \boldsymbol{e}_n), \quad \boldsymbol{v} \in \Lambda^{n-k}(C^m).$
0902.0431_E01140	192	■ 1.000	*:\Lambda^k\left(C^n\right)\rightarrow\Lambda^{n-k}\left(C^n\right)	$* : \Lambda^k(C^n) \rightarrow \Lambda^{n-k}(C^n)$	$* : \Lambda^k(C^n) \rightarrow \Lambda^{n-k}(C^n)$
0902.0431_E01141	192	■ 1.000	*^2\boldsymbol{u}=(-1)^{k(n-k)}\boldsymbol{u},\quad\boldsymbol{u}\in\Lambda^k(C^m).	$*^2\boldsymbol{u} = (-1)^{k(n-k)}\boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^m).$	$*^2\boldsymbol{u} = (-1)^{k(n-k)}\boldsymbol{u}, \quad \boldsymbol{u} \in \Lambda^k(C^m).$

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0902.0431_E01142	192	■ 1.000	$\mathbf{A}\left(\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\boldsymbol{x}_{\{k\}}\right)=\mathbf{A}\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\mathbf{A}\boldsymbol{x}_{\{k\}},\quad\mathbf{A}\mathbf{1}=\mathbf{1}.$	$A\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=A\boldsymbol{x}_1\wedge\cdots\wedge A\boldsymbol{x}_k,\quad A\mathbf{1}=\mathbf{1}.$	$A(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k)=A\boldsymbol{x}_1\wedge\cdots\wedge A\boldsymbol{x}_k,\quad A\mathbf{1}=\mathbf{1}.$
0902.0431_E01143	193	■ 1.000	$\mathbf{D}\left(\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\boldsymbol{x}_{\{k\}}\right)=\sum_{j=1}^k\boldsymbol{x}_{\{1\}}\wedge\cdots\wedge\mathbf{D}\boldsymbol{x}_{\{j\}}\wedge\cdots\wedge\boldsymbol{x}_{\{k\}},\quad\mathbf{D}\mathbf{1}=\mathbf{0}.$	$D\left(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k\right)=\sum_{j=1}^k\boldsymbol{x}_1\wedge\cdots\wedge D\boldsymbol{x}_j\wedge\cdots\wedge\boldsymbol{x}_k,\quad D\mathbf{1}=\mathbf{0}.$	$D(\boldsymbol{x}_1\wedge\cdots\wedge\boldsymbol{x}_k)=\sum_{j=1}^k\boldsymbol{x}_1\wedge\cdots\wedge D\boldsymbol{x}_j\wedge\cdots\wedge\boldsymbol{x}_k,\quad D\mathbf{1}=\mathbf{0}.$
0902.0431_E01146	193	■ 1.000	$\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=\left(D,\boldsymbol{u},\boldsymbol{v}\right),$	$[(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2)]=(D,\boldsymbol{u},\boldsymbol{v}),$	$[(D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2)]=(D,\boldsymbol{u},\boldsymbol{v}),$
0902.0431_E01156	195	■ 1.000	$\boldsymbol{e}_{\{I\}}=\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\in\Lambda^3\left(C^9\right).$	$\boldsymbol{e}_I=\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\in\Lambda^3\left(C^9\right).$	$\boldsymbol{e}_I=\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k\in\Lambda^3(C^9).$
0902.0431_E01158	195	■ 1.000	$D=\frac{1}{3}\left(E_{ii}+E_{jj}+E_{kk}\right)-E_{ll},\quad l\neq i,j,k$	$D=\frac{1}{3}\left(E_{ii}+E_{jj}+E_{kk}\right)-E_{ll},\quad l\neq i,j,k$	$D=\frac{1}{3}(E_{ii}+E_{jj}+E_{kk})-E_{ll},\quad l\neq i,j,k.$
0902.0431_E01159	195	■ 1.000	$\begin{aligned} &\left(\mathbf{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}},\mathbf{0}\right)=\left[\left(D,\mathbf{0},\mathbf{0}\right),\left(\mathbf{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}},\mathbf{0}\right)\right]\in\mathfrak{a}, \\ &\left(\mathbf{0},\mathbf{0},\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\right)=\left[\left(D,\mathbf{0},\mathbf{0}\right),\left(\mathbf{0},\mathbf{0},-\boldsymbol{e}_{\{i\}}\wedge\boldsymbol{e}_{\{j\}}\wedge\boldsymbol{e}_{\{k\}}\right)\right]\in\mathfrak{a}. \end{aligned}$	$\begin{aligned} (0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0) &= [(D,0,0),(0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0)]\in\mathfrak{a}, \\ (0,0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k) &= [(D,0,0),(0,0,-\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k)]\in\mathfrak{a}. \end{aligned}$	$\begin{aligned} (0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0) &= [(D,0,0),(0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k,0)]\in\mathfrak{a}, \\ (0,0,\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k) &= [(D,0,0),(0,0,-\boldsymbol{e}_i\wedge\boldsymbol{e}_j\wedge\boldsymbol{e}_k)]\in\mathfrak{a}. \end{aligned}$
0902.0431_E01161	196	■ 1.000	$B_{\{8\}}\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=60\left(\operatorname{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{2\}}\right)+\left(\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{1\}}\right)\right).$	$B_8\left((D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2)\right)=60\left(\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right)\right).$	$B_8((D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2))=60(\operatorname{tr}(D_1D_2)+(\boldsymbol{u}_1,\boldsymbol{v}_2)+(\boldsymbol{u}_2,\boldsymbol{v}_1)).$
0902.0431_E01162	196	■ 1.000	$B\left(\left(D_{\{1\}},\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{1\}}\right),\left(D_{\{2\}},\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{2\}}\right)\right)=\operatorname{tr}\left(D_{\{1\}}D_{\{2\}}\right)+\left(\boldsymbol{u}_{\{1\}},\boldsymbol{v}_{\{2\}}\right)+\left(\boldsymbol{u}_{\{2\}},\boldsymbol{v}_{\{1\}}\right).$	$B\left((D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2)\right)=\operatorname{tr}\left(D_1D_2\right)+\left(\boldsymbol{u}_1,\boldsymbol{v}_2\right)+\left(\boldsymbol{u}_2,\boldsymbol{v}_1\right).$	$B((D_1,\boldsymbol{u}_1,\boldsymbol{v}_1),(D_2,\boldsymbol{u}_2,\boldsymbol{v}_2))=\operatorname{tr}(D_1D_2)+(\boldsymbol{u}_1,\boldsymbol{v}_2)+(\boldsymbol{u}_2,\boldsymbol{v}_1).$
0902.0431_E01163	196	■ 1.000	$B_{\{8\}}(R,R)=120,\quad B(R,R)=2.$	$B_8(R,R)=120,\quad B(R,R)=2.$	$B_8(R,R)=120,\quad B(R,R)=2.$
0902.0431_E01164	196	■ 1.000	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$	$\tau\widetilde{\lambda}(D,\boldsymbol{u},\boldsymbol{v})=\left(-\tau^tD,-\tau\boldsymbol{v},-\tau\boldsymbol{u}\right).$
0902.0431_E01166	196	■ 1.000	$\left\langle R_1,R_2\right\rangle =60\left(\operatorname{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$	$\langle R_1,R_2\rangle =60\left(\operatorname{tr}\left(D_1\tau^tD_2\right)+\left(\boldsymbol{u}_1,\tau\boldsymbol{u}_2\right)+\left(\boldsymbol{v}_2,\tau\boldsymbol{v}_1\right)\right).$	$\langle R_1,R_2\rangle =60(\operatorname{tr}(D_1\tau^tD_2)+(\boldsymbol{u}_1,\tau\boldsymbol{u}_2)+(\boldsymbol{v}_2,\tau\boldsymbol{v}_1)).$
0902.0431_E01167	196	■ 1.000	$E_{\{8\}}=\left\{\alpha\in\operatorname{Aut}\left(\mathfrak{e}_{\{8\}}^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$	$E_8=\left\{\alpha\in\operatorname{Aut}\left(\mathfrak{e}_8^C\right)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$	$E_8=\left\{\alpha\in\operatorname{Aut}(\mathfrak{e}_8^C)\mid\left\langle\alpha R_1,\alpha R_2\right\rangle=\left\langle R_1,R_2\right\rangle\right\}$

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0902.0431_E01169	197	■ 1.000	$\left(E_8\right)^{w_3}=\left\{\alpha \in E_8 \mid w_3 \alpha=\alpha w_3\right\} .$	$(E_8)^{w_3}=\left\{\alpha \in E_8 \mid w_3 \alpha=\alpha w_3\right\} .$	$(E_8)^{w_3}=\left\{\alpha \in E_8 \mid w_3 \alpha=\alpha w_3\right\} .$
0902.0431_E01170	197	■ 1.000	$\varphi(A)(D, \boldsymbol{u}, \boldsymbol{v})=\left(A D A^{-1}, A \boldsymbol{u},{ }^t A^{-1} \boldsymbol{v}\right) .$	$\varphi(A)(D, \boldsymbol{u}, \boldsymbol{v})=\left(A D A^{-1}, A \boldsymbol{u},{ }^t A^{-1} \boldsymbol{v}\right) .$	$\varphi(A)(D, \boldsymbol{u}, \boldsymbol{v})=\left(A D A^{-1}, A \boldsymbol{u},{ }^t A^{-1} \boldsymbol{v}\right) .$
0902.0431_E01177	198	■ 1.000	$\left(\boldsymbol{a} \wedge \boldsymbol{b}\right) \wedge \boldsymbol{c}=\boldsymbol{a} \wedge\left(\boldsymbol{b} \wedge \boldsymbol{c}\right) .$	$\left(\boldsymbol{a} \wedge \boldsymbol{b}\right) \wedge \boldsymbol{c}=\boldsymbol{a} \wedge\left(\boldsymbol{b} \wedge \boldsymbol{c}\right) .$	$\left(\boldsymbol{a} \wedge \boldsymbol{b}\right) \wedge \boldsymbol{c}=\boldsymbol{a} \wedge\left(\boldsymbol{b} \wedge \boldsymbol{c}\right) .$
0902.0431_E01184	199	■ 1.000	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=-\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{z}+\frac{4}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v})=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v}),$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=-\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{z}+\frac{4}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v})=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v}),$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=-\boldsymbol{x} \wedge \boldsymbol{y} \wedge \boldsymbol{z}+\frac{4}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v})=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y})(\boldsymbol{z}, \boldsymbol{v}),$
0902.0431_E01185 (i)	199	■ 1.000	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{z} .$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{z} .$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{z}=\left(\boldsymbol{y}, \boldsymbol{z}\right) \boldsymbol{x}-\frac{1}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{z} .$
0902.0431_E01191	200	■ 1.000	$\begin{aligned} & \left(\boldsymbol{a} \times \boldsymbol{b}\right) \boldsymbol{v}=\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{v}+\frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}) \\ & =-\left(\boldsymbol{a}_1, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}\right)+\left(\boldsymbol{a}_2, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}\right)+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}), \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{a} \times \boldsymbol{b}\right) \boldsymbol{v}=\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{v}+\frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}) \\ & =-\left(\boldsymbol{a}_1, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}\right)+\left(\boldsymbol{a}_2, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}\right)+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}) \end{aligned}$	$\begin{aligned} & \left(\boldsymbol{a} \times \boldsymbol{b}\right) \boldsymbol{v}=\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{v}+\frac{3}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}) \\ & =-\left(\boldsymbol{a}_1, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_2, \boldsymbol{b}\right)+\left(\boldsymbol{a}_2, \boldsymbol{w}\right)\left(\boldsymbol{x} \wedge \boldsymbol{a}_1, \boldsymbol{b}\right)+\frac{2}{5}(\boldsymbol{a}, \boldsymbol{b})(\boldsymbol{v}, \boldsymbol{w}), \end{aligned}$
0902.0431_E01194	201	■ 1.000	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{a}=\left(\boldsymbol{y}, \boldsymbol{a}_1\right) \boldsymbol{x} \wedge \boldsymbol{a}_2-\left(\boldsymbol{y}, \boldsymbol{a}_2\right) \boldsymbol{x} \wedge \boldsymbol{a}_1-\frac{2}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{a} .$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{a}=\left(\boldsymbol{y}, \boldsymbol{a}_1\right) \boldsymbol{x} \wedge \boldsymbol{a}_2-\left(\boldsymbol{y}, \boldsymbol{a}_2\right) \boldsymbol{x} \wedge \boldsymbol{a}_1-\frac{2}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{a} .$	$\left(\boldsymbol{x} \times \boldsymbol{y}\right) \boldsymbol{a}=\left(\boldsymbol{y}, \boldsymbol{a}_1\right) \boldsymbol{x} \wedge \boldsymbol{a}_2-\left(\boldsymbol{y}, \boldsymbol{a}_2\right) \boldsymbol{x} \wedge \boldsymbol{a}_1-\frac{2}{5}(\boldsymbol{x}, \boldsymbol{y}) \boldsymbol{a} .$
0902.0431_E01198	201	■ 1.000	$\left[(C, D),\left(E_{45}, 0\right)\right]=\left(5 E_{45}, 0\right) \in \mathfrak{g}_{01} \cap \mathfrak{a},$	$\left[(C, D),\left(E_{45}, 0\right)\right]=\left(5 E_{45}, 0\right) \in \mathfrak{g}_{01} \cap \mathfrak{a},$	$\left[(C, D),\left(E_{45}, 0\right)\right]=\left(5 E_{45}, 0\right) \in \mathfrak{g}_{01} \cap \mathfrak{a},$
0902.0431_E01199	202	■ 1.000	$\mathfrak{a} \supset\left[\mathfrak{g}_1, \mathfrak{g}_{-1}\right] \ni\left[e_1 \otimes\left(e_1 \wedge e_2\right), e_1 \otimes\left(e_1 \wedge e_3\right)\right]=\left(0,-E_{23}\right),$	$\mathfrak{a} \supset\left[\mathfrak{g}_1, \mathfrak{g}_{-1}\right] \ni\left[e_1 \otimes\left(e_1 \wedge e_2\right), e_1 \otimes\left(e_1 \wedge e_3\right)\right]=\left(0,-E_{23}\right),$	$\mathfrak{a} \supset\left[\mathfrak{g}_1, \mathfrak{g}_{-1}\right] \ni\left[e_1 \otimes\left(e_1 \wedge e_2\right), e_1 \otimes\left(e_1 \wedge e_3\right)\right]=\left(0,-E_{23}\right),$

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0902.0431_E01200	202	■ 1.000	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$	$\operatorname{ad}(T) \operatorname{ad}(S_{1523}) \operatorname{ad}(S_{1415}) \operatorname{ad}(S_{1314}) \operatorname{ad}(S_{1213}) R = (-E_{12}, 0) \in \mathfrak{g}_{01} \cap \mathfrak{a}.$
0902.0431_E01203	202	■ 1.000	$B_8(R, R) = 120, \quad B(R, R) = 2.$	$B_8(R, R) = 120, \quad B(R, R) = 2.$	$B_8(R, R) = 120, \quad B(R, R) = 2.$
0902.0431_E01204	202	■ 1.000	$\begin{aligned} & \tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ &= (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}). \end{aligned}$	$\tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ = (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}).$	$\tau \widetilde{\lambda}(C, D, \mathbf{x} \otimes \mathbf{a}, \mathbf{b} \otimes \mathbf{y}, \mathbf{c} \otimes \mathbf{z}, \mathbf{w} \otimes \mathbf{d}) \\ = (-\tau^t C, -\tau^t D, \tau \mathbf{w} \otimes \tau \mathbf{d}, \tau \mathbf{c} \otimes \tau \mathbf{z}, \tau \mathbf{b} \otimes \tau \mathbf{y}, \tau \mathbf{x} \otimes \tau \mathbf{a}).$
0902.0431_E01205	203	■ 1.000	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2)$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2)$	$\langle R_1, R_2 \rangle = -B_8(R_1, \tau \widetilde{\lambda} R_2).$
0902.0431_E01207	203	■ 1.000	$E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$	$E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$	$E_8 = \{ \alpha \in \operatorname{Aut}(\mathfrak{e}_8^C) \mid \langle \alpha R_1, \alpha R_2 \rangle = \langle R_1, R_2 \rangle \}$
0902.0431_E01208	203	■ 1.000	$z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$	$z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$	$z_5(C, D, g_1, g_2, g_{-2}, g_{-1}) = (C, D, \zeta(g_1), \zeta^2(g_2), \zeta^3(g_{-2}), \zeta^4(g_{-1})).$
0902.0431_E01209	203	■ 1.000	$(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$	$(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$	$(E_8)^{z_5} = \{ \alpha \in E_8 \mid z_5 \alpha = \alpha z_5 \}.$
0902.0431_E01212	203	■ 1.000	$\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$	$\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$	$\langle \varphi(A) R_1, \varphi_1(A) R_2 \rangle = \langle R_1, R_2 \rangle.$
0902.0431_E01213	204	■ 1.000	$\varphi(A, B) = \varphi_1(A) \varphi_2(B).$	$\varphi(A, B) = \varphi_1(A) \varphi_2(B).$	$\varphi(A, B) = \varphi_1(A) \varphi_2(B).$
0902.0431_E01215	204	■ 1.000	$\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$	$\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$	$\begin{aligned} & \mathfrak{e}_{8(8)} = \mathfrak{e}_{7(7)} \oplus \mathfrak{P}' \oplus \mathfrak{P}' \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(7)} = (\mathfrak{e}_7^C)^{\tau\gamma} \right), \\ & \mathfrak{e}_{8(-24)} = \mathfrak{e}_{7(-25)} \oplus \mathfrak{P} \oplus \mathfrak{P} \oplus \mathbf{R} \oplus \mathbf{R} \oplus \mathbf{R}, \quad \left(\text{where } \mathfrak{e}_{7(-25)} = (\mathfrak{e}_7^C)^\tau \right). \end{aligned}$
0902.0431_E01219	204	■ 1.000	$z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$	$z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$	$z(E_{8(8)}) = \{1\}, \quad z(E_{8(-24)}) = \{1\}.$

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0902.0431_E00341	55	—	—	—	$\begin{aligned} \lambda_0 - \lambda_1 &= \alpha_1 \\ \lambda_0 - \lambda_2 &= \alpha_1 + \alpha_2 \\ -\lambda_0 + \lambda_3 &= \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_1 - \lambda_2 &= \alpha_2 \\ -\lambda_1 + \lambda_3 &= \alpha_1 + \alpha_2 + 2\alpha_3 + 2\alpha_4 \\ -\lambda_2 + \lambda_3 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + 2\alpha_4 \\ \lambda_0 + \lambda_1 &= \alpha_1 + 2\alpha_2 + 2\alpha_3 \\ \lambda_0 + \lambda_2 &= \alpha_1 + \alpha_2 + 2\alpha_3 \\ \lambda_0 + \lambda_3 &= 2\alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_1 + \lambda_2 &= \alpha_2 + 2\alpha_3 \\ \lambda_1 + \lambda_3 &= \alpha_1 + 3\alpha_2 + 4\alpha_3 + 2\alpha_4 \\ \lambda_2 + \lambda_3 &= \alpha_1 + 2\alpha_2 + 4\alpha_3 + 4\alpha_4, \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_1 + 2\alpha_2 + 3\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_4 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_1 + 2\alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_1 + \alpha_2 + 2\alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) &= \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) &= \alpha_2 + 2\alpha_3 + \alpha_4. \end{aligned}$

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0902.0431_EQ0758	126	—	—	—	$\begin{aligned} \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 3 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 3 & 2 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 1 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(-\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 1 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 3 & 4 & 3 & 2 & 2 \\ \frac{1}{2}(-\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 0 & 0 & 1 & 2 & 1 & 1 & 1 \\ -\mu_1 - \frac{2}{3}\nu &= 1 & 2 & 3 & 4 & 2 & 1 & 2 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
8982.0431_EQ0940	156	—	—	—	$ \begin{aligned} &P = (F_1(1), 0, 0, 0) \\ &\xrightarrow{f} \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\xrightarrow{D} (D\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge D\mathbf{e}_4 \wedge \mathbf{e}_5 + \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge D\mathbf{e}_5) \\ &\quad + (D\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge D\mathbf{e}_6 \wedge \mathbf{e}_4 + \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge D\mathbf{e}_4) \\ &= \left(b_{12}\mathbf{e}_1 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 + \left(b_{22} + \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{b}_{23}\mathbf{e}_3 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{l}_{23}\mathbf{e}_6 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 \right. \\ &\quad + l_{11}\mathbf{e}_2 \wedge \mathbf{e}_1 \wedge \mathbf{e}_5 + l_{31}\mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_5 + \left(c_{11} - \frac{\nu}{3} \right) \mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_5 - \bar{c}_{13}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \\ &\quad + l_{12}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_1 + l_{32}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_3 + \left(c_{22} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{23}\mathbf{e}_2 \wedge \mathbf{e}_4 \wedge \mathbf{e}_6 \Big) \\ &\quad + \left(b_{13}\mathbf{e}_1 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + b_{23}\mathbf{e}_2 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 + \left(b_{33} + \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{l}_{32}\mathbf{e}_5 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \right. \\ &\quad + l_{13}\mathbf{e}_3 \wedge \mathbf{e}_1 \wedge \mathbf{e}_4 + l_{23}\mathbf{e}_3 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4 + c_{23}\mathbf{e}_3 \wedge \mathbf{e}_5 \wedge \mathbf{e}_4 + \left(c_{33} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 \\ &\quad \left. + l_{11}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_1 + l_{21}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_2 + \left(c_{11} - \frac{\nu}{3} \right) \mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_4 - \bar{c}_{12}\mathbf{e}_3 \wedge \mathbf{e}_6 \wedge \mathbf{e}_5 \right) \\ &\xrightarrow{f^{-1}} \left(\begin{array}{ccc} 0 & h(b_{13}, c_{13}) & 0 \\ * & h(b_{23} + \bar{c}_{23}) & h(b_{22} - c_{33} - \frac{\nu}{3}, -b_{33} + c_{33} + \frac{\nu}{3}) \\ h(\bar{c}_{12}, \bar{b}_{12}) & * & -h(\bar{b}_{23} + c_{23}) \end{array} \right) \\ &\quad \left(\begin{array}{ccc} -h(l_{23} + l_{32}) & h(l_{13}, \bar{l}_{31}) & * \\ * & 0 & -h(l_{11}) \\ h(l_{21}, \bar{l}_{12}) & * & 0 \end{array} \right) \\ &\quad \left(\begin{array}{ccc} 0 & & \\ & & -h(\bar{l}_{23} + \bar{l}_{32}) \end{array} \right) \end{aligned} $

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_E01028	172	—	—	—	$\begin{aligned} \lambda_2 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \lambda_3 + \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \lambda_0 - \frac{1}{2}(\mu_2 - \mu_3) &= 2 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \lambda_1 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ \lambda_2 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ \lambda_3 - \frac{1}{2}(\mu_2 - \mu_3) &= 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 2 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 1 & 2 & 2 & 2 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 3 & 4 & 3 & 2 & 1 & 0 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_3 - \mu_1) &= 1 & 2 & 2 & 1 & 0 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 2 & 1 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) + \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 3 & 4 & 3 & 3 & 2 & 1 & 2 \\ \frac{1}{2}(\lambda_0 - \lambda_1 + \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 0 & 1 \\ \frac{1}{2}(\lambda_0 - \lambda_1 - \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 0 & 0 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 1 & 2 & 2 & 1 & 1 & 1 & 1 \\ \frac{1}{2}(\lambda_0 + \lambda_1 + \lambda_2 + \lambda_3) - \frac{1}{2}(\mu_1 - \mu_2) &= 1 & 2 & 4 & 4 & 3 & 2 & 1 & 2 \\ \mu_1 + \frac{2}{3}\nu &= 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ \mu_2 + \frac{2}{3}\nu &= 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ -\mu_3 - \frac{2}{3}\nu &= 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{aligned}$